

BC-6800Plus
Auto Hematology Analyzer
Operator's Manual



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WARNING

- **It is important for the hospital or organization that employs this equipment to carry out a reasonable service/maintenance plan. Neglect of this may result in machine breakdown or personal injury.**



CAUTION

- **Be sure to operate the analyzer under the situation specified in this manual; otherwise, the analyzer will not work normally and the analysis results will be unreliable, which would damage the analyzer components and cause personal injury.**

NOTE

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Table of Contents

1 Safety Information	1 - 1
1.1 Labels and Symbols on the System	1 - 1
1.2 Safety Warning Symbols and Messages	1 - 13
1.3 Safety Precautions	1 - 13
1.3.1 General Safety Precautions	1 - 13
1.3.2 Safety Precautions on Analyzer Transportation and Installation	1 - 14
1.3.3 Safety Precautions on Reagents	1 - 15
1.3.4 Safety Precautions on Operation	1 - 15
1.3.5 Safety Precautions on Maintenance	1 - 16
1.3.6 Laser Warning	1 - 17
1.3.7 When the Instrument Operation is Stopped for Maintenance, Transportation or Service	1 - 17
1.3.8 Software and Network Safety	1 - 18
2 Using this Manual	2 - 1
2.1 Who Should Read This Manual	2 - 1
2.2 How to Find Information	2 - 1
2.3 Conventions Used in This Manual	2 - 2
3 Understanding Your System	3 - 1
3.1 Intended Use	3 - 1
3.2 Parameters	3 - 1
3.2.1 Blood Sample Test Parameters, Histograms, and Scattergrams	3 - 1
3.2.2 Body Fluid Sample Test Parameters, Histograms, and Scattergrams	3 - 6
3.3 Supported Test Panels	3 - 7
3.4 Product Description	3 - 8
3.4.1 Product Structure and Components	3 - 8
3.4.2 Side View of the Analyzer	3 - 8
3.4.3 Back View of the Analyzer	3 - 9
3.4.4 Right View of the Analyzer	3 - 11
3.4.5 Left View of the Analyzer	3 - 12
3.4.6 Operator Panel	3 - 13
3.5 Accessories List of the Analyzer	3 - 14
3.6 Sample Tubes and Sample Racks	3 - 14
3.7 Software System	3 - 14
3.7.1 Functionality	3 - 15
3.7.2 Software version	3 - 19
3.7.3 Performance and Efficiency	3 - 19
3.7.4 Compatibility	3 - 20
3.7.5 Usability	3 - 20
3.7.6 Reliability, Known Limitations, and Precautions	3 - 21
3.7.7 Portability	3 - 21
3.7.8 Validity	3 - 21
3.7.9 Efficiency	3 - 21
3.7.10 Satisfaction	3 - 21
3.7.11 Freedom from Risk	3 - 21
3.7.12 Context Coverage	3 - 22
3.7.13 Input and Output Data	3 - 22
3.8 Reagents, Controls and Calibrators	3 - 22
3.8.1 Reagent	3 - 23
3.8.2 Controls and Calibrators	3 - 23
4 Understanding the System Principles	4 - 1
4.1 Introduction	4 - 1
4.2 WBC Measurement	4 - 1
4.2.1 Laser Flow Cytometry + SF CUBE Cell Analysis Technology	4 - 1

4.2.2 Derivation of WBC-Related Parameters	4 - 3
4.3 HGB Measurement	4 - 6
4.3.1 Colorimetric Method	4 - 6
4.3.2 Derivation of HGB-Related Parameters	4 - 6
4.4 RBC/PLT Measurement	4 - 7
4.4.1 Sheath Fluid Impedance Method	4 - 7
4.4.2 SF CUBE Cell Analysis Technology	4 - 7
4.4.3 Measurement Principle of Platelets in DIFF Channel (PLT-H)	4 - 7
4.4.4 RBC-Related Parameters	4 - 8
4.4.5 PLT-Related Parameters	4 - 9
4.4.6 Reticulocyte Parameters	4 - 10
4.5 Erythrocyte Sedimentation Rate Measurement	4 - 10
4.6 Body Fluid Parameters	4 - 10
4.7 Rinse	4 - 11
5 Installing and Connecting the System	5 - 1
5.1 Precautions on Analyzer Installation	5 - 1
5.1.1 Space Requirements	5 - 1
5.1.2 Power Requirements	5 - 1
5.1.3 Environment Requirements	5 - 2
5.1.4 Moving and Installing the Analyzer	5 - 2
5.2 Connecting the Analyzer System	5 - 4
5.2.1 Connecting Power Supply	5 - 4
5.2.2 Connecting the Pneumatic Unit	5 - 5
5.2.3 Connecting the Reagents	5 - 6
5.2.4 Connecting the Peripherals	5 - 8
6 Customizing Your Analyzer	6 - 1
6.1 Access Levels for Different Accounts	6 - 1
6.2 Saving Settings after Changes	6 - 1
6.3 Setting up a Printing Template	6 - 1
6.4 Setting up for Communication Methods	6 - 1
6.4.1 Setting up Communication Protocol	6 - 1
6.4.2 Setting up Transmission Mode	6 - 2
6.5 Setting up Barcode Systems	6 - 2
6.6 Setting Date/Time	6 - 3
6.7 Setting up Laboratory Information	6 - 4
6.8 Managing User Accounts	6 - 4
6.8.1 Changing Passwords	6 - 4
6.8.2 Adding a New Account	6 - 4
6.8.3 Deleting Accounts	6 - 5
6.9 Setting up Auto-loading Mode	6 - 5
6.9.1 Customizing Your Analyzer How to Proceed When Blood/Analysis Mode Inquiry Failed	6 - 6
6.9.2 How to Handle Sample ID When There is Sample Tube Vacancy	6 - 6
6.9.3 Display Summary after Autoloading Finishes	6 - 7
6.9.4 Setting Sample Tube Loading Methods	6 - 7
6.9.5 Customizing Your Analyzer How to Proceed When out of the Analyzer Capability	6 - 7
6.9.6 Setting up the Rules for "Stop analysis when consecutive samples are flagged for abnormal results"	6 - 7
6.10 Setting up Methods for Acquiring Sample Information	6 - 9
6.10.1 Setting up the Entry Method of Next Sample ID in Open-Vial Mode Analysis	6 - 9
6.10.2 Setting up for the First Sample after Startup	6 - 10
6.10.3 Setting up Auto-Scan Rack No. in Auto-Loading Mode	6 - 10
6.11 Setting up for Parameters	6 - 10
6.11.1 Setting up Parameter Unit	6 - 10
6.11.2 Setting up Parameter Reference Ranges	6 - 11
6.11.3 Setting up Microscopic Parameters	6 - 12
6.12 Setting up the Probe Cleanser Maintenance Time	6 - 13

6.12.1 Performing Maintenance at Specified Time	6 - 13
6.12.2 Performing Maintenance at Appointed Time	6 - 14
6.13 Setting up Gains	6 - 14
6.14 Setting up Flag Alarm Sensitivity	6 - 14
6.15 Setting up Flag Alarm Rules	6 - 15
6.16 Setting up for Ext. WBC/PLT Tests	6 - 16
6.17 Setting up for Other Settings	6 - 16
6.18 Setting up the Waiting Time before the Analyzer Enters Standby Status	6 - 18
6.19 Setting up Auto Startup/Shutdown	6 - 18
7 Operating Your Analyzer	7 - 1
7.1 Overview	7 - 1
7.1.1 Introduction to the Count Screen	7 - 2
7.1.2 Sorting Test Panel on "Request Sample" Screen	7 - 2
7.2 Preparations before Operations	7 - 3
7.3 Startup and Login	7 - 4
7.3.1 Starting up the Analyzer	7 - 4
7.3.2 Auto Startup	7 - 4
7.3.3 Switching Login Account	7 - 4
7.4 Daily Quality Control (QC)	7 - 5
7.5 Preparing Samples	7 - 5
7.5.1 Preparing Whole Blood Samples	7 - 5
7.5.2 Preparing Predilute Samples	7 - 6
7.5.3 Preparing Body Fluid Samples	7 - 7
7.5.4 Placing Barcode Labels	7 - 7
7.6 Analyzing Samples in OV Mode	7 - 9
7.6.1 Setting up Analysis Orders in OV Mode	7 - 9
7.6.2 Performing Sample Analysis in OV Mode	7 - 9
7.7 Analyzing Samples in Auto-Loading Mode	7 - 13
7.7.1 Setting up Analysis Orders in Auto-Loading Mode	7 - 13
7.7.2 Performing Sample Analysis in Auto-Loading Mode	7 - 14
7.7.3 Stop Count	7 - 15
7.7.4 Performing Auto Re-exam	7 - 15
7.8 Analyzing STAT Samples	7 - 16
7.9 Entering and Exiting Standby Status	7 - 16
7.10 Shutting down the Analyzer	7 - 17
7.10.1 Manual Shutdown	7 - 17
7.10.2 Auto Shutdown	7 - 17
8 Reviewing Sample Results	8 - 1
8.1 Introduction	8 - 1
8.2 Reviewing Sample Results	8 - 1
8.2.1 Entering the "Table Review" Screen	8 - 1
8.2.2 Entering the "Graph" Screen	8 - 6
8.3 Flags of Analysis Results	8 - 8
8.3.1 Parameter Flags	8 - 8
8.3.2 Flags of Abnormal Blood Cell Differential or Morphology	8 - 9
9 Using the QC Program	9 - 1
9.1 Introduction	9 - 1
9.2 L-J QC	9 - 1
9.2.1 Setting up L-J QC Files	9 - 1
9.2.2 Running L-J QC Analysis	9 - 4
9.2.3 Reviewing L-J QC Results	9 - 6
9.2.4 Deleting QC Records	9 - 9
9.2.5 Transmitting Data	9 - 9

9.2.6 Exporting Data	9 - 10
9.2.7 Editing QC Results	9 - 10
9.2.8 Reviewing the Parameter Results of Control Samples	9 - 11
9.3 X-B QC	9 - 11
9.3.1 Validity Determination for X-B QC Samples	9 - 11
9.3.2 Setting up X-B QC Rules	9 - 11
9.3.3 Reviewing X-B QC Results	9 - 12
9.3.4 Deleting QC Records	9 - 14
9.3.5 Exporting Data	9 - 14
9.4 When QC Results are Out of Range	9 - 15
9.4.1 Troubleshooting	9 - 15
9.4.2 Analyzing the Causes	9 - 15
10 Calibrating Your Analyzer	10 - 1
10.1 Overview	10 - 1
10.2 When to Calibrate	10 - 1
10.3 Preparation before Calibration	10 - 1
10.4 Manual Calibration	10 - 2
10.5 Calibrating with Calibrators	10 - 3
10.6 Calibrating with Fresh blood Samples	10 - 4
10.7 Verifying Calibration Factors	10 - 6
10.8 Calibration History	10 - 6
11 Using the Printing Function	11 - 1
11.1 Setting up Print Templates	11 - 1
11.2 Printing Sample Result Report	11 - 2
11.2.1 Printing Current Sample Result Report	11 - 2
11.2.2 Printing from the Table Review Interface	11 - 2
11.2.3 Printing from the Graph Review Screen	11 - 3
11.2.4 Printing RUO Parameters Results	11 - 3
11.2.5 Printing Microscopic Parameter Results	11 - 3
11.3 Printing QC Result Reports	11 - 4
11.3.1 Printing L-J QC Result from L-J QC Table Screen	11 - 4
11.3.2 Printing L-J QC Result from L-J QC Graph Screen	11 - 4
11.3.3 Printing X-B QC Graph from the X-B QC Graph Screen	11 - 4
11.3.4 Printing X-B QC Parameters Results Report from the X-B QC Table Tab	11 - 4
11.4 Printing Manual Calibration Factors	11 - 5
12 Servicing Your Analyzer	12 - 1
12.1 Introduction	12 - 1
12.2 When and Why to Perform the Maintenance	12 - 2
12.3 Replacing Reagents	12 - 4
12.3.1 Replacing an Old Reagent with a New Reagent	12 - 4
12.3.2 Entering Reagent Information	12 - 6
12.3.3 Replacing the Reagents in the Fluidics and Priming with Reagents	12 - 7
12.4 Replacing Waste Container	12 - 7
12.5 When Two DS Diluent Containers are Connected at the Same Time	12 - 7
12.5.1 Replacing the used-up DS diluent container	12 - 9
12.5.2 Replacing both DS Diluent containers at the same time	12 - 10
12.6 Probe Cleanser Maintenance	12 - 11
12.6.1 Daily Probe Cleanser Maintenance	12 - 11
12.6.2 Probe Cleanser Maintenance to Parts and Components	12 - 11
12.7 Auto-Cleaning the Parts and Components	12 - 11
12.8 Manually Cleaning the Parts and Components	12 - 12
12.8.1 Opening and Closing the Front Cover	12 - 12
12.8.2 Cleaning the Waste Collecting Trays	12 - 13

12.8.3 Cleaning the SRV	12 - 14
12.8.4 Cleaning the Probe Wipe	12 - 15
12.8.5 Cleaning the Analyzer Cover	12 - 15
12.9 Performing Packing-up Procedure	12 - 16
12.10 Performing Touchpad Calibration	12 - 16
12.11 Reviewing and Exporting Logs	12 - 16
12.11.1 Viewing Logs	12 - 17
12.11.2 Exporting Logs	12 - 17
13 Troubleshooting	13 - 1
13.1 Introduction	13 - 1
13.2 Checking Analyzer Status	13 - 1
13.3 Error Messages and Solutions	13 - 2
A Index	A - 1
B Technical Information	B - 1
B.1 Reagents	B - 1
B.2 Recommended Sample Tube Types	B - 1
B.3 Parameters	B - 2
B.4 Sampling Features	B - 4
B.4.1 Sample Volumes Required for Each Analysis	B - 4
B.4.2 Minimum Sample Volume	B - 4
B.4.3 Throughput	B - 5
B.5 Performance Specifications	B - 5
B.5.1 Background/Blank Count Requirements	B - 5
B.5.2 Linearity Range	B - 6
B.5.3 Repeatability	B - 6
B.5.4 Carryover	B - 9
B.5.5 Correlation	B - 9
B.5.6 Accuracy	B - 11
B.5.7 Precision	B - 13
B.5.8 Mode Comparison	B - 15
B.5.9 Sample Stability	B - 16
B.5.10 Interference Study	B - 18
B.5.11 Analytical Measuring Intervals	B - 19
B.5.12 Flags of Sensitivity and Specificity	B - 21
B.5.13 Reference Intervals	B - 21
B.5.14 Limitations	B - 24
B.6 Input/Output Devices	B - 25
B.6.1 Computer	B - 25
B.6.2 Keyboard (Optional)	B - 26
B.6.3 Mouse (Optional)	B - 26
B.6.4 External bar-code scanner (optional)	B - 26
B.6.5 Printer (optional)	B - 26
B.6.6 USB Device (Optional)	B - 26
B.7 Interfaces	B - 26
B.8 Electronic Interface Specifications	B - 26
B.9 Power supply	B - 27
B.10 Fuse	B - 27
B.11 EMC Description	B - 28
B.12 Sound Level	B - 28
B.13 Environment Requirements	B - 28
B.14 Dimensions and Weight	B - 29
B.15 Contraindications	B - 29
B.16 Barcode Specifications	B - 29
B.17 Safety Classification	B - 32

B.18 Software Version	B - 32
C Accessories	C - 1
D Transmission	D - 1
E Radio Regulatory Compliance	E - 1
F References	F - 1
G Glossary	G - 1
H Maintenance Logs	H - 1

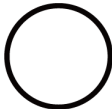
1 Safety Information

1.1 Labels and Symbols on the System

You may find the following symbols on package or on the analyzer:

CAUTION

- During the daily use of the instrument, especially the cleaning process, the operator shall ensure the intactness of the labels.

When you see...	It means...
	Caution Note: Indicates the need for the user to consult the instructions for use for important cautionary information such as warnings and precautions that cannot, for a variety of reasons, be presented on the medical device itself.
	Biological risks
	Warning, laser beam
	Caution, hot surface
	PROTECTIVE CONDUCTOR TERMINAL
	(Off) Power
	(On) Power
	Computer Network

When you see...	It means...
	Alternating current
	<i>In vitro</i> diagnostic medical device
	Temperature limit
	Humidity limitation
	Atmospheric pressure limitation
	SERIAL NUMBER
	DATE OF MANUFACTURE
	MANUFACTURER
	Fragile, handle with care
	This way up
	Keep dry

When you see...	It means...
	Do not roll
	Stacking limit by mass
	Standby
	Biological risks (on the sample tube of the waste container cap assembly) Do not replace the waste container when the power indicator is flickering!
	THE FOLLOWING DEFINITION OF THE WEEE LABEL APPLIES TO EU MEMBER STATES ONLY: THE USE OF THIS SYMBOL INDICATES THAT THIS PRODUCT SHOULD NOT BE TREATED AS HOUSEHOLD WASTE. BY ENSURING THAT THIS PRODUCT IS DISPOSED OF CORRECTLY, YOU WILL HELP PREVENT BRINGING POTENTIAL NEGATIVE CONSEQUENCES TO THE ENVIRONMENT AND HUMAN HEALTH. FOR MORE DETAILED INFORMATION WITH REGARD TO RETURNING AND RECYCLING THIS PRODUCT, PLEASE CONSULT THE DISTRIBUTOR FROM WHOM YOU PURCHASED THE PRODUCT.
	European Conformity
	Authorized Representative in European Union
	Unique device identifier
	Radio Frequency Identification

The general meaning assigned to geometrics shapes, safety colors and contrast colors for safety signs are as follows:

Geometric shape	Meaning	Meaning	Meaning	Meaning
	Warning	Yellow	Black	Black
	Warning	Yellow	Black	Black

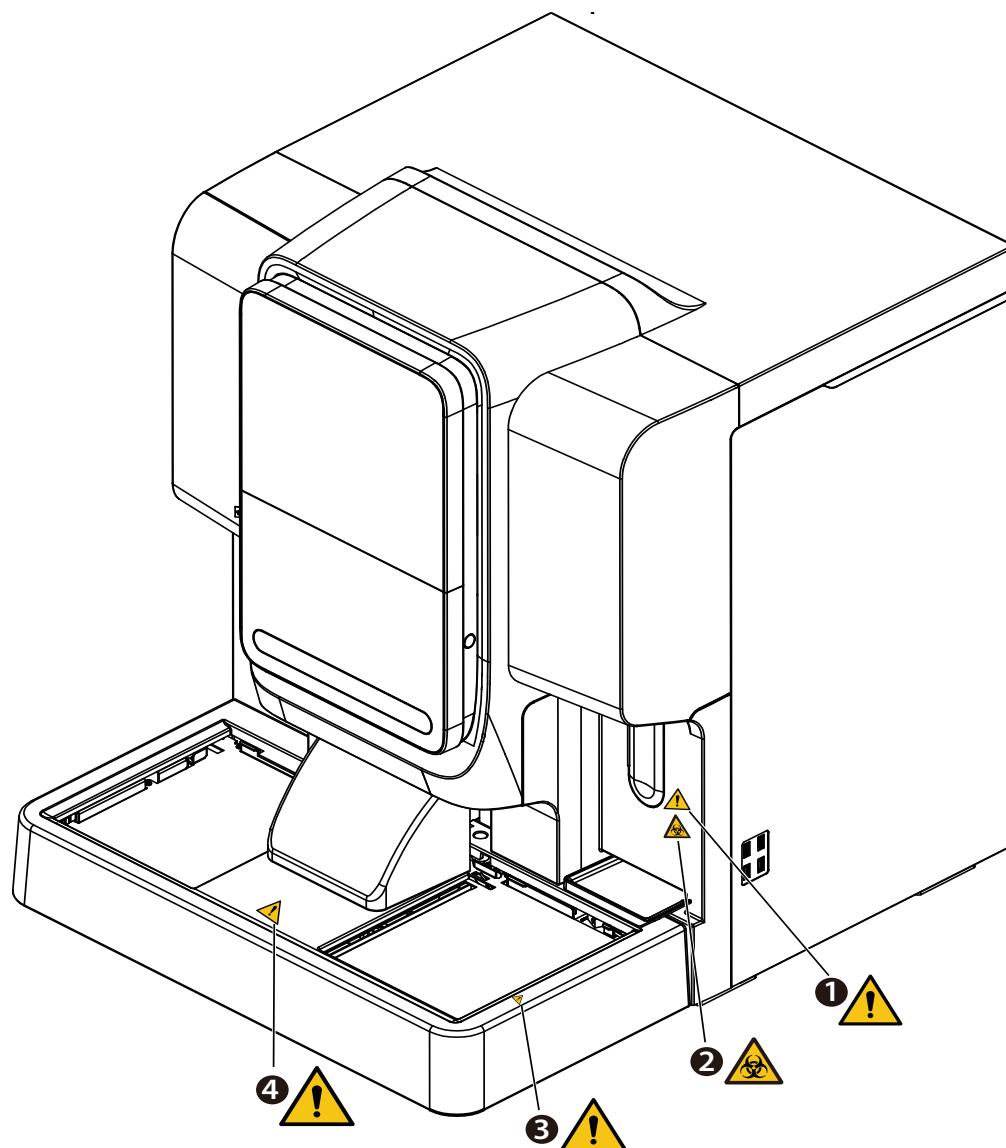


Figure 1-1 Warning labels on the front of the analyzer

Table 1-1 Description of warning labels on the front of the analyzer

	Labels	Description
1		WARNING BIOHAZARD
2		WARNING Take caution when working around to avoid pricking.
3		Note 1. Test tubes must come into the rack smoothly. If not, shape the sample tube by replacing the labels. 2. Make sure cap the tubes securely. 3. Make sure the left tray is not full and the detection sensor is not blocked, and then start the analysis.
4		WARNING Be aware for pinching.

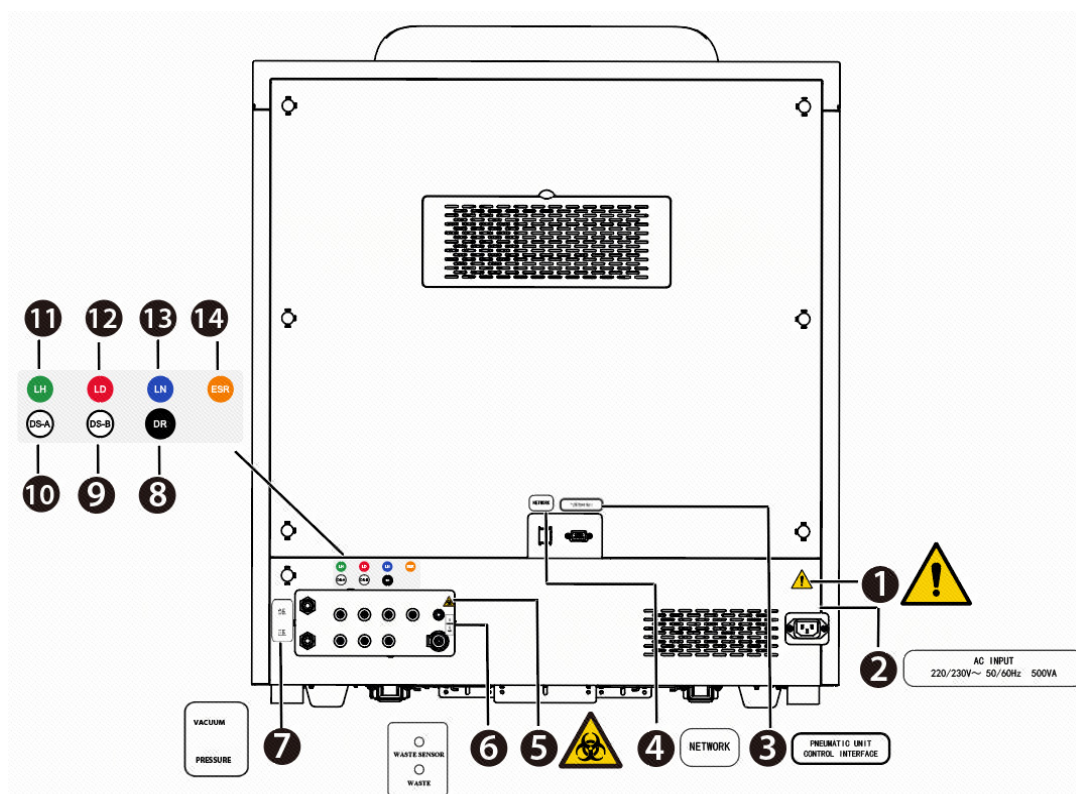
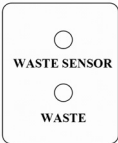


Figure 1-2 Labels on the back of the analyzer

Table 1-2 Description of the labels on the back of the analyzer

	Labels	Description
1		WARNING 1. CONNECT ONLY TO A PROPERLY EARTH GROUNDED OUTLET. 2. TO AVOID ELECTRIC SHOCK, DISCONNECT POWER CORD PRIOR TO REMOVING OR REPLACING FUSE. 3. REPLACE FUSE ONLY WITH THE TYPE AND RATING SPECIFIED.
2		Indicating the power supply inlet
3		Indicating the pneumatic control interface
4		WARNING Biological risk
5		Indicating the waste sensor interface/waste outlet
6		Indicating the vacuum outlet/the pressure inlet
7		Indicating the DR Diluent inlet
8		Indicating the DS-B Diluent inlet
9		Indicating the DS-A Diluent inlet

	Labels	Description
10		Indicating the LH Lyse inlet
11		Indicating the LD Lyse inlet
12		Indicating the LN Lyse inlet
13		Indicating the ESR Solution Reagent inlet

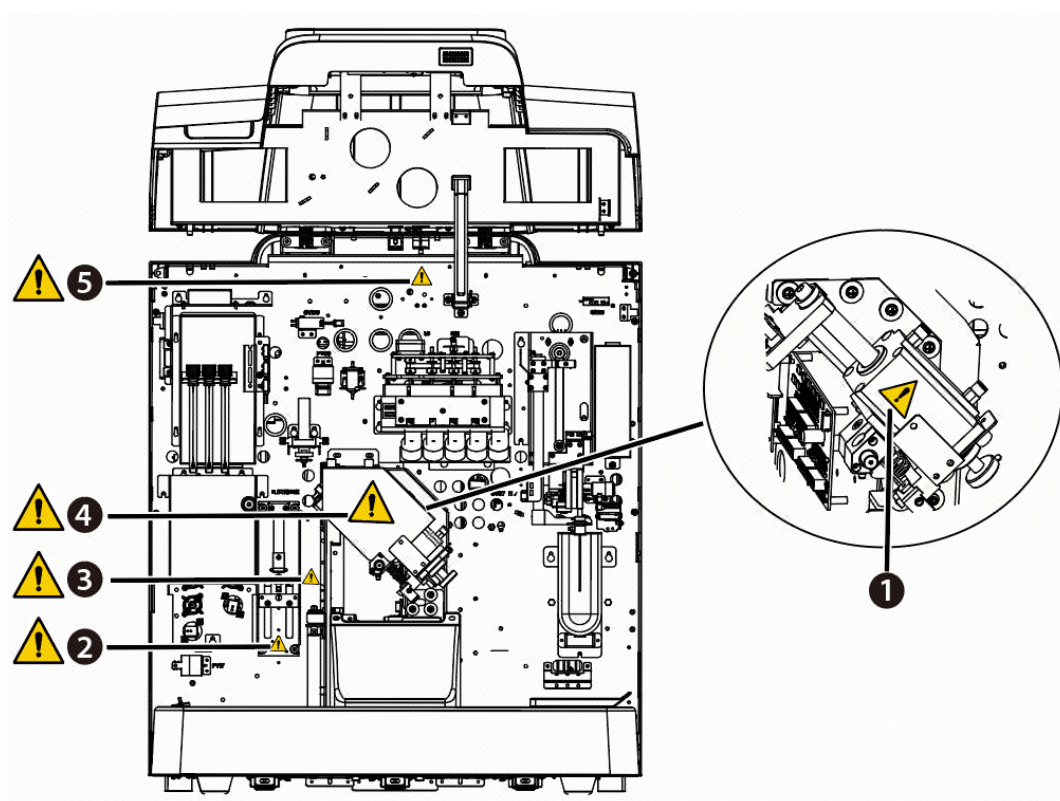


Figure 1-3 Warning labels when the front cover opens

Table 1-3 Description of warning labels when the front cover opens

	Labels	Description
1		WARNING (when the protective cover of the pierce probe is removed) To avoid personal injury, do not put your hand into the analyzer.
2		WARNING Mechanical risk
3		WARNING Mechanical risk
4		WARNING Mechanical risk may exist if the protective cover is removed.
5		WARNING To avoid personal injury, after you lift the cover, be sure to fix it with the stop bar exactly as the above figure shows.

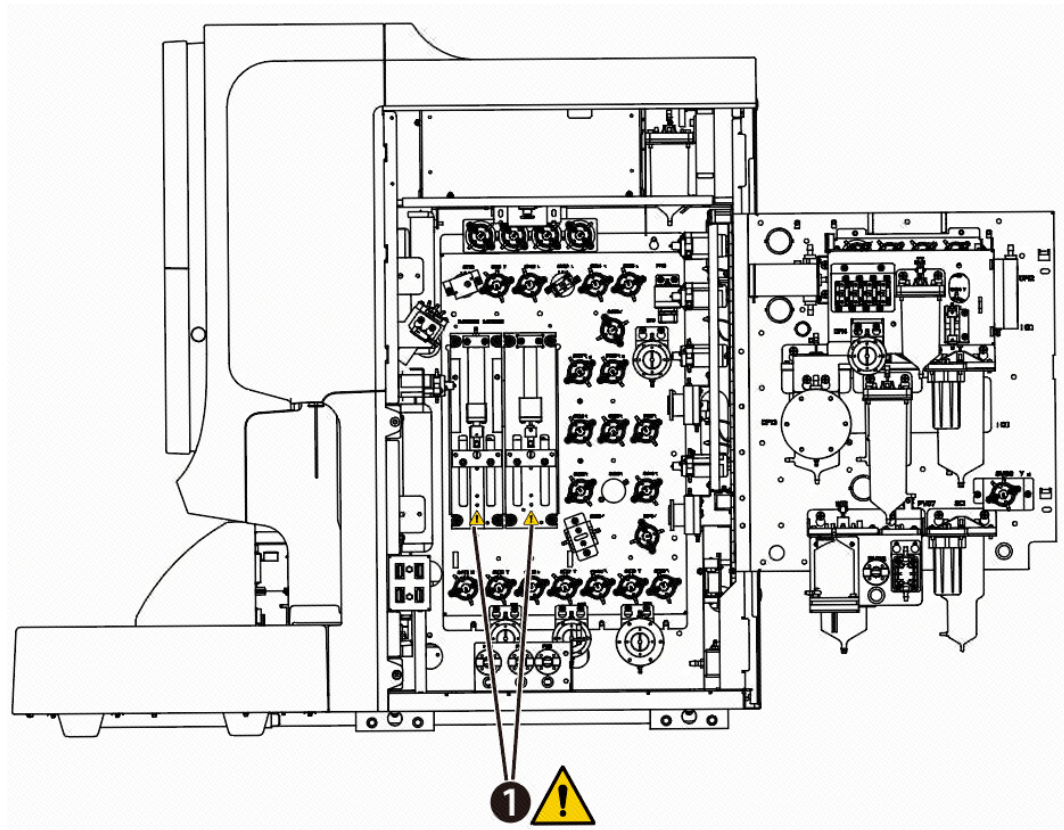
**Figure 1-4 Warning labels on the moving parts**

Table 1-4 Description of warning labels on the moving parts

	Label	Description
1		WARNING Mechanical risk

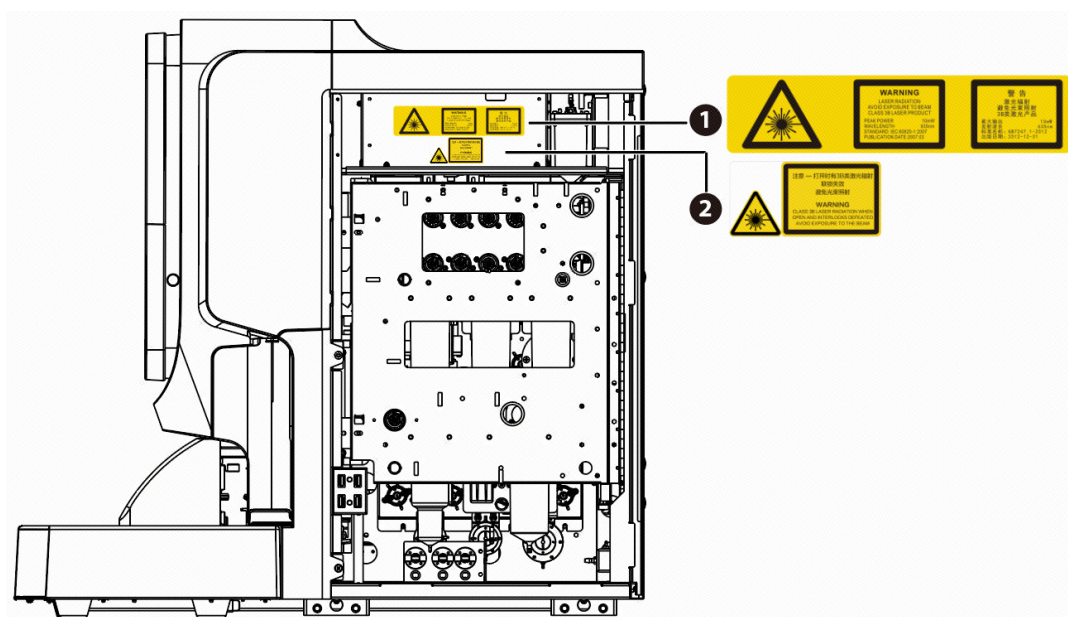


Figure 1-5 Laser warning labels

Table 1-5 Description of the laser warning labels

	Labels	Description
1	  	WARNING • LASER RADIATION • AVOID EXPOSURE TO THE BEAM • CLASS 3B LASER PRODUCT • MAXIMUM OUTPUT: 15mW • WAVELENGTH: 635nm • STANDARD: IEC 60825-1:2014 • PUBLICATION DATE: 2014.05
2	 	WARNING • CLASS 3B LASER RADIATION WHEN OPEN AND INTERLOCKS DEFEATED • AVOID EXPOSURE TO THE BEAM

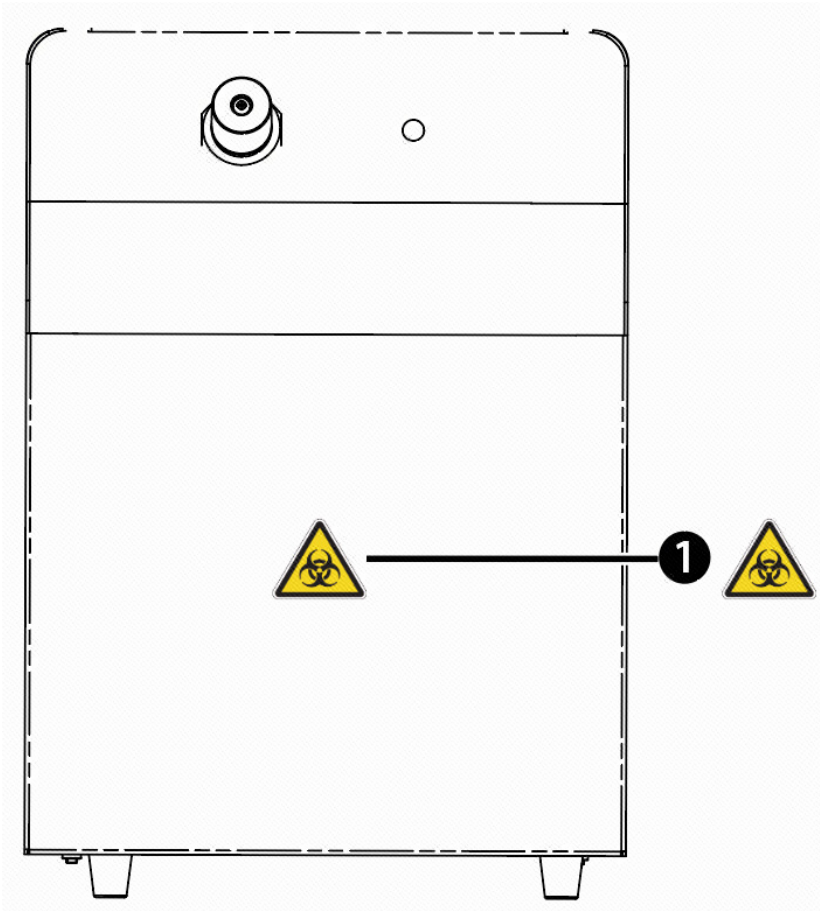


Figure 1-6 Warning label on the front of the pneumatic unit

Table 1-6 Description of warning label on the front of the pneumatic unit

	Label	Description
1		WARNING Biological risk

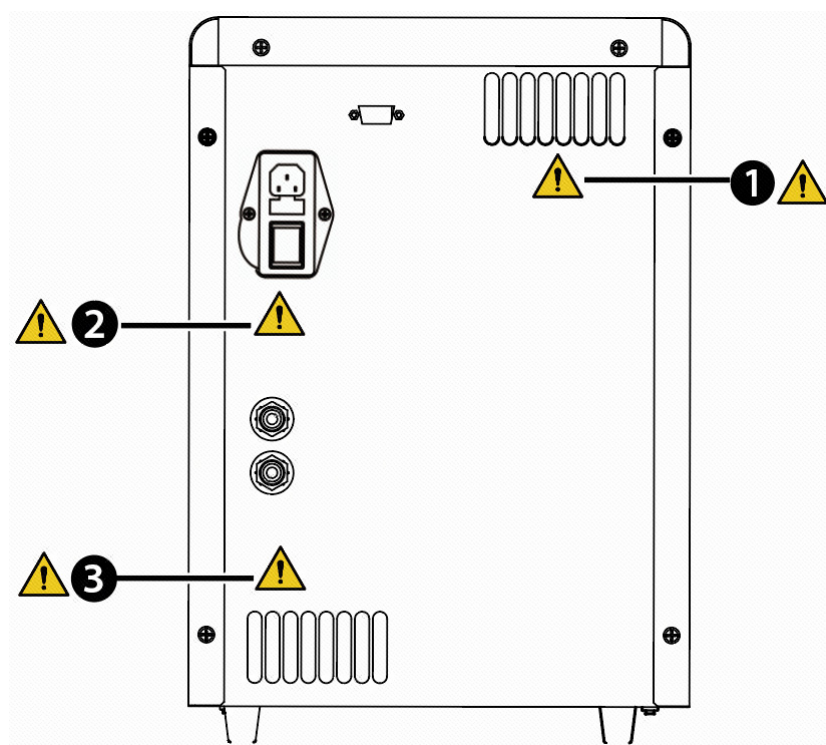


Table 1-7 Warning labels on the back of the pneumatic unit

Table 1-8 Description of warning labels on the back of the pneumatic unit

	Labels	Description
1		WARNING To avoid damaging the pneumatic unit, do not block the vent at its back.
2		WARNING 1. CONNECT ONLY TO A PROPERLY EARTH GROUNDED OUTLET. 2. TO AVOID ELECTRIC SHOCK, DISCONNECT POWER CORD PRIOR TO REMOVING OR REPLACING FUSE. 3. REPLACE FUSE ONLY WITH THE TYPE AND RATING SPECIFIED.
3		WARNING To avoid damaging the pneumatic unit, do not block the vent at its back.

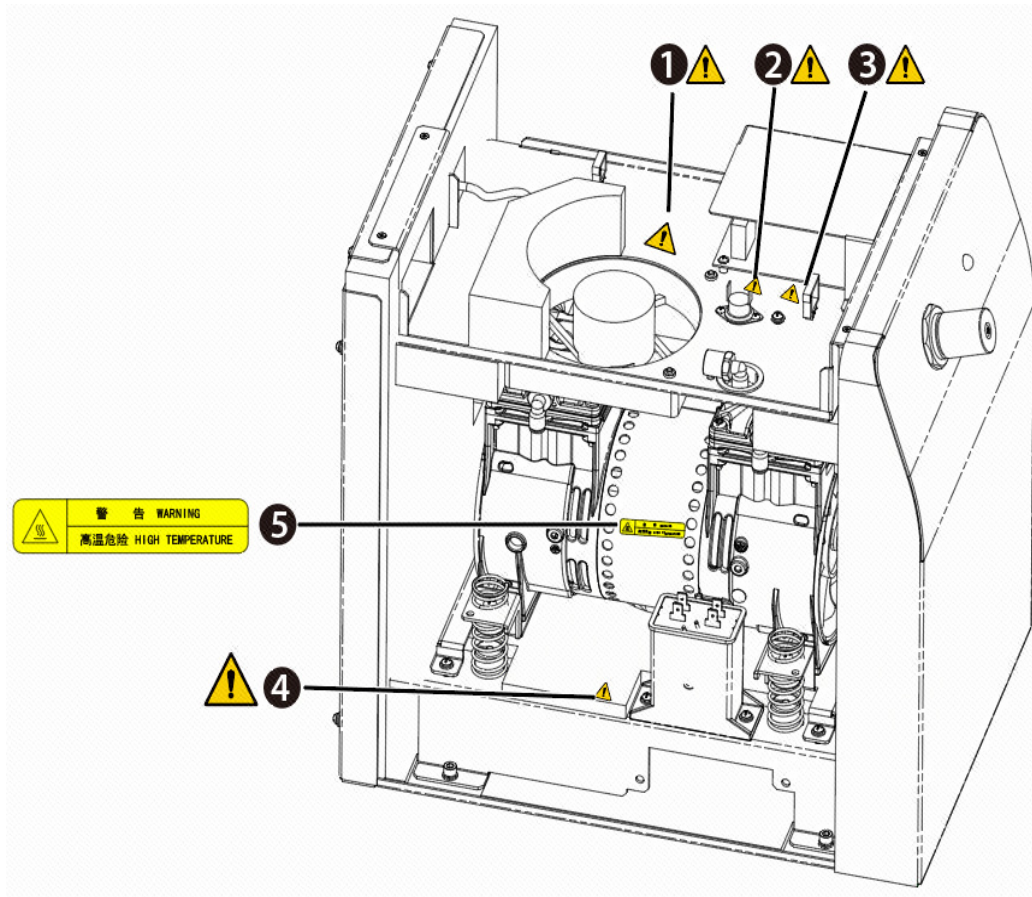


Figure 1-7 Warning labels in the pneumatic unit

Table 1-9 Description of warning labels in the pneumatic unit

	Labels	Description
1		WARNING To avoid personal injury, do not put hand into the fan.
2		WARNING Risk of electric shock. Do not touch. Cut off the power before maintenance.
3		WARNING Risk of electric shock. Do not touch. Cut off the power before maintenance.
4		WARNING Risk of electric shock. Do not touch. Cut off the power before maintenance.

	Labels	Description
5		WARNING HIGH TEMPERATURE

1.2 Safety Warning Symbols and Messages

You will find the following symbols in this manual:

When you see...	It means...
 BIOHAZARD	Read the statement below the symbol. The statement is alerting you to a potentially biohazardous condition.
 WARNING	Read the statement below the symbol. The statement is alerting you to an operating condition that can cause death, serious personnel injury or property damage.
 CAUTION	Read the statement below the symbol. The statement is alerting you to an operating condition that can cause minor personnel injury, system failure/ damage, and property damage.
NOTE	Read the statement below the symbol. The statement is alerting you to an operating condition that requires your attention.

1.3 Safety Precautions

1.3.1 General Safety Precautions

BIOHAZARD

- All the samples, controls, calibrators, wastes and areas contacted them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab coat, goggles, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.
- Our company does not claim the validity of the listed chemicals in infection control. For effective control of infection, please consult the Infection Prevention Department of the hospital or the epidemic professionals.
- Be sure to dispose of reagents, waste, samples, consumables, etc. according to government regulations.
- Discard the system according to government regulations.

** WARNING**

- Only install fuses of specified type and specification to prevent from fire hazard. If there is any problem with the fuse, contact our Customer Service Department or your local distributor.
 - Any serious incident that has occurred in relation to the device shall be reported to the manufacturer and the competent authority of the country in which the user and/or the patient is established.
 - To avoid fire, make sure you use the fuse of specified type and model. If the fuse of the system is damaged, contact our Customer Service Department or your local distributor for replacement.
-

** CAUTION**

- Be sure to operate the instrument under the situation specified in this manual. If the instrument is used in a manner not specified by the manufacturer, the protection provided by the instrument may be impaired.
 - Be sure to use the specified external devices only.
 - External equipment connected to the analyzer and digital interfaces must be authorized and complied with relevant safety and EMC standards (e.g. IEC 60950 Safety of Information Technology Equipment Standard (Class B)). Any persons who connects additional equipment to the signal input or output ports and configures an IVD system, is responsible for ensuring that the system works normally and complies within the safety and EMC requirements. If you have any questions, consult the technical service department of your local representative.
 - Make sure all the safety measures are adopted. It is prohibited to disable any safety devices or sensors.
 - Keep your clothes, hairs and hands away from the moving parts to avoid injury. Do not touch the moving parts.
 - Remove the waste container cap and replace the waste container only when the power indicator is not flickering, in order not to make the waste overflow from the container.
 - The pusher will push the rack into the autoloader. Be sure your hand is away from the sample rack before starting the autoloader.
 - Use the analyzer under operating conditions specified in the manual. Otherwise, the analyzer will not work normally, the analysis results may be unreliable, the system components may be damaged, or even cause personal injury.
 - This analyzer is only intended for use by test professionals, doctors and testers trained by our agents.
 - Make sure the sample probe contacts the sample tube bottom when aspirating sample. Otherwise, the aspirated volume may be imprecise.
 - Remove the waste container cap and replace the waste container only when the power indicator is not flickering, in order not to make the waste overflow from the container.
 - Use the reagents, controls and calibrators specified by the manufacturer only. Using other reagents, controls and calibrators, may lead to analyzer damage and inaccurate measurement, control and calibration results.
 - Only use the accessories and consumables manufactured or recommended by our company to achieve the promised system performance and safety. For more information, contact our Customer Service Department or your local distributor.
 - When power cut occurs suddenly, turn off the instrument immediately.
 - The instrument must be within the calibration validity period; otherwise, the measurement results may be inaccurate.
-

1.3.2 Safety Precautions on Analyzer Transportation and Installation

** WARNING**

- Before turning on the instrument, make sure the input voltage meets the requirements.
-

- Do not position the device so that it is difficult to operate the disconnecting device.

CAUTION

- Unpacking, installation or transportation by personnel not authorized or trained by our company may cause personal injury or damage your instrument. Do not unpack, transport or install your instrument without the presence of our company authorized personnel.
 - The installation, authorization, upgrade and modification of the system software must be performed by personnel authorized by our company. Make sure to install only our company authorized software.
 - Using pinboard may bring electrical interference and the analysis results may be unreliable. Place the analyzer near the electrical outlet to avoid using the pinboard.
 - Use the power cord provided by the manufacturer. Using the power cord other than provided by the manufacturer may lead to system damage or unqualified smear output.
 - Check if the reagent sample tubes are properly connected before using the system. Otherwise, the results may be inaccurate.
-

1.3.3 Safety Precautions on Reagents

BIOHAZARD

- All the samples, controls, calibrators, wastes and areas contacted with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab coat, goggles, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.
 - Be sure to dispose of reagents, waste, samples, consumables, etc. according to government regulations.
-

CAUTION

- The reagents are irritating to eyes, skin and mucous membranes. Wear proper personal protective equipment (e.g. gloves, lab coat, glasses, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.
 - If reagents come into contact with your skin or your eyes, rinse with plenty of water immediately, and consult a physician if necessary.
 - Use the reagents, controls and calibrators specified by the manufacturer. Store and use them as instructed by instructions for use of the reagents. Using other reagents, controls and calibrators, may lead to analyzer damage and inaccurate measurement, control and calibration results.
 - Use the reagents, controls and calibrators within the validity period stated in their instructions for use. Otherwise an inaccurate result may occur.
 - Dilute the sample when the analyte content in the sample exceeds the upper limit of the linearity range.
 - To ensure measurement accuracy, do not mix the new container of reagent with the residue in the replaced container, and prevent reagents from contamination following safe laboratory procedures.
-

1.3.4 Safety Precautions on Operation

BIOHAZARD

- Broken collection sample tubes may cause biohazard and/or personal injury. Exercise extreme caution when loading the collection sample tubes to the rack or removing the collection sample tubes from the rack, be sure not to break the sample tubes.
-

** CAUTION**

- To attain accurate analysis results, make sure the sample volume meets requirements. Otherwise, the results may be inaccurate.
 - Use sample tubes matching adapter specifications. Otherwise, the results may be inaccurate.
 - After mixing the capillary sample with the diluent, be sure to wait at least for 3 minutes, and mix the sample again before running the sample. Otherwise, the results may be inaccurate.
 - Be sure to mix any sample that has been prepared for a while before running it. Otherwise, the result may be inaccurate.
 - To attain accurate analysis results, make sure the sample volume meets requirements. Otherwise, the results may be inaccurate.
 - Be sure to use the specified disposable products like blood collection sample tubes. Do not reuse disposal products, otherwise the result may be inaccurate.
 - Check the firmness of all the doors and covers before running the system, and make sure they will not open or get loose during analysis. Exercise caution when open/close, install/un-install the doors and covers to avoid dropping to the ground.
 - After an abnormal shutdown, when you have changed the diluent, lyse or dye, run a background test to see if the results meet the requirement. Otherwise the result may be inaccurate.
-

1.3.5 Safety Precautions on Maintenance

** BIOHAZARD**

- There are 2 ways of waste connection, one is using waste container, the other is discharging waste directly. If the waste is discharged using waste container, make sure the pickup sample tube of the waste container cap assembly is above the waste container, and the sample tube is smooth and not bent. If the waste is discharged directly, make sure the waste tube is at a lower position than the waste outlet on the analyzer.
 - After replacing the reagent container/bag, check the tubing connected to the cap assembly and make sure it is not bent over. Otherwise it may cause personal injury or incorrect measurement results.
-

** CAUTION**

- Before cleaning the floating blood barrier and the blood barrier bracket, make sure the analyzer is shut down.
- Improper maintenance may impair the analyzer. Operators must follow the instruction of this Operator's Manual to perform maintenance operations. For problems not mentioned in this manual, contact our Customer Service Department for service advice.
- Only parts supplied by our company can be used for maintenance. For any question, contact our Customer Service Department.
- If you accidentally spill hazardous material (for example, reagents or samples) on the instrument, clean the instrument with specified disinfectant. Do not use material that could corrupt metal (for example, hydrogen peroxide). Wear proper personal protective equipment (e.g. gloves, lab coat, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.
- Perform regular cleaning on the covers of the analyzer.
- The sterilization may damage the system to some extent. It is recommended to perform sterilization only when necessary according to your laboratory protocol. Remember to clean the instrument before sterilizing.
- Do not use any cleaning agents that could cause hazards as a result of a reaction with parts of the analyzer or with materials inside the analyzer. If there is any doubt about the compatibility of the cleaning agents with parts of the equipment or with material contained in it, please contact our Customer Service Department or the local distributor.

- Only use the accessories and consumables manufactured or recommended by our company to achieve the promised system performance and safety. For more information, contact our Customer Service Department or your local distributor.
- If any of the pipes or fluidic components are worn out, stop using the analyzer and contact our Customer Service Department immediately for inspection or replacement.
- After maintenance, be sure to check the system status and confirm that it is safe before delivering it to the user.
- Before maintaining, transporting or servicing the instrument, clean and sterilize the instrument cover as well as the parts and components with biological risks (such as the sampling probe). Remind the persons who handle the instrument of the related risks.

1.3.6 Laser Warning

NOTE

- Class 3B laser radiation when open and interlocks defeated
- Avoid exposure to the beam
- Standard name: IEC 60825-1:2014
- Publication date: 2014.05

1.3.6.1 Laser parameters

- Maximum output: 15mW
- Wavelength: 635nm

1.3.6.2 Warning information

WARNING

- This product is a CLASS 1 embedded laser product. When opened or when the interlock defeated, there are Class 3B laser radiation. For laser parameters, refer to **1.3.6.1 Laser parameters**. Users must not open, disassemble or damage the interlock device. Otherwise may be exposed to laser radiation. For any questions, contact our Customer Service Department.
- The debugging of parts of the laser shall be performed by our Customer Service Department or local distributor. Do not use, adjust or operate the device in a manner not specified as instructed, otherwise the exposure to the laser radiation may occur. During assembly, maintenance and debugging, do not stare at the beam and you must wear laser protective device (e.g. safety goggles).

1.3.6.3 Maintenance plan of product consistency

NOTE

- If the laser requires any maintenance, it will be delivered to our company by our Customer Service Department and its maintenance will be performed by our professional personnel.

Use the QC program as instructed by **9 Using the QC Program** and perform service as instructed by **12 Servicing Your Analyzer** to ensure the analyzer works normally. In case of any error, handle it according to the troubleshooting instructions. If the error still exists, contact our Customer Service Department or local distributor.

1.3.7 When the Instrument Operation is Stopped for Maintenance, Transportation or Service

CAUTION

- In case of power failure, power off the analyzer immediately.

- Before maintaining, transporting or servicing the instrument, clean and sterilize the instrument cover as well as the parts and components with biological risks (such as the sampling probe). Remind the persons who handle the instrument of the related risks.

1.3.8 Software and Network Safety

NOTE

- The analyzer software uses closed-loop operating system, meaning that the applications of the analyzer works in exclusive mode and are free from other application's disturbance. Users can only operate the software interface but cannot directly access the operating system or install software.
- Data transmission must be performed in a closed network or virtual network. The network must be isolated.
- The installation, authorization, upgrade and modification of the system software must be performed by personnel authorized by our company.
- When the instrument connects to an external computer, install anti-virus software on the computer and scan for viruses and update patches periodically. Do not use it for unintended purposes.
- When the instrument connects to an external computer, install the original edition software to prevent the computer from being infected by virus.
- Make sure to install only our company authorized software.
- Users have the responsibility to protect the network authentication information, such as password and user information, from being obtained by unauthorized personnel.

The analyzer software is developed based on the following off-the-shelf software.

Name	Model	Version	Title	Manufacturer
Operating System	OS Core	3.18.24	Linux embedded operating system	Linux Kernel Organization, Inc

2 Using this Manual

This chapter explains how to use your Operator's Manual of BC-6800Plus Automatic Hematology Analyzer (hereby referred to as "the analyzer"). The manual is shipped with your product and describes the purpose, functions, and operations of the product. Read this manual carefully before operating your product to ensure that the product can be operated strictly as instructed in this manual to bring its performance to full swing and guarantee operator safety.

NOTE

- **All illustrations in this manual are provided as examples only. They may not necessarily reflect your analyzer setup or data displayed.**
- **This manual describes the use, functions and operation methods of the product based on the most complete configuration; and some of the content may not be applicable to your product. Contact our Customer Service Department if you have any questions.**

2.1 Who Should Read This Manual

This manual is intended to be read by clinical laboratory professionals to:

- learn about the analyzer hardware and software;
- set up system parameters;
- perform daily operating task;
- perform system maintenance and troubleshooting

2.2 How to Find Information

This operator's manual comprises 13 chapters and 8 appendices. Refer to the table below to find the information you need.

If you want to ...	See...
learn about the safety messages of the analyzer	1 Safety Information
learn about the intended use and parameters of the analyzer	3 Understanding Your System
learn about the hardware and software of the analyzer	3 Understanding Your System
learn about how the analyzer works (its measurement principles and processes)	4 Understanding the System Principles
learn about the installation requirements of the analyzer	5 Installing and Connecting the System
learn about how to define/adjust system settings	6 Customizing Your Analyzer
learn about how to collect, prepare and analyze the samples	7 Operating Your Analyzer
learn about how to use the analyzer to perform your daily operating tasks	7 Operating Your Analyzer
learn about how to review the saved analysis results	8 Reviewing Sample Results
learn about basic requirements of quality control and quality control methods of the hematology analyzers	9 Using the QC Program
learn about how to calibrate the analyzer	10 Calibrating Your Analyzer
learn about how to use the quality control programs of the analyzer	11 Using the Printing Function

If you want to ...	See...
learn about how to maintain/service the analyzer	12 Servicing Your Analyzer
learn about the troubleshooting methods of the analyzer	13 Troubleshooting
learn about the technical specifications of the analyzer	B Technical Information

2.3 Conventions Used in This Manual

This manual uses certain typographical conventions to clarify meaning in the text:

Format	Meaning
[xx]	All capital letters enclosed in [] indicate a key name (either on the pop-up keyboard or the external keyboard), such as [ENTER].
"xx"	Letters included in " " indicate text you can find on the screen of the analyzer.
Refer to XX	XX is the referenced section
	Revision bar indicates update to previous version.

3 Understanding Your System

3.1 Intended Use

The Auto Hematology Analyzer is a quantitative, automated hematology analyzer for In Vitro Diagnostic Use in clinical laboratories; it provides Complete Blood Count, Leukocyte 5-part Differential, Hemoglobin Concentration Measurement, Reticulocyte and Nucleated Red Blood Cell Measurement, and Erythrocyte Sedimentation Rate (ESR) Measurement for blood samples, as well as Body Fluid analysis.

ESR can be used as aids to diagnosis of inflammation diseases.

NOTE

- The subtypes of body fluid supported by BC-6800Plus include cerebrospinal fluid, serous cavity fluid (pleural fluid and ascitic fluid) and synovial fluid. You may not get accurate results if you analyze body fluid samples other than specified above.

3.2 Parameters

The analyzer is used for quantitative determination of the following report parameters and Research Use Only (RUO) parameters of blood sample tests and body fluid sample tests, and provides histograms and scattergrams.

3.2.1 Blood Sample Test Parameters, Histograms, and Scattergrams

Table 3-1 Blood Sample Test Report Parameters

Groups	Name	Abbreviation
WBC group	White Blood Cell count	WBC
	Basophils number	Bas#
	Basophils percentage	Bas%
	Neutrophils number	Neu#
	Neutrophils percentage	Neu%
	Eosinophils number	Eos#
	Eosinophils percentage	Eos%
	Lymphocytes number	Lym#
	Lymphocytes percentage	Lym%
	Monocytes number	Mon#
	Monocytes percentage	Mon%
	Immature Granulocyte number	IMG#
	Immature Granulocyte percentage	IMG%
	DIFF scattergram, mean neutrophil distribution-side scatter intensity	Neu-X
	DIFF scattergram, mean neutrophil distribution-side fluorescent light intensity	Neu-Y
	DIFF scattergram, mean monocyte distribution-side scatter intensity	Mon-X

Groups	Name	Abbreviation
RET group	Reticulocyte percentage	RET%
	Reticulocyte number	RET#
	Reticulocyte Hemoglobin Expression	RHE
	Immature reticulocyte fraction	IRF
	Low fluorescent ratio	LFR
	Middle fluorescent ratio	MFR
	High fluorescent ratio	HFR
	Mean corpuscular volume of reticulocyte	MCVr
	Mean corpuscular hemoglobin of reticulocyte	MCHr
	Expression of MCHr - MCHm	DELTA-CH
RBC group	Red Blood Cell count	RBC
	Hemoglobin Concentration	HGB
	Mean Corpuscular Volume	MCV
	Mean Corpuscular Hemoglobin	MCH
	Mean Corpuscular Hemoglobin Concentration	MCHC
	Mean hemoglobin distribution width of red blood cell	HDW
	Percentage of mature red blood cell with a low corpuscular hemoglobin	HYPO-CH%
	Percentage of mature red blood cell with a high corpuscular hemoglobin	HYPER-CH%
	Mean corpuscular hemoglobin of mature red blood cell	MCHm
	Red Blood Cell Distribution Width - Coefficient of Variation	RDW-CV
	Red Blood Cell Distribution Width - Standard Deviation	RDW-SD
	Hematocrit	HCT
	Microcyte percentage	Micro%
	Macrocyte percentage	Macro%
	Nucleated red blood cell number	NRBC#
	Nucleated red blood cell percentage	NRBC%

Groups	Name	Abbreviation
Platelet group	Platelet count	PLT
	Optical platelet count	PLT-O
	Platelet count hybrid	PLT-H
	Platelet count- impedance	PLT-I
	Mean Platelet Volume	MPV
	Platelet Distribution Width	PDW
	Plateletcrit	PCT
	Immature Platelet Fraction	IPF
	Platelet-large cell ratio	P-LCR
	Platelet-large cell count	P-LCC
	Immature platelet count	IPF#
ESR	Erythrocyte sedimentation rate	ESR

Table 3-2 Blood Sample Test Research Use Only (RUO) Parameters

Name	Abbreviation
High fluorescent Cell number	HFC#
High fluorescent Cell percentage	HFC%
Optical Red Blood Cell count	RBC-O
Optical hemoglobin concentration	HGB-O
Optical white blood cell count	WBC-O
White blood cell count -DIFF	WBC-D
Total nucleated cell counts-DIFF	TNC-D
Immature eosinophil percentage	IME%
Immature eosinophil number	IME#
High forward scatter NRBC ratio	H-NR%
Low forward scatter NRBC ratio	L-NR%
Neutrophil-to-lymphocyte ratio	NLR
Platelet-to-lymphocyte ratio	PLR
White blood cell count-WNB	WBC-N
Total nucleated cell counts-WNB	TNC-N
Infected Red Blood Cell count	InR#
Infected Red Blood Cell permillage	InR‰
NEU# Minus IMG#	Neu#&
NEU% Minus IMG%	Neu%&
LYM# Minus HFC#	Lym#&
LYM% Minus HFC%	Lym%&

Name	Abbreviation
Microcyte count	Micro#
Macrocyte count	Macro#
Hyperchromic Red Blood Cell Percentage	HYPER%
Hypochromic Red Blood Cell Percentage	HYPO%
Mean Reticulocyte Volume	MRV
RET scattergram, mean reticulocyte distribution-forward scatter intensity	RET-Y
RET scattergram, mean reticulocyte distribution-side fluorescent intensity	RET-X
RET scattergram, mean immature reticulocyte fraction distribution-forward scatter intensity	IRF-Y
RET scattergram, mean immature reticulocyte fraction distribution-side fluorescent intensity	IRF-X
RET scattergram, mean red blood cell distribution-forward scatter intensity	RET-RBC-Y
RET scattergram, mean red blood cell distribution-side fluorescent intensity	RET-RBC-X
RBC Fragment count	FRC#
RBC Fragment percentage	FRC%
Platelet Distribution Width Standard Deviation	PDW-SD
Reticulocyte Production Index	RPI
High fluorescent Immature Platelet Fraction	H-IPF
Mean Platelet Component Concentration	MPC
Mean Platelet Matter Content	MPM
DIFF scattergram, mean neutrophil distribution- forward scatter intensity	Neu-Z
DIFF scattergram, mean lymphocyte distribution- side scatter intensity	Lym-X
DIFF scattergram, mean lymphocyte distribution-side fluorescent intensity	Lym-Y
DIFF scattergram, mean lymphocyte distribution- forward scatter intensity	Lym-Z
DIFF scattergram, mean monocyte distribution-side fluorescent light intensity	Mon-Y
DIFF scattergram, mean monocyte distribution- forward scatter intensity	Mon-Z
DIFF scattergram, neutrophil side scatter distribution width	Neu-XW
DIFF scattergram, neutrophil side fluorescent light distribution width	Neu-YW
DIFF scattergram, neutrophil forward scatter distribution width	Neu-ZW
DIFF scattergram, lymphocyte side scatter distribution width	Lym-XW
DIFF scattergram, lymphocyte side fluorescent light distribution width	Lym-YW
DIFF scattergram, lymphocyte forward scatter distribution width	Lym-ZW
DIFF scattergram, monocyte side scatter distribution width	Mon-XW
DIFF scattergram, monocyte side fluorescent light distribution width	Mon-YW

Name	Abbreviation
DIFF scattergram, monocyte forward scatter distribution width	Mon-ZW
Dimorphic population, smaller distribution RBC count	SRBC
Dimorphic population, larger distribution RBC count	LRBC
Dimorphic population, smaller distribution mean corpuscular volume	SMCV
Dimorphic population, larger distribution mean corpuscular volume	LMCV
Highest Frequency MCV	HF-MCV
Corrected erythrocyte sedimentation rate	ESR-Corr.
Surface area	SA
Amplitude	AMP
Aggregation index	AI
Minimum	MIN
Aggregation half time	T1/2
Immature platelet fraction- DIFF	IPF-D
Reticulocyte percentage- DIFF	RET%-D
Reticulocyte count- DIFF	RET#-D
Immature reticulocyte fraction- DIFF	IRF-D
Low fluorescent ratio- DIFF	LFR-D
Middle fluorescent ratio- DIFF	MFR-D
High fluorescent ratio- DIFF	HFR-D
High fluorescence lymphocytes percentage	HF-Lym%
High fluorescence lymphocytes count	HF-Lym#
Ultra-high fluorescence lymphocytes percentage	UHF-Lym%
Ultra-high fluorescence lymphocytes count	UHF-Lym#
Monocytes and neutrophils distribution information	MND

NOTE

- **RUO parameters and scattergrams are for research purpose only. They cannot be used for diagnosis purpose.**
- **To print RUO parameter results on report, refer to 11.2.4 Printing RUO Parameters Results.**

Table 3-3 Blood Sample Test Histograms

Name	Abbreviation
Red Blood Cell Histogram	RBC Histogram
Platelet Histogram	PLT Histogram
Platelet Hybrid Histogram	PLT-H Histogram

Table 3-4 Blood Sample Test Scattergrams

Name	Abbreviation
Differential Scattergram	DIFF Scattergram
Basophils and Nucleated Red Blood Cells Scattergram	WNB Scattergram
Reticulocyte Scattergram	RET Scattergram
Optical Platelet Scattergram	PLT-O Scattergram
Reticulocyte - Extension Scattergram	RET-EXT Scattergram
Platelet Hybrid Scattergram	PLT-H Scattergram

Table 3-5 Blood Sample Test Research Use Only (RUO) Scattergrams

Name	Abbreviation
Red blood cell volume hemoglobin concentration fluorescence signal scattergram	RBC VHF Scattergram
Red blood cell scattergram	RBC Scattergram

3.2.2 Body Fluid Sample Test Parameters, Histograms, and Scattergrams

Table 3-6 Body Fluid Sample Test Report Parameters

Groups	Parameter Name	Abbreviation
WBC group (6)	White Blood Cell count-body fluid	WBC-BF
	Total nucleated cell counts-body fluid	TC-BF#
	Mononuclear cell number	MN#
	Mononuclear cell percentage	MN%
	Polymorphonuclear cell number	PMN#
	Polymorphonuclear cell percentage	PMN%
RBC group (1)	Red Blood Cell count-body fluid	RBC-BF

Table 3-7 Body Fluid Sample Test Research Use Only (RUO) Parameters

Parameter name	Abbreviation
Eosinophils number- body fluid	Eos-BF#
Eosinophils percentage- body fluid	Eos-BF%
Neutrophils number- body fluid	Neu-BF#
Neutrophils percentage- body fluid	Neu-BF%
Lymphocytes number- body fluid	Lym-BF#
Lymphocytes percentage- body fluid	Lym-BF%
Monocytes number- body fluid	Mon-BF#
Monocytes percentage- body fluid	Mon-BF%
High fluorescent cell number- body fluid	HFC-BF#

Parameter name	Abbreviation
High fluorescent cell percentage- body fluid	HFC-BF%
Red Blood Cell count-body fluid	RBC-BF(R)

NOTE

- RUO parameters and scattergrams are for research purpose only. They cannot be used for diagnosis purpose.
- To print RUO parameter results on report, refer to 11.2.4 *Printing RUO Parameters Results*.

Table 3-8 Body Fluid Sample Test Histogram

Name	Abbreviation
Red Blood Cell Histogram	RBC Histogram

Table 3-9 Body Fluid Sample Test Scattergrams

Name	Abbreviation
Differential Scattergram	DIFF Scattergram

3.3 Supported Test Panels

Table 3-10 Description of supported test panels

Test panels	Meaning	Description
CBC	CBC	Complete blood count
CD	CBC+DIFF	Complete blood count + WBC 5-part differentiation tests
CDR	CBC+DIFF+Ret	Complete blood count + WBC 5-differentiation tests + Reticulocyte-related tests
CR	CBC+Ret	Complete blood count+Reticulocyte-related tests
Ret	Ret	Reticulocyte-related tests
ESR	ESR	Erythrocyte sedimentation rate tests
CBC+ESR	CBC+ESR	Complete blood count + Erythrocyte sedimentation rate tests
CD+ESR	CBC+DIFF+ESR	Complete blood count + WBC 5-part differentiation tests + Erythrocyte sedimentation rate tests
CDR+ESR	CBC+DIFF+Ret+ESR	Complete blood count + WBC 5-part differentiation tests+ Reticulocyte-related tests + Erythrocyte sedimentation rate tests
CDR/PLT-8X	CBC+DIFF+Ret/PLT-8X, using the multiple sampling method (PLT 8X mode)	Complete blood count + WBC 5-part differentiation tests+ Reticulocyte-related tests, using the multiple sampling method (PLT 8X)
CR/PLT-8X	CBC+Ret/PLT-8X, using the multiple sampling method (PLT 8X mode)	Complete blood count+Reticulocyte-related tests/PLT-8X, using the multiple sampling method (PLT 8X mode)

Test panels	Meaning	Description
PLT-O	PLT-O, using the multiple sampling method (PLT 8X mode)	Optical platelet count, using the multiple sampling method (PLT 8X mode)
CD/PLT-O	CBC+DIFF/PLT-O, using the multiple sampling method (PLT 8X mode)	Complete blood count + WBC 5-part differentiation tests/Optical platelet count, using the multiple sampling method (PLT 8X mode)

3.4 Product Description

3.4.1 Product Structure and Components

The BC-6800Plus Auto Hematology Analyzer consists of sample processing unit, data management unit, the result output unit and the accessories.

3.4.2 Side View of the Analyzer



CAUTION

- Do not use anything sharp on the touch screen or strike on it.
- Clean the touch screen with clean and soft cloth, as well as neutral cleanser or ethanol rather than chemical solution, acid or alkali solution.

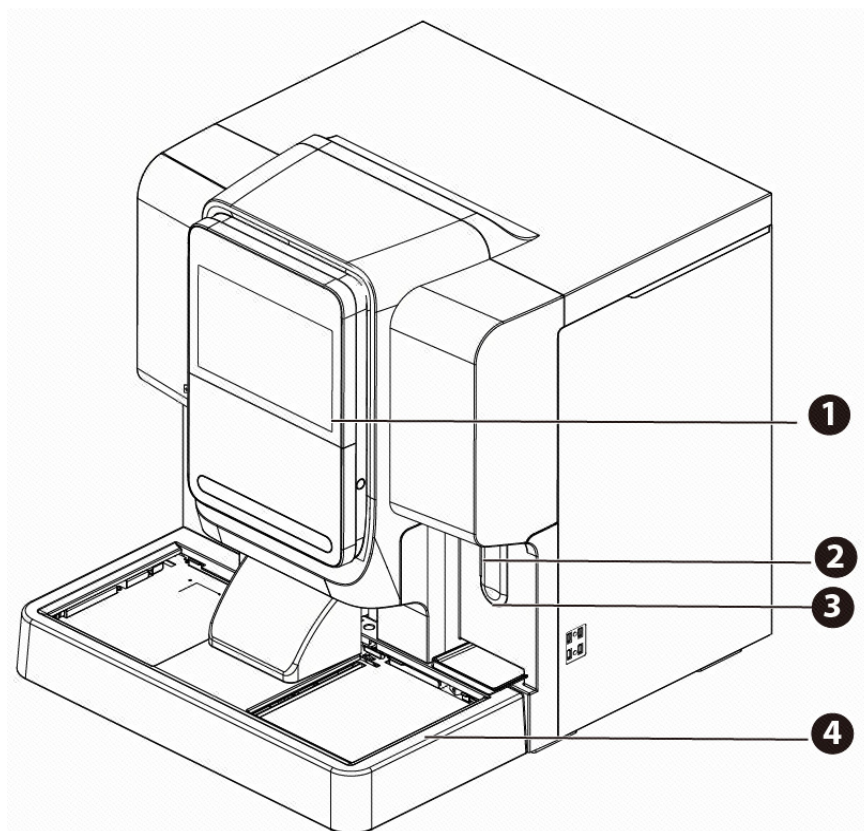
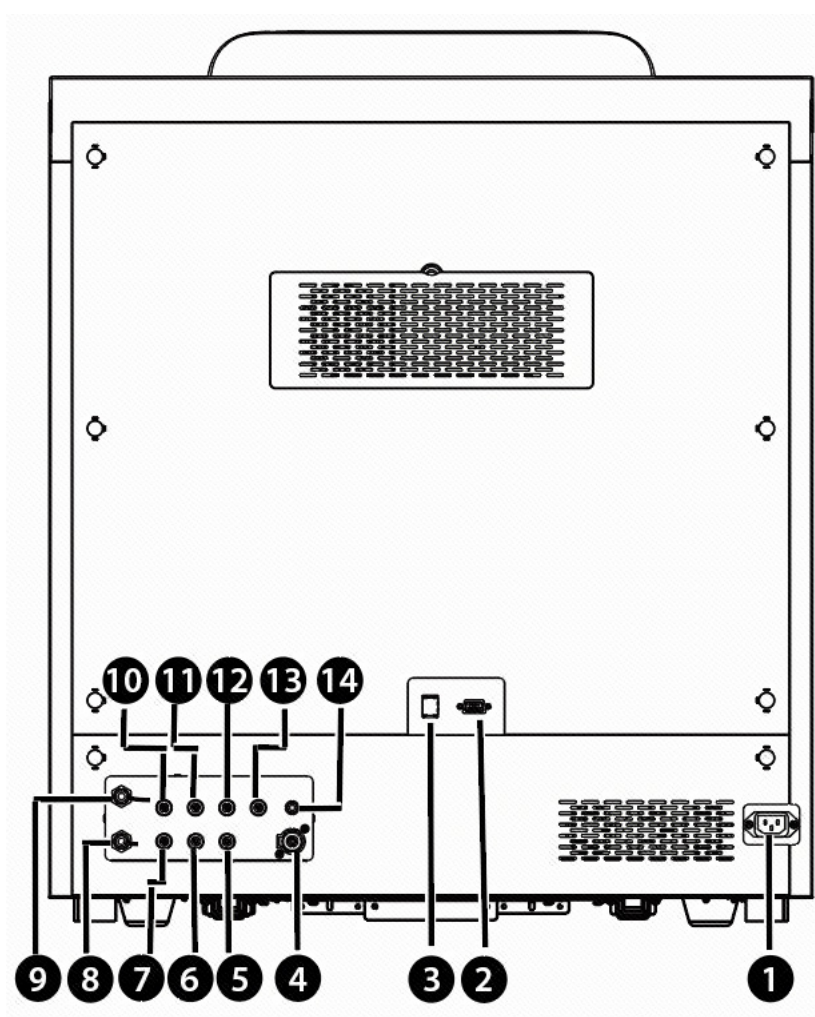


Figure 3-1 Side view of the analyzer

Table 3-11 Description of the side view of the analyzer

	Name	Function	Description
1	Touch Screen	Used to operate the instrument and display information.	/
2	Sample probe	The sample probe aspirates samples and adds diluent.	/
3	[Aspirate] key	The [Aspirate] key is behind the sample probe. Press the key to start analysis or add diluent.	/
4	Autoloader	Place sample racks on the autoloader, and the samples are loaded.	/

3.4.3 Back View of the Analyzer

**Figure 3-2 Back view of the analyzer****Table 3-12 Description of the back view of the analyzer**

	Name	Function	Description
1	Power socket	Connect the power cords.	For power supply and voltage specifications, refer to B.9 Power supply .

	Name	Function	Description
2	Pneumatic unit control port	You can use the pneumatic unit control port to startup or shutdown the pneumatic unit.	/
3	Network interface	Connect the network wire to enable LIS (Laboratory Information System).	For network interfaces specifications, refer to B.8 Electronic Interface Specifications .
4	Waste liquid outlet	Connect the waste container (when waste liquid is discharged into a container) or the waste tube (when the waste liquid is discharged directly).	For more information, refer to 5.2 Connecting the Analyzer System .
5	DR Diluent inlet	Connect the DR Diluent container.	/
6	DS Diluent inlet B	Connect the DS Diluent container.	- Depending on the analyzer configuration, the analyzer may use a single container of DS diluent, or use two containers of DS diluent at the same time. - For the single DS diluent configuration, connect the DS diluent container to DS diluent inlet A (position 7 as in Figure 3-2 Back view of the analyzer). - For the two DS diluent configuration, connect one DS diluent container to DS diluent inlet A (position 7 as in Figure 3-2 Back view of the analyzer), and the other to DS diluent inlet B (position 6 as in Figure 3-2 Back view of the analyzer). - For more information about how to connect and replace DS diluent containers, refer to 12.5 When Two DS Diluent Containers are Connected at the Same Time.
7	DS Diluent inlet A	Connect the DS Diluent container.	
8	Pressure interface	/	/
9	Vacuum interface	/	/
10	LH Lyse inlet	Connect the LH Lyse container.	/
11	LD Lyse inlet	Connect the LD Lyse container.	/
12	LN Lyse inle	Connect the LN Lyse container.	/
13	ESR Solution Reagent inlet	Connect the ESR Solution Reagent container.	/
14	Waste sensor interface	Connect the cap assembly of the waste container.	- If waste is discharged into a container, connect the floater sensor to the interface to detect whether the waste container is full. - If waste is discharged directly, the interface does not need to be connected.

3.4.4 Right View of the Analyzer

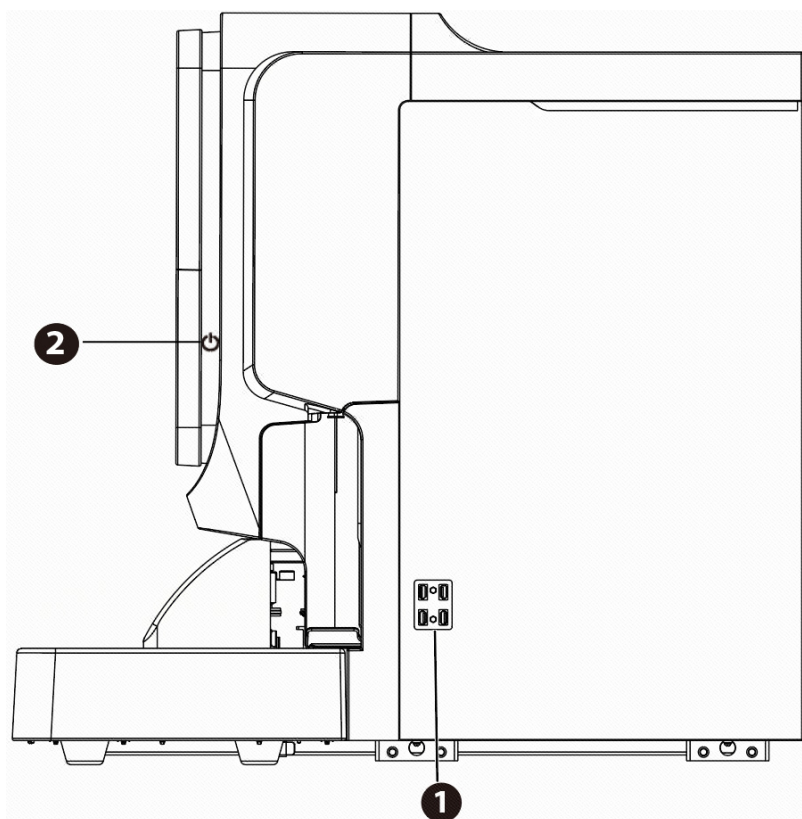


Figure 3-3 Right view of the analyzer

Table 3-13 Description of the right view of the analyzer

	Name	Function	Description
1	USB interfaces	Connect USB, external barcode scanner and etc.	- There are four USB interfaces on the right of the analyzer. - For USB interface specifications, refer to B.8 Electronic Interface Specifications.
2	[Standby] key	Press to turn on/off the power supply of the analyzing unit.	/

3.4.5 Left View of the Analyzer

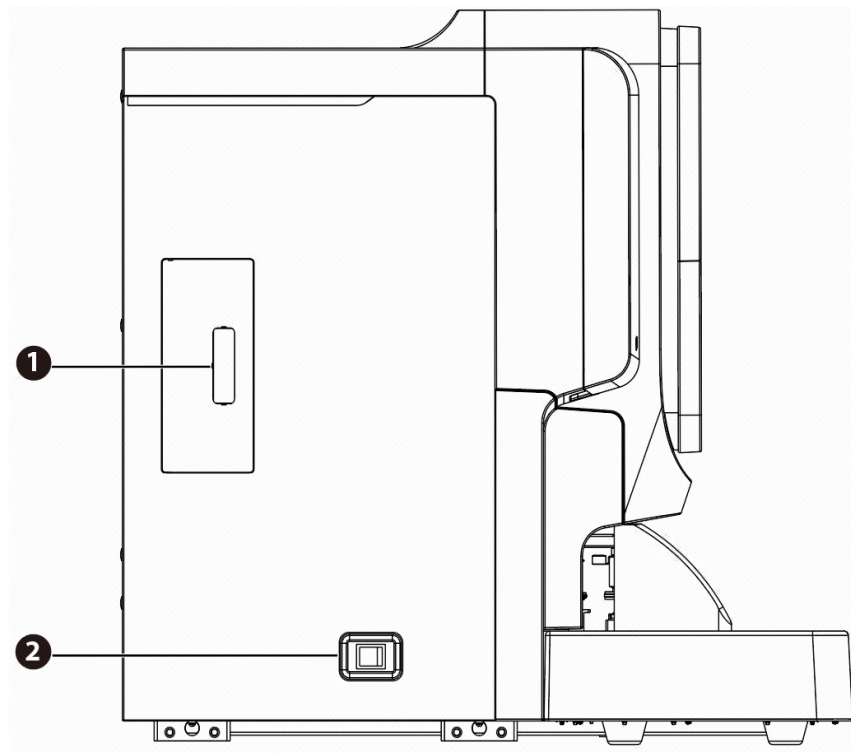


Figure 3-4 Left view of the analyzer

Table 3-14 Description of left view of the analyzer

	Name	Function	Description
1	Valve module lock	/	/
2	Power switch	You can use the power switch to start up or shut down the analyzer.	/

3.4.6 Operator Panel

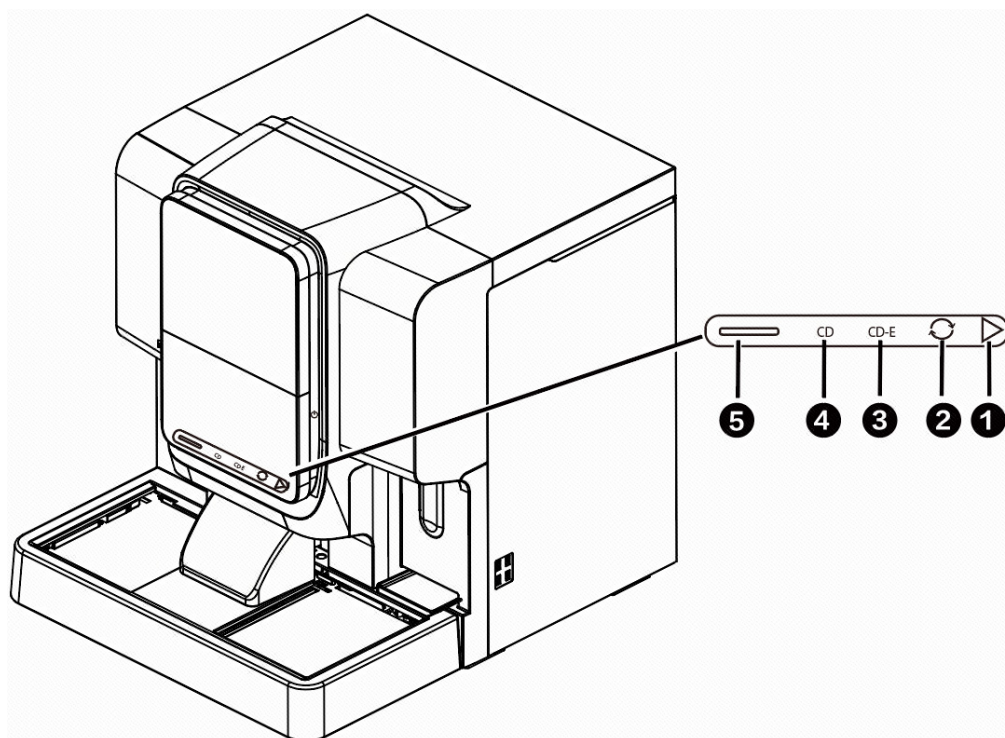


Figure 3-5 Operator panel of the analyzer

Table 3-15 Description of operator panel of the analyzer

	Name	Functions	Description
1	[Count] key	Press the key to start analysis in open-vial sample presenting mode; or to start to dispense diluent when preparing predilute samples.	/
2	[Sampling mode] switch key	Press the key to quickly switch between OV (open-vial) sampling mode and AL (auto-loading) mode.	/
3	CD-E key	Press the key to switch to the CD+ESR mode.	This key lights up after the analyzer is switched to the CD+ESR mode.
4	CD key	Press the key to switch to the CD mode.	This key lights up after the analyzer is switched to the CD mode.
5	Status indicator	The indicator indicates the analyzer status.	• Lighting in steady blue: the analyzer is ready for tests. • Flicker in blue: the analyzer is running. • Lighting in steady red: the analyzer encounters error. • Off: the analyzer is powered off.

3.5 Accessories List of the Analyzer

The analyzer has the following configured or optional accessories.

	Configured	Optional
Sample racks	✓	
Reagent container cap assembly	✓	
Waste tube assembly	✓	
Main power cords	✓	
External barcode scanner		✓
Autoloader built-in barcode scanner		✓
Printer		✓
Sample transportation track		✓

CAUTION

- **Only use the accessories and consumables manufactured or recommended by our company to achieve the promised system performance and safety. For more information, contact our Customer Service Department or your local distributor.**
- **Use the manufacturer-provided power cord, which matches the power supply of the instrument. Using the power cord other than provided by the manufacturer may damage the analyzer or lead to unreliable analysis results.**

NOTE

- **For any questions about the configured/optional accessories, consult your sales representative.**
- **The accessories actually provided with the product depend on your product configuration.**

3.6 Sample Tubes and Sample Racks

Applicable sample tube size: 11-13mm (external diameter)×65-78mm/80-83mm (height without cap/height with cap)

For the open-vial sampling mode, the height of sample tube without cap should be no larger than 85mm.

3.7 Software System

NOTE

- **For any problems not mentioned in this manual, contact our Customer Service Department.**
- **This system is only intended for use by clinical laboratory professionals, doctors and testers trained by our company or your local distributor.**

The analyzer software is "Mindray's Auto Hematology Analysis Software". The Auto Hematology Analyzer is a quantitative, automated hematology analyzer for In Vitro Diagnostic Use in clinical laboratories; it provides Complete Blood Count, Leukocyte 5-part Differential, Hemoglobin Concentration Measurement, Reticulocyte and Nucleated Red Blood Cell Measurement, and Erythrocyte Sedimentation Rate (ESR) Measurement for blood samples, as well as Body Fluid analysis.

ESR can be used as aids to diagnosis of inflammation diseases.

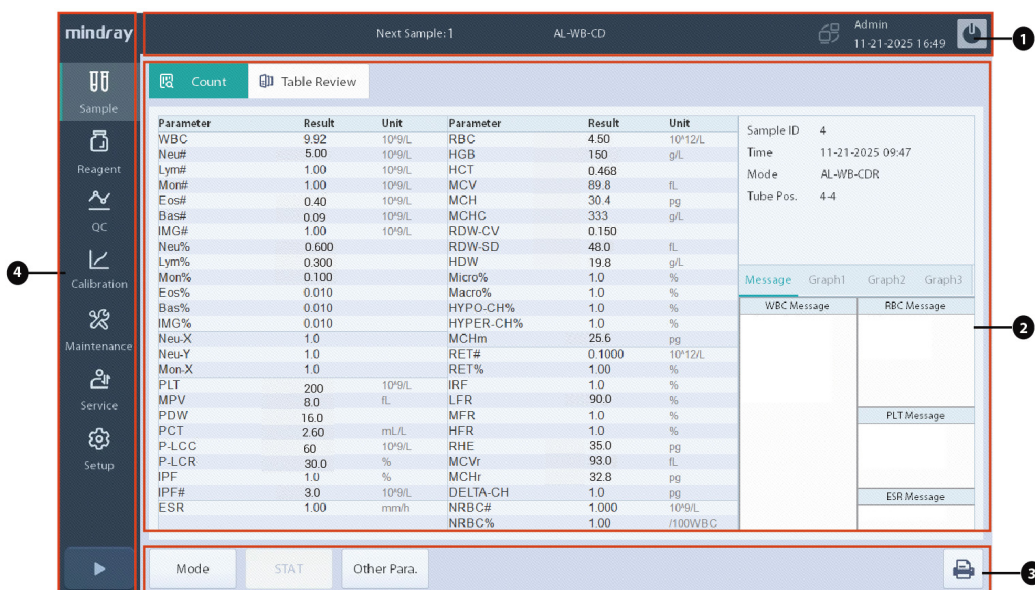


Figure 3-6 Software interface

Table 3-16 Software interface introduction

	Name	Functions	Description
1	Auxiliary information area	Display the auxiliary information related to the current screen.	- Display the current system time. - For example, the “Count” screen displays the next sample ID and test mode.
2	Operation area	Display contents of the screen.	/
3	Function button area	Display function buttons according to different screens.	For example, the “Count” screen displays function buttons related to sample analysis as well as the sample analysis results.
4	Menu	Select an menu item to enter the corresponding screen.	/

3.7.1 Functionality

NOTE

- When the maximum number has been reached, the newest result will overwrite the oldest. Be sure to backup important data to an external storage medium on a regular basis.

3.7.1.1 Overview of software functions

The software supports the following operations.

Starting up and shutting down

The analyzer supports functions of analyzer startup/shutdown and user login/logout.

Sample analysis

The analyzer supports sample analysis and outputs the test results.

Data management

The analyzer supports the function of storing, reviewing, searching, validating and exporting the test results.

Quality Control

The analyzer provides at least two methods for quality control.

Calibration

The analyzer provides at least two methods for calibration.

Automatic maintenance

The analyzer can support probe cleanser maintenance function and standby function depending on the actual use situation.

Manual maintenance

The analyzer supports the fluidics maintenance function such as replacing reagent, cleaning, emptying and priming.

Information prompt

The analyzer provides error prompt, operation prompt and displays analyzer status.

Error reporting and troubleshooting

The analyzer prompts/gives alarm for abnormal status of the analyzer and supports removing errors.

Communicating with LIS

The analyzer can communicate with LIS by data transmission interface.

Data printing

The analyzer supports data and report printing function.

Data inputting

The analyzer supports information inputting by keyboard, barcode scanner or USB driver.

Keeping logs

The analyzer keeps logs that record operations and errors for users to export.

Interface navigation

The analyzer supports interface navigation function that allows the user to switch the interface easily.

Providing histogram

The analyzer provides RBC and PLT histogram.

Providing scattergram

The analyzer provides differential scattergram.

Sample mode

The analyzer supports the sample analysis in WB mode, PD mode and BF mode.

Test report parameter

The analyzer outputs the test report parameters.

Body fluid sample type

The analyzer supports the subtypes of body fluid including cerebrospinal fluid, pleural fluid, ascitic fluid and synovial fluid samples.

ESR measurement

The user can select the ESR function or clear the selection of ESR function.

Self-test function

The analyzer supports self-test function.

Sampling function

The analyzer has two sample presentation modes: OV mode (Open-vial mode) and AL mode (autoloading mode).

Restrictions on use

- Restrictions on user: the password length for the user to access screen should be set within 12 characters.
- Technical limitation: the actual storage capacity of sample records should not be lower than claimed.

3.7.1.2 Software functions by access levels

The analyzer has following functions.

Table 3-17 Software functions by access levels

Level 1 Menu	Level 2 Menu	Level 3 Menu	Functions	Administrator's Level	Operator's level
Sample	Count	/	Mode	√	√
			STAT	√	√
			Other Para.	√	√
	Table Review	/	Search	√	√
			Graph	√	√
			CV	√	√
			Edit Info.	√	×
			Delete	√1.	×
Reagent	Reagent Overview	/	Replace Reag.	√	√
	Replacement Records	/	/	√	√
QC	Overview	L-J QC	Setup	√	√
			Count	√	√
			L-J QC Table	√	√
			L-J QC Graph	√	√
		X-B QC	Setup	√	√
			X-B QC Graph	√	√
			X-B QC Table	√	√
Calibration	Calibration Analysis	Manual	/	√	√
		Calibrator	/	√	×
			/	√	×
		Fresh Blood	/	√	×
			/	√	×
	Calibration History	Details	/	√	×
		Go to	/	√	×
		Export	/	√	×
Service	Debug & Self-Test	Self-Test	/	√	√
		Touchpad Cal.	/	√	√
	Version Info.	More	/	√	√
	Status	Statistics	Detail	√	√

Level 1 Menu	Level 2 Menu	Level 3 Menu	Functions	Administrator's Level	Operator's level
		Temp.&Pressure	/	√	×
		Floater Status	/	√	×
		Sensor	/	√	×
		Voltage & Current	/	√	√
	Log	All Logs	Details	√	√
			Go to	√	√
			Export	√	√
		General Logs	Details	√	√
			Go to	√	√
			Export	√	√
		Setup Adjustment	Details	√	×
			Go to	√	×
			Export	√	×
		Error Information	Details	√	×
			Go to	√	×
			Export	√	×
Maintenance	Probe Cleanser Maint.	/	/	√	√
	Maintenance	Fluidics	/	√	√
		Flow Cell	/	√	√
		SRV	/	√	√
		Apertures	/	√	√
		Unclog Aperture	/	√	√
	Cleaning	Clean Fluidics	/	√	√
		Remove Bubble	/	√	√
		Flow Cell Flush	/	√	√
	Fluidics	/	Pack-up	√	√
	Reagent	/	Replace Reagent	√	√
			Prime Reagent	√	√
Setup	Basic Settings	Lab Setup	User Management	√	√
			Lab Info. Setup	√	√

Level 1 Menu	Level 2 Menu	Level 3 Menu	Functions	Administrator's Level	Operator's level
			Auto Startup/Shutdown	√	√
			Date/Time Setup	√	√
		System Setup	Print Setup	√	√
			Reagent Setup	√	√
			Communication	√	√
			Gain Setup	√	√
			Auxiliary Setup	√	√
	Other Settings	Loading Setup	Barcode	√	√
			Auto-loading	√	√
			Manual Sampling	√	√
		Test Setup	Ext. WBC/PLT Tests	√	√
			Flag Alarm Sensitivity	√	√
			Flag Rules Setup	√	√
		Para. Setup	Parameter Unit Setup	√	√
			Ref. Range Setup	√	√
			Microsco. Para. Setup	√	√
		Maintenance	Daily Maint.	√	√

NOTE

- "√" indicates that the access level supports the corresponding function, while "x" indicates that the access level does not support corresponding the function.

3.7.2 Software version

Software Version: 01

3.7.3 Performance and Efficiency

For more information about the maximum sample records, refer to **8 Reviewing Sample Results**.

3.7.3.1 Minimum configuration requirements

- Hardware configuration: processor main frequency is no less than 1G Hz, the storage capacity is no less than 4G.
- Display resolution: 1280 × 800.
- Software environment: Linux
- Network: LAN

3.7.4 Compatibility

For requirements of input/output devices, refer to **B.6 Input/Output Devices**.

The software of the analyzer is developed based in the following Off-the-Shelf software.

Table 3-18 Off-the-shelf software

Name	Model	Version	Software	Manufacturer
Operating system	OS Core	3.18.24	Linux embedded operating system	Linux Kernel Organization, Inc

3.7.5 Usability

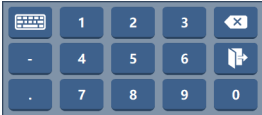
The analyzer is intended to be used by skilled and trained operators.

3.7.5.1 Software controls and basic operations

Table 3-19 Description of software operations

Action	Meaning
Select	Select the options and focus an input entry.
Slide	Slide your finger up/down/left/right on the touch screen to page up/down/left/right.

Table 3-20 Description of software controls of the main unit

Controls	Examples	Meaning
Button		Select the button to perform operations or open a dialog box.
Pop-up keyboard		On the main unit software, select the input box, and a keyboard appears. - The pop-up keyboard appears as a numeric keypad or a QWERTY keypad. - The numeric keypad is used for digit input, and the QWERTY keypad is used for entering all characters, including digit. - On the numeric keypad, select  to switch to the QWERTY keypad. - On the QWERTY keypad, select  to switch to the numeric keypad. - On the numeric keypad, select  to close the keypad.
Input boxes		Select the input box and the pop-up keyboard appears. Then you can enter desired information.
Radio button	When selected: 	When there are a set of choices arranged with radio buttons, only one choice can be selected.
	When not selected: 	/

Controls	Examples	Meaning
Check box		When there are a set of choices arranged with check boxes, multiple options can be selected.
		/
Pull-down list		Specify an option from the pull-down list.
Tab		Select tab to enter the corresponding tab.

3.7.6 Reliability, Known Limitations, and Precautions

After each analysis, the analysis results are stored to the database automatically.

When the maximum number has been reached, the newest result will overwrite the oldest. You must back up the important data into external storage device regularly.

Important parameters of the software can be restored to factory value. You can restore parameters on the parameter setup screen. For any question, contact our Customer Service Department.

3.7.7 Portability

Software installation, connection, and configuration should be performed by our after-sales service personnel.

NOTE

- For minimum configuration, refer to 3.7.3.1 *Minimum configuration requirements*.

3.7.8 Validity

For communicating with laboratory information management system (LIS), software supports specific communication protocols. For more information, refer to **D Transmission**.

3.7.9 Efficiency

The product is verified to allow at least one end user to simultaneously access the analyzer software.

3.7.10 Satisfaction

For feedback and questions about the software, contact our Customer Service Department.

3.7.11 Freedom from Risk

Follow the instructions below to safeguard device stability and data security.

CAUTION

- Be sure to use the specified devices only.
- The use of mobile storage devices and computer for unintended purposes may result in risks such as virus attack and information leakage.
- The user should ensure the data safety of the USB devices connecting to the analyzer.

- **Accessory equipment connected to the analogue and digital interfaces must be complied with the relevant Safety and EMC standards (e.g., IEC 60950 Safety of Information Technology Equipment Standard and CISPR 22 EMC of Information Technology Equipment Standard (CLASS B)). Any person, who connects additional equipment to the signal input or output ports and configures an IVD system, is responsible for ensuring that the system work normally and complies with the safety and EMC requirements. If you have any problem, consult the technical services department of your local representative.**

Ensure the following requirements are met to ensure network security.

NOTE

- **The analyzer software uses a closed-loop operating system, meaning that the applications of the analyzer works in exclusive mode and are free from other application's disturbance. Users can only operate the software interface but cannot directly access the operating system or install software.**
- **Data transmission must be performed in a closed network or virtual network. The network must be isolated.**
- **The installation, authorization, upgrade and modification of the system software must be performed by our personnel authorized. Make sure to install only our company authorized software.**
- **When the product connects to an external computer, install anti-virus software on the computer and scan for viruses and update patches periodically.**
- **When the product connects to an external computer, install the original edition software to prevent the computer from being infected by virus.**
- **Users have the responsibility to protect the network authentication information, such as password and user information, from being obtained by unauthorized personnel.**

3.7.12 Context Coverage

For communicating with laboratory information management system (LIS), software supports specific communication protocols. For more information, refer to **D Transmission**.

3.7.13 Input and Output Data

Input data: patient information, software commands;

Output data: analysis results (including parameters and graphs)

3.8 Reagents, Controls and Calibrators

As the analyzer, reagents, controls and calibrators are components of a system, performance of the system depends on the combined integrity of all components. It is recommended that the operator use the Mindray-specified reagents (see **B.1 Reagents**) which are formulated specifically for the fluidic system of your analyzer in order to provide optimal system performance. Do not use the analyzer with reagents from multiple suppliers. In such use, the analyzer may not meet the performance specified in this manual and may provide unreliable results. All references related to reagents in this manual refer to the reagents specifically formulated for this analyzer.



BIOHAZARD

- **All the samples, controls, calibrators, wastes and areas contacted them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab coat, goggles, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory. Wear proper personal protective instrument (e.g. gloves, lab coat, glasses, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.**
- **Be sure to dispose of reagents, waste, samples, consumables, etc. according to government regulations.**

CAUTION

- Use the reagents, controls and calibrators specified by the manufacturer only. Using other reagents, controls and calibrators, may lead to system damage and inaccurate measurement, control and calibration results.

NOTE

- For details about the models, intended use, test principles, main components, storage conditions, expiration dates and applicable instruments of the reagents, controls and calibrators, please refer to the instructions of the reagents, controls and calibrators.
- The reagents should be properly used and stored as per the corresponding instructions.
- For any doubts on the reagents, controls and calibrators, consult Customer Service Department.

3.8.1 Reagent

The reagents used with the analyzer are listed below.

Table 3-21 Reagents

	Name
HGB channel	LH Lyse
WNB channel	LN Lyse
	FN Dye
DIFF channel	LD Lyse
	FD Dye
RET channel	DR diluent
	FR Dye
ESR channel	ESR Solution Reagent
Diluent	DS Diluent
Components cleaning	Probe Cleanser

3.8.2 Controls and Calibrators

The controls and calibrators are used for the analysis quality control and calibration of the analyzer.

The controls are suspension of stimulated human blood, specifically manufactured to monitor and evaluate the analysis precision of the analyzer. The controls are prepared with three levels, namely low, normal and high. Daily use of all levels verifies the operation of the analyzer and ensures reliable results are obtained. The calibrators are commercially prepared whole-blood products used to calibrate some parameters (WBC, RBC, HGB, MCV and PLT etc.) of analyzer to build the metrological traceability of analysis results. For the use and storage of controls and calibrators, please refer to the Instruction for Use of each product.

All references related to “controls” and “calibrators” in this manual refer to the controls and calibrators specifically formulated for this analyzer by Mindray. You must buy those controls and calibrators from Mindray or Mindray-authorized distributors.

For the controls and calibrators used with the analyzer, refer to the table below:

Table 3-22 Controls and calibrators

	Name
Quality control for routine blood tests except for RET tests	BC-6D Hematology Control (High, Low, and Normal Levels) BR60 Hematology Control (High, Low, and Normal Levels)
Quality control for RET tests	BC-RET Hematology Control (High, Low, and Normal Levels)
Quality control for ESR tests	BC-6D Hematology Control (Low and Normal Levels)
Quality control for body fluid tests	BC-BF Hematology Control (High, Normal and Low Levels)
Calibration for routine blood tests	SC-CAL PLUS Hematology Calibrator
Calibration for ESR tests	SC-CAL PLUS Hematology Calibrator

4 Understanding the System Principles

4.1 Introduction

The principles used by the analyzer for measurement are:

- sheath flow impedance method, laser scatter and SF Cube cell analysis technology (3D analysis using information from scatter of laser light at two angles and fluorescence signals) for cell differentiation and counting;
- the colorimetric method for HGB measurement.

4.2 WBC Measurement

4.2.1 Laser Flow Cytometry + SF CUBE Cell Analysis Technology

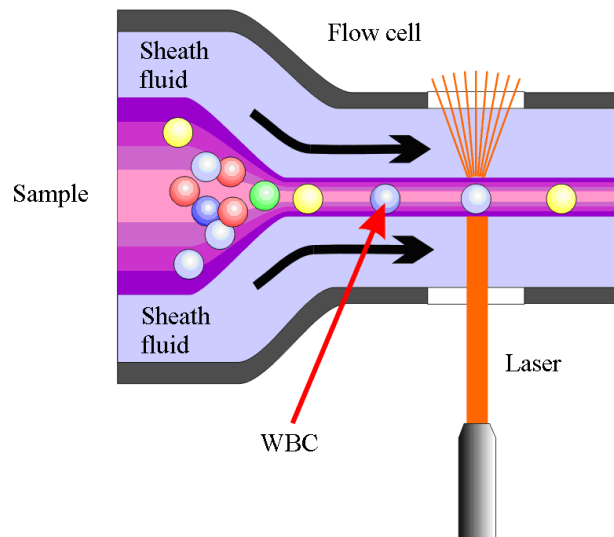


Figure 4-1 Laser flow cytometry

In normal peripheral blood, white blood cells can be classified into 5 categories: lymphocytes, monocytes, neutrophils, eosinophils and basophils. Analyzing all types of white blood cells will provide a great deal of useful information for the clinical diagnosis of diseases. Under the influence of certain diseases, the peripheral blood may contain various abnormal cells apart from the five subpopulations of normal cells, such as atypical lymphocytes, immature cells, etc. Most of these abnormal cells are different kinds of immature cells in the cell generation process. But what they have in common is they contain a great deal of nucleic acid (DNA and RNA), the content of which decreases as the cell gets maturer. Therefore, normal cells and immature cells can be differentiated by detecting the content of nucleic acid in the cells.

Body fluid refers to the fluid in side body cavities except blood vessels. There are many sub-types of body fluid, among which the most commonly seen sub-types are cerebrospinal fluid, serous cavity fluid (pleural fluid and ascitic fluid) and synovial fluid. Both cerebrospinal fluid and serous cavity fluid are colorless and transparent in normal case, but in abnormal cases, there could be increase of cells (including leukocytes and erythrocytes). Leukocytes in body fluid can be categorized into mononuclear cells (MN) and polymorphonuclear cells (PMN). The analysis of the cells in body fluid can provide useful information for clinical diagnosis.

This analyzer adopts the SF Cube cell analysis technology to recognize and detect the immature cells in blood accurately besides doing WBC 5-part differentiation, as well as identify the nucleated cells in body fluid.

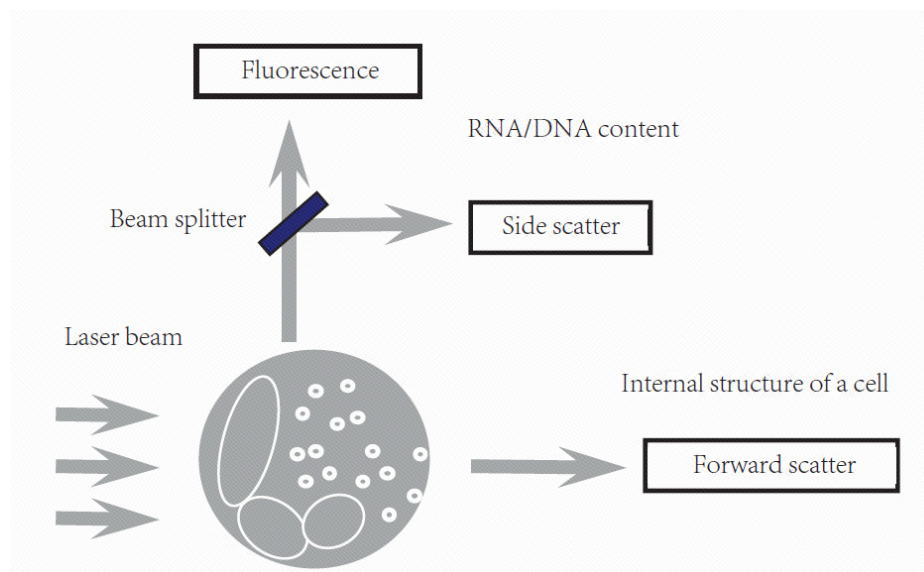


Figure 4-2 SF Cube Technology

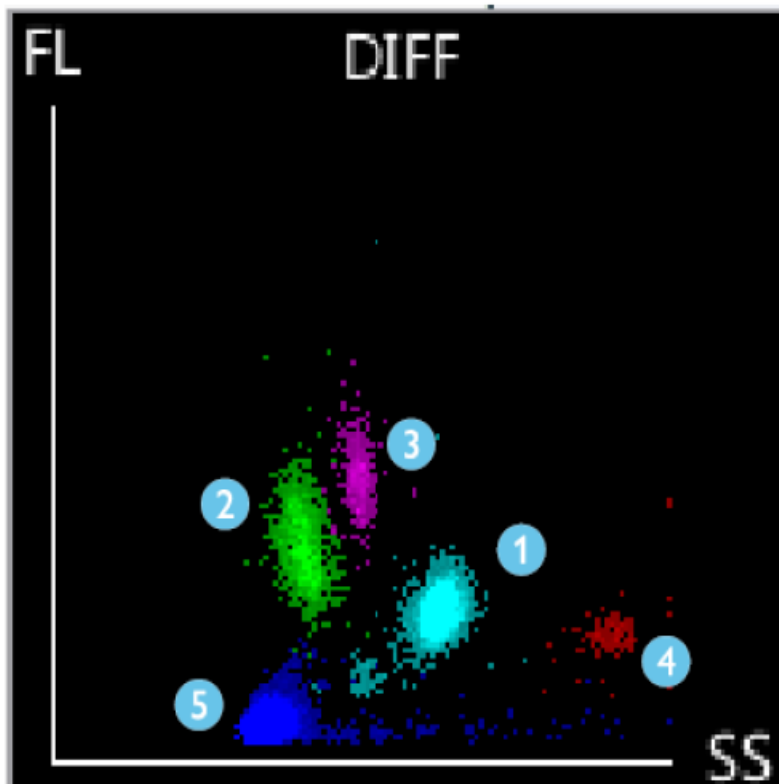
The analyzer adopts the SF Cube cell analysis technology in its DIFF and WNB channels. The RBCs are lysed and the WBC subpopulations are made different in size and complexity by the lyse; the nucleic acid substances in WBCs are marked by the new asymmetric cyanine fluorescent substance. Due to the different content of nucleic acid in different WBC subpopulations, maturity stages or abnormal development status, the volume of fluorescent dye staining the nucleic acid substances can be different; The low-angle light scatter reflects cell size, the high-angle light scatter reflects intracellular granularity, and the intensity of fluorescent signal reflects the degree that the cell is stained. By sensing the difference in signal in three dimensions of the cells processed with lyse, the DIFF channel differentiates the subpopulations of WBCs (lymphocytes, monocytes, neutrophil and eosinophils), as well as identifies and flags abnormal cells like immature granulocytes, abnormal lymphocytes and blast cells. The WNB channel tests basophils and nucleated red blood cells and counts the WBCs through different reagents.

The lymphocytes are smaller in size with the nucleus taking most part of them. Lymphocytes have a high nucleus-to-cytoplasm ratio, but their nucleic acid content is low, therefore they are at a lower position in the direction of fluorescence and side scatter. The monocytes are larger in size, with high nucleus-to-cytoplasm ratio and high nucleic acid content, and less complex in structure, therefore they are at a higher position in the direction of fluorescence, and have stronger side scatter. The neutrophils and basophils are larger in size, and have medium nucleus-to-cytoplasm ratio and low nucleic acid content, therefore they are at a lower position in the direction of fluorescence, but they have stronger side scatter. The characteristics of the eosinophils are similar to those of the neutrophils, but they contain a lot of alkaline grains, so they have very strong side scatter. The blast cells, atypical lymphocytes and immature granulocytes have high nucleic acid content, so they are at a higher position in the direction of fluorescence on the scattergram.

In body fluid samples, the mononuclear cells (MN) are less complex in intracellular granularity, so the side scatter is weaker, while polymorphonuclear cells are more complex in intracellular granularity, so the side scatter is stronger.

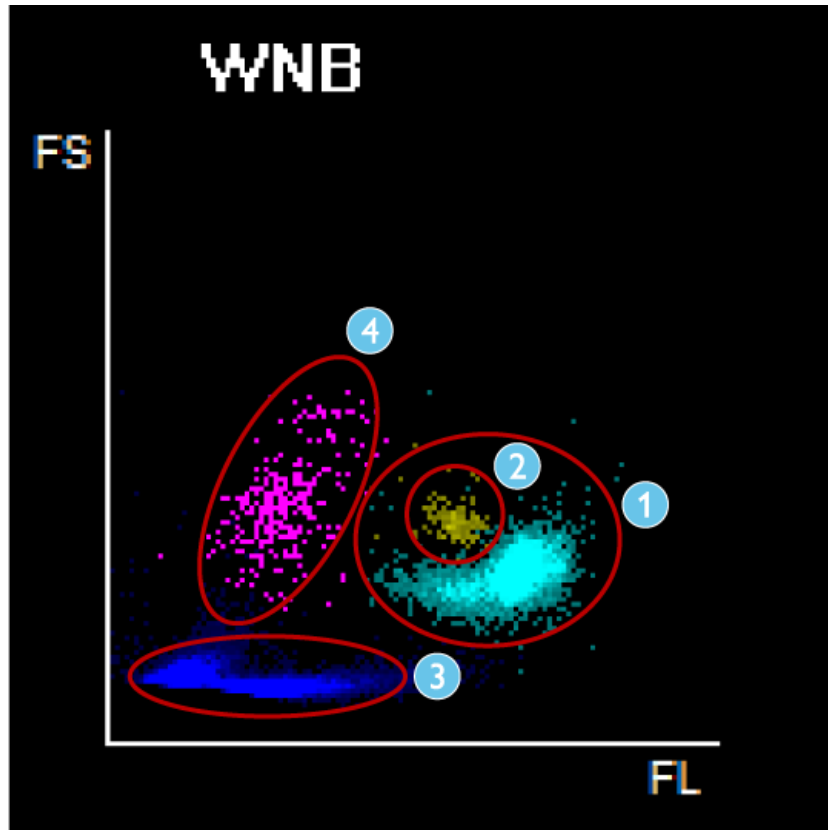
4.2.2 Derivation of WBC-Related Parameters

DIFF Scattergram



1. Neu and Bas area
2. Lym area
3. Mon area
4. Eos area
5. Ghost area

WNB Scattergram



- 1. WBC area
- 2. Bas area
- 3. Ghost area
- 4. NRBC area

Parameter	Name	Formula/Test Methods	Unit
WBC	WBC count	WBC = Sum of all particles in the WBC region in the WNB channel	10 ⁹ /L
Bas#	Basophils number	Bas# = Sum of all particles in the BASO region in the WNB channel	10 ⁹ /L
Bas%	Basophils percentage	$\text{Bas\%} = \frac{\text{Bas\#}}{\text{WBC}} \times 100\%$	%
Neu#	Neutrophils number	Neu# = WBC x Neu%	10 ⁹ /L
Neu%	Neutrophils percentage	$\text{Neu\%} = \frac{\text{Particles in the Neu region in the DIFF channel}}{\text{Sum of all particles in DIFF channel except those in Ghost region}} \times 100\%$	%
Eos#	Eosinophils number	Eos# = WBC x Eos%	10 ⁹ /L
Eos%	Eosinophils percentage	$\text{Eos\%} = \frac{\text{Particles in the Eos region in the DIFF channel}}{\text{Sum of all particles in DIFF channel except those in Ghost region}} \times 100\%$	%
Lym#	Lymphocytes number	Lym# = WBC x Lym%	10 ⁹ /L
Lym%	Lymphocytes percentage	$\text{Lym\%} = \frac{\text{Particles in the Lym region in the DIFF channel}}{\text{Sum of all particles in DIFF channel except those in Ghost region}} \times 100\%$	%
Mon#	Monocytes number	Mon# = WBC x Mon%	10 ⁹ /L
Mon%	Monocytes percentage	$\text{Mon\%} = \frac{\text{Particles in the Mon region in the DIFF channel}}{\text{Sum of all particles in DIFF channel except those in Ghost region}} \times 100\%$	%
IMG#	Immature Granulocyte number	IMG# = WBC x IMG%	10 ⁹ /L
IMG%	Immature Granulocyte percentage	$\text{IMG\%} = \frac{\text{Particles in the IMG region in the DIFF channel}}{\text{Sum of all particles in DIFF channel except those in Ghost region}} \times 100\%$	%
Neu-X	DIFF scattergram, mean neutrophil distribution-side scatter intensity	Measured based on the DIFF scattergram; it is the mean side scatter light intensity of the neutrophils area in the DIFF channel	—
Neu-Y	DIFF scattergram, mean neutrophil distribution-side fluorescent light intensity	Measured based on the DIFF scattergram; it is the mean fluorescent light intensity of the neutrophils area in the DIFF channel	—
Mon-X	DIFF scattergram, mean monocyte distribution-side scatter intensity	Measured based on the DIFF scattergram; it is the mean side scatter intensity of the monocytes area in the DIFF channel	—

4.3 HGB Measurement

4.3.1 Colorimetric Method

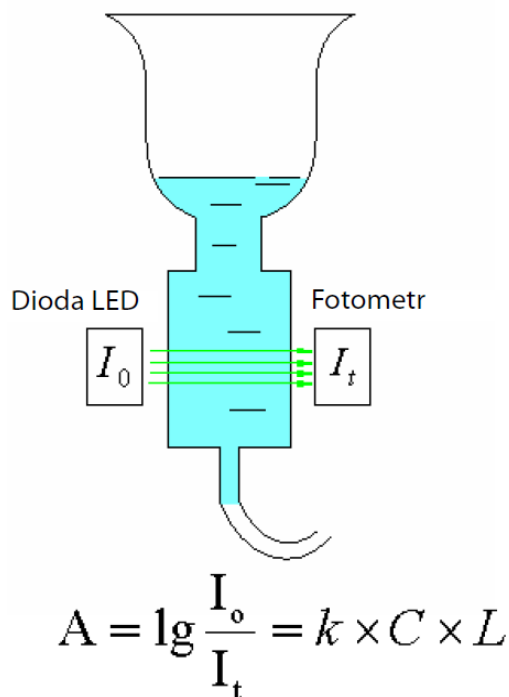


Figure 4-3 Colorimetric Method

According to the Lambert-Beer Principle, when a beam of monochromatic light passes through a well-proportioned non-scattering light-absorbing solution, the absorbance A is proportional to the product of the thickness L and the concentration C . The sample in the HGB channel acts as the light absorbing substance after being treated by reagent, therefore the HGB concentration can be measured by measuring the absorbance.

4.3.2 Derivation of HGB-Related Parameters

The HGB is calculated per the following equation and expressed in g/L.

Parameter	Name	Formula/Test Methods	Unit
HGB	Hemoglobin Concentration	$\text{HGB} = \text{Constant} \times \ln \left(\frac{\text{Blank Photocurrent}}{\text{Sample Photocurrent}} \right)$	g/L

4.4 RBC/PLT Measurement

4.4.1 Sheath Fluid Impedance Method

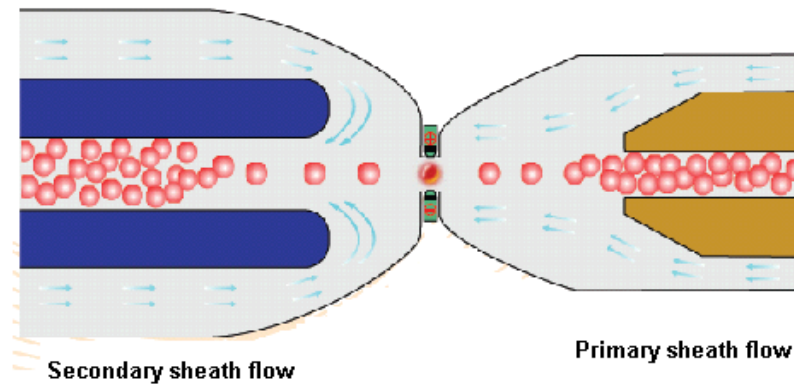


Figure 4-4 Sheath Fluid Impedance Method

This analyzer uses the sheath impedance method to count RBC/PLT. Under the action of "focusing" of the fluid, the RBC and PLT line up to pass through the aperture of the gemstone and generate corresponding pulses according to the Coulter principle. The backend processing unit amplifies the collected electrical pulses, compares them with the channel voltage threshold corresponding to the normal RBC/PLT volume range, and calculates the number of electrical pulses whose amplitudes fall in the RBC/PLT channel. All the collected pulses are thus classified based on the reference voltage thresholds of different channels, and the number of the pluses in the RBC/PLT channel indicates the number of the RBC/PLT particles. The number of cells in each channel range divided according to the pulse voltage amplitude determines the cell volume distribution. The 2-dimensional distribution diagram (scattergram), with the X-axis representing cell volume, and Y-axis representing relative cell numbers, shows the distribution of cell population.

Compared with the common simple impedance method, the sheath fluid impedance method has higher test efficiency, less reagent consumption, better signal quality, and more accurate test results.

4.4.2 SF CUBE Cell Analysis Technology

The RET channel also adopts the SF CUBE Cell Analysis Technology. The general measurement principle in RET channel is similar to that of WNB and DIFF channels, only that in the RET channel, the RBCs are not lysed, but are spherized by RET diluent. Then the nucleic acid of the spherized RBCs and the PLTs are stained by fluorescent dyes.

4.4.3 Measurement Principle of Platelets in DIFF Channel (PLT-H)

Traditional impedance method counts platelets by sorting and detecting the pulses fall in the PLT channel per cell size. However, as microcytes and fragments are similar in size with platelets with relatively large size, when there are microcytes/fragments present, the impedance method may deliver not so accurate PLT results. To solve the problem, the analyzer lyses the erythrocytes in the DIFF channel, so the count of platelets in relatively large sizes in DIFF channel will not be affected by erythrocytes. By combining the large-size platelet count result in DIFF channel and the small-size platelet count result in the impedance channel, the analyzer provides more accurate PLT results.

4.4.4 RBC-Related Parameters

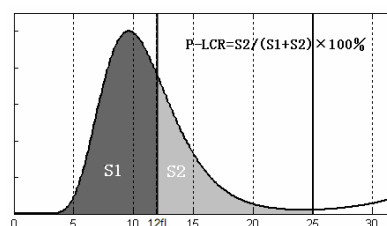
Parameter	Name	Formula/Test Methods	Unit
RBC	Red Blood Cell count	The analyzer provides the number of red blood cells (RBC) directly by counting the red blood cells passing through the aperture.	$10^{12}/L$
MCV	Mean Corpuscular Volume	Calculated based on the red blood cell histogram	fL
HCT	Hematocrit	$HCT = \frac{RBC \times MCV}{10}$	%
MCH	Mean Corpuscular Hemoglobin	$MCH = \frac{HGB}{RBC}$	pg
MCHC	Mean Corpuscular Hemoglobin Concentration	$MCHC = \frac{HGB}{HCT} \times 100$	g/L
RDW-CV	Red Blood Cells Distribution Width - Coefficient of Variation	RBC Histogram	%
RDW-SD	Red Blood Cells Distribution Width - Standard Deviation	Standard deviation of RBC distribution	fL
NRBC%	Nucleated Red Blood Cell percentage	$NRBC\% = \frac{\text{Particles in the NRBC region of the WNB channel}}{\text{Sum of all particles in the WBC region of the WNB channel}} \times 100\%$	%
NRBC#	Nucleated Red Blood Cell count	$NRBC\# = WBC \times NRBC\%$	$10^9/L$
MCHm	Mean corpuscular hemoglobin of mature red blood cell	Calculated based on the hemoglobin distribution of mature red blood cells in the RET channel.	pg
HDW	Mean hemoglobin distribution width of red blood cell	Calculated based on the hemoglobin distribution of red blood cells in the RET channel.	g/L
HYPO-CH%	Percentage of mature red blood cell with a low corpuscular hemoglobin	Calculated based on the hemoglobin distribution of mature red blood cells in the RET channel. It is the ratio of the number of mature red blood cells whose hemoglobin is less than 27 pg to the total number of mature red blood cells.	%
HYPER-CH%	Percentage of mature red blood cell with a high corpuscular hemoglobin	Calculated based on the hemoglobin distribution of mature red blood cells in the RET channel. It is the ratio of the number of mature red blood cells whose hemoglobin is more than 31 pg to the total number of mature red blood cells.	%
Micro%	Microcyte percentage	Calculated based on the red blood cell histogram. It is the ratio of the number of red blood cells whose volume is less than 60 fL to the total number of mature red blood cells.	%
Macro%	Macrocyte Percentage	Calculated based on the red blood cell histogram. It is the ratio of the number of red blood cells whose volume is more than 120 fL to the total number of mature red blood cells.	%

4.4.5 PLT-Related Parameters

Parameter	Name	Formula/Test Methods	Unit
PLT	Platelet count	- For analyzers configured with only one PLT test report parameter, platelet number (PLT) is measured directly by counting the platelets passing through the aperture. - For analyzers configured with PLT-I, PLT-H and PLT-O test report parameters, the analyzer offers different technologies/methods to report PLT results. - The PLT count may come from PLT-I, PLT-H and PLT-O channel based on the analyzer configuration or user-defined test panel.	$10^9/L$
MPV	Mean Platelet Volume	Mean Platelet Volume (MPV) is calculated based on the PLT histogram.	fL
PDW	Platelet Distribution Width	Platelet distribution width is derived from the platelet histogram, and is reported as 10 geometric standard deviation (10 GSD).	/
PCT	Plateletcrit	$PCT = \frac{PLT \times MPV}{10000}$	%
P-LCR	Platelet-large cell ratio	P-LCR is derived from the platelet histogram, it represents the ratio of the number of platelets with a size over 12fL to the total platelets number.	%
P-LCC	Platelet-large cell count	$P-LCC = PLT \times P-LCR$	$10^9/L$
IPF	Immature Platelet Fraction*	Derived based on PLT-O scattergram: $IPF = \frac{\text{Liczba niedojrzałych płytek krwi w kanale optycznym}}{\text{Suma wszystkich cząstek płytkowych w kanale optycznym}} \times 100\%$	%
IPF#	Immature Platelet Count	Calculated based on the optical platelet scattergram.	$10^9/L$
PLT-I	Platelet count-Impedance	Platelet number (PLT) is measured directly by counting the platelets passing through the aperture.	$10^9/L$
PLT-H	Platelet count-hybrid	PLT results derived from impedance channel, which are corrected by the DIFF channel large size platelet results.	$10^9/L$
PLT-O	Optical count platelet	PLT results derived from the RET channel	$10^9/L$

NOTE

- Platelet-Large Cell Ratio (P-LCR) is derived from the platelet histogram; it represents the ratio of the number of platelets with a size over 12fL to the total platelets number. The ratio is represented in %. In the following figure, S2 represents the number of larger platelet cells, and S1+S2 represents the total PLT count.**



4.4.6 Reticulocyte Parameters

Parameter	Name	Formula/Test Methods	Unit
RET%	Reticulocyte percentage	$\text{RET}\% = \frac{\text{Number of cells in the reticulocyte region}}{\text{Number of cells in mature RBC region} + \text{Number of cells in RET region}} \times 100\%$	%
RET#	Reticulocyte number	$\text{RET}\# = \text{RBC} \times \text{RET}\%$	$10^{12}/\text{L}$
HFR	High Fluorescent Reticulocyte Ratio	$\text{HFR} = \frac{\text{Number of cells in HFR region}}{\text{Number of cells in RET region}} \times 100\%$	%
MFR	Middle Fluorescent Ratio	$\text{MFR} = \frac{\text{Number of cells in MFR region}}{\text{Number of cells in RET region}} \times 100\%$	%
LFR	Low Fluorescent Ratio	$\text{LFR} = \frac{\text{Number of cells in LFR region}}{\text{Number of cells in RET region}} \times 100\%$ Note: Sum of all particles in the reticulocyte region = Sum of all particles in the LFR region + Sum of all particles in the MFR region + Sum of all particles in the HFR region	%
IRF	Immature Reticulocyte Fraction	$\text{IRF} = \text{MFR} + \text{HFR}$	%
RHE	Reticulocyte Hemoglobin Expression	Calculated based on the light scatter information of RET.	pg
MCVr	Mean corpuscular volume of reticulocyte	Calculated based on the reticulocyte volume distribution in the RET channel.	fL
MCHr	Mean corpuscular hemoglobin of reticulocyte	Calculated based on the hemoglobin distribution of reticulocytes in the RET channel.	pg
DELTA-CH	Expression of MCHr-MCHm	$\text{DELTA-CH} = \text{MCHr} - \text{MCHm}$	pg

4.5 Erythrocyte Sedimentation Rate Measurement

The analyzer uses the photometric method to measure the aggregation of erythrocytes within a specified time period and calculates the sedimentation rate. RBCs separated in the whole blood sample first aggregate and form Rouleaux formation; then the aggregation falls. The sedimentation rate depends on the degree of aggregation. As the measuring beam is subject to the RBC aggregation state, specifically manifested in that its light transmittance increases with the aggregation degree. As a result, the aggregation degree of RBCs can be obtained by measuring the change in the light transmittance of whole blood samples over time, and then the erythrocyte sedimentation rate is calculated.

4.6 Body Fluid Parameters

Body fluid refers to the fluid in side body cavities except blood vessels. There are many sub-types of body fluid, among which the most commonly seen sub-types are cerebrospinal fluid, serous cavity fluid (pleural fluid and ascitic fluid) and synovial fluid. Both cerebrospinal fluid and serous cavity fluid are colorless and transparent in normal case, but in abnormal cases, there could be increase of cells (including leukocytes and erythrocytes).

Leukocytes in body fluid can be categorized into mononuclear cells (MN) and polymorphonuclear cells (PMN). The analysis of the cells in body fluid can provide useful information for clinical diagnosis.

Parameter	Name	Formula/Test Methods	Unit
WBC-BF	White Blood Cell count-body fluid	WBC-BF = Sum of all particles in the DIFF channel except those in Ghost region and HFR	$10^9/L$
TC-BF#	Total nucleated cell counts-body fluid	TC-BF# = Count of all particles in the DIFF channel except those in Ghost region	$10^9/L$
MN%	Mononuclear cell percentage	$MN\% = \frac{\text{Particles in MN region of DIFF channel}}{WBC - BF} \times 100\%$	%
MN#	Mononuclear cell number	$MN\# = WBC-BF \times MN\%$	$10^9/L$
PMN%	Polymorphonuclear cell percentage	$PMN\% = \frac{\text{Particles in PMN region of DIFF channel}}{WBC - BF} \times 100\%$	%
PMN#	Polymorphonuclear cell number	$PMN\# = WBC-BF \times PMN\%$	$10^9/L$
RBC-BF	Red Blood Cell count-body fluid	The analyzer provides the number of red blood cells in body fluid (RBC-BF) directly by counting the red blood cells passing through the aperture.	$10^{12}/L$

4.7 Rinse

After each analysis cycle, all elements of the analyzer that the sample runs through are rinsed to ensure no residue is left.

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5

Installing and Connecting the System

5.1 Precautions on Analyzer Installation

CAUTION

- The installation, authorization, upgrade and modification of the system software must be performed by the personnel authorized by our company. Make sure to install only our company authorized software.
 - When the system is delivered in the manufacturer's packaging, the transportation, storage, unpacking, installation, disposal, and other handling process should be performed by personnel authorized or trained by the manufacturer. Otherwise, personal injury or system damage may be caused. Do not perform the process mentioned above without the presence of personnel authorized by the manufacturer.
-

NOTE

- The safety of any system incorporating the equipment is the responsibility of the assembler of the system.
 - The instrument has been tested before it is shipped from the factory. The instrument was carefully packaged and labels were placed to tell the carrier how to treat this instrument. When you receive your instrument, carefully inspect the carton. If you see any signs of mishandling or damage, contact our Customer Service Department or your local distributor immediately.
 - After you open the package, check the integrity of the product according to the packing list. If you find any part missing, contact our Customer Service Department or your local distributor.
-

5.1.1 Space Requirements

NOTE

- For information about the analyzer dimensions and weight, see *B.14 Dimensions and Weight*.
-

Check the site for proper space allocation. In addition to the space required for the system itself, arrange for:

- proper height to place the analyzer;
- at least 500mm on each side of the analyzer, which is the preferred access to perform service procedures;
- at least 600mm above the system;
- at least 150 mm to each side of the pneumatic unit;
- at least 250 mm behind the analyzer and the pneumatic unit;
- The diluent container must be placed within 1.0 meters reach under the main unit, the lyse container must be placed in a plane of the same level with the main unit or within 850mm's reach under the main unit.
- Desktop or platform must be capable to bear the weight of the analyzer.

5.1.2 Power Requirements

For the power requirements, refer to *B.9 Power supply*.

WARNING

- Before turning on the analyzer, make sure the input voltage meets the requirements.
-

⚠ CAUTION

- **Using pinboard may bring electrical interference and the analysis results may be unreliable. Place the analyzer near the electrical outlet to avoid using the pinboard.**
 - **Use the manufacturer-provided power cord, which matches the power supply of the instrument. Using the power cord other than provided by the manufacturer may damage the analyzer or lead to unreliable analysis results.**
 - **When installing the instrument, ensure that the power switch is in close proximity to the equipment and within easy reach of you.**
 - **Make sure the instrument is properly grounded.**
 - **Make sure the analyzer is switched off before connecting the power cord.**
 - **Do not place heavy objects on the power cord or pull the power cord by force. Otherwise, a fire or electric shock may occur due to a short circuit or rupture of the power cord.**
 - **In case of an abnormal odor or smoke, immediately turn off the main switch and disconnect the power cord, and contact our Customer Service Department or the local distributor. If the analyzer is still operated under this condition, fire, electric shock or personal injury may occur.**
-

5.1.3 Environment Requirements

NOTE

- **For normal working condition and storage environment requirements of the analyzer, refer to *B.13 Environment Requirements*.**
-
- Install the analyzer in a place not exposed to the splashing water.
 - The environment should be as free as possible from dust, mechanical vibrations, sources of noises and electrical interference;
 - The electromagnetic environment should be evaluated prior to operation of the device.
 - Do not use this device in proximity to sources of strong electromagnetic radiation (e.g. Unshielded intentional RF sources), as these can interfere with proper operation.
 - Do not place the analyzer near brush-type motors, flickering fluorescent lights, and electrical contacts that regularly open and close;
 - Avoid exposure to direct sunlight or to the source of heat or drafts;
 - Do not use the analyzer in a working environment with conductive or combustible gases;
 - The environment shall be well ventilated;
 - Do not place the analyzer on a slope;
 - Connect only to a properly earth grounded outlet;
 - Only use this analyzer indoors.

5.1.4 Moving and Installing the Analyzer

Moving and installation of the analyzer shall be conducted by personnel authorized by our company. Do not move or install your analyzer without the presence of personnel authorized by our company.

⚠ CAUTION

- **Unpacking, installation or transportation by personnel not authorized or trained by our company may cause personal injury or damage your instrument. Do not unpack, transport or install your instrument without the presence of our company authorized personnel.**
 - **When installing or using the system, make sure at least the two inside supporters of the autoloader are on the supporting table of the analyzer.**
-

NOTE

- **After the autoloader is installed, do not lay too much pressure to it or transport the analyzer by holding the autoloader.**

Follow the following steps to move the analyzer:

1. Insert the handle bars horizontally into the four handling holes at the bottom plate of the main unit to the limit positions respectively, as shown in the figure below.



Figure 5-1 Inserting the handle bar

2. Rotate the handles to the vertical position, as shown in the following figure. Pull the handles out to check whether they are firm. If yes, the main unit can be moved.



Figure 5-2 Handling position of the handle bar

NOTE

- **To avoid accidents (such as analyzer titling), the analyzer must be lifted by two persons with the handles.**

- Sufficient space must be reserved for the main unit to ensure that the two inside supporters of the autoloader can be placed onto the deck panel of the main unit.
- Avoid colliding with the main unit, especially the inlets on the back of the main unit.

5.2 Connecting the Analyzer System

CAUTION

- Be sure to use the specified external devices only.
- External equipment connected to the analyzer and digital interfaces must be authorized and complied with relevant safety and EMC standards (e.g. IEC 60950 Safety of Information Technology Equipment Standard (Class B)). Any persons who connects additional equipment to the signal input or output ports and configures an IVD system, is responsible for ensuring that the system works normally and complies within the safety and EMC requirements. If you have any questions, consult the technical service department of your local representative.

5.2.1 Connecting Power Supply

The power socket locates on the back of the analyzer.

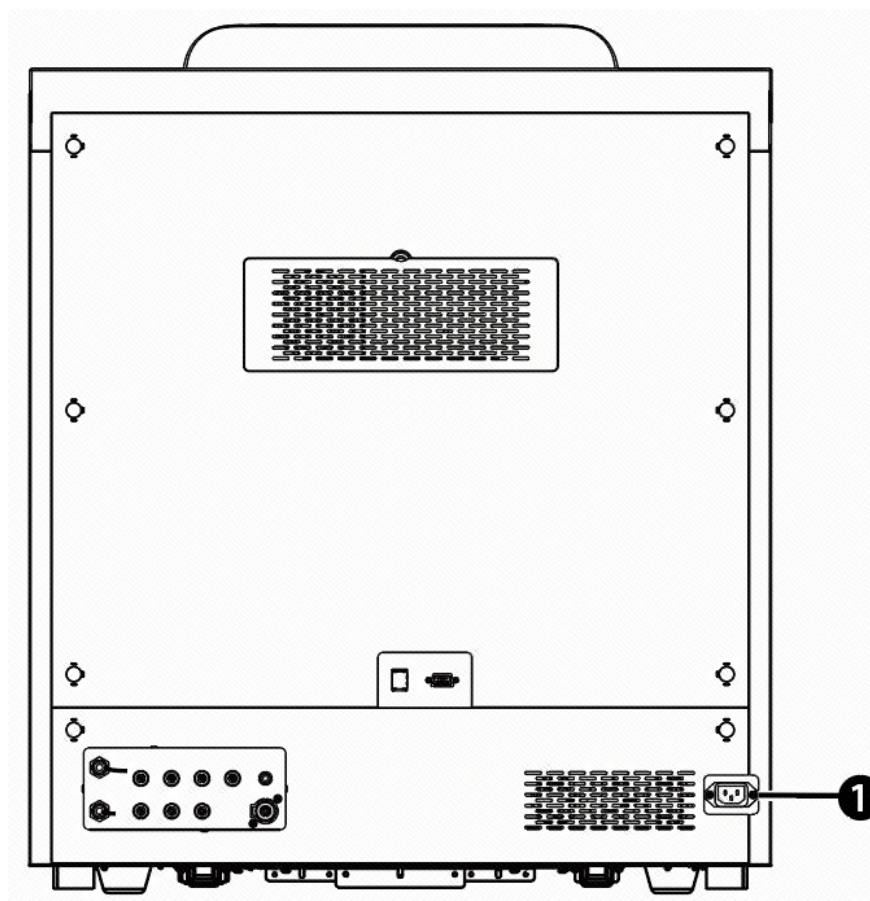


Figure 5-3 Power supply connection

	Name	Description
1	Power socket	Connect the power supply

5.2.2 Connecting the Pneumatic Unit

Connect the pneumatic unit to the analyzer as shown in the following figure. The operator shall make sure they are connected properly.

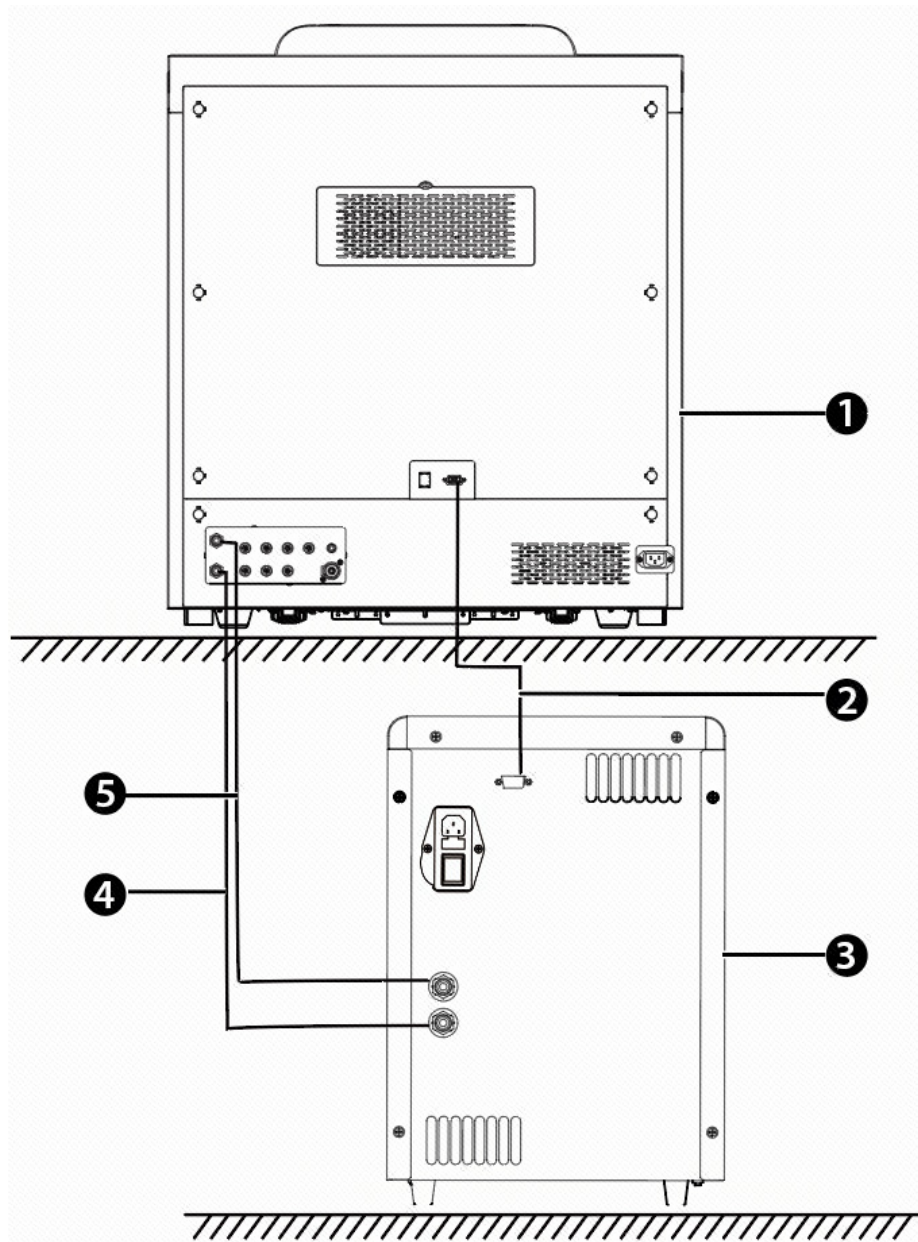


Figure 5-4 Connecting the Pneumatic Unit

	Name
1	Main unit
2	Control cable
3	Pneumatic unit
4	Pressure
5	Vaccum

5.2.3 Connecting the Reagents

NOTE

- Check if the reagent sample tubes are properly connected before using the system. Otherwise, the results may be inaccurate.
- When connecting the reagents, make sure the color of the reagent container cap assembly is the same as that of label for the reagent inlet to which it is connected and as that of the label for the reagent tubing.

Connect the reagent containers to the analyzer as shown in the following figure.

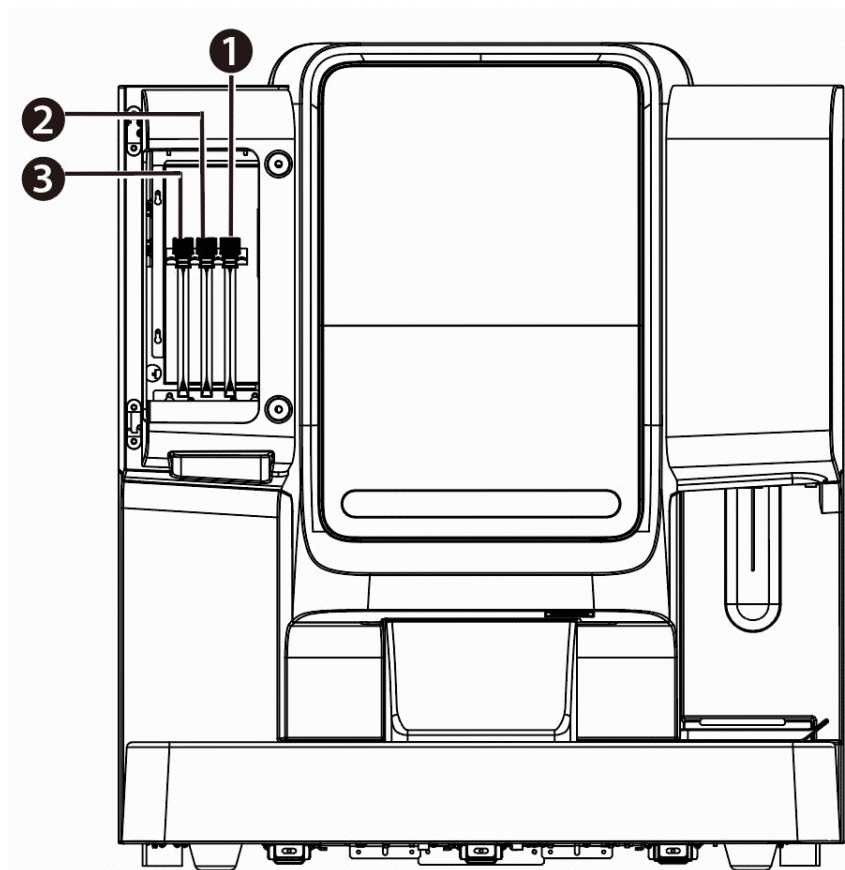


Figure 5-5 Reagent connection - connecting fluorescent dyes

	Reagent Connection
1	FR Dye
2	FD Dye
3	FN Dye

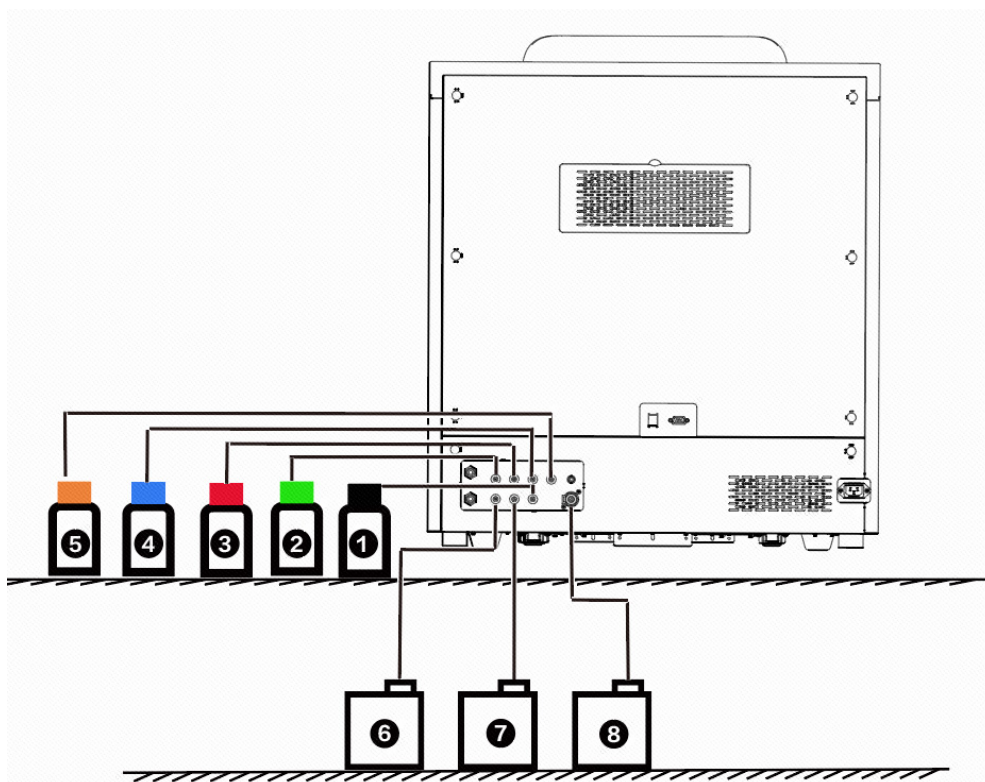


Figure 5-6 Connecting reagents - lysates and diluents

	Name	Reagent Connection	Description
1	DR diluent inlet DR Diluent inlet	Connect the DR diluent	/
2	LH lyse inlet LH Lyse inlet	Connect the LH lyse Connect the LH Lyse	/
3	LD lyse inlet Lyse inlet	Connect the LD lyse Lyse	/
4	LN lyse inlet Lyse inlet	Connect the LN Lyse lyse	/
5	ESR Solution Reagent inlet ESR Solution Reagent inlet	Connect the ESR Solution Reagent Connect the ESR Solution Reagent	/

	Name	Reagent Connection	Description
6	DS Diluent A inlet	Connect the DS Diluent A	- Depending on the analyzer configuration, the analyzer may use a single container of DS diluent, or use two containers of DS diluent at the same time. - For the single DS diluent configuration, connect the DS diluent container to DS diluent inlet A (position 6 as in Figure 5-6 Connecting reagents - lysates and diluents). - For the two DS diluent configuration, connect one DS diluent container to DS diluent inlet A (position 6 as in Figure 5-6 Connecting reagents - lysates and diluents), and the other to DS diluent inlet B (position 7 as in Figure 5-6 Connecting reagents - lysates and diluents) - For more information about how to connect and replace DS diluent containers, refer to 12.5 When Two DS Diluent Containers are Connected at the Same Time.
7	DS Diluent B inlet	Connect the DS Diluent B	
8	Waste liquid outlet	/	- If waste is discharged into a container, the floating sensor interface must be connected to detect whether the waste container is full. If waste is discharged into a container, the floating sensor interface must be connected to detect whether the waste container is full. - If waste is discharged directly, the floating sensor interface does not need to be connected. If waste is discharged directly, the floating sensor interface does not need to be connected.

When you are using the waste direct-discharge method, follow below requirements:

- The inner diameter of the common pipes $\geq 2"$;
- The common pipes to the analyzer must include both horizontal and vertical sections of pipes. The vertical pipe should be in the length of 300mm to avoid foams spilling over;
- If more than one analyzer is installed in your laboratory, the distance between the vertical pipe to this analyzer and the vertical pipe to the neighboring analyzer must be ≥ 300 mm;
- The outlet of the vertical pipe to this analyzer must be open;
- Position the common pipes at an angle, with its upper-stream section is 50mm higher than the lower-stream section, so that the waste automatically flows down into the sewage, refer to **Figure 5-7 Pipe installation requirements for waste direct-discharge**;
- The direct-discharge pipe of the analyzer must not be bent, and is properly fixed into the common pipe.

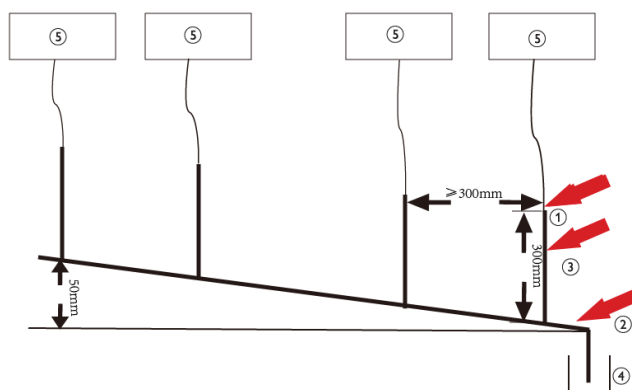


Figure 5-7 Pipe installation requirements for waste direct-discharge

	Name
1	Vertical section of the analyzer
2	Common pipe
3	Vertical pipe
4	Sewer
5	Analyzer

5.2.4 Connecting the Peripherals

Check and make sure the peripherals are connected properly and firmly.

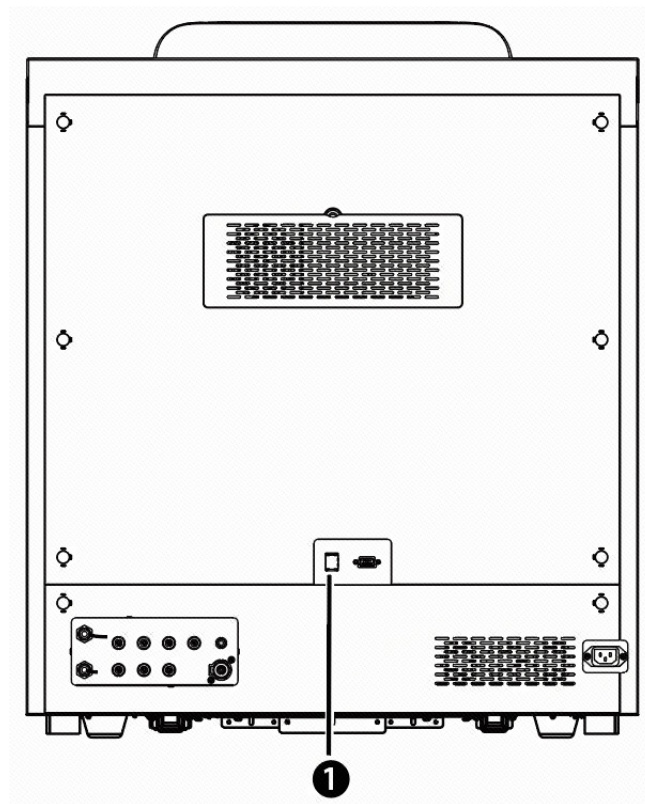


Figure 5-8 Connecting the peripherals-1

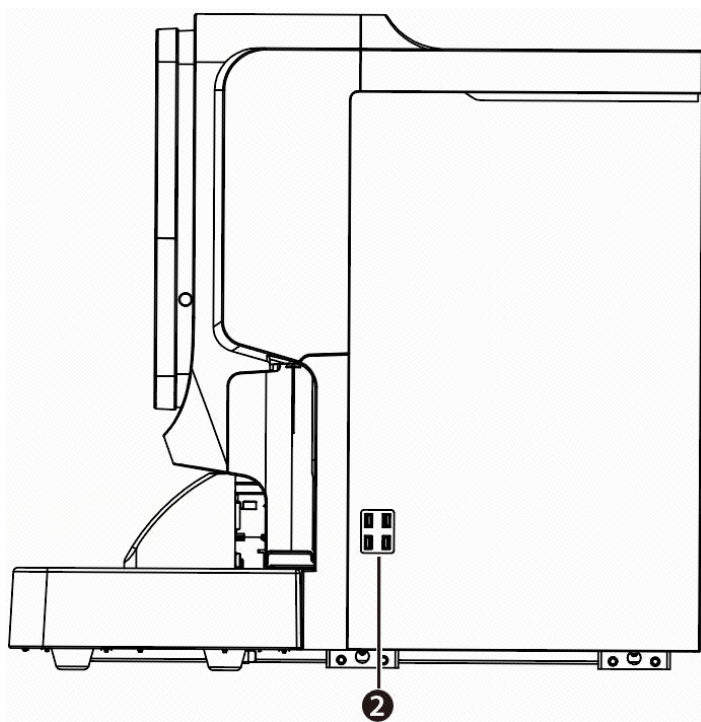


Figure 5-9 Connecting the peripherals-2

Table 5-1 Description of connecting to the peripherals

	Interfaces	Description
1	Network interface (1)	Connect the network to enable LIS (Laboratory Information System).
2	USB interface (4)	Complied with USB 2.0 standard; connect to devices such as the printer and scanner through the USB interfaces as needed.

CAUTION

- Be sure to use the specified external devices only, and keep away from water.

NOTE

- The user should ensure the data safety of the USB flash drive connecting to the analyzer.
- When the analyzer connects to an external computer, install anti-virus software on the computer and scan for viruses and update patches periodically. Do not use it for unintended purposes.

6 Customizing Your Analyzer

The analyzer is a flexible laboratory instrument that can be tailored to your work environment. You can customize the software options as introduced in this chapter.

6.1 Access Levels for Different Accounts

For the security of the settings and data, two access levels are provided to the operators of the analyzer: “**General User**” and “**Administrator**”. The administrator access level provides the operator with access to more functions or settings, some of which can be configured to be accessible to operators.

NOTE

- For description of setting the access levels, refer to *Table 3-17 Software functions by access levels*.

6.2 Saving Settings after Changes

After you have changed or modified analyzer settings, follow below steps to save the changes.

1. Select “**Setup**” to select the setting item you want to change.
2. Make necessary changes on the setting screen.
3. Select another button on the software interface.
- √ A dialog box displays asking if you want to save the change.
4. Select “**Yes**”.
- √ The new setting is saved.

6.3 Setting up a Printing Template

For setting a printing template, refer to *11.1 Setting up Print Templates*.

6.4 Setting up for Communication Methods

Administrators can set up the following communication settings:

- **Protocol Setup**
- **Transmission Mode**

NOTE

- Before customizing the communication options, make sure the network wire is firmly connected to the analyzer.

6.4.1 Setting up Communication Protocol

Follow the instructions below:

1. Select “**Setup**”–“**Basic Settings**” to go to “**Basic Settings**” screen.
2. In the “**System Setup**” area, select “**Communication**” to go to “**Communication**” screen.
3. Set up the communication protocol in accordance with the real needs in your laboratory.

See below for setting descriptions

Items	Description	Note
IP Address	Enter the correct IP address	For your network settings, consult our Customer Service Department or your network administrator.
Subnet Mask	Enter correct subnet mask	
Default Gateway	Enter the correct gateway	
ACK Synchronous Transmission	Select "ACK Synchronous Transmission" to enable the ACK synchronous transmission.	When "ACK Synchronous Transmission" is enabled, enter the ACK overtime in the "ACK Overtime" field (10 seconds by default).

NOTE

- If you want to modify other setting items, contact our Customer Service Department.

6.4.2 Setting up Transmission Mode

Follow the instructions below:

1. Select **"Setup"**–**"Basic Settings"** to go to **"Basic Settings"** screen.
2. In the **"System Setup"** area, select **"Communication"** to go to **"Communication"** screen.
3. Set up the transmission mode in accordance with the real needs in your laboratory.

See below for setting descriptions:

Items	Description	Note
Auto Retransmit	When enabled, if the analyzer does not receive any ACK response during the specified time, the analyzer automatically re-transmits the results. The analyzer may automatically re-transmit the result for up to 5 times.	• It is available when you select LIS as the data channel. • This function becomes available only when "ACK Synchronous Transmission" is selected.
Auto Communicate	When enabled, the analyzer will automatically transmit sample results to LIS after it finishes the test.	It is available when you select LIS as the data channel.
Transmit as Print Bitmap Data	Set up whether the test data is transmitted as Bitmap data.	It is available when you select LIS as the data channel.
Communicate L-J QC Results as Sample Results	When enabled, the L-J QC results will be transmitted to LIS as sample results.	It is available when you select LIS as the data channel.
Data channel	■ LIS ■ labXpert	If you want to change the settings of "Data channel" , contact our Customer Service Department.
Scattergram transmitted as/ Histogram transmitted as	Select from the pull-down list: ■ Not to be transmitted ■ Bitmap ■ Data	It is available when you select LIS as the data channel.

6.5 Setting up Barcode Systems

Users logged-in as administrators can set up barcode system as needed.

Analyzer supports the following barcode systems:

- **CODE128**
- **CODE39**
- **ITF**
- **CODE93**
- **CODABAR**
- **UPC/EAN/JAN**

For the length limits and check bit of different barcode systems, see **B.16 Barcode Specifications**.

Follow the instructions below:

1. Select **"Setup" – "Other Settings"** to go to **"Other Settings"** screen.
2. In the **"Loading Setup"** area, select **"Barcode"** to go to **"Barcode"** screen.
3. Select desired barcode system to go to the corresponding setup screen.
4. Select **"Apply"** to apply the barcode system.
5. In the **"Digits"** area, select the number of digits to be used.
6. (Optional) If you are using **"CODE39"**, **"ITF"**, and **"CODABAR"** with check digit, and select **"Check Bit"**.
7. (Optional) Set other barcod systems if needed.

NOTE

- When the ITF, CODE39, and CODABAR with check bits are used as barcodes, the actual length of the checked barcodes should be the length of the original barcode plus one check bit. For example, when the original barcode is 9 bits in length and "Check Bit" is selected, the barcode length should be "10".
- For barcode systems using check bits, it is recommended to use check bit in barcode labels if possible to reduce the rate of misreading.
- The barcode systems and length limits set on the analyzer shall be those used in your laboratory.
- Do not select the barcode systems that are not used in the laboratory, otherwise it may increase the rate of misreading.
- Do not use the barcodes longer than 20 digits. Barcodes longer than 20 digits will not be read correctly.

6.6 Setting Date/Time

You can set up date and time on **"Date/Time Setup"** screen.

1. Select **"Setup" – "Basic Settings"** to go to **"Basic Settings"** screen.
2. In the **"Lab Setup"** area, select **"Date/Time Setup"** to go to **"Date/Time Setup"** screen.
3. Set the date and time as necessary.

Items	Description	Note
Date	Confirm the analyzer displays current date. If not, enter current date.	/
Time	Confirm the analyzer displays current time. If not, enter current time.	• The analyzer uses the 24-hour clock system.
Date Format	Select the desired date format from the pull-down list, including: • YYYY-MM-DD • YYYY/MM/DD • MM-DD-YYYY • MM/DD/YYYY • DD-MM-YYYY • DD/MM/YYYY	/

6.7 Setting up Laboratory Information

You can enter laboratory information on the “**Lab Info. Setup**” tab.

Follow the instructions below:

1. Select “**Setup**”– “**Basic Settings**” to go to “**Basic Settings**” screen.
2. In the “**Lab Setup**” area, select “**Lab Info. Setup**” to go to “**Lab Info. Setup**” screen.
3. Enter laboratory information as needed.

6.8 Managing User Accounts

You can set up user accounts and passwords as needed, including:

- Changing the passwords
- Adding users
- Deleting users

6.8.1 Changing Passwords

NOTE

- **Users at either administrator's or operator's level can only change the passwords for the users currently logged in.**

Follow the instructions below:

1. Select “**Setup**”– “**Basic Settings**” to go to “**Basic Settings**” screen.
 2. In the “**Lab Setup**” area, select “**User Management**” to go to “**User Management**” screen.
 3. Select the current user, and then select “**Modify Password**”.
 4. Enter “**Old Password**”, “**New Password**” and “**Confirm Password**”.
 5. Select “**OK**”.
- ✓ The new password is activated.

6.8.2 Adding a New Account

The administrator can add a new account in the “**User Management**” interface.

Follow the instructions below:

1. Select “**Setup**”– “**Basic Settings**” to go to “**Basic Settings**” screen.
2. In the “**Lab Setup**” area, select “**User Management**” to go to “**User Management**” screen.
3. Select “**New**”, and “**Add User**” dialog box displays.
4. Set the user information as necessary.

Items	Description	Notes
User ID	Enter the user name used for logging in the analyzer.	• It cannot be left blank; • You can enter 1-12 characters; • You can enter numbers, alphabets or symbols like ~@#%-_+={}[]; • Case sensitive.

Items	Description	Notes
Name	Enter the user's actual name.	- It cannot be left blank; - You can enter 1-20 characters; - All entries are allowable; - The first and last characters cannot be left blank; - Case sensitive.
Password	Enter the password used for logging in the analyzer.	- It cannot be left blank; - You can enter 0-12 characters: - You can enter numbers, alphabets and symbols like ~`!@#\$%^&*()-_+={}[]\ ;:',./<>; - Case sensitive.
Confirm Password	Re-enter the password you have set.	/
Access Level	Select the access level. - Administrator's access level: view all users' information; - General user's access level: only view current user's information.	- You can set up to 30 general users and up to 20 administrators; - For description of setting access levels, refer to Table 3-17 Software functions by access levels.

5. Select **"OK"**.

√ The new user account is added.

6.8.3 Deleting Accounts

Users logged in as administrators can delete any account under its own accounts.

NOTE

- You cannot delete a built-in user account.**
- You cannot delete the current account.**

Follow the instructions below:

- Select **"Setup"–"Basic Settings"** to go to **"Basic Settings"** screen.
- In the **"Lab Setup"** area, select **"User Management"** to go to **"User Management"** screen.
- Select a user to be deleted and select **"Delete"**.
- Select **"Yes"** in the dialog box.

√ The selected user is deleted.

6.9 Setting up Auto-loading Mode

Administrators may set up the following functions regarding the autoloading of samples in accordance with the real needs in your laboratory.

- How to proceed when blood/analysis mode inquiry failed
- How to set up the next sample IDs when there is sample tube vacancy
- Whether to display the summary after auto-loading finishes
- Whether to automatically load sample racks on the autoloader
- How to proceed when out of the analyzer capability
- Stop analysis when consecutive samples are flagged for abnormal results

6.9.1 Customizing Your Analyzer How to Proceed When Blood/Analysis Mode Inquiry Failed

When the analyzer has connected to LIS/labXpert, blood/analysis mode inquiry may fail, if:

- No sample order is found in the LIS/labXpert
- Network connection is abnormal

NOTE

- **If the analyzer still fails to acquire analysis mode from LIS after troubleshooting, contact our Customer Service Department.**

If the analyzer fails to acquire analysis mode, it will proceed the sample order in the manner preset. On the **"Auto-loading"** screen, you can set the method how to proceed this situation.

Follow the instructions below:

1. Select **"Setup"–"Other Settings"** to go to **"Other Settings"** screen.
2. In the **"Loading Setup"** area, select **"Auto-loading"** to go to **"Auto-loading"** screen.
3. In the **"When Blood/Analysis Mode Inquiry Failed"** field, see below for setting description:

Options	Description	Note
Skip after	When enabled, if blood /analysis mode inquiry still fails during the specified time, it will skip the sample. You can run the sample again after troubleshooting.	Set up the waiting time in second (n) (default setting is 10 seconds and n is no longer than 15 seconds).
Use specified mode	When enabled, select a mode from the pull down list and run the analysis in the specified mode.	/

4. Exit the current screen and a **"Prompt"** dialog box displays.
5. Select **"Yes"** to save current settings.

6.9.2 How to Handle Sample ID When There is Sample Tube Vacancy

You can enable this function to deal with the situation that no proper barcode is available or the analyzer cannot scan the barcode when the analyzer works in AL mode.

Follow the instructions below:

1. Select **"Setup"–"Other Settings"** to go to **"Other Settings"** screen.
2. In the **"Loading Setup"** area, select **"Auto-loading"** to go to **"Auto-loading"** screen.
3. In the **"When there is tube vacancy, sample ID"** area, see below for setting description.

Options	Description	Note
increases	When this option is selected, the sample ID will increase by the sample sample tube position, even if there is no sample tube vacancy.	/
does not increase	When this option is selected, the sample ID will not increase.	/

4. Exit the current screen and a **"Prompt"** dialog box displays.
5. Select **"Yes"** to save current settings.

NOTE

- **When you select "Auto Increase" in the "Entry of next sample ID" field, if the analyzer detects there is a sample tube vacancy, it will dispose of the sample ID in the manner preset.**

6.9.3 Display Summary after Autoloading Finishes

When the autoloading finishes, the analyzer can display the autoloading summary dialog box, including **“Finished blood samples”, “Rack vacancies”, “ID reading errors”** and etc.

If you want to review the statistics after autoloading finishes, you can enable this function in AL mode.

Follow the instructions below:

1. Select **“Setup”–“Other Settings”** to go to **“Other Settings”** screen.
2. In the **“Loading Setup”** area, select **“Auto-loading”** to go to **“Auto-loading”** screen.
3. Select **“Display summary after autoloading finished”**.
4. Exit the current screen and a **“Prompt”** dialog box displays.
5. Select **“Yes”** to save current settings.

6.9.4 Setting Sample Tube Loading Methods

You can set up if the analyzer will automatically load sample racks placed on the autoloader.

Follow the instructions below:

1. Select **“Setup”–“Other Settings”** to go to **“Other Settings”** screen.
2. In the **“Loading Setup”** area, select **“Auto-loading”** to go to **“Auto-loading”** screen.
3. Specify the method of loading samples as instructed below.

Automatically start sample loading	1. Select “Automatically load tube racks on the autoloader”. 2. In the “Wait” edit box, set the waiting time (setup range: [1, 30] seconds). ✓ The analyzer waits for the specified time, and then automatically starts to load samples.
Manually start sample loading	Clear the selection of “Automatically load tube racks on the autoloader” . Manually press the [Count] key on the front cover or select  on the software to start sample loading.

4. Exit the current screen and a **“Prompt”** dialog box displays.
5. Select **“Yes”** to save current settings.

6.9.5 Customizing Your Analyzer How to Proceed When out of the Analyzer Capability

You can enable this function if the whole blood sample is analyzed in the auto-loading mode.

When the function is enabled and the blood sample test is out of the analyzer capability, the analyzer will proceed the test with the supported routine blood count tests.

Follow the instructions below:

1. Select **“Setup”–“Other Settings”** to go to **“Other Settings”** screen.
2. In the **“Loading Setup”** area, select **“Auto-loading”** to go to **“Auto-loading”** screen.
3. Select **“When out of analyzer capability, the sample will be analyzed using the supported routine blood count tests.”**.
4. Exit the current screen and a **“Prompt”** dialog box displays.
5. Select **“Yes”** to save current settings.

6.9.6 Setting up the Rules for “Stop analysis when consecutive samples are flagged for abnormal results”

After the **“Stop analysis when consecutive samples are flagged for abnormal results”** function is enabled, when the consecutive sample results or QC samples trigger the flagged rules preset under the auto-loading mode, the analyzer stops auto-loading analysis, and reports the error of **“Consecutive samples are flagged for abnormal results”**.

Follow below instruction:

1. Select **"Setup" – "Other Settings" – "Auto-loading"** to enter the **"Auto-loading"** setup screen.
 2. Select **"Stop analysis when consecutive samples are flagged for abnormal results"**. And then select the **"Setup"**.
- ✓ A dialog box displays.
- Enter a number to the **"Number of abnormal samples"** text field. (The default value: 5, setup range: [2, 20])
When the number of samples triggering either abnormal sample result rules reaches the defined number, the analyzer automatically stops auto-loading analysis.
 - Set up the abnormal sample result rules.

Options	Description	Note
Aspiration abnormal or clogging	Select the check box to enable this function.	When the analyzer reports " Aspiration Abn ", "Asp.Abn/Abn.Sample" or "Clog", the analyzer counts the relevant sample as one "abnormal" sample.
WBC result invalid or $\leq$	Select the check box to enable this function and you can enter the threshold value.	When the WBC result is not reliable and is screened on the report (displayed as "*****" on the report), or the result is lower than the defined threshold, the analyzer counts the relevant sample as one "abnormal" sample. The default threshold value is 0.10, the allowed range is [0.00,99999.99].
RBC result invalid or $\leq$	Select the check box to enable this function and you can enter the threshold value.	When the RBC result is not reliable and is screened on the report (displayed as "*****" on the report), or the result is lower than the defined threshold, the analyzer counts the relevant sample as one "abnormal" sample. The default threshold value is 0.02, the allowed range is [0.00,9999.99].
HGB result invalid or $\leq$	Select the check box to enable this function and you can enter the threshold value.	When the RBC result is not reliable and is screened on the report (displayed as "*****" on the report), or the result is lower than the defined threshold, the analyzer counts the relevant sample as one "abnormal" sample. The default threshold value is 0.02, the allowed range is [0,999].
PLT result invalid or $\leq$	Select the check box to enable this function and you can enter the threshold value.	When the PLT result is not reliable and is screened on the report (displayed as "*****" on the report), or the result is lower than the defined threshold, the analyzer counts the relevant sample as one "abnormal" sample. The default threshold value is 5, the allowed range is [0,9999].

Options	Description	Note
WBC DIFF parameter results invalid	Select the check box to enable this function and you can enter the threshold value.	When any WBC DIFF parameters are not reliable and are screened on the report (displayed as "****" on the report), the analyzer counts the relevant sample as one "abnormal" sample.

NOTE

- When more than one rule is enabled, when the results of a sample have triggered any of the enabled rules, the sample will be counted as an abnormal sample.

3. Select **"OK"** to complete the setting, and exit the dialog box.

6.10 Setting up Methods for Acquiring Sample Information

The analyzer can acquire the sample information automatically and you can set the methods of acquiring sample information.

6.10.1 Setting up the Entry Method of Next Sample ID in Open-Vial Mode Analysis

"Setting of the next sample" function is applicable to the situation that you should enter the sample ID manually in open-vial mode analysis.

NOTE

- This software function only applies to open-vial mode analysis.

You can set the entry method of the next sample ID as **"Auto Increase"** or **"Manual Entry"**, and you can enter the sample information more effectively.

Follow the instructions below:

1. Select **"Setup"** – **"Basic Settings"** to go to **"Basic Settings"** screen.
2. In the **"System Setup"** area, select **"Auxiliary Setup"** – **"Get Sample Information"** to go to **"Get Sample Information"** screen.
3. In the **"Setting of the next sample"** field, see below for setting description.

Options	Description	Notes
Auto Increase	When you select "Auto Increase" , you only need to entry an ID for the first sample, and the subsequent sample IDs will automatically increase by 1 based on the previous one.	/
Manual Entry	When you select "Manual Entry" , you need to enter each sample ID manually.	/
Prefix Length	Define "Prefix Length" of the sample ID.	• The maximum of prefix length is no more than 20 digits. • The field becomes available when "Auto Increase" is selected. • The sample ID prefix will not automatically increase. For example, if you set "Prefix Length" to 6, the first 6 figures of the first sample ID will be applied to all sample IDs in the same batch and will not automatically increase.

6.10.2 Setting up for the First Sample after Startup

Set up the test panel for the first sample after startup in accordance with the real needs in your laboratory.

Follow the instructions below:

1. Select **"Setup"**–**"Basic Settings"** to go to **"Basic Settings"** screen.
2. In the **"System Setup"** area, select **"Auxiliary Setup"**–**"Get Sample Information"** to go to **"Get Sample Information"** screen.
3. In the **"Setting of the first sample after startup"** field, see below for setting descriptions.

Items	Description	Note
First sample after startup	• Custom: analyze the sample using specified test mode. • Run the suspended sample after restart: analyze the sample using the test mode of the last sample before last shutdown.	/
Mode	Select a test mode from pull-down list.	You need to specify a test mode if you have selected "Custom" for "First sample after startup" .
Sample ID	Enter the ID (1 by default) of the first sample after startup.	• If you have selected "Custom" in the "First sample after startup" field, you need to specify the sample ID. • If you specify "Run the suspended sample after restart" in the "First sample after startup" field, the startup sample ID will be increased by based on the last one before shutdown.

6.10.3 Setting up Auto-Scan Rack No. in Auto-Loading Mode

When **"Auto-Scan rack No."** is selected, the analyzer will automatically scan the rack No. in auto loading mode analysis.

NOTE

- This setting is available to the analyzer equipped with a built-in barcode scanner.

Follow the instructions below:

1. Select **"Setup"**–**"Other Settings"** to go to **"Other Settings"** screen.
2. In the **"System Setup"** area, select **"Auxiliary Setup"**–**"Get Sample Information"** to go to **"Get Sample Information"** screen
3. In the **"Get Sample Information"** area, **"Auto-Scan rack No."** is selected by default. If you want to disable this function, you should manually clear the selection of **"Auto-Scan rack No."**.

6.11 Setting up for Parameters

Administrators can customize the parameters.

6.11.1 Setting up Parameter Unit

Administrators can set up unit systems and parameter units.

Follow the instructions below:

1. Select **"Setup" – "Other Settings"** to go to **"Other Settings"** screen.
 2. In the **"Para. Setup"** area, select **"Parameter Unit Setup"** to go to **"Parameter Unit Setup"** screen.
 3. (Optional) If necessary, select the unit system from the **"Unit System:"** pull-down list.
 4. Select the column of the parameter for which you want to change the unit. The available units for the parameter displays on the right side of the tab.
 5. Select the desired unit for the parameter in the **"Unit option:"** field.
- ✓ The parameter unit refreshes.

NOTE

- **General users can only review the unit system and the parameter units.**
- **Select "Default" to restore the default units for all parameters.**

6.11.2 Setting up Parameter Reference Ranges

The **"Reference Range Setup"** tab provides 5 factory reference groups for your selection. In addition, you can set up to 10 custom reference groups. Users at administrator's level may select and customize reference ranges and reference groups.

6.11.2.1 Creating a new reference group

Follow the instructions below:

1. Select **"Setup" – "Other Settings"** to go to **"Other Settings"** screen.
 2. In the **"Para. Setup"** area, select **"Reference Range Setup"** to go to **"Reference Range Setup"** screen. Select **"New"** to create a new reference group.
 3. Set up **"Reference group"**, **"Lower limit of age (>)"**, **"Upper limit of age (<=)"** and **"Gender"** on the right side of the screen.
 4. Set up the upper and lower limits for each parameter.
 5. Select **"Return"** and a **"Prompt"** dialog box displays. And then select **"Yes"** to save current settings.
- ✓ The newly added reference group displays in the **"Reference Range Setup"** screen.

6.11.2.2 Setting up default reference group

You can set the selected reference group as the default reference group. When the sample result does not match any reference group, the analyzer will select the default reference group as the reference group of the parameter.

Follow the instructions below:

1. Select **"Setup" – "Other Settings"** to go to **"Other Settings"** screen.
2. In the **"Para. Setup"** area, select **"Reference Range Setup"** to go to **"Reference Range Setup"** screen and then select a reference group.
3. Select **"Set to Default"**.
4. Exit current tab and a **"Prompt"** dialog box displays. And then select **"Yes"** to save current settings.

6.11.2.3 Selecting a method for matching reference group

You can set up whether the analyzer matches the customized reference group first.

Follow the instructions below:

1. Select **"Setup" – "Other Settings"** to go to **"Other Settings"** screen.
2. In the **"Para. Setup"** area, select **"Reference Range Setup"** to go to **"Reference Range Setup"** screen.

- See below for setting descriptions.

Methods	Description
Match customized reference group first	When "Match customized ref. group first" is selected, the analyzer first searches the matching reference group in the custom reference groups based on the age and gender conditions.
Match built-in reference group first	When "Match customized ref. group first" is not selected, the analyzer first finds the matching built-in reference group in the factory reference groups based on the age/gender conditions.

- Exit current tab and a **"Prompt"** dialog box displays. And then select **"Yes"** to save current settings.

6.11.2.4 Modifying/deleting reference groups

If a reference group is not applicable, modify or delete them.

Follow the instructions below:

- Select **"Setup"–"Other Settings"** to go to **"Other Settings"** screen.
- In the **"Para. Setup"** area, select **"Reference Range Setup"** to go to **"Reference Range Setup"** screen.
- See below for setting descriptions

Options	Description	Notes
To edit reference groups	Select "Edit" to select the reference group to be edited and edit it.	- For the 5 factory reference groups, you can only edit the "Lower Limit" and "Upper Limit" of the parameters. - For the custom reference groups, you can edit the group name, limits of ages, the gender information and "Lower Limit" and "Upper Limit" of parameters results.
To delete reference groups	Select the custom reference group to be deleted, and select "Delete" .	- You can only delete custom reference groups. - You cannot delete factory reference groups.

6.11.3 Setting up Microscopic Parameters

NOTE

- The analyzer does not have a microscopic examination module. The microscopic parameter setting function provides a tool to record microscopic parameter result.**

By default, the **"Microscopic Para. Setup"** tab displays 22 microscopic parameters of the blood cell analysis.

Administrators may add new parameters, edit or delete existing microscopic parameters on the screen.

Follow the instructions below:

- Select **"Setup"–"Other Settings"** to go to **"Other Settings"** screen.
- In the **"Para. Setup"** area, select **"Microscopic Para. Setup"** to go to **"Microscopic Para. Setup"** screen.
- Define the settings as needed.

See below for setting description.

To add a new microscopic parameter(s)	- Select "New" to add a row. - Enter the new parameter name in the "Microscopic Parameter Name" column.
To edit existing microscopic parameter(s)	Select a parameter in the table to edit the name.

To delete microscopic parameter(s)	Select a row in the table, and select "Delete" to delete the parameter.
------------------------------------	--

NOTE

- The analyzer can save at most 40 microscopic parameters.

6.12 Setting up the Probe Cleanser Maintenance Time

Administrators may define the maintenance policy for Probe Cleanser maintenance on the **"Maintenance"** screen. You can select one of the following policies for Probe Cleanser maintenance:

- **Daily Maintenance**
- **Maintenance at Appointed Time**

NOTE

- You can select appropriate policy for the Probe Cleanser maintenance according to your laboratory situation. For recommended policy, contact our Customer Service Department.
- The modified Probe Cleanser maintenance setting will become effective after you restart the analyzer.
- It is recommended that you choose the idle time of the analyzer as the Probe Cleanser maintenance time.

6.12.1 Performing Maintenance at Specified Time

You can perform the maintenance at specified time once a day. On the **"Daily Maint."** screen, you can define the maintenance start time (the **"Start at"** time) and the prompt interval (the **"Remind every"** interval). The analyzer prompts for the Probe Cleanser maintenance at the defined maintenance start time every day.

At the due time, you can choose to postpone the maintenance if the time is inconvenient. The analyzer will then prompts for the maintenance at the defined prompt interval until the maintenance is performed.

Follow the instructions below:

1. Select **"Setup"**–**"Other Settings"** to go to **"Other Settings"** screen.
2. In the **"Maintenance"** area, select **"Daily Maint."** to go to **"Daily Maint."** screen.
3. See below for setting description:

Options	Description	Note
Start at	Setup the Probe Cleanser maintenance start time. After setup, the prompt of Probe Cleanser maintenance displays before each daily analysis. • Select "Yes" to start the Probe perform the Cleanser maintenance. • Select "No" to delay the Probe perform the Cleanser maintenance.	• Setup range: [00:00, 23:59].
Remind every	If you select "No" when the prompt of Probe Cleanser maintenance displays, it will display again after the time interval reaches.	• Setup range:[5, 60] minutes. • You can delay the Probe Cleanser maintenance for most 3 times. When the prompt of Probe Cleanser maintenance displays in the 4th time, you must perform the maintenance.

4. Exit current tab and a **"Prompt"** dialog box displays. And then select **"Yes"** to save current settings.

6.12.2 Performing Maintenance at Appointed Time

This option suits the laboratories that have relatively large number of samples (more than 500 samples a day). You can choose to perform the Probe Cleanser maintenance at the appropriate time(s) (for example, when the number of the tested samples is close to 500).

When you choose **"Maintenance at Appointed Time"**, you can appoint several maintenance time points in a day.

At the due time, the analyzer prompts for the Probe Cleanser maintenance. You can cancel the maintenance if the time is not convenient.

Follow the instructions below:

1. Select **"Setup"**–**"Other Settings"** to go to the **"Other Settings"** screen.
2. In the **"Maintenance"** field, select the **"Daily Maint."**
3. On the **"Probe Cleanser Maint. Time"** tab, select **"Maintenance at Appointed Time"**.
4. Set up/edit the time for appointed maintenance.

Options	Description
Time	Select a "Time" edit box, enter/edit the appointed Probe Cleanser maintenance time.
In Use	Select "In Use" for the appointed time to make it effective.

NOTE

- You can set up at most 10 time points for appointed Probe Cleanser maintenance.
- The **"Time"** is in the 24-hour format.

6.13 Setting up Gains

When the analyzer reports the blank voltage abnormal error, and you cannot remove the error by pressing the **"Remove Error"** button, adjust the gains to correct the blank voltage.

Follow the instructions below:

1. Select **"Setup"**–**"Basic Settings"** to go to **"Basic Settings"** screen.
2. In the **"System Setup"** area, select **"Gain Setup"** to go to **"Gain Setup"** screen.
3. In the **"Set"** text box, you can adjust the gains until the blank voltage is in the appropriate range.

NOTE

- When you modify the default gain, the blank voltage will change accordingly.
4. If necessary, repeat above procedure to adjust the voltages for other modes.
 5. Select other tab on the software, and select **"Yes"** to save the settings in the **"Prompt"** dialog box.

6.14 Setting up Flag Alarm Sensitivity

The analyzer reports messages for abnormal blood cell morphology. For more information, refer to **8.3.2 Flags of Abnormal Blood Cell Differential or Morphology**.

During sample analysis, the analyzer evaluates and scores the possibility of the presence of all types of abnormal blood cell morphology. When the score for a certain type of abnormal blood cell morphology exceeds the set threshold, the analyzer reports the messages accordingly.

Administrators may set up the flag alarm threshold values. The higher the threshold value, the lower the alarm sensitivity of the flag.

Follow the instructions below:

1. Select **"Setup"**–**"Other Settings"** to go to **"Other Settings"** screen.

2. In the “**Test Setup**” area, select “**Flag Alarm Sensitivity**” to go to “**Flag Alarm Sensitivity**” screen.
3. Define the flag alarm threshold values in the “**Value (0-100)**” edit boxes as needed.

NOTE

- The allowed range for all flag alarm threshold values is [0-100].
- The flag alarm setting may vary depending on the model configuration. For details, see the screen of the model you purchase.
- The default threshold for “**Infected RBC?**” is 100; and the default thresholds for all other flags are 40.

6.15 Setting up Flag Alarm Rules

The analyzer provides the following types of flags for analysis results.

Flag Message	Description	Criteria
Neutropenia	Neu# low	Neu# < $1.00 \times 10^9/L$
Neutrophilia	Neu# high	Neu# > $11.00 \times 10^9/L$
Lymphopenia	Lym# low	Lym# < $0.80 \times 10^9/L$
Lymphocytosis	Lym# high	Lym# > $4.00 \times 10^9/L$
Monocytosis	Mon# high	Mon# > $1.50 \times 10^9/L$
Eosinophilia	Eos# high	Eos# > $0.70 \times 10^9/L$
Basophilia	Bas# high	Bas# > $0.20 \times 10^9/L$
Leukocytopenia	Low WBC count	WBC < $2.50 \times 10^9/L$
Leucocytosis	High WBC count	WBC > $18.00 \times 10^9/L$
NRBC Present	NRBCs are detected in the WNB channel	NRBC% > 1% and NRBC# > $0.01 \times 10^9/L$
Anisocytosis	Anisocytosis	RDW-CV > 22% or RDW-SD > 64fL
Hypochromia	Hypochromia	MCHC < 290 g/L
Microcytosis	MCV low	MCV < 70fL
Macrocytosis	MCV high	MCV > 110fL
Erythrocytosis	RBC high	RBC > $6.5 \times 10^{12}/L$
Anemia	Anemia	HGB < 90g/L
Reticulocytosis	RET high	RET% > 5.00% or RET# > $0.2000 \times 10^{12}/L$ or presence of abnormally distributed dots in RET sensitive region of the DIFF channel. Note: when the tests for the sample include RET tests, the message is reported from RET channel; otherwise, the message is reported from DIFF channel.
Thrombocytopenia	PLT low	PLT < $60 \times 10^9/L$
Thrombocytosis	PLT high	PLT > $600 \times 10^9/L$
Pancytopenia	WBC, RBC and PLT low	WBC < 4.0×10^9 and RBC < 3.5×10^{12} and PLT < 100×10^9

Administrator can set up the flag alarm rule.

Follow the instructions below:

1. Select **"Setup"–"Other Settings"** to go to **"Other Settings"** screen.
2. In the **"Test Setup"** area, select **"Flag Rules Setup"** go to **"Flag Rules Setup"** screen.
3. See below for setting descriptions.

Items	Description
Default	When selected, the analyzer restores an edited rule to default setting.
All to Default	When selected, the analyzer restores all edited rule to default setting.
Import	Insert a USB storage device that saves flag rules to the analyzer, and select "Import" to import the flag rules from the USB storage device to the analyzer.
Export	Insert a USB storage device to the analyzer and select "Export" to export the flag rules from the analyzer to the USB storage device.
Flag Rules	Select what you want to define in the flag rules column, and then set up the rules in the "Flag Rules" field.

4. Select another button, and save the settings as prompted.

6.16 Setting up for Ext. WBC/PLT Tests

For low PLT/WBC samples, the analyzer provides an **"Ext. WBC/PLT Tests"** solution. When the function is enabled, the analyzer automatically predicts WBC and PLT results during sample tests and when the predicted WBC/PLT results are lower than the customized thresholds, the analyzer automatically re-run the samples using DIFF-3X re-exam method (for low WBC samples) or PLT-8X re-exam method (for low PLT samples).

Options	Description
DIFF-3 times re-exam	The analyzer automatically predicts the WBC test results during sample analysis process. If the predicted WBC result is lower than the set threshold, the analyzer automatically prolongs the test duration in the DIFF channel to three times that of a normal test, so as to acquire particles that are three times those of a normal test.
PLT-8 times re-exam	The analyzer automatically predicts the PLT test results during sample analysis process. If the predicted PLT result is lower than the set threshold, the analyzer automatically prolongs the test duration in the PLT-O channel to eight times that of a normal test, so as to acquire particles that are eight times those of a normal test.

Follow the instructions below:

1. Select **"Setup"–"Other Settings"** to go to **"Other Settings"** screen.
2. In the **"Test Setup"** area, select **"Ext. WBC/PLT Tests"** to go to **"Ext. WBC/PLT Tests"** screen.
3. Select **"Rerun low WBC samples"** or **"Rerun low PLT samples"** as needed and set the threshold.

NOTE

- The default lower threshold of WBC is $2.00 \times 10^9/L$ and the default lower threshold of PLT is $50 \times 10^9/L$.
- The **"Ext. WBC/PLT Tests"** function is not available to predilute (PD) and body fluid (BF) sample tests.

6.17 Setting up for Other Settings

You can perform the following settings on the **"Other Settings"** screen.

- **Predilute mode prompt**
- **Pop-up keyboard**
- **Enable waste direct discharge**

- **Enable fluorescent reagent detect sensor**
- **Monitor reagent expiration date**
- **Enable the blood sensor**
- **Flags**
- **Alarm Volume**
- **Enable auto exit STAT**
- **Auto replace reagent after setup**

1. Select "**Setup**" – "**Basic Settings**" – to enter the "**Basic Settings**" screen.
2. In the "**System Setup**" area, select "**Auxiliary Setup**" to go to "**Other Settings**" screen.
3. Define the settings as needed.

See below for setting descriptions

Options	Description	Note
Predilute mode prompt	When this function is enabled, if the analyzer starts the analysis using predilute mode, a dialog box will pop up to remind you that the current analysis mode is predilute.	/
Pop-up keyboard	When this function is enabled, select any editable area on tab, and then the pop-up keyboard will display for you to input information.	When you are using an external keyboard, you can clear the selection of " Pop-up keyboard " to disable the function.
Enable waste direct discharge	Enable this function when you are not using waste containers, but directly discharge the waste.	/
Enable fluorescent reagent detect sensor	When the function is enabled, the analyzer gives alarms when a fluorescent reagent is in low volume.	/
Enable the blood sensor	When the function is enabled, the analyzer detects and monitors sample volume, and reports alarm when sample aspiration is abnormal.	/
Monitor reagent expiration date	When the function is enabled, the analyzer gives warning when a reagent gets expired.	/
Flags	Select from the pull-down list to define the suspect, high and low flags (default: "R" for suspect, "H" for high and "L" for low).	Suspect: R, r
		High: H, h, ↑
		Low: L, l, ↓
Alarm Volume	Select proper volume of alarm sound.	The options are: Low, Medium, High, Max
Enable auto exit STAT	During the STAT analysis mode, when the user does not continue to run another STAT sample during a defined period of time, the analyzer automatically exits the STAT mode, and resumes the autoloading analysis that has been paused.	- The default time for "Wait for" is 2 minutes. - The allowed range for the time is [2,5] minutes

Options	Description	Note
Auto replace reagent after setup	- When this function is enabled, if you replace the reagents and enter new reagent barcode information on the "Reagent Setup" screen, the analyzer will automatically replace the reagents in the tubing. - When this function is not enabled, after new reagents are replaced and new barcode information is entered, you must manually select "Replace" to replace the reagents in the tubing.	- The default waiting time is 30 seconds, setup range: [0, 100] seconds. - For more information about reagent replacement, refer to 12.3 Replacing Reagents.

6.18 Setting up the Waiting Time before the Analyzer Enters Standby Status

Administrators may set the wait time before the analyzer entering the standby status on the **"Daily Maint.Standby"** setup screen.

When the time for which the analyzer is free from fluidic operations reaches the defined wait time, the analyzer automatically enters the standby status.

1. Select **"Setup"**–**"Other Settings"** to go to **"Other Settings"** screen.
2. In the **"Maintenance"** area, select **"Daily Maint.Standby"** to go to **"Daily Maint.Standby"** screen.
3. In the **"Standby"** tab, follow the instructions below.

Set the waiting time before the analyzer enters standby status	In the "Wait" edit box, enter the waiting time (setup range: [1, 30] minutes). ✓ When the time for which the analyzer is free from fluidic operations reaches that you have set, the analyzer will automatically enter standby status.
Set the auto exit	1. Select "Auto Exit" to enable this function. 2. Set the time interval of auto exiting standby status (allowable time interval is no longer than 12 hours). ✓ The analyzer will immediately exit the standby status during specified time.

4. Exit the current tab to save the settings.

6.19 Setting up Auto Startup/Shutdown

Administrators may set up the auto startup and shutdown time for the analyzer on the **"Auto Startup/Shutdown"** screen.

When you have set the auto startup/shutdown time, the analyzer automatically starts and shuts down at the set time.

Follow the instructions below:

1. Select **"Setup"**–**"Basic Settings"** to go to **"Basic Settings"** screen.
2. In the **"Lab Setup"** area, select **"Auto Startup/Shutdown"** to go to **"Auto Startup/Shutdown"** screen.
3. See below for setting description.

Setting auto startup time	1. Select the check box of the time you want to start up the analyzer. 2. Define the auto startup complete time in the "Complete Time" edit box. ✓ For example, if you set the "Complete Time" for auto startup to 8:00 on Monday, the analyzer automatically begins the start procedure at 7:40, and completes the procedure at 8:00.
---------------------------	--

Setting auto shutdown time	1. Select the check box of the time you want to shut down the analyzer. 2. Define the auto shutdown time in the "Time" field. ✓ For example, if you set the "Time" for auto shutdown to 17:00 on Monday, the analyzer automatically starts the shutdown procedure at 17:00.
----------------------------	--

4. Exit the current tab to save the settings.

NOTE

- **To use the auto startup function, do not power off the Analyzer.**

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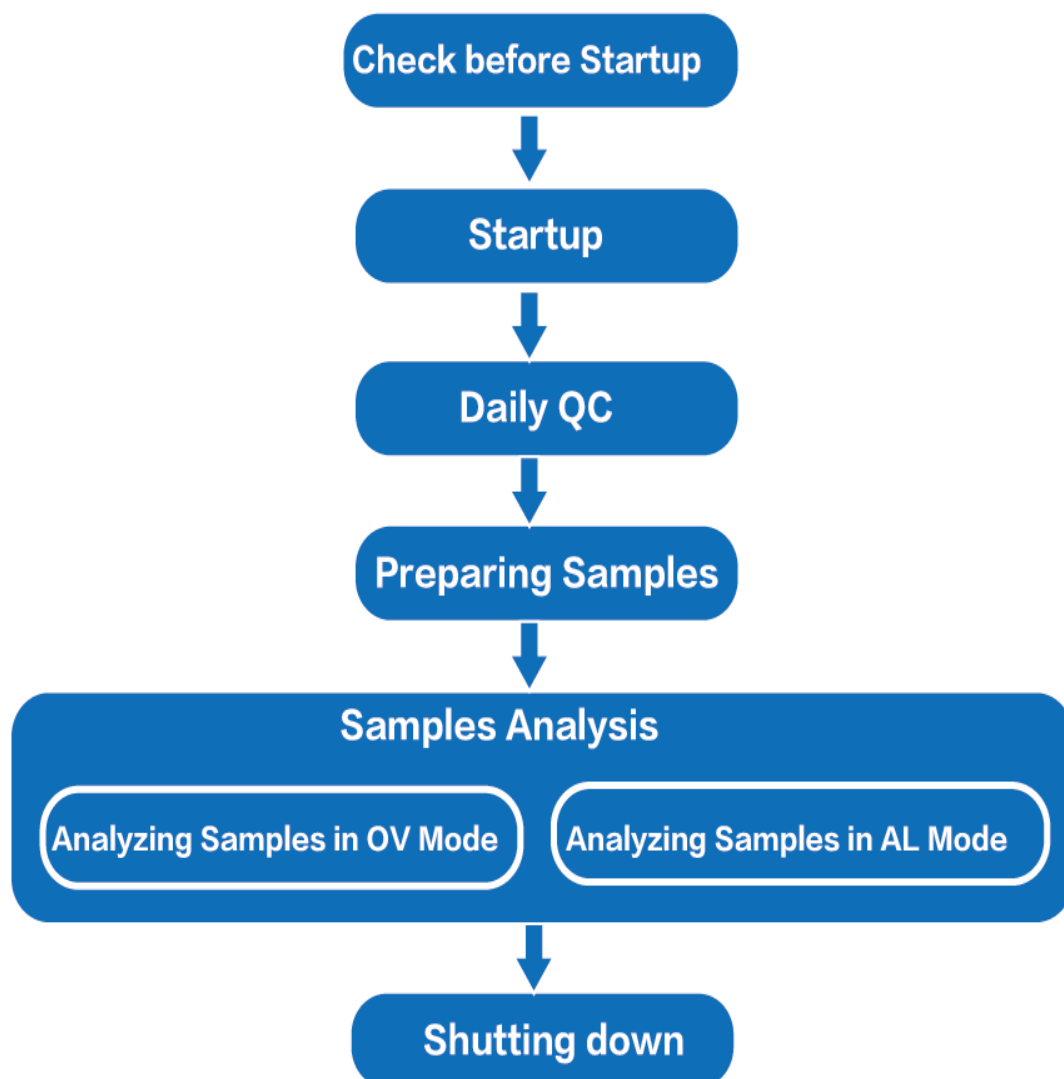
7

Operating Your Analyzer

7.1 Overview

This chapter provides step-by-step procedures for operating your analyzer on a daily basis.

A flow chart indicating the common daily operating process is presented below.



NOTE

- When the analyzer has fault or needs your attention, it beeps to alarm.

7.1.1 Introduction to the Count Screen

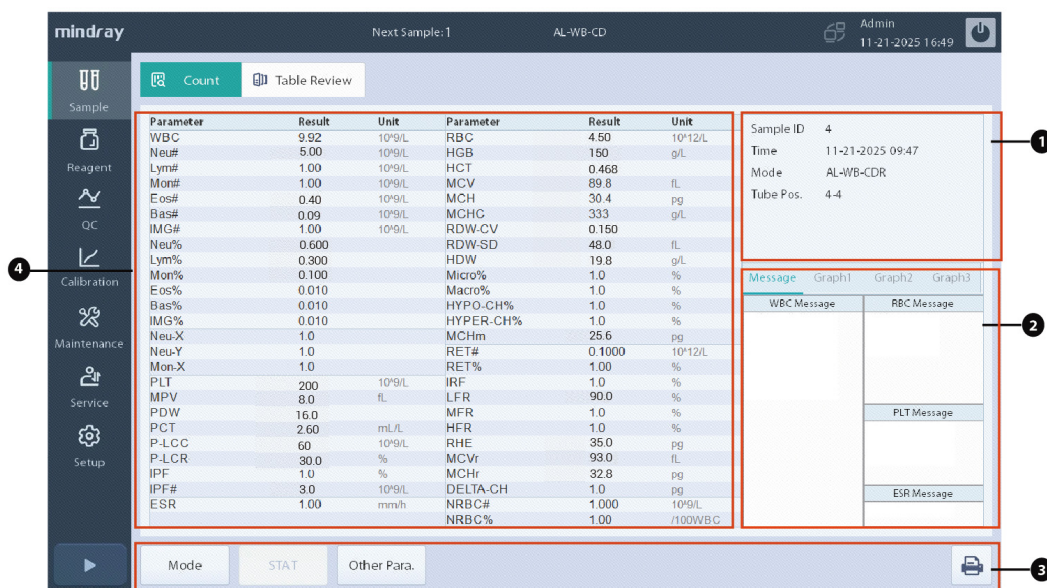


Figure 7-1 “Count” screen introduction

Table 7-1 Description of the count screen

		Description
1	Sample information area	The area displays sample information.
2	Messages and graphs area	Select the “ Message ” tab to review the abnormal sample, abnormal blood cell differential or morphology messages for current sample, select “ Graph1 ” and “ Graph2 ” to review the scattergram and histogram.
3	Button area	• “Mode”: select the button to open a “Request Sample” dialog box. • “STAT”: select “STAT” to run the STAT sample analysis when in auto-loading mode. • “Other parameters”: when sample analysis completes, select “Other parameters” button to review the RUO parameters results of current sample, or edit microscopic parameter. • button: select this button to print current sample results.
4	Report parameter result area	The area displays the results of report parameters.

7.1.2 Sorting Test Panel on “Request Sample” Screen

You can select up to 5 most used tests to display on top of the test selection area, so you can quickly select them when needed.

Follow below instruction:

1. Select “**Count**” – “**Mode**”.
✓ The “**Request Sample**” dialog box displays.
2. Select “**Mode Sort**”.
✓ The dialog box displays.
3. Select up to 5 most used tests.
4. Select “**OK**” to save the selection, or select “**Cancel**” to exit the dialog box.

7.2 Preparations before Operations

BIOHAZARD

- All the samples, controls, calibrators, reagents, wastes and areas contacted them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab coat, glasses, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.
- Be sure to dispose of reagents, waste, samples, consumables, etc. according to government

CAUTION

- The reagents are irritating to eyes, skin and mucous membranes. Wear proper personal protective equipment (e.g. gloves, lab coat, glasses, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.
- If reagents come into contact with your skin, rinse with plenty of water, and consult a physician if necessary; if reagents come into contact with your eyes, rinse with plenty of water immediately and consult a physician.
- Check and make sure the reagent sample tubes are properly connected before using the system. Otherwise, the results may be inaccurate.
- Keep your clothes, hairs and hands away from the moving parts to avoid injury. Do not touch the moving parts.

NOTE

- Before using the analyzer, confirm that the analyzer is in the normal working status; the indicator on the analyzer stays in green, and the screen does not display temperature, liquid level or other related alarms.

Perform the following checks before turning on the analyzer.

■ Checking the waste container status

Check and make sure the waste container (not supplied) is not full before starting up the analyzer.

To replace waste container, refer to **12.4 Replacing Waste Container**.

■ Checking the reagent status

After installing a new container of reagent, keep it still for a while before use.

Check whether the reagents are expired and frozen. And keep them still for 24 hours after a long-distance transportation.

Check whether there are enough reagents for the tests of the day. If the reagents run out during analysis, the analyzer will pause working automatically and prompt the operator to replace the reagents. Follow the instruction to replace the reagents.

For safety precautions on reagents, refer to **1.3.1 General Safety Precautions**; for use on reagents, controls and calibrators, refer to their instructions for use; for connecting and replacing the reagents, refer to **5.2.3 Connecting the Reagents** and **12.3 Replacing Reagents**.

■ Checking fluidic sample tubes and power cords

Check and make sure the reagent and waste tubes are properly connected and not bent.

Check and make sure the power cord of the analyzer is properly plugged into the power outlet. For power requirements, refer to **B.9 Power supply**.

■ Checking the printer (optional)

Check and make sure the printers are properly installed, and have enough paper. Check the printer power cord is properly plugged into the power outlet and the power cord is connected properly.

■ Checking the keyboard and mouse

Check whether the keyboard and mouse are properly connected to the analyzer.

7.3 Startup and Login

7.3.1 Starting up the Analyzer

WARNING

- Before turning on the analyzer, make sure the input voltage meets the requirement.

Start up the analyzer according to the actual condition:

Analyzer Status	Operation
When the power switch is turned off	Turn on the power switch on the back of the analyzer • The power switch lights in green
When the power switch is turned on	Press the [Standby] key on the side of the analyzer

✓ The system automatically performs the self-test procedure, background test cycle and initializes the system. When the self-test and startup initialization procedures complete, the analyzer software displays the main interface.

CAUTION

- Check the firmness of all the doors and covers before running the system, and make sure they will not open or get loose during analysis. Exercise caution when open/close, install/un-install the doors and covers to avoid dropping to the ground.

NOTE

- Time needed for initializing the fluidic systems depends on how the analyzer was previously shut down. Generally the startup process takes about 10 minutes.
- Before using the analyzer, check and ensure that the analyzer is in normal operation status, that is, the indicator on the analyzer is in steady green, and no temperature alarm, liquid level alarm, or any other alarm is reported.
- If any error occurs during initialization (e.g., the background test results exceeds the range of background/blank count), the alarm will display. For handling it, refer to *13 Troubleshooting*.
- Background check of the analyzer measures the particle interference and electrical interference.
- If the first background result obtained during the fluidic initialization process is out of the specified range, the analyzer will perform background check again.
- Since the pneumatic unit is controlled by the analyzer, the power switch of the pneumatic unit can be kept in the ON position.
- The sample ID corresponds to background check result is "0".
- The analyzer does not flag background check result with H/L or suspect flag.

7.3.2 Auto Startup

If the auto startup function is enabled, the analyzer will start up at the specified time.

For the setup of auto startup, refer to **6.19 Setting up Auto Startup/Shutdown**.

7.3.3 Switching Login Account

If needed, follow below instructions to switch the login account.

1. Select " " in the upper right corner of the software.
- ✓ A dialog box displays.
2. Select "**Logout**".

- √ The **"Logout"** dialog box displays.
3. Select **"OK"**.
4. In the dialog box, enter the new **"User ID"** and **"Password"**, and then select  .

7.4 Daily Quality Control (QC)

Before running any samples, run the controls to ensure reliable results of the analyzer. See Chapter **9 Using the QC Program** for details.

7.5 Preparing Samples



BIOHAZARD

- All the samples, controls, calibrators, reagents, wastes and areas contacted them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab coat, glasses, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.



CAUTION

- Do not reuse disposal products, otherwise the result may be inaccurate.
- Sample agglutination may result in inaccurate analysis results. Check the samples to see if there is any agglutination, if yes, process the samples according to your laboratory's protocols.
- To attain accurate analysis results, make sure the sample volume meets requirements. Otherwise, the results may be inaccurate.
- The samples stored under refrigeration condition (2°C–8°C) must be kept at room temperature for at least 15 minutes before analysis. Otherwise, the result may be inaccurate.
- Before analyzing samples in evacuated collection sample tubes, make sure the sample tube cap is clean before loading sample.
- Samples may volatilize if the sample tube cap is opened for too long time, which will result in incorrect analysis results.
- Be sure to use the Mindray-specified disposable products including evacuated blood collection sample tube, anticoagulant collection sample tubes and capillary sample tubes etc.

NOTE

- If the 2-way LIS/HIS mode is selected, then after the sample ID is entered/scanned and saved, all the corresponding information will be obtained from the LIS/HIS, and then the analyzer starts running per the obtained information. Once the running is finished, the result, graph and sample/patient information will be uploaded to the LIS/HIS.

7.5.1 Preparing Whole Blood Samples



CAUTION

- For the whole blood samples to be tested for WBC differential or PLT counts, store them at the room temperature and run them within 8 hours after collection.
- The samples not to be tested for WBC differential, MCV or PLT counts can be stored in a refrigerator at the temperature of 2°C–8°C for 24 hours after collection. The refrigerated samples must be kept at room temperature for at least 30 minutes before analysis.
- For ESR test samples, when stored at the room temperature, they shall be analyzed within 6 hours after collection; when stored under refrigerator condition (2°C–8°C), they shall be analyzed within 24 hours after collection.

NOTE

- **Capillary samples cannot be tested for ESR.**
- **Perform the analysis at least 5 minutes after collecting the capillary blood sample and it is recommended to complete the analysis within 1 hour.**
- **You cannot perform ESR tests using open-vial (OV) mode.**

1. Use clean K₂EDTA or K₃EDTA (1.5–2.2mg/mL blood) anticoagulant collection sample tubes to collect venous blood samples.
2. Mix the sample immediately according to your laboratory's protocol.

7.5.2 Preparing Predilute Samples **CAUTION**

- **After mixing the capillary sample with the diluent, be sure to wait at least for 3 minutes, and mix the sample again before running the sample.**
- **Be sure to run the prediluted samples within 30 minutes after the dilution; otherwise the results may be unreliable.**
- **Be sure to mix any sample that has been prepared for a while before running it.**
- **If there is much prediluted sample adhering to the sample tube cap or the sample tube wall, the analyzer may not be able to aspirate sufficient volume of the sample. When preparing prediluted samples, slightly tap the bottom of the EP sample tubes with a finger. There should be no less than 150µL of prediluted sample settled at the sample tube bottom.**

NOTE

- **You can dispense diluent into the sample tube by a pipette.**
- **Be sure to evaluate predilute stability based on your laboratory's sample population and sample collection techniques or methods.**

Prepare the prediluted samples in the ratio of 40:140 (blood sample: diluent)

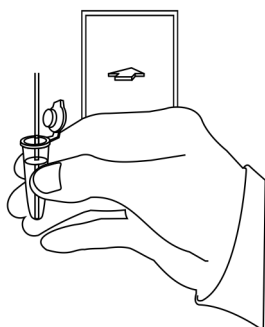
1. Select **"Mode"** on the **"Count"** screen.
- ✓ The **"Request Sample"** dialog box displays.
2. Select **"OV-PD"** and then select **"OK"**.
 - ✓ The analyzer switches to predilute analysis mode.
 - ✓ When the analyzer is on the predilute mode, the software status bar lights in orange.
3. Select **"Diluent"** in the pop-up window.
 - ✓ The analyzer prepares for dispensing diluent. When the preparation completes, a dialog box displays.

NOTE

- **The number in the "Cancel" button shows the number of diluent dispensation after your last tapping of the "Diluent" button. For example, if the number "2" displays in the "Cancel" button, it means the user has tapped the "Diluent" button twice.**

4. Prepare diluent.

Prepare a clean centrifugal sample tube and place a sample probe into its bottom vertically as instructed by the figure below. This can avoid fluid spillage, hangover or any bubble.



5. Press the **[Aspirate]** key to add diluent.
6. Remove the centrifugal sample tube when the buzzer sounds.
7. After the prediluted sample is prepared, tap the **"Cancel"** button to quit dispensing the diluent.
 ✓ The analyzer cleans the sample probe and quits the dispensing the diluent.

NOTE

- The analyzer dispenses 140µL of diluent a time.
- If more portions of diluent are needed, repeat step 3-5.
- You can also add diluent by pipette into the sample tube.

8. Add capillary blood to the diluent, close the sample tube cap and shake the sample tube to mix the sample.

7.5.3 Preparing Body Fluid Samples

The subtypes of body fluid supported by the analyzer include cerebrospinal fluid and serous cavity fluid (pleural fluid and ascitic fluid).

CAUTION

- To ensure sample stability, process serous cavity fluid (pleural fluid and ascitic fluid) samples and synovial fluid samples with EDTA anticoagulant. It is not recommended to process cerebrospinal fluid samples with anticoagulant.
- To attain accurate analysis results, it is recommended to complete the body fluid samples within 1 hour after collection.
- Misleading results may occur if the body fluid sample has floccules and clots. Follow your laboratory protocol to deal with such samples.

NOTE

- Slightly mix the body fluid samples. Do not use too much force.

Follow the steps below:

1. Use clean collection sample tubes or syringes to collect body fluid samples.
2. Mix the sample according to your laboratory's protocol.

7.5.4 Placing Barcode Labels

To read the barcode correctly, you must place the barcode label on the sample tube properly. For samples to be tested in auto-loading mode, you must place the barcode labels properly (as shown in **Figure 7-2 Where to**

place the barcode label), and in the right region of the sample sample tube (as shown in *Figure 7-3 How to place the barcode label*)

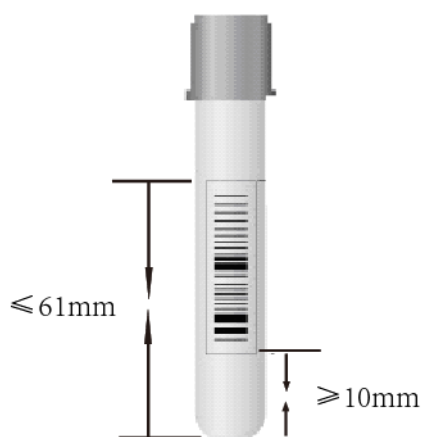


Figure 7-2 Where to place the barcode label

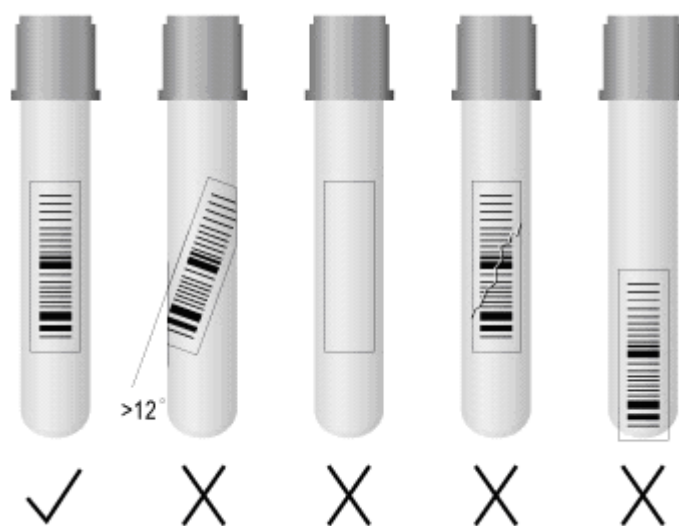


Figure 7-3 How to place the barcode label

7.6 Analyzing Samples in OV Mode

7.6.1 Setting up Analysis Orders in OV Mode

Follow below instructions to set analysis orders.

1. Select **"Mode"** on the **"Count"** screen.
✓ The **"Request Sample"** dialog box displays.
2. Manually enter the Sample ID, or use an external barcode scanner to scan the barcode label on the sample tube to enter Sample ID into the **"Sample ID"** field.

NOTE

- If you have set the **"Entry of next sample ID"** to **"Auto Increase"**, you only need to enter an ID for the first sample. The subsequent sample IDs will automatically increase by 1 based on the previous one.
- If you have set the **"Entry of next sample ID"** to **"Manual Entry"**, you need to enter each sample ID manually.
- For more information of the sample ID entry method under OV mode, refer to **6.11 Setting up for Parameters**.

3. Select the desired sampling mode and blood sample mode.
 ■ **OV-WB**
 ■ **OV-PD**
 ■ **OV-BF**
4. Set up the tests to run.
 ■ If the analyzer connects to LIS/labXpert, select **"Auto acquire"**. The analyzer will automatically inquire the order from LIS/labXpert.
 ■ If the analyzer does not connect to LIS/labXpert, confirm that **"Auto acquire"** is not selected, and manually set up the tests to run.
5. Select **"OK"**.
 ✓ The analyzer saves the mode settings and return to the **"Count"** screen.

NOTE

- If you have set Sample ID to **"auto increase"**, you only need to enter the sample ID for the first sample. The subsequent sample IDs will automatically increase by 1 based on the previous one. Refer to **6.10.1 Setting up the Entry Method of Next Sample ID in Open-Vial Mode Analysis**.
- When you have selected **"Auto acquire test mode"**, but the analyzer does not found the test order for the entered Sample ID on the 2-way LIS/HIS or on the labXpert software, it tests the sample using CD mode.
- When sample type or sample presentation mode changes, the analyzer will switch mode automatically.
- Refer to **3.3 Supported Test Panels** for supported test panels.
- 20 characters are allowed for sample ID maximally (including prefix); the ID must end with a number, and must not be consisted of **"0"** only.

7.6.2 Performing Sample Analysis in OV Mode

CAUTION

- After starting analysis, do not open the front cover of the analyzer.
- Before running samples, make sure the analyzer background results are in acceptable range.
- Make sure the sample probe contacts the sample tube bottom when aspirating sample. Otherwise, the aspirated volume may be imprecise.

- If the analyzer detects clogging or bubbles during the analysis, the corresponding error message will be displayed in the error message area and the results of all related parameters will be invalidated. See Chapter 13 *Troubleshooting* for solutions.

NOTE

- If you start sample analysis immediately after selecting work mode, then the default reference range is "General". After the analysis finishes, the analyzer will flag results per the reference range "General".
- When the ambient temperature is out of the specified operating range, the analyzer will alarm you for abnormal ambient temperature and the analysis results may be unreliable. See Chapter 13 *Troubleshooting* for solutions.

7.6.2.1 Analyzing samples in OV-WB mode

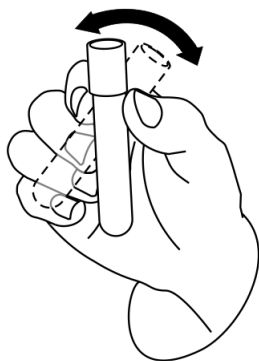
Make sure the sample probe tip does not touch the sample tube wall; otherwise the blood sample may spill.

Check the following before analysis:

- Sample(s) prepared in accordance with the laboratory protocol (refer to **7.5.1 Preparing Whole Blood Samples**).
- Analysis orders are set (refer to **7.6.1 Setting up Analysis Orders in OV Mode**).
- Be sure that the entered sample ID, rack No., sample tube No. and the analysis mode are strictly in accordance with the sample to be run.
- The analyzer is ready to run samples (i.e. the analyzer indicator stays in green).
- The upper area of the software main interface displays the mode as "OV-WB".

1. Mix the sample.

Shake the sample tube as instructed by the figure below to mix the sample thoroughly.



2. Present the sample tube uncapped, and make sure the sample probe can aspirate the well mixed sample.
3. Press the [**Aspirate**] key on the analyzer front cover again to start sample analysis.
 - ✓ The sample probe automatically aspirates sample.

When you hear the beep, remove the sample sample tube.

- ✓ The analyzer automatically analyzes the sample, the analyzer indicator is flickering in green.
- ✓ When the analysis completes, the analyzer indicator returns to its ready status (stay in green).
- ✓ The screen displays the current sample results, histograms, scattergrams and flags (if any).

NOTE

- When the analyzer has fault or needs your attention, it beeps to alarm.

7.6.2.2 Analyzing samples in OV- PD mode

Check the following before analysis:

- Sample(s) prepared in accordance with the laboratory protocol (refer to **7.5.2 Preparing Predilute Samples**).
 - Analysis orders are set (refer to **7.6.1 Setting up Analysis Orders in OV Mode**).
 - Be sure that the entered sample ID, and the analysis mode are strictly in accordance with the sample to be run.
 - The analyzer is ready to run samples (i.e. the analyzer indicator stays in green).
 - The upper area of the software main interface displays the mode as “**OV-PD**”.
1. Shake the capped sample tube of the prediluted sample to mix it thoroughly.
 2. Open the cap of the sample tube carefully and present the sample to the sample probe.
 3. Press the **[Aspirate]** key.
- ✓ If “**Predilute mode prompt**” is enabled, a dialog box will pop up to remind you that the current analysis mode is predilute.

NOTE

- **For more information about the enabling or disabling predilute mode prompt, see 6.17 Setting up for Other Settings.**

4. Press the **[Aspirate]** key on the analyzer front cover again to start sample analysis.
 - ✓ The sample probe automatically aspirates sample.
5. When you hear the beep, remove the sample tube.
 - ✓ The analyzer automatically analyzes the sample, the analyzer indicator is flickering in green.
 - ✓ When the analysis completes, the analyzer indicator returns to “**Ready**” status (stay in green).
 - ✓ The screen displays the current sample results, histograms, scattergrams and flags (if there is).

NOTE

- **When the analyzer has fault or needs your attention, it beeps to alarm.**

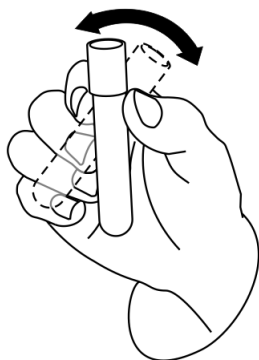
7.6.2.3 Analyzing samples in OV-BF mode

The analyzer only supports body fluid sample analysis under OV mode.

Check the following before analysis:

- Sample(s) prepared in accordance with the laboratory protocol (refer to **7.5.3 Preparing Body Fluid Samples**).
 - Analysis orders are set (refer to **7.6.1 Setting up Analysis Orders in OV Mode**).
 - The entered sample ID and the analysis mode are strictly in accordance with the sample to be run.
 - The analyzer is ready to run samples (i.e. the analyzer indicator stays in green).
 - The upper area of the software main interface displays the mode as “**OV-BF**”.
1. Mix the sample.

- ◆ Shake the sample tube as instructed by the picture below to mix the sample thoroughly.



2. Place the sample tube uncapped to the sample probe.
3. Press the **[Aspirate]** key on the analyzer front cover again to start sample analysis.
 - ✓ The sample probe automatically aspirates sample.
4. When you hear the beep, remove the sample tube.
 - ✓ The analyzer automatically analyzes the sample, the analyzer indicator is flickering in green.
 - ✓ When the analysis completes, the analyzer indicator returns to "Ready" status (stay in green).
 - ✓ The screen displays the current sample results, histograms, scattergrams and flags (if there is).



CAUTION

- When the room temperature is out of the temperature range for the normal-operation, the parameter results may not be accurate.
- Misleading results may occur if the body fluid sample has floccules, clots, cryoglobulin, leukocyte clumps, bacteria, or high protein. Follow your laboratory protocol to deal with such samples.

NOTE

- When the analyzer has fault or needs your attention, it beeps to alarm.

7.6.2.4 Non-stop sample analysis under OV-WB mode

The analyzer supports non-stop analysis of samples under OV-WB mode. You can set analysis order for a second sample as soon as the analyzer finishes sample aspiration for the previous sample, instead of waiting until the previous sample analysis completes. In this way, the time for sample analyses is reduced.

Check the following before analysis:

- Sample(s) are prepared in accordance with laboratory protocols.
 - The upper area of the software main interface displays the correct analysis mode.
 - The analyzer is ready to run samples (i.e. the analyzer indicator stays in green).
1. Present a well-mixed sample uncapped to the sample probe for aspiration.
 2. Press the **[Aspirate]** button on the analyzer front cover to start sample analysis.
 3. After the sample probe finishes sample aspiration for the first sample, the analyzer beeps twice. Remove away the sample safely.
 4. (Optional) If needed, modify the sample information and the analysis information for the next sample.
 - On the **"Count"** screen, select **"Mode"**, and set up the sample ID, sample mode and analysis mode as necessary.
 - When using an external barcode scanner, use the scanner to scan the sample ID of the next sample.
 - ✓ The analyzer software displays the information for the next sample.
 5. Confirm whether the sample ID, sample mode and analysis mode are correct. If necessary, select the **"Mode"** button to edit the information.

- When the analyzer beeps, and the status indicator on the front cover flickers in yellow, present the second sample uncapped to the sample probe.

NOTE

- If you have set the sample ID entry method to “Auto Increase”, you do not need to enter sample ID. For the entry method of next sample ID under open-vial analysis mode, see 6.11 *Setting up for Parameters*.
- When the analyzer has fault or needs your attention, it beeps to alarm.
- If you have set the sample ID entry method to “Auto Increase”, you may choose not to set analysis mode for the following samples. The analyzer will run the samples in the same mode of the previous sample.

- Press the [Aspirate] key to start sample analysis.
✓ The sample probe automatically aspirates sample.
- If necessary, repeat steps 2 to 6 to run other samples.

7.7 Analyzing Samples in Auto-Loading Mode

7.7.1 Setting up Analysis Orders in Auto-Loading Mode

You can set up analysis orders on the main unit of the analyzer. When the analyzer connects to LIS/labXpert, you can acquire order information from LIS/labXpert.

Follow below instructions:

- On the “Count” screen, select “Mode”.
✓ The “Request Sample” dialog box displays.
- Select “AL-WB” mode.
- Enter the sample ID, rack No., and the sample rack No. as needed.

You can enter the information in two ways:

Scan the sample ID by the built-in barcode scanner	- Select “Auto scan”. (Optional) Edit “Tube No.”, if needed.
Manually scan or enter the sample ID	- Clear the selection of “Auto scan”. (Optional) if needed, edit “Tube No.” and “Sample ID”.

NOTE

- If your analyzer configured with a rotatory barcode scanner, you can select “Auto scan”, and the analyzer will automatically inquire the sample ID. When you select “Auto scan”, make sure the barcode labels on samples are intact and readable.
 - If “Auto scan” is not selected, you should enter the sample ID for the first sample. The subsequent sample IDs will automatically increase by 1 based previous one.
 - 20 characters are allowed for sample ID maximally (including prefix); the ID must end with a number, and must not be consisted of “0” only.
 - Refer to 3.3 *Supported Test Panels* for supported test panels.
- Select tests to run.

When the analyzer connects to LIS/labXpert, you can inquire orders from LIS/labXpert.	Select “Auto acquire”. ✓ The analyzer will automatically inquire orders from LIS/labXpert.
Manually select tests to run	- Clear the selection of “Auto acquire”. - Select the desired tests.

5. Select **"OK"**.
6. The analyzer saves the settings and returns to the **"Count"** screen.

7.7.2 Performing Sample Analysis in Auto-Loading Mode



BIOHAZARD

- **Broken collection sample tubes may cause personal injury and/or biohazard. Exercise caution when loading the collection sample tubes to the rack or getting the collection sample tubes from the rack, be sure not to break the sample tubes.**



CAUTION

- **Repeated piercing the cap of the evacuated blood collection sample tube may damage the cap, and the scraps produced may result in inaccurate analysis results. It is recommended that each sample tube is pierced for no more than 3 times.**
- **Do not run prediluted samples under autoloading mode. Otherwise the results may be inaccurate.**

NOTE

- **If abnormal power interruption occurs after analysis is started, you should remove the racks manually and open the front cover to see if there are dropped sample tubes to be removed.**
- **Before analysis, make sure the entered sample ID, rack No., sample tube No. and the analysis mode are strictly in accordance with the sample to be run.**

Check the following before analysis:

- Sample(s) prepared in accordance with the laboratory protocol (refer to **7.5.1 Preparing Whole Blood Samples**).
 - Analysis orders are set (refer to **7.7.1 Setting up Analysis Orders in Auto-Loading Mode**).
 - Make sure the entered sample ID, rack No., sample tube No. and the analysis mode are strictly in accordance with the sample to be run.
 - The analyzer is ready to run samples (i.e. the analyzer indicator stays in green).
 - The upper area of software main interface displays the mode as **"AL-WB"**.
 - When you have selected **"Auto-Scan sample ID"** as the sample ID entry method, make sure the barcode labels are clear, complete and are properly pasted on the sample tubes.
1. Place the sample rack with the sample tubes on the right tray of the autoloader, with the side opening facing the analyzer.
- When **"Automatically load tube racks on the autoloader"** is enabled, the analyzer will load and start the analysis automatically.
 - When **"Automatically load tube racks on the autoloader"** is not enabled, select  on the software or press **[Count]** key on the front cover of the analyzer to start sample analysis.

NOTE

- **For setting sample tube loading methods in auto-loading mode, refer to 6.9.4 Setting Sample Tube Loading Methods.**
- **If abnormal power interruption occurs after analysis is started, you should remove the racks manually and open the front cover check if there are dropped sample tubes.**

- ✓ The analyzer will automatically run the sample. During this process, the analyzer indicator is flickering in green.
 - ✓ After every analysis cycle, the results will be saved to the sample database.
2. (Optional) When you have selected **"Display summary after autoloading finished"**, a summary is presented. Review the autoloading summary.

NOTE

- **This function is only available when you have selected “Display summary after autoloading finished.” For more information, refer to 6.9.3 Display Summary after Autoloading Finishes.**

The summary indicates the quantities of routine samples and controls that have been analyzed, the number of empty sample tube positions on sample racks, the number of samples for which the sample ID was not read correctly, the number of sample racks with invalid rack No., the instances of sample clogging, and the cases of abnormal aspiration.

- When the number of empty sample tube positions, the number of samples with invalid sample ID, number of sample racks with invalid rack No., the instances of sample clogging, and the cases of abnormal aspiration are not “0”, select “**Details...**” to check the sample ID, test time and sample tube position of the corresponding samples.
 - Select “**OK**” to exit the dialog box.
- When the analysis completes, all sample racks are moved to the left tray of the autoloader automatically, and then you can safely remove the racks.

NOTE

- **When you have selected “Auto-Scan sample ID” as sample ID entry method, but the sample ID scanned is invalid, then “Inv.test time_sample tube position” will be the sample ID.**
- **If more racks are needed when running the samples, you should load the rack from the right of the autoloader while removing the completed rack from the left of the autoloader in time.**
- **If “Auto-scan rack No.” is selected and the rack No. scanned is invalid, then “??” will be the rack No.. For enabling or disabling “Auto-scan rack No.”. Refer to 6.10.3 Setting up Auto-Scan Rack No. in Auto-Loading Mode.**

7.7.3 Stop Count

If you need to stop auto-loading analysis during the analysis process, select .

- ✓ The analyzer will stop running when it completes the current sample analysis.
- ✓ Sample racks will be automatically transported to the unloading tray (the left tray of the autoloader).

7.7.4 Performing Auto Re-exam

When “**Auto-exam**” function is enabled on the labXpert software, the analyzer automatically transfers the samples back for re-exam.

If a sample is re-examined, the graph review screen of the sample for the second test (re-exam) displays “**Re-exam Result**”.

NOTE

- **When re-exam rules are enabled, take away the sample racks only when they are completely unloaded to the unload tray.**
- **When the “Auto-exam” function is enabled, if you want to add more sample racks during an autoloading analysis cycle, always place the new sample racks in the area behind the sample rack pushers (shadowed area shown below). Otherwise the path for the returning re-exam sample racks may be blocked.**

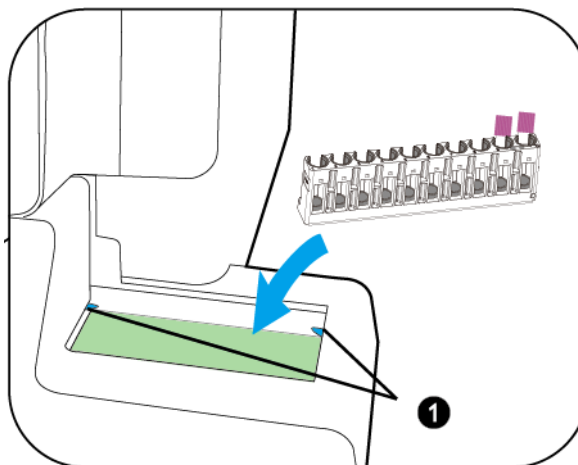


Figure 7-4 Putting sample racks to the rear shadow area of the rack pusher

	Name
1	Sample rack pusher

7.8 Analyzing STAT Samples

Follow the instructions to insert STAT samples when auto-loading analysis is in process.

- On the **"Count"** screen, select **"STAT"**.
 - ✓ After the current sample analysis has been completed, the analyzer will pause the auto-loading process and convert from the AL mode to OV mode.
 - ✓ The **"Request Sample"** dialog box displays.
- Set up the analysis order.
 - Enter the sample information and select the desired sample presentation mode, sample type and tests to be run.
 - Select **"OK"** to save the settings and exit the **"Request Sample"** dialog box.
- Select  on the main interface or press the [Count] key on the front cover to start STAT sample analysis.
- After the analyzer finishes analyzing all STAT samples, select **"Exit STAT"**.
 - ✓ The analyzer switches back to auto-loading mode, then you can proceed with analysis in auto-loading mode.

NOTE

- For the information of setting analysis order under OV mode, refer to 7.6.1 *Setting up Analysis Orders in OV Mode*.
- For the information of running samples under OV mode, refer to 7.6.2 *Performing Sample Analysis in OV Mode*.

7.9 Entering and Exiting Standby Status

When the analyzer is free from fluidic operations for a period of time, the analyzer automatically enters the standby status.

When the analyzer is standby, the status bar at the bottom of the main unit software screen is in gray.

When the user starts sample tests, or performs any operation that initiates fluidic system actions or moving parts actions, the analyzer automatically exits standby.

NOTE

- Refer to **6.18 Setting up the Waiting Time before the Analyzer Enters Standby Status** for how to edit waiting time before entering the standby mode.

7.10 Shutting down the Analyzer

CAUTION

- In daily use of the analyzer, it is recommended that you press [Standby] key on the front cover of the analyzer to turn off the analyzing unit, while keeping the main power on.
- You must perform the shutdown procedure as instructed in the Operator's Manual. Otherwise the interior fluidic sample tubes may be contaminated, which may cause inaccurate results.
- Do not start up the analyzer immediately after shutdown. Wait at least 10 seconds. Otherwise, the analyzer may be damaged.

7.10.1 Manual Shutdown

Follow the instructions to shut down the analyzer.

1. In the upper right corner of the screen, select .
- √ A dialog box displays.
2. Select **"Shutdown"**.
- √ The **"Shutdown"** dialog box displays.
3. Select **"OK"** to perform the shutdown procedure.
4. (Optional) When a dialog box displays requiring you to perform Probe Cleanser maintenance, place the Probe Cleanser under the sampling probe as instructed, and select **"OK"** to start Probe Cleanser maintenance.
5. After Probe Cleanser maintenance completes, the touch screen will black out and message will be displayed instructing you to turn off the power of the analyzer.

NOTE

- For daily Probe Cleanser maintenance, refer to **12.6 Probe Cleanser Maintenance**.

- √ The software screen displays **"The analyzer is about to power off"**.
- 6. (Optional) Empty the waste container and dispose of the waste properly.
- 7. Turn off the power switch on the back of the analyzer.

7.10.2 Auto Shutdown

If **"Auto Startup/Shutdown"** function is enabled, the analyzer will be shut down at the specified time. For the setup of auto shutdown, refer to **6.19 Setting up Auto Startup/Shutdown**.

NOTE

- To enable the **"Auto Shutdown"** function, do not power off the analyzer.

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8

Reviewing Sample Results

8.1 Introduction

After every analysis cycle, the analyzer automatically saves the analysis results into the sample database. Up to 100,000 analysis results can be stored in the analyzer.

NOTE

- When the maximum number has been reached, the newest result will overwrite the oldest.

8.2 Reviewing Sample Results

The analyzer supports two ways of reviewing sample results:

- Table Review
- Graph Review

NOTE

- Screenshots in this manual are for reference only. The screen displayed on your system may be different, depending on your actual configuration.

8.2.1 Entering the “Table Review” Screen

Select “Sample”–“Table Review” to go to “Table Review” interface.

The “Table Review” screen displays the sample information in sequence of sample analysis time (from top to bottom, latest to oldest), including sample test information, parameter results, and patient data.

8.2.1.1 Browsing sample records

Select  /  /  /  /  /  to turn to the left/right column (page);

Select  /  /  /  and  /  to turn to the previous/next row (page).

8.2.1.2 Selecting/clearing selection of records

- Select a unselected record to select it.
 - Select a selected record to clear the selection.
 - Select records that you want to select one by one to select multiple records.
- √ The selected sample records are highlighted.

8.2.1.3 Searching for sample records

You can search for sample records that match defined conditions.

1. Select “Sample”–“Table Review” to go to “Table Review” screen.
 2. Select “Search” in the tool bar.
- √ A dialog box displays.
3. Define the searching conditions.

- To search for samples on the current day by sample status

Options	Description
Not Validated Today	Select " Not Validated Today " to search for the samples on the current day that are not validated.
Not Printed Today	Select " Not Printed Today " to search for the samples on the current day that are not printed.
Not Transmitted Today	Select " Not Transmitted Today " to search for the samples on the current day that are not transmitted.

- To search for sample records that match defined conditions

You can define one or more searching conditions.

When you have defined more than one searching conditions, the analyzer will search for the sample records that match all defined conditions.

Item	Description	Remarks
Sample ID	Enter " Sample ID " as a search condition, the analyzer searches for and displays all sample records whose sample ID includes the entered sample ID.	For example, if you enter "1235" into the " Sample ID " field, the analyzer will search for all sample records whose sample ID includes "1235".
Patient ID	Enter " Patient ID " as a search condition, the analyzer searches for and displays all sample records whose patient ID includes the entered patient ID.	For example, if you enter "1235" into the " Patient ID " field, the analyzer will search for all sample records whose patient ID includes "1235".
Last Name	Enter patient last name as a search condition. The analyzer will search for all sample records whose patient name matches the defined name.	/
Date	Define the period in the " Date " as a search condition. The analyzer will search for all sample records tested between the define time period.	The " Date " range is set to the current date by default.
Sample No.	Define the range of sample No.. The analyzer searches and displays all sample records within the selected range.	/
Sample State	Select " Not validated ", " Not printed " and " Not transmitted ", and the analyzer searches from all sample records that match the selected state.	You can select more than one sample state, and the analyzer searches from all sample records that match the either of the selected states.
Auto select searched record	When you have selected " Auto select searched record ", the analyzer will highlight and display all searched results that match the defined conditions.	/

4. Select "**OK**".
- ✓ All searched results display on the screen.
5. (Optional) Select "**Return**" to return to the "**Table Review**" screen.

8.2.1.4 Editing sample information

1. Select "**Sample**"–"**Table Review**" to go to "**Table Review**" screen.
2. Select the sample records of which you want to edit the sample information.

- √ The selected sample record is highlighted.
- 3. Select **"Edit Info."**
- √ A dialog box displays.
- 4. See below for setting description.

Item	Description	Notes
Sample ID	Scan the sample ID by a barcode scanner or enter the sample ID manually.	/
Patient ID	Enter the patient ID.	/
Last Name	Enter the patient name.	/
Gender	Select from the pull-down list: • Male • Female	When "Date of Birth" or "Age" is selected, the analyzer will automatically match the reference group for the patient by the gender and age.
Date of Birth	Enter the birth date of the patient.	/
Age	Manually enter the patient age.	When you enter the birth date of the patient, the analyzer will automatically calculate the age of the patient and match the reference group by the gender and age.
Patient Type	Select from the pull-down list: • Outpatient • Inpatient • Medical Examination • STAT	/
Ref. Group	Select the reference group from the pull-down list.	For reference groups, refer to 6.11.2 Setting up Parameter Reference Ranges .
Department	Enter the department that delivers the sample.	/
Bed No.	If necessary, enter the bed No. of the patient.	/
Draw Time	Enter the time when the sample is drawn from the patient.	/
Delivery Time	Enter the time when the sample is delivered.	/
Clinician	Enter the name of the clinician.	/
Time	Enter the time when the sample is examined.	"Time" is automatically generated by the system.
Mode	Display the sample type, sampling mode and analysis mode of the sample in test.	"Mode" is automatically generated by the system.
Operator	Display the account ID of the operator.	"Operator" is automatically generated by the system.
Validated By	Display which account ID validates the results.	"Validated By" is automatically generated by the system.
Comments	Enter the comments.	/

- 5. Edit the information as needed.

6. Select **"OK"** to save the entered information.

NOTE

- For the setup of reference groups, refer to **6.11.2 Setting up Parameter Reference Ranges**.

8.2.1.5 Validating /canceling validating sample results

Administrators can validate/cancel validating the sample results.

1. Select **"Sample"**–**"Table Review"** to go to **"Table Review"** interface.
2. Select the sample results you want to validate or cancel validating.
- ✓ The selected sample record(s) is/are highlighted.
3. Select  "or"  to validate or cancel validating the sample results.

NOTE

- You cannot validate invalid sample results.

8.2.1.6 Exporting sample results

Administrators can export the sample records to a USB flash drive.

Before exporting sample records, make sure you have insert a USB flash drive safely to the USB interfaces on the analyzer.



CAUTION

- The user should ensure the data safety of the USB flash drive connecting to the analyzer.

Follow below instructions:

1. Select **"Sample"**–**"Table Review"** to go to **"Table Review"** screen.
2. Select the sample(s) you want to export.

Export selected sample records	1. Select one or more sample records that you want to export. 2. Select . 3. In the "Export Range" field, select "Selected records".
Export all sample results	1. Select . 2. In the "Export Range" field, select "All records".

3. (Optional) In the **"Export"** dialog box, select what you want to export. See below for the setting description.

Options	Description	Notes
Sample data	When selected, you can export the patient information, sample test information, sample results, etc.	/
Graph data	When selected, you can export histogram, scattergram, etc..	/
Flags	When selected, you can export the sample results with flags (if any).	For description of sample flags, refer to 8.3.1 Parameter Flags .
Other parameters	When selected, you can export information as follows: • RUO parameters • Reference results	For description of "Other parameters" , refer to 8.2.2.6 Reviewing reference results .

4. Select **"OK"**.
- √ The analyzer exports the selected sample records to the USB flash drive.

8.2.1.7 Calculating CV value

Follow the instruction to calculate the CV values of sample results.

1. Select **"Sample"–"Table Review"** to go to **"Table Review"** screen.
 2. Select the sample records of which you want to calculate CV value.
 3. Select **"CV"**.
- √ The screen displays the Mean value, SD value and the CV value for each parameter.

NOTE

- **To calculate CV values, select at least 3 samples.**

8.2.1.8 Communicating with LIS

You can set the analyzer to transmit the sample results to LIS automatically after it finishes the analysis.

NOTE

- **For the setup about LIS, refer to 6.4.1 Setting up Communication Protocol.**
- **Prior to transmitting sample results to LIS, make sure the network connection is good.**

Follow below instructions:

1. Select **"Sample"–"Table Review"** to go to **"Table Review"** screen.
2. Select the sample(s) you want to communicate.

Transmit selected record	1. Select ". 2. Select "Selected records".
Transmit all records	1. Select ". 2. Select "All records".

3. Select **"OK"**.

8.2.1.9 Deleting sample records

The administrators can delete sample records on the **"Table Review"** screen.

Follow below instructions:

1. Select **"Sample"–"Table Review"** to go to **"Table Review"** screen.
2. Select the sample(s) you want to delete.

Delete the selected records	1. Select one or multiple samples records you want to delete. 2. Select "Delete". 3. On the "Delete" dialog box, select "Selected records".
Delete current all records	1. Select "Delete". 2. On the "Delete" dialog box, select "All records".

3. Select **"OK"**.
- √ All records or the selected records are deleted.

8.2.2 Entering the “Graph” Screen

Follow below instructions:

1. Select “**Sample**”–“**Table Review**” to go to “**Table Review**” screen.
2. Select one or more sample records of which you want to review the graph data.
√ The selected sample record is highlighted.
3. Select “**Graph**” to go to “**Graph**” screen. Operations on the “**Graph**” screen:
 - Select “**Previous**” or “**Next**” to view previous or next sample.
 - Select “**Table Review**” to go to the “**Table Review**” screen.

8.2.2.1 Reviewing RUO parameters results

Follow below instructions:

1. Select “**Sample**”–“**Table Review**” to go to “**Table Review**” screen.
2. Select the samples of which you want to review the RUO parameter results.
√ The selected sample record is highlighted.
3. Select “**Graph**” to go to “**Graph**” screen.
4. Select “**Other Para.**”
√ “**Other Para.**” screen displays.
5. Select “**RUO Para.**” to view “**RUO Para.**” results.

NOTE

- The RUO parameters cannot be used for diagnostic purpose.
- When the analysis is made in PLT-8X mode or the system is triggered to make an analysis in PLT-8X mode, the PLT-O parameter result displayed in the parentheses has one more decimal place than the normal parameter result.

8.2.2.2 Entering, reviewing and printing microscopic parameter results

Before you can review microscopic parameter results, make sure that the microscopic parameter category has been set.

NOTE

- For information on how to define microscopic parameters, refer to 6.11.3 *Setting up Microscopic Parameters*.

1. Select “**Sample**”–“**Table Review**” to go to “**Table Review**” screen.
2. Select the sample you want to review the results of microscopic parameters.
√ The selected sample record is highlighted.
3. Select “**Graph**” to go to “**Graph**” screen.
4. Select “**Other Para.**”
√ The “**Other Para.**” dialog displays.
5. Select “**Microscopic Para.**” to review the results of “**Microscopic Para.**” or select “**Result**” edit box to enter the microscopic parameter results.
6. (Optional) If necessary, select “**Print**” on the “**Microscopic Para.**” screen to print microscopic parameter results.

NOTE

- For information on how to define microscopic parameters, refer to 6.11.3 *Setting up Microscopic Parameters*.

- **You cannot edit the microscopic parameter results of a validated sample.**
-

8.2.2.3 Reviewing special information

Follow below instructions:

1. Select **"Sample"–"Table Review"** to go to **"Table Review"** screen.
2. Select the samples of which you want to review the special information.
- √ The selected sample record is highlighted.
3. Select **"Graph"** to go to **"Graph"** screen.
4. Select **"Special Info."**
- √ The screen displays the information relevant to the analyzer when analyzing the current sample.
5. (Optional) select the **"Error Information"** pull-down list to review the error lists (if there are) when analyzing the current sample.

8.2.2.4 Reviewing traceability information

The **"Traceability"** tab displays the information of the reagents and controls relevant to the selected sample test.

Follow below instructions:

1. Select **"Sample"–"Table Review"** to go to **"Table Review"** screen.
2. Select the samples of which you want to review the traceability information.
- √ The selected sample record is highlighted.
3. Select **"Graph"** to go to **"Graph"** screen.
4. Select **"Traceability"**.
- √ The **"Traceability"** tab displays the information of the reagents and controls when analyzing the current sample.

8.2.2.5 Editing sample results

Administrators can edit analysis results.

1. Select **"Sample"–"Table Review"** to go to **"Table Review"** screen.
2. Select the sample record of which you want to edit the analysis results.
- √ The selected sample record is highlighted.
3. Select **"Graph"** to go to **"Graph"** screen.
4. Select **"Edit Result"**.
- √ The **"Edit Result"** dialog box displays.
5. Select the edit box of the parameter you want to edit and modify the parameter result.
6. Select **"OK"** to close the dialog box.
- √ The analyzer saves the new parameter results.
- √ The parameters result that has been manually modified will be flagged with an **"E"**. A parameter result modified due to the manual modification of another parameter result is flagged with an **"e"**.

NOTE

- **You cannot edit the validated sample results and background test results.**
-

Follow below instructions to restore the edited results to its original analysis results:

1. Select **"Sample"–"Table Review"** to go to **"Table Review"** screen.
2. Select the sample record of which you want to edit the analysis results.
- √ The selected sample record is highlighted.

3. Select **"Graph"** to go to **"Graph"** screen.
4. Select **"Edit Result"**.
- ✓ The **"Edit Result"** dialog box displays.
5. Select **"Restore"**.
- ✓ The edited results will be changed back into the initial values.
- ✓ The **"E"** and **"e"** flags disappear on the **"Table Review"** or **"Graph"** tab.

NOTE

- **The analyzer stores original analysis results of the latest 2,000 samples whose results have been modified.**

8.2.2.6 Reviewing reference results

When the system decides a parameter result is not reliable (for example, the DIFF parameter results of certain abnormal samples), the software will not display the results in values, but as **"*****"**. The estimated results will be provided on the reference results tab. Decide if the reference results can be used for report based on a comprehensive consideration regarding information, like historical results, microscopic exam results or re-exam results.

Follow below instructions:

1. Select **"Sample"** – **"Table Review"** to go to **"Table Review"** screen.
2. Select the sample you want to review the reference results.
- ✓ The selected sample record is highlighted.
3. Select **"Graph"** to go to **"Graph"** screen.
4. Select **"Other Para."**.
- ✓ **"Other Para."** tab displays.
5. Select **"Reference results"** to review **"Reference results"** of the sample.

8.3 Flags of Analysis Results

The analyzer provides two types of flags for analysis results:

- parameter flags;
- flags of abnormal blood cell differential or morphology

8.3.1 Parameter Flags

The analyzer provides the following parameter flags.

Flags	Messages	Meaning
"H" and "L" (default) or "h" and "l" or "↑" and "↓"	High and low result flags	The analysis result exceeds the upper or lower limit of the reference range, but still within the display range
"R" (default) or "r"	Suspicious flag	The analysis result is suspicious
"&"	Algorithm-rectified flag	The analysis result is rectified by the algorithm of the instrument
"@" (for complete blood count tests)	Out of linearity range flag	The analysis result is out of the linearity range

"@" (for ESR tests)	Out of parameter result report range flag	The analysis result is out of the report range (0~140 mm/h) of parameter result
"++++"	Out of parameter result display range flag	The analysis result is out of the parameter result display range
"*****"	Screened results	When the analyzer deems that some parameter results are not reliable (for example, the DIFF results for certain types of abnormal samples), the results of the parameters would be screened from the report screen and display as "*****".

NOTE

- The results of background check will not be flagged for abnormal parameters, abnormal blood cell differential or morphology.
- For the linearity range of each parameter, refer to *B.5.2 Linearity Range*.
- For the display ranges of the major parameters, refer to *B.5.1 Background/Blank Count Requirements*.

8.3.2 Flags of Abnormal Blood Cell Differential or Morphology

NOTE

- Abnormal cells may not necessarily trigger the flags during the analysis process, it is recommended that reexamination is conducted per the operation instruction of your laboratory.

The analyzer reports the flags for the following abnormal blood cell differential or morphology.

Flag Message	Indication	Criteria
WBC Abn. Scattergram	WBC scattergram abnormal	The distribution of DIFF channel scattergram is abnormal
WNB Abn. Scattergram	WNB channel scattergram is abnormal	The distribution of WNB channel scattergram is abnormal
Neutropenia	Neu# low	Neu# < $1.00 \times 10^9/L$
Neutrophilia	Neu# high	Neu# > $11.00 \times 10^9/L$
Lymphopenia	Lym# low	Lym# < $0.80 \times 10^9/L$
Lymphocytosis	Lym# high	Lym# > $4.00 \times 10^9/L$
Monocytosis	Mon# high	Mon# > $1.50 \times 10^9/L$
Eosinophilia	Eos# high	Eos# > $0.70 \times 10^9/L$
Basophilia	Bas# high	Bas# > $0.20 \times 10^9/L$
Leukocytopenia	Low WBC count	WBC < $2.50 \times 10^9/L$
Leucocytosis	WBC count high	WBC > $18.00 \times 10^9/L$
NRBC Present	NRBC detected in the WNB channel	NRBC% > 1% and NRBC# > 0.01
Blasts?	Possible presence of blast cells	Presence of excessive dots in blast sensitive region of the scattergram
Abn. Lymph/blast?	Possible presence of abnormal lymphocytes or blasts	Presence of excessive dots in abnormal lymphocyte/blast sensitive region of the scattergram

Flag Message	Indication	Criteria
Immature Gran?	TPossible presence of immature granulocytes	Presence of excessive dots in immature granulocyte sensitive region of the scattergram
Left Shift?	Possibility of left shift	Obvious scattering in the left shift area on the scattergram
Atypical Lymph?	Possible presence of atypical lymphocytes	Presence of excessive dots in atypical lymphocyte sensitive region of the scattergram
RBC Lyse Resistance?	Presence of abnormally distributed dots in WBC sensitive region of the DIFF scattergram	Atypical scattering in WBC sensitive region of the DIFF or WNB scattergram.
RBC Histogram Abn.	Abnormal distribution of RBC histogram	Abnormal distribution of RBC histogram
RET Scattergram Abn.	Abnormal distribution of RET scattergram	The distribution of RET scattergram is abnormal
Dimorphic Population	RBC displays dimorphic population	Two or more wave peaks on the RBC histogram
Reticulocytosis	RET high	RET% > 5% or RET#>0.20×10 ¹² /L or presence of abnormally distributed dots in sensitive region of the DIFF RET channel Note: Flags for test panels that include RET tests are reported by RET channel. Flags for CD test panel are reported by DIFF channel.
Anisocytosis	Anisocytosis	RDW-CV> 22 or RDW-SD> 64fL
Microcytosis	MCV low	MCV< 70fL
Macrocytosis	MCV high	MCV> 110fL
Hypochromia	Hypochromia	MCHC<290
Anemia	Anemia	HGB< 90g/L
Erythrocytosis	RBC high	RBC > 6.5×10 ¹² /L
RBC Agglutination?	RBC results possibly inaccurate	Calculate and compare special parameters
Turbidity/HGB Interfere?	Abnormal HGB or there may be interference	Calculation and comparison of special parameters.
Iron Deficiency?	May indicate iron deficiency anemia	Calculation and comparison of special parameters
Fragments?	Possible presence of RBC fragments	Presence of abnormally distributed dots in sensitive region of the RET channel or abnormal or the distribution of PLT-H and PLT-I histogram is abnormal. Note: Flags for test panels that include RET tests are reported by RET channel. Flags for CD test panels are reported by DIFF channel (PLT-H)
PLT Scattergram Abn.	Abnormal distribution of PLT scattergram	The distribution of PLT scattergram is abnormal
PLT Histogram Abn.	Abnormal distribution of PLT histogram	The distribution of PLT histogram is abnormal
Thrombocytopenia	PLT low	PLT<60×10 ⁹ /L

Flag Message	Indication	Criteria
Thrombocytosis	PLT high	$PLT > 600 \times 10^9/L$
PLT Clump?	There may be PLT clump	Calculation and comparison of special parameters
Pancytopenia	WBC, RBC and PLT low	$WBC < 4.0$ and $RBC < 3.5$ and $PLT < 100$
Lipid Particles?	Possible presence of lipid particles	Presence of excessive dots in lipid particle sensitive region of the scattergram
Infected RBC?	Possible presence of infected RBC	Presence of excessive dots in infected RBC sensitive region of the scattergram
WBC Fragments?	Possible presence of WBC fragments.	Presence of abnormally distributed dots in WBC fragment sensitive region
PLT-H Histogram Abn.	The PLT-H histogram is abnormal.	The PLT-H histogram is abnormal.
ESR Analysis Abn.	The ESR channel may be abnormal and the ESR measurement may be inaccurate.	The message is reported as the analyzer analyzes and monitors the measurements of the ESR channel.

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9

Using the QC Program

9.1 Introduction

Quality Control (QC) consists of strategies and procedures that measure the precision and stability of the analyzer. The results imply the reliability of the sample results.

QC involves measuring materials with known, stable characteristics at frequent intervals. Analysis of the results with statistical methods allows the inference that sample results are reliable. Mindray recommends you run the QC program daily with normal, low and high level controls.

A new lot of controls should be analyzed in parallel with the current lot prior to their expiration dates.

This may be accomplished by running the new lot of controls for five days using any empty QC files. The QC files calculate the mean, standard deviation and coefficient of variation for each selected parameter. The instrument-calculated means of these ten runs should be within the expected ranges published by the manufacturer.

This analyzer provides 2 QC programs: L-J QC and X-B QC.

CAUTION

- **Use the controls and reagents specified by the manufacturer only. Store and use the controls and reagents as instructed by their instructions for use.**
-

9.2 L-J QC

L-J QC is named after S. Levey and E.R. Jennings, who in 1950 introduced statistical control into the clinical laboratories.

On a L-J QC graph, quality control data is plotted to show whether the instrument is working well.

Laboratories may set up allowable deviations (by standard deviations (SD) or coefficient of variation (CV%)) from the targets for the control based on their real scenario. QC points are then plotted so the operators may easily see how far the actual QC results are from their targets. The x-axis indicates the QC date and time; and the y-axis indicates the targets as well as the defined limits. Lines run across the graph at the target as well as at the upper and lower limits to either side of the target value for the control. The QC points' distance from the target value is measured in SD or CV%.

You can select one of the two ways below to run controls:

- Run controls under the **"QC"** screen.
- Put controls together with normal samples, and run the controls under the sample analysis screen.

9.2.1 Setting up L-J QC Files

You can set up QC files on the QC file setup screens.

Follow the instructions below:

1. Select **"QC"** – **"Overview"** to go to **"Overview"** screen.
2. In the **"L-J QC"** area, select **"Setup"** – **"New"** to go to **"New"** screen.

See below for setting description.

Options	Description	Note
Lot No.	Find the Lot No. of controls on the vial labels of the controls	The lot No. shall not be empty and up to 16 digits can be entered. You can enter characters, numbers, letters and special characters. The Lot No. cannot be null. Enter the lot No. of the controls by one of the following ways: Manual entry, or using external barcode scanner.
Level	Levels of the controls "High", "Low", "Normal"	/
Exp. Date	Expiration dates of the controls	The expiration date shall not be earlier than the current system date.
Mode	The sampling and sample modes in QC tests: ■ AL-WB ■ OV-WB ■ OV-PD ■ OV-BF	/
Test Panel	The available test panels include: "CD", "CDR", "Ret", "ESR", CDR+ESR and "CD+ESR", .	/
Type	The controls types include: • Mindray • Others	When you are using BR60 controls, select "Others"; When you are using BC-6D or BC-RET controls, select "Mindray".
Target	The target values for the QC parameters Find the targets in the target sheet of the controls	/
Limit	The allowed deviation limit for each QC parameter Find the limits information in the target sheet of the controls	The limits are represented by SD or by CV. Tap "Set Limits" and select to represent the deviations "By SD(#)" or "By CV(%)" .
QC Sample ID	If you are used to analyzing control samples together with blood samples, you can set a unique ID for the control. The analyzer will recognize the sample as control when it reads the unique ID. After the analysis completes, the results will be saved into the QC file of the QC sample ID.	• You can enter letters, digits and all other characters on the keyboard (including special characters) for QC sample ID. Chinese and other languages are not supported (e.g. Japanese, Korean, etc.) • If you are using external barcode scanner to scan sample IDs, make sure the QC sample IDs set on the QC file screen are the same as that of the Lot No. labels on the control vials. • If you are manually entering sample IDs, make sure the QC sample IDs set on the QC file screen are the same as the sample IDs you entered on the "Mode" screen.
Communication ID	If you are using 2-way LIS, LIS identifies QC results by the communication ID set here.	/

9.2.1.1 Setting up new L-J QC file

Before running a new lot of controls, you must set up a QC file for each lot of controls.

1. Select **"QC"**–**"Overview"** to go to the **"Overview"** screen.
2. In the **"L-J QC"** area, select **"Setup"** to go to the L-J QC file setting screen.
3. Select **"New"** to create a new L-J QC file.
4. Enter the necessary QC file information.

You shall enter the required QC file information using one of the following ways.

- Reading the information provided by the manufacturer
 - a Insert the USB flash drive that saves the QC files to the USB interface on the analyzer.
 - b Select **"Import File"** and import the QC file as instructed by the software.
- Manually enter the necessary QC file information.

NOTE

- **For the setup of the L-J QC file, refer to 9.2.1 Setting up L-J QC Files.**
- **The user should ensure the data safety of the USB flash drive connecting to the analyzer.**

5. Customize **"QC Sample ID"** and **"Communication ID"**.
6. (Optional) If necessary, set the QC file to **"In Use"** from the **"In Use"** pull-down list.

NOTE

- **You can select or clear the selection "In Use" option in the QC file table tab to activate or deactivate the QC files.**
- **For files having the same "QC Sample ID" and presentation mode, only one of them can be "In Use".**
- **For files having the same QC type and level, only one of them can be "In Use".**

7. Save the QC file.
 - a Select **"Return"** or other buttons on the tab.
- √ A dialog box displays.
 - b select **"Yes"** to save the new QC file.

9.2.1.2 Editing L-J QC files

You can only edit empty L-J QC files.

NOTE

- **You cannot edit the QC files that already have QC data.**

Follow below instructions:

1. Select **"QC"**–**"Overview"** to go to the **"Overview"** screen.
2. In the **"L-J QC"** area, select **"Setup"** to go to the L-J QC file setting tab.
3. Select the QC file to edit.
- √ The **"*"** mark displays next to the **"File No."** of the selected QC file.
4. Select **"Edit"** to edit the L-J QC file.
5. Edit the QC file as necessary.

NOTE

- **For the setup of the L-J QC file, refer to 9.2.1 Setting up L-J QC Files .**

6. (Optional) If necessary, set the QC file to **"In Use"** from the **"In Use"** pull-down list.

NOTE

- You can select or clear the selection of “In Use” option on the QC file table tab to activate or deactivate the QC files.
- For files having the same “QC Sample ID” and presentation mode, only one of them can be “In Use”.
- For files having the same QC type and level, only one of them can be “In Use”.

7. Save the QC file.
 - a Select “Return” or other buttons on the tab.
- ✓ A dialog box displays.
 - b select “Yes” to save the new QC file.

9.2.1.3 Deleting L-J QC file

Administrators may delete the selected or all QC records in the QC file.

1. Select “QC”–“Overview” to go to the “Overview” screen.
2. In the “L-J QC” area, select “Setup” to go to the L-J QC file setting tab.
3. Delete the selected or all QC records.

Delete all records	1. Select “Delete”. 2. In the “Delete” dialog box, select “All records”.
Delete selected records	1. Select one QC file. 2. Select “Delete”. 3. In the “Delete” dialog box, select “Selected records”.

9.2.2 Running L-J QC Analysis

You can select one of the two ways below to run controls.

- Running controls on the QC count screen.
- Put controls together with normal samples, and run the controls on the “Count” screen.

When running the QC analysis, check the following:

- Make sure you have set up a suitable and correct QC file for the control to be run, and the QC file is “In Use”. Make sure you have prepared the controls in accordance with your laboratory protocols and with the requirements in the Instruction for Use of the controls.
- Make sure the analysis system is without error.

9.2.2.1 Running controls on the QC count screen

Follow below instructions:

1. Select “QC”–“Overview” to go to the “Overview” screen.
2. In the “L-J QC” area, select “Count” to go to QC count screen.
3. From the “File No.” pull-down list, select the QC file for the control to be run.
4. Make sure the QC file information displayed on the screen is correct. Make sure the level, QC mode of the control to be run is the same with the current QC file, and the control is not expired.

NOTE

- The expiry date of expired controls is displayed in red.

5. Prepare the control as instructed by instructions for use of the controls. Paste the barcode labels on the control vials (for details about pasting barcode labels, refer to **7.5.4 Placing Barcode Labels**).

6. Run QC analysis as instructed below.

To run QC samples using AL-WB mode	1. Place the sample tube with prepared controls into the rack. 2. Place the rack loading controls on the right tray of the autoloader, and leave one side with the opening facing the analyzer. 3. Select "Count" to start the QC analysis. ✓ The analyzer will automatically run the sample. 4. When the running finishes, you can remove the rack from the left of the autoloader safely.	For more information about analyzing samples in AL mode, refer to 7.7 Analyzing Samples in Auto-Loading Mode.
To run QC samples using OV-WB mode	1. Present the prepared WB QC sample uncapped to the sample probe for aspiration. 2. Select "Count" to start the QC analysis. ✓ The sample probe automatically aspirates sample. 3. When you hear the beep, remove the sample tube. ✓ The analyzer will automatically run the sample. ✓ When analysis finishes, the QC results will be displayed in the current screen and be saved in the QC file automatically.	• For more information about analyzing samples in OV modes, refer to 7.6 Analyzing Samples in OV Mode.

✓ When analysis finishes, the QC results will display on the current tab and be saved in the QC file automatically.

NOTE

- **When the analyzer has fault or needs your attention, it beeps to alarm.**
- **Up to 372 QC results can be saved in each QC file.**

4. (Optional) Repeat the above steps to perform the QC analysis with different levels.

9.2.2.2 Placing controls together with normal samples, and running the controls on the "Count" screen

After setting special **"QC Sample ID"** for a control under the QC setup screen, you can put the control together with normal samples, and run it on the **"Count"** screen.

Check the following before analysis:

- Make sure you have prepared the controls in accordance with your laboratory protocols, and the requirements in the Instruction for Use of the controls.
- The analyzer is ready to run samples (i.e. the analyzer indicator stays in green).
- Make sure the analyzer is without error.
- Make sure you have set up a suitable and correct QC file for the control to be run, and the QC file is **"In Use"**.

Note when you set sample ID for the control:

- If you are using external barcode scanner to scan sample IDs, make sure the QC sample IDs set on the QC file screen are the same as that of the Lot No. labels on the control vials.
- If you are manually entering sample IDs, make sure the QC sample IDs set on the QC file screen are the same as the sample IDs you entered on the **"Mode"** screen.

Follow below instructions:

1. Select **"Mode"** on the **"Count"** screen.

✓ The **"Mode"** dialog box displays.
2. Select the desired analysis mode.

NOTE

- Make sure the analysis mode is the same with that you set in the QC file for the control.

3. Manually enter the Sample ID, or use an eternal barcode scanner to scan the barcode label on the sample tube to enter Sample ID into the "Sample ID" field.

NOTE

- Make sure the Sample ID you entered is the same with the QC sample ID you set in the QC file for the control.

4. Run the samples in accordance with the normal sample analysis procedure.
 ✓ After the analysis, the QC results will be automatically saved to the corresponding QC file.

NOTE

- For the normal sample analysis procedures, refer to 7.6 Analyzing Samples in OV Mode or 7.7 Analyzing Samples in Auto-Loading Mode.
- Up to 372 QC results can be saved in each QC file.

9.2.3 Reviewing L-J QC Results

After QC analysis, you can review the QC results by the following methods.

- Graph
- Table Review

9.2.3.1 Reviewing QC results by Graph Review

1. Select "QC" – "Overview" to go to the "Overview" screen.
2. In the "L-J QC" area, select "L-J QC Graph" to go to the QC graph tab.
3. Select file No. from the "File No." pull-down list.

Introduction to the L-J Blood QC Graph

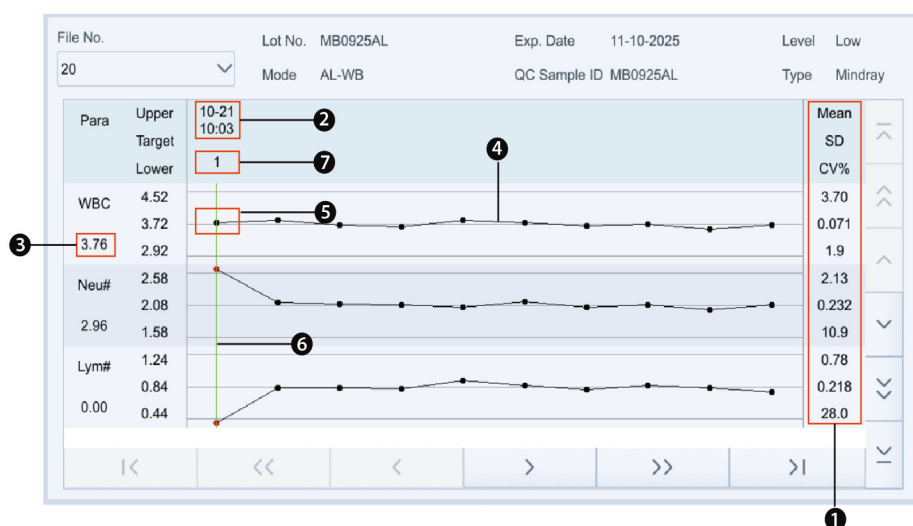


Figure 9-1 L-J blood sample QC graph

	Description
1	The Mean, SD and CV% of all the QC results of each parameter in the current graph.
2	The saving date and time of the QC points on the green line.

	Description
3	The operator who run the QC analysis and obtained the QC points on the green line.
4	The line connecting all QC points of the same parameter to show the trend. The QC points in each graph are displayed from left to right according to the sequence from the earliest to the latest.
5	Currently selected QC point. The analysis result of the selected QC point is displayed under the parameter. A black QC point indicates the value is within the limit; a red QC point indicates the value is out of the limit.
6	The green vertical line is used to identify the QC points of the same analysis, all of which are displayed on the line when you select one of them.
7	The sequence number of the QC point on the green line among all the QC points in the current QC file.

Introduction to the L-J BF QC Graph

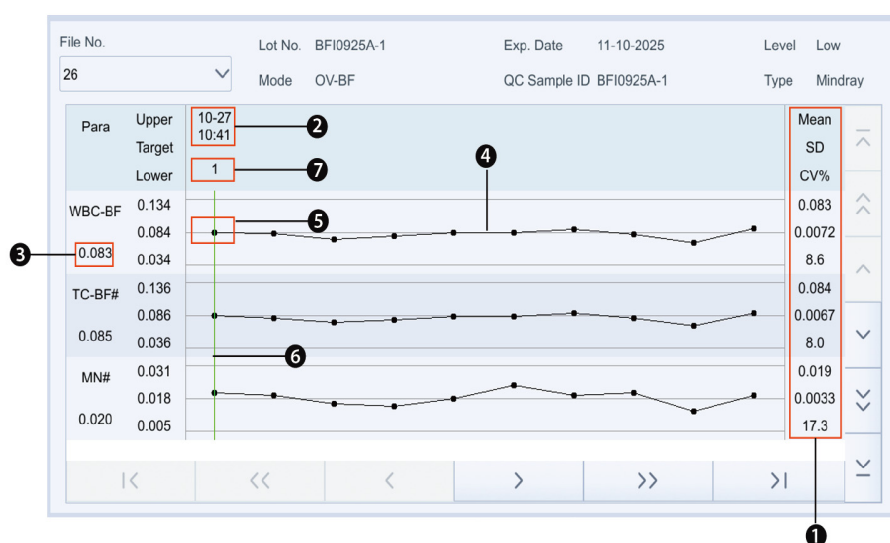


Figure 9-2 L-J BF QC graph

	Description
1	The Mean, SD and CV% of all the QC results of each parameter in the current graph.
2	The saving date and time of the QC points on the green line.
3	The operator who run the QC analysis and obtained the QC points on the green line.
4	The line connecting all QC points of the same parameter to show the trend. The QC points in each graph are displayed from left to right according to the sequence from the earliest to the latest.
5	Currently selected QC point. The analysis result of the selected QC point is displayed under the parameter. A black QC point indicates the value is within the limit; a red QC point indicates the value is out of the limit.
6	The green vertical line is used to identify the QC points of the same analysis, all of which are displayed on the line when you select one of them.
7	The sequence number of the QC point on the green line among all the QC points in the current QC file.

Entering the Causes for the Outliers

A red QC point indicates the value is out of the limit.

If necessary, follow below instruction to enter the causes for the outliers.

1. Select "QC"—"Overview" to go to the "Overview" screen.

2. In the **"L-J QC"** area, select **"L-J QC Graph"** to go to the QC graph screen.
3. Select file No. from the **"File No."** pull-down list.
4. On **"QC Graph"** screen, move the green line to the desired QC point, and select the **"Outliers"** button.
- ✓ A dialog box displays the QC results, targets and limits of all the parameters.
5. Enter the causes for outliers.

Select the proper causes of outliers or select **"Others"** and enter the causes in the edit box.

NOTE

- You may enter up to 200 characters when manually entering the causes of the outliers.

6. Select **"OK"** to save the causes of the outliers and exit the dialog box.

9.2.3.2 QC Table Review

9.2.3.3 Reviewing QC results by QC table

1. Select **"QC" – "Overview"** to go to the **"Overview"** screen.
2. In the **"L-J QC"** area, select **"L-J QC Table"** to go to the QC table screen and then select file No. from the **"File No."** pull-down list.
3. Select the desired QC file to review.

Introduction to the L-J blood sample QC table

File No.	Lot No.	MB0925AL	Exp. Date	11-10-2025	Level	Low
20	Mode	AL-WB	QC Sample ID	MB0925AL	Type	Mindray
	Date	Time	WBC	Neu#	Lym#	Mon#
Target	/	/				
Limit	/	/				
15*	11-10-2025	10:06	3.74	2.13	0.82	0.12
14	11-08-2025	10:01	3.75	2.07	0.90	0.12
13	11-07-2025	09:46	3.64	2.09	0.80	0.12
12	11-06-2025	10:01	3.58	1.97	0.82	0.11
11	11-05-2025	09:53	3.66	2.07	0.83	0.11
10	11-04-2025	09:52	3.68	2.09	0.78	0.12

Figure 9-3 L-J blood sample QC table

	Description
1	The sequence number of the QC results saved in the QC file (earliest to the latest from top to down).
2	QC test results.
3	QC parameters (displayed in the same order as that on the graph tab).

Introduction to the L-J body fluid sample QC table

	Date	Time	WBC-BF	TC-BF#	MN#	MN%
Target	/	/				
Limit	/	/				
13*	11-10-2025	09:59	0.093	0.093	0.024	25.6
12	11-08-2025	09:53	0.090	0.090	0.024	26.4
11	11-07-2025	09:41	0.092	0.092	0.018	20.0
10	11-06-2025	09:56	0.089	0.089	0.019	21.6
9	11-05-2025	09:45	0.069	0.070	0.013	19.5
8	11-04-2025	09:46	0.081	0.082	0.020	24.3

Figure 9-4 L-J body fluid sample QC table

	Description
1	The sequence number of the QC results saved in the QC file (earliest to the latest from top to down).
2	QC test results.
3	QC parameters (displayed in the same order as that on the graph tab).

9.2.4 Deleting QC Records

Administrators can delete selected or all QC records in the QC file.

Follow below instructions.

1. Select **"QC"**–**"Overview"** to go to the **"Overview"** screen.
2. In the **"L-J QC"** area, select **"L-J QC Table"** to go to the QC table screen, select file No. from the **"File No."** pull-down list and select the desired QC records.

Delete all records	1. Select "Delete". 2. On the "Delete" dialog box, select "All records".
Delete the selected records	1. Select one or multiple records to be deleted. 2. Select "Delete". 3. On the "Delete" dialog box, select "Selected records".

3. Select **"OK"**.

√ All records or the selected records are deleted.

9.2.5 Transmitting Data

You can transmit QC data to an external data management software or LIS System.

Before transmitting QC data, make sure the network is properly connected.

Follow below instructions.

1. Select **"QC"**–**"Overview"** to go to the **"Overview"** screen.
2. In the **"L-J QC"** area, select **"L-J QC Table"** to go to the QC table screen, and then select file No. from the **"File No."** pull-down list.
3. Select the desired QC file.

- ✓ The "*" mark displays next to the file No. of the selected QC file.
- 4. Select "".
- ✓ The data of the selected QC file will be transmitted by default.

NOTE

- **The QC data saved in the process of transmission will not be transmitted.**
- **If auto-communication is enabled and a sample is run during the transmission of the QC data, then only when the QC data transmission finished will the auto-communication of the sample result start.**

9.2.6 Exporting Data

Administrators can export selected or all QC records to an external USB flash drive.

Before exporting data, make sure the USB flash drive is firmly connected to the USB interface on the side of the analyzer.

NOTE

- **The user should ensure the data safety of the USB flash drive connecting to the analyzer.**

■ Exporting selected records

Follow below instructions:

1. Select "QC"-"Overview" to go to the "Overview" screen.
2. In the "L-J QC" area, select "L-J QC Table" to go to the QC table tab, and select file No. from the "File No." pull-down list.
3. Select the QC record(s) you want to export.

Export all records.	1. Select . 2. In the "Export Range" field, select "All records".
Export selected records	1. Select one or more sample records that you want to export. 2. Select . 3. In the "Export Range" field, select "Selected records".

4. Select "OK".
- ✓ The analyzer exports QC data to the USB flash drive automatically.

9.2.7 Editing QC Results

If needed, administrators can edit the QC results in the QC count screen.

Follow below instructions:

1. Select "QC"-"Overview" to go to the "Overview" screen.
2. In the "L-J QC" area, select "Count" to go to L-J QC count screen.
3. Select file No. from the "File No." pull-down list.
4. (Optional) If necessary, select "Previous" or "Next" to switch the QC sample.
5. Select "Edit Result" to go to "Edit Result" screen.
6. Modify the parameters results as required.
7. Select "OK" to go to QC count screen.
- ✓ The modified results are saved to the count screen.
- ✓ The parameter results that have been manually modified is flagged with an "E". A parameter result that has been modified due to the manual modification of another parameter result is flagged with an "e".
8. (Optional) If needed, select "Restore".

√ The edited results will be changed back into the initial values.

9.2.8 Reviewing the Parameter Results of Control Samples

If necessary, you can review the test results, RUO parameter results, histograms, scattergrams, and other information about a QC analysis on the QC count screen.

Follow below instructions:

1. Select **"QC"**–**"Overview"** to go to the **"Overview"** screen.
 2. In the **"L-J QC"** area, select **"Count"** to go to L-J QC count screen.
 3. Select file No. from the **"File No."** pull-down list.
 4. (Optional) If necessary, select **"Previous"** or **"Next"** to switch the QC sample.
- √ The tab displays the results, scattergram, and histogram of the QC sample test.
5. Select **"RUO Para."** to go to **"RUO Para."** screen.
- √ The results of **"RUO Para."** of QC analysis displays in the screen.

9.3 X-B QC

The X-B analysis is a weighted moving average analysis that uses values obtained from patient samples. It uses the 3 red cell indices, MCV, MCH and MCHC to indicate the hematology instrument performance.

It is recommended the X-B QC be activated when the sample volume of your laboratory is greater than 100 samples per day. Effective use of X-B requires randomization of samples and a normal cross section of patients to prevent skewing of indices. It observes the trend of QC results in the reference range formed by the specified target and limits.

The analyzer implements X-B QC on the 3 parameters: MCV, MCH and MCHC, each group of samples for X-B analysis consists of 20-200 sample results obtained from normal analysis of both WB and PD modes. The analyzer can save up to 1,000 X-B QC results. When the saved QC results have reached the maximum number, the newest result will overwrite the oldest.

9.3.1 Validity Determination for X-B QC Samples

In X-B QC, sample results conforming to any of the following conditions will be considered as invalid and cannot be used in the QC calculation.

- Sample results exceeding the linearity range;
- Background results;
- Sample results not conforming to the rules set in the **"Sample Validity Setup"**;
- L-J QC results
- Calibration data.

9.3.2 Setting up X-B QC Rules

Administrators may set up X-B QC rules on the XB QC setup screen.

1. Select **"QC"**–**"Overview"** to go to the **"Overview"** screen.
2. In the **"X-B QC"** area, select **"Setup"** to go to the X- B QC file setting screen.
3. Set up X-B QC rules as needed.

See below for setting descriptions.

Items	Description	Note
X-B QC On/Off	Select "On" or "Off" to enable/disable the X-B QC	When you select "Off" , the analyzer does not run X-B QC program.

Items	Description	Note
Samples/Batch	The amount of samples to be included in calculating for an X-B QC point.	Setup range: 20-200; For example, when " Samples/Batch " is set to 20, when there are 20 valid samples, the analyzer start to run X-B QC.
Target/Limits Setup	Enter the target/limit for each parameter	- Do not leave any of the targets and limits for the QC parameters blank. - The units of target/limits of all parameters are the same as those on the "Parameter Unit Setup" screen.
Sample Validity Setup	Set the upper and lower limits of the 4 parameters in the sample validity setup area.	"Sample Validity Setup" is to set up the ranges of valid RBC, MCV, MCH and MCHC results. Only when the results of all these four parameters are within the specified ranges, the sample results can be used for X-B QC calculation. - In the sample validity setup, the upper limit shall be no smaller than the lower limit. Otherwise, there will be prompted message asking you to revise. - The valid ranges of the RBC parameters are their linearity ranges; the valid ranges of other parameters are their display ranges. - All the entries shall be numbers with only one decimal point. The length of the number entered cannot be longer than the length of the text box. - The units of lower and upper limits of all parameters are the same as those on the "Parameter Unit Setup" screen.
Restore Default	Tap " Restore Default " to restore the targets and limits to the default values.	/
Set Limits	Tap " Set Limits " and select " By SD " or " By CV (%) "	- When "By SD(#)" is selected, the limits are displayed in the format of SD value; - When "By CV (%)" is selected, the limits are displayed in the format of CV percent

After editing X-B setup, the system will start X-B QC automatically.

After every 20~200 valid sample results (determined by your setting for "Samples/Batch", the system will perform the X-B calculation once automatically. You can review the result in X-B QC graph or X-B QC table.

9.3.3 Reviewing X-B QC Results

After QC analysis, you can review the QC results by the following methods.

- **Graph**
- **Table Review**

9.3.3.1 Reviewing QC results by QC Graph

1. Select "**QC**"-"**Overview**" to go to the "**Overview**" screen.

- In the "X-B QC" area, select "X-B QC Graph" to go to X-B QC graph screen.

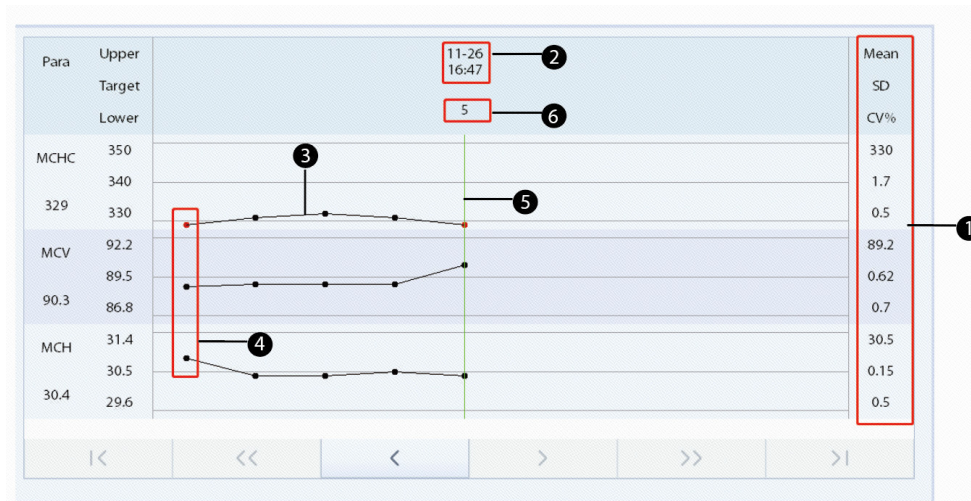


Figure 9-5 X-B QC graph

	Description
1	The Mean, SD and CV% of all the QC results of each parameter in the current graph
2	The saving date and time of the QC points on the green line.
3	The line connecting all QC points of the same parameter to show the trend. The QC points in each graph are displayed from left to right according to the sequence from the earliest to the latest.
4	Currently selected QC point. The analysis result of the selected QC point is displayed under the parameter. A black QC point indicates the value is within the limit; a red QC point indicates the value is out of the limit.
5	The green vertical line is used to identify the QC points of the same analysis, all of which are displayed on the line when you select one of them.
6	The sequence number of the QC point on the green line among all the QC points in the current QC file.

9.3.3.2 QC Table Review

- Select "QC"-"Overview" to go to the "Overview" screen.
- In the "X-B QC" area, select "X-B QC Table" to go to "X-B QC Table" screen.

	Date	Time	MCV	MCH	MCHC
Target	/	/	89.5	30.5	340
Limit	/	/	2.7	0.9	10
5	11-26-2024	16:47	90.3	30.4	L 329
4	11-26-2024	16:47	89.0	30.5	331
3	11-26-2024	16:47	89.0	30.4	332
2	11-26-2024	16:47	89.0	30.4	331
1	11-26-2024	16:47	H 93.2	30.8	L 328

Figure 9-6 X-B QC table

	Description
1	The sequence number of the QC results saved in the QC file (earliest to the latest from bottom to top).
2	QC result
3	QC parameters (displayed in the same order as the graph tab)
4	QC flag: results beyond the set limits are marked with a symbol ("H" for beyond the high limit and "L" for beyond the low limit by default).

9.3.4 Deleting QC Records

Administrators can delete selected or all QC records in the QC file.

Follow the instructions below:

1. Select **"QC"–"Overview"** to go to the **"Overview"** screen.
2. In the **"X-B QC"** area, select **"X-B QC Table"** to go to **"X-B QC Table"** screen.
3. Select the QC record(s) you want to delete.

Delete all records	1. Select "Delete". 2. In the "Delete" dialog box, select "All records".
Delete the selected records	1. Select one or multiple records to be deleted. 2. Select "Delete". 3. In the "Delete" dialog box, select "Selected records".

4. Select **"OK"**.

✓ All records or the selected records are deleted.

9.3.5 Exporting Data

Administrators can export selected or all QC records to an external USB flash drive.

Before exporting data, make sure the USB flash drive is firmly connected to the USB interface on the side of the analyzer.

NOTE

- **The user should ensure the data safety of the USB flash drive connecting to the analyzer.**

Follow the instructions below:

1. Select **"QC"–"Overview"** to go to the **"Overview"** screen.
2. In the **"X-B QC"** area, select **"X-B QC Table"** to go to **"X-B QC Table"** screen.
3. Select the QC record(s) you want to export.

Export all records.	1. Select . 2. In the "Export Range" field, select "All records".
Exporting the selected records	1. Select one or more sample records that you want to export. 2. Select . 3. In the "Export Range" field, select "Selected records".

4. Select **"OK"**.

✓ The analyzer exports QC data to the USB flash drive automatically.

9.4 When QC Results are Out of Range

9.4.1 Troubleshooting

When a QC result falls outside the control range, follow below steps to solve the problem.

- Analyze the cause of outliers, take corrective measures, and verify effectiveness of the corrective measures.
- If the corrective measures fail, report it according to the laboratory protocol.
- Contact our Customer Service Department.

9.4.2 Analyzing the Causes

Retest the samples with outliers. If the retest result still contains outliers and the trend is the same as that of the previous result with outliers, find out the cause of outliers by sequence referring to **Figure 9-7 Analyze the cause of QC outliers**.

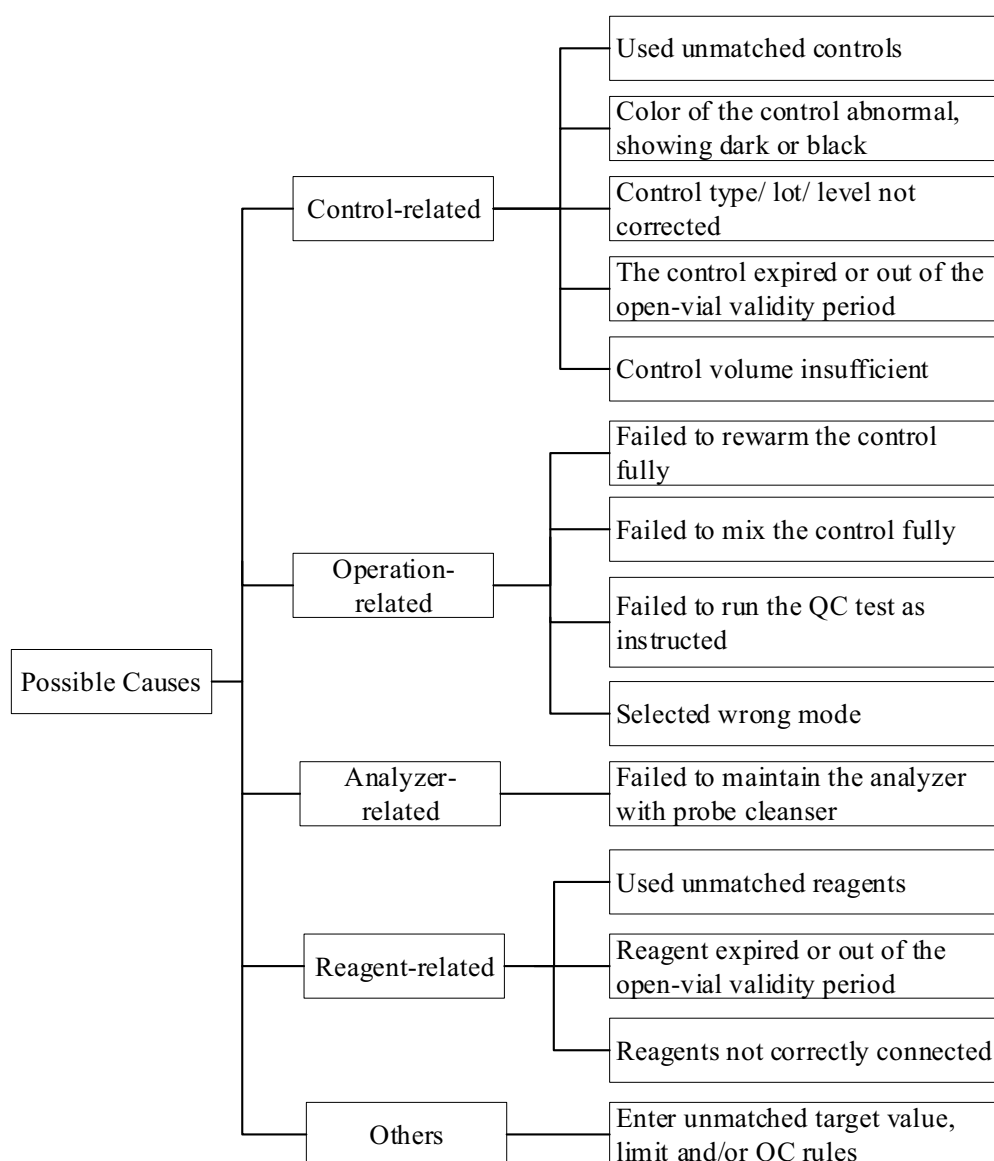


Figure 9-7 Analyze the cause of QC outliers

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10 Calibrating Your Analyzer

10.1 Overview

Calibration is a procedure to standardize the analyzer by determining its deviation under certain specified conditions. In order to get accurate sample analysis results, you should calibrate the analyzer per the procedure below when necessary.

The analyzer supports the following calibration programs:

- Manual calibration
- Calibration with calibrators
- Calibration with fresh blood samples
- ESR (perform the calibration under the guidance of a service engineer)



CAUTION

- **Use the reagents, controls and calibrators specified by the manufacturer. Store and use the reagents as instructed by instructions for use of the reagents.**
 - **Using controls, calibrators and reagents of other manufacturers may lead to damage to the analyzer and incorrect control, calibration and analysis results.**
-

10.2 When to Calibrate

Your analyzer is calibrated at the factory just before shipment. It is electronically stable and does not require frequent recalibration if you operate and maintain it as instructed by this manual. It is recommended that you run the calibration program every half year. You only need to recalibrate this analyzer, if:

- You are going to use this analyzer for the first time (usually done by a representative authorized by our company when installing the analyzer).
- A major analytical component (including sampling probe, syringe, etc.) has been changed.
- You are going to re-use the analyzer after a long-term storage.
- The quality control results indicate there may be a problem.



CAUTION

- **All of the measured parameters must be calibrated before readings of this analyzer can be used as valid analysis results.**
-

10.3 Preparation before Calibration

Before calibration, check and make sure the analyzer works properly and enough reagents have been prepared for the calibration. You need to start over the calibration if the reagents run out during the process.

It is recommended that you create a log table for your analyzer. This log table should contain all necessary information that is pertinent to your analyzer. Suggested items that you may want to include in the log table are: calibration date, supplier of calibrator, lot number, expected results and limits, and result of background check.

Before calibration, follow the CLSI (Clinical & Laboratory Standards Institute) standards or your laboratory protocol to do tests, and make sure the analyzer's background (blank count) results, repeatability results and carryover results are all within the specified ranges.

If any of the above items is not within the range, check if the analyzer is in error. Remove the errors (if there are) and check again. If the error still exists, contact our Customer Service Department.

NOTE

- Be sure to mix any calibrators that have been prepared for a while before running it.
- For information of blank count, repeatability and carryover range, refer to *B.5 Performance Specifications*.

10.4 Manual Calibration**NOTE**

- Common users can only review calibration factors on the manual calibration screen. You cannot modify the calibration factors.

You can use the calibrators specified by our company or value-assigned fresh blood samples to perform manual calibration. If you are using value-assigned fresh blood samples, make sure the samples meet the following requirements:

- has normal erythrocyte, leukocyte, and platelet morphology
- the test results are within corresponding reference ranges
- has sufficient sample volume for the whole calibration procedure

Follow below instructions.

1. Select a calibrator or a value-assigned fresh blood sample which meets the sample requirements for manual calibration, and run the sample consecutively for no less than 3 times under whole blood or predilute mode.
2. Calculate the CV values and the Mean values for the tests.

NOTE

- You may review the CV and Mean values through the "Table Review" screen. For more information, refer to *8.2.1.7 Calculating CV value*.

3. Check if the CV values are in the acceptable reference ranges specified.

NOTE

- When the CV value for any parameter exceeds the acceptable range, check if the analyzer is in error. If there are errors, remove the errors and test again. If the error still exists, contact our Customer Service Department.

4. Select "Calibration"-"Calibration Analysis"-"Manual" to go to "Manual" screen.
5. Calculate the new calibration factors for the parameters per the following equation. The calculated factors should show 2 decimal places.

$$\text{New calibration factor} = \frac{\text{Current calibration factor} \times \text{Reference value}}{\text{Mean}}$$

For example: suppose the WBC reference value of a calibrator or fresh blood sample under WB mode is 8.40, and the current calibration factor under WB mode is 98.90%.

Run the sample in WB mode for 10 consecutive times (n = 10). The WBC results of the ten runs are 8.10, 8.10, 8.10, 8.10, 8.30, 8.30, 8.20, 8.00, 8.10, and 8.30. The obtained CV is 1.5% and Mean is 8.16. which meet the requirements.

Whereas:

$$\text{New Calibration Factor} = \frac{98.90\% \times 8.40}{8.16} = 101.81\%$$

6. The calculated calibration factors shall be within the valid range(75.00% to 125.00%).

In case of an invalid calibration factor, try to find out the reason (e.g. calibration material not thoroughly mixed, dis-operation, etc.). Then recalibrate the analyzer and recalculate the calibration factors.

7. Enter the new calibration factors into the factor cell of the parameter that require calibration.

✓ The date cells automatically display the date when the new calibration factors are entered.

If the entered calibration factors are invalid, the factors will be highlighted in red.

NOTE

- **The entered calibrator factor should be from 75.00% to 125.00%, and it only shows two decimal places.**

8. Save calibration factors.

- a Select the other button on the tab.

✓ A dialog box displays

- b Select **"Yes"** to save new calibration factors.

9. If the entered calibration factors are invalid, a dialog box will display prompting **"Invalid entry!"** when you are switching to another tab screen.

- a If you want to re-do the calibration, select **"No"**, and perform the calibration procedures again.

- b Select **"Yes"** to close the dialog box and switch to another tab screen without saving the changes; the original calibration factors and dates will remain the same.

10.5 Calibrating with Calibrators



BIOHAZARD

- **All the samples, controls, calibrators, wastes and areas contacted them are potentially biohazardous. Wear proper personal protective instrument (e.g. gloves, lab coat, glasses, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.**

Administrators can perform the calibration with the calibrator. You must use calibrators specified by our company for calibration.

Follow below instructions:

NOTE

- **When you are using calibrators to calibrate the analyzer, you can only perform the calibration with fresh blood samples only in OV-WB mode.**

1. Make sure you have prepared the calibrators in accordance with your laboratory protocols and with the requirements in the Instruction for Use of the calibrators.

NOTE

- **Be sure to mix any calibrator that has been prepared for a while before running it.**

2. Select **"Calibration"–"Calibration Analysis"–"Calibrator"**.

3. Set up calibrator information.

Enter the **"Lot No."**, **"Exp. Date"** and the target value for each parameter.

See below for setting descriptions.

Options	Description
Lot No.	Refer to the calibrator container.
Exp. Date	The date should be the date that container is opened + the open-vial stability days or the expiration date stated on the label of the calibrator, whichever is earlier.

Options	Description
Target	Find the parameter targets on the target sheet of the corresponding calibrator.

4. Place the prepared calibrator uncapped under the sample probe.
 5. Perform the calibrator calibration tests.
 - a Press the **[Aspirate]** key on the analyzer front cover to start calibration count.
- ✓ The analyzer starts the calibration procedure.

NOTE

- **To obtain valid calibration factors, you need 3-10 valid calibration results.**

- b After the sample probe finishes sample aspiration for the calibrator, the analyzer beeps. Move away the calibrator sample safely.
 - c After the software screen shows the test results for the first analysis, present the calibrator sample to the sample probe again.
 - d Repeat step a) to c) for the remaining calibration tests.
- ✓ The analyzer automatically calculates the calibration factors for the parameters.
6. (Optional) You may select to use which calibration results to calculate the calibration factors.

Select at least 3 groups of calibration results to calculate the calibration factors. The calibration results will be invalid, if:

- there is a parameter whose calibration data is out of its linearity range but still within the display range;
- there is a parameter whose calibration data is out of the display range, then the non-numeric parameter values "****" will be displayed in the list;

When there are invalid calibration results, a dialog box will display. Select **"Yes"** to close the dialog box, and the data will be deleted from the table without saving automatically.

7. Save calibration factors.
 - a Select another button on the screen.
- ✓ Based on the calculated calibration factors, a dialog box will display.
- b Save the calibration factor or exit the tab as prompted.

NOTE

- **When the analyzer has fault or needs your attention, it beeps to alarm.**
- **The calibration factors of the unchecked parameters will not be saved.**

10.6 Calibrating with Fresh blood Samples

Administrators can perform the calibration with the value-assigned fresh blood samples.

If you are using value-assigned fresh blood samples, make sure the samples meet the following requirements:

- has normal erythrocyte, leukocyte, and platelet morphology
- the test results are within corresponding ranges
- has sufficient sample volume for the whole calibration procedure

NOTE

- **When you are using fresh blood samples to calibrate, you can only perform the calibration with fresh blood samples only in OV-WB mode.**

Follow below instructions:

1. Prepare sufficient fresh blood samples as instructed in **7.5.1 Preparing Whole Blood Samples**.

2. Select "**Calibration**"–"**Calibration Analysis**"–"**Fresh Blood**" to go to "**Fresh Blood**" screen.
3. Select a sample ID from the "**Current Sample ID:**" pull-down list.
4. Enter the targets into the "**Target**" boxes.
5. Place the prepared fresh blood sample uncapped under the sample probe.
6. Perform the calibration test.

NOTE

- **To obtain valid calibration factors, you need 3-10 valid calibration results.**

- a Press the [**Aspirate**] key on the analyzer front cover to start calibration count.
- √ The analyzer starts the calibration procedure.
- b After the sample probe finishes sample aspiration for the sample, the analyzer beeps. Move away the sample safely.
- c After the software screen shows the test results for the first analysis, present the sample to the sample probe again.
- d Repeat step a)-c) to conduct the remaining calibration tests.
- √ The analyzer automatically calculates the calibration factors for the parameters.
7. (Optional) You may select to use which calibration results to calculate the calibration factors.
- √ Select at least 3 groups of calibration results to participate the calculation of calibration factors. The analyzer automatically calculates the calibration factors for the parameters.

The calibration results will be invalid, if:

- there is a parameter whose calibration data is out of its linearity range but still within the display range;
- there is a parameter whose calibration data is out of the display range, then the non-numeric parameter values "****" will be displayed in the list.

When there are invalid calibration results, a dialog box will display. Select "**OK**" to close the dialog box, and the data will be deleted from the table without saving automatically.

The calculated calibration factors are invalid, if:

- The calculated calibration factor is out of the valid range (75%-125%).
- The CV% of any calibration parameter is out of the repeatability range.

When there are invalid factors, a dialog box will display in the tab page. Select "**OK**" to close the message box, and the data will be deleted from the table without saving automatically.

8. Repeat step 3-7 to test the remaining samples.
9. Obtain the mean calibration factor.

To obtain mean calibration factor, at least 3 groups of valid calibration factors of fresh blood samples are required.

- a Select "**Calculate**" to calculate the calibration factors of the fresh blood sample.
- b Select at least 3 groups of calibration results to calculate the mean calibration factors.
- √ The analyzer automatically calculate the mean calibration factors.

The calculated mean calibration factors are invalid, if:

- the deviation of absolute value between the calibration factors included in calculating the mean and the original calibration factors reaches or exceeds 5%;
- calibration factors are out of the valid range (75% to 125%).

10. Save calibration factors.
 - a Select another button on the screen.
- √ Based on the calculated calibration factors, a dialog box will display.
 - b Save the calibration factor or exit the tab as prompted.

NOTE

- The calibration factors of the unchecked parameters will not be saved.

10.7 Verifying Calibration Factors

Verify the calibration factors after calibration.

Verify the calibration factors using any of the following methods:

- run the calibration at least 3 times, and check if the results are within the allowed range.
- run three levels (high, normal and low levels) of hematology control for at least 3 times each, and check if the results are within the allowed range.
- run at least 3 fresh blood samples from normal patients, each sample for at least 3 times, and check if the results are within the allowed range.

10.8 Calibration History

Administrators can view the calibration history.

Select “**Calibration**” – “**Calibration History**” to go to “**Calibration History**” screen.

Follow below instructions:

Options	Description
Details	Select one calibration record and then select “ Details ” to view the details of the calibration.
Go to	Select “ Go to ” to view the calibration records in the specified date.
Export	Select “ Export ” to export the specified records or all records to the USB flash drive.

NOTE

- The user should ensure the data safety of the USB flash drive connecting to the analyzer.

11 Using the Printing Function

You can set up the print templates for sample results report, graphs, QC results, QC graphs, manual calibration factors etc., and print them using the print templates.

11.1 Setting up Print Templates

You can customize the print settings on the **"Print Setup"** tab.

Follow below instructions:

1. Select **"Setup"**–**"Basic Settings"** to go to **"Basic Settings"** screen.
2. In the **"System Setup"** area, select **"Print Setup"** to go to the **"Print Setup"** screen.
3. Select **"Print Setup"**, **"Printing Content"** and **"Auto print after sample analysis"** to set printing as required.

See below for setting descriptions:

Options	Description	Note
Print device	Select a print device from the pull-down list.	For questions about the print device and print drive settings, contact the customer service personnel of your print device supplier.
Printer driver	Select a proper printer driver from the pull- down list: • Auto Identification • PCL6 • Raster print	
Paper	Select the desired paper type from the pull-down list: • A4 • A5	/
Blood Sample Report Title/Body Fluid Sample Report Title	1. Select "Blood Sample Report Title" or "Body Fluid Sample Report Title" from the pull-down list. 2. Enter the desired title in the edit box. ✓ The report tile set here will appear on the printed report.	/
Blood Sample Report Template/Body Fluid Sample Report Template	1. Select "Blood Sample Report Template" or "Body Fluid Sample Report Template" from the pull-down list. 2. Select the desired template format from the pull-down list: • One page with histogram • One page without histogram	• When "One page with histogram" is selected, the print results include the parameter results and graphs. • When "One page without histogram" is selected, the print results include only the parameter results.
Para. Language	Select a parameter display language from the pull-down list: • English abbreviation	• When "English abbreviation" is selected, the printed report displays the English parameter names.

Options	Description	Note
Copies	Enter the number of report copies to be printed when you select "Print" . The default value is 1 copy, and the setting range is [1 to 20].	/
Printing Content	Select the options you want to display on the printed report: • Print flags of edited result • Print high/low result flags • Print suspect flags • Print flags of revised result • Print out of linearity range flag • Print flags • Print reference range • Monochrome Print • Print RUO Parameters	• For the description of setting up high/low results as well as suspect results flags, refer to 6.14 Setting up Flag Alarm Sensitivity. • For the description of setting up reference ranges, refer to 6.11.2.2 Setting up default reference group.
Auto print after sample analysis	Enable one or more desired auto print settings: • Auto print after analysis • Auto print after validating • Auto print after QC count	• When "Auto print after analysis" is selected, the analyzer automatically prints the sample results after each analysis. • When "Auto print after validating" is selected, the analyzer automatically prints the validated sample results. • When "Auto print after QC count" is selected, the analyzer automatically prints the QC count results after each QC count.

11.2 Printing Sample Result Report

NOTE

- **The analyzer can print at most 700 records at one time.**

Prior to printing the sample test report, check and make sure the followings:

- The printer has been set and connected properly;
- There is sufficient paper in the printer.

11.2.1 Printing Current Sample Result Report

When a cycle of sample analysis finishes, select  on the **"Count"** screen.?

- ✓ The analyzer automatically prints the results of the current sample according to the print setup.

11.2.2 Printing from the Table Review Interface

Follow below instructions:

1. Select **"Sample"**–**"Table Review"** to go to **"Table Review"** screen.
 2. (Optional) Select one or multiple sample records to be printed in the test results report. If you want to print all sample records, skip this step.
 3. Select .
- ✓ A dialog box for printing displays.

4. Select the print pattern.

See below for setting descriptions.

Items		Description
Print pattern	Table print	When " Table print " is selected, only sample results will be printed. Graphs like histograms and scattergrams will not be printed.
	Graph print	When " Graph print " is selected, sample results as well as the graphs (histograms or scattergrams) will be printed.
Print record	Selected records	Select " Selected records " to print the selected sample records.
	All records	Select " All records " to print all sample records.

5. Select "**OK**".

√ The analyzer prints the test results report of the selected sample.

11.2.3 Printing from the Graph Review Screen

Follow below instructions:

1. Select "**Sample**"–"**Table Review**" to go to "**Table Review**" screen.

2. Select one or more sample records to print.

√ The selected sample record is highlighted.

3. Select "**Graph**" to go to "**Graph**" screen.

4. Select .

√ The analyzer automatically prints the results of the current samples according to the print setup.

11.2.4 Printing RUO Parameters Results

Follow below instructions:

1. Select "**Setup**"–"**Basic Settings**" to go to "**Basic Settings**" screen.

2. In the "**System Setup**" area, select "**Print Setup**" to go to "**Print Setup**" screen.

3. Select "**Printing Content**" and then select "**Print RUO Parameters**".

NOTE

- If you select to print RUO parameters, the statement "****For research use only, not for diagnostic use.**" will be printed on the report. The statement cannot be edited.

4. Select another button on the tab to save the settings and exit "**Printing Content**" screen.

√ The report will display the RUO parameter results.

11.2.5 Printing Microscopic Parameter Results

Follow below instructions.

1. Select "**Sample**"–"**Table Review**" to go to "**Table Review**" screen.

2. Select one or more sample records of which you want to review the microscopic parameters.

√ The selected sample record(s) is/are highlighted.

3. Select "**Graph**" to go to "**Graph**" screen.

4. Select "**Other Para.**"–"**Microscopic Para.**".

5. Select "**Print**" to print microscopic parameter results.

11.3 Printing QC Result Reports

Before printing a QC test result report, check and make sure the followings:

- The printer has been set and connected properly;
- There is enough paper in the printer;
- To print the L-J results, confirm that there are results in the L-J QC file;
- To print the X-B QC results, confirm that X-B QC is enabled and there are valid X-B results.

11.3.1 Printing L-J QC Result from L-J QC Table Screen

Follow below instructions:

1. Select **"QC" – "Overview"** to go to the **"Overview"** screen.
 2. In the **"L-J QC"** area, select **"Setup"** to go to the L-J QC file setting screen.
 3. Select a QC file in which the test results need to be printed.
 4. Select **"L-J QC Table"** to go to the QC Table screen of the specified QC file.
 5. Select  .
- ✓ The analyzer prints all QC test results of the selected QC file.

NOTE

- When the QC test results are printed from the QC table screen, the printed report does not include QC graph.

11.3.2 Printing L-J QC Result from L-J QC Graph Screen

Follow below instructions:

1. Select **"QC" – "Overview"** to go to the **"Overview"** screen.
 2. In the **"L-J QC"** area, select **"Setup"** to go to the L-J QC file setting screen.
 3. Select a QC file in which the QC graph needs to be printed.
 4. Select **"L-J QC Graph"** to go to the QC graph screen of the specified QC file.
 5. Select  .
- ✓ The analyzer prints the QC graph of the specified QC file.

11.3.3 Printing X-B QC Graph from the X-B QC Graph Screen

Follow below instructions:

1. Select **"QC" – "Overview"** to go to the **"Overview"** screen.
 2. In the **"X-B QC"** area, select **"X-B QC Graph"** to go to the X- B QC graph screen.
 3. Select  .
- ✓ The analyzer prints the X-B QC graph.

11.3.4 Printing X-B QC Parameters Results Report from the X-B QC Table Tab

Follow below instructions:

1. Select **"QC" – "Overview"** to go to the **"Overview"** screen.
 2. In the **"X-B QC"** area, select **"X-B QC Graph"** to go to **"X-B QC Graph"** screen.
 3. Select  .
- ✓ The analyzer prints the X-B QC parameter results report.

11.4 Printing Manual Calibration Factors

Prior to printing the manual calibration factors, check and make sure the followings:

- The printer has been set up and connected correctly;
- There is enough paper in the printer.

Follow below instructions:

1. Select **"Calibrate"**–**"Calibration Analysis"**–**"Manual"**.
 2. Select  .
- ✓ The analyzer prints the manual calibration factors.

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12 Servicing Your Analyzer

12.1 Introduction

Preventive and corrective maintenance procedures are required to keep the analyzer in a good operating condition. This analyzer provides multiple maintenance functions for this purpose.

This chapter introduces how to use the provided functions to maintain and troubleshoot your analyzer.



BIOHAZARD

- All the samples, controls, calibrators, wastes and areas contacted them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab coat, goggles, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.
 - Broken collection sample tubes may cause personal injury and/or biohazard. Avoid reversing the collection sample tube when loading, exercise caution when loading the collection sample tubes to the rack or getting the collection sample tubes from the rack, and be sure not to break the sample tubes.
 - Mindray does not claim the validity of the listed chemicals in infection control. For effective control of infection, please consult the Infection Prevention Department of the hospital or the epidemic professionals.
-



CAUTION

- If you accidentally spill hazardous material (for example, reagents or samples) on the instrument, clean the instrument with specified disinfectant. Wear proper personal protective equipment (e.g. gloves, lab coat, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.
 - The cleaning and sterilization may damage the instrument to some extent. It is recommended to perform sterilization only when necessary according to your laboratory protocol. Remember to clean the instrument before sterilizing.
 - Do not use any decontamination or cleaning agents which could cause a HAZARD as a result of a reaction with parts of the instrument or with material contained in it.
 - Improper maintenance may damage the analyzer. Operators must follow the instruction of this Operator's Manual to perform maintenance operations. For problems not mentioned in this manual, contact our Customer Service Department for service advice.
 - Only parts supplied by Mindray can be used for maintenance. For any questions, contact our Customer Service Department.
 - Before maintaining, transporting or servicing the instrument, clean the instrument cover as well as the parts and components with biological risks (such as the sampling probe). Remind the persons who handle the instrument of the related risks.
 - When you have replaced the diluent, lyse or dye after an abnormal shutdown, run a background test to check and make sure the results meet the requirement. Otherwise an inaccurate result may occur.
 - Unpacking, transportation or installation by our personnel not authorized or trained by our company may cause personal injury or damage to your analyzer. Do not unpack, transport or install your analyzer without the presence of personnel authorized by our company.
-

12.2 When and Why to Perform the Maintenance

Table 12-1 Maintenance of parts and components

Item/Software access		Maintenance Time	Purpose
Daily Maint.	/	Every day, when software prompts	To maintain the normal working of the analyzer
Probe Cleanser Maintenance (Maintenance-Maintenance)	Fluidics	As needed	To clean and soak specified components
	Flow Cell		
	SRV		
	Apertures		
Unclog Aperture (Maintenance-Aperture)	Unclog Aperture	When aperture clogging is reported	Unclog the aperture
Auto-cleaning the parts and components (Maintenance-Cleaning)	Clean Fluidics	When all the blank count results exceed their limits	To clean the fluidic system
	Remove Bubbles	When abnormal distribution is found in scattergrams	To remove bubbles from the flow cell
	Flow Cell Flush	When abnormal distribution is found in scattergrams	To remove bubbles from the flow cell
Fluidics (Service-Maintenance-Fluidics)	Pack-up	When the analyzer is to be left idle for a long time (more than 10 days)	Drain the residual reagent from fluidics

Table 12-2 Replacing and priming with reagents

Item/Software access		Maintenance Time	Purpose
Replace Reagent (Maintenance-Reagent)	Replace the Diluent	After new reagents are installed	Replace the residue reagent in the sample tubes
	Replace the LH Lyse		
	Replace the LN Lyse		
	Replace the LD Lyse		
	Replace the DR Diluent		
	Replace the FN Dye		
	Replace the FD Dye		
	Replace the FR Dye		
	Replace ESR Solution Reagent		

Item/Software access		Maintenance Time	Purpose
Prime Reagent (Maintenance-Reagent)	Priming with DS Diluent	when reagents are contaminated the analyzer reports there are bubbles in the WBC/RBC channels	Replace the residue reagent in the sample tubes Remove the bubbles in the WBC/RBC channels
	Prime with LH Lyse		
	Prime with LN Lyse		
	Prime with LD Lyse		
	Prime with DR Diluent		
	Prime with FN Dye		
	Prime with FD Dye		
	Prime with FR Dye		
	Prime with ESR Solution Reagent		

Table 12-3 Manual cleaning

Procedure	When to Clean	Why to Clean	Consumables and Tools You May Need
Cleaning the waste collecting trays	As needed	To remove the buildup in the trays	Distilled water, dry cloth
Cleaning the SRV	Every 2 months (of operation)	To remove blood residue or other buildup in the SRV	Distilled water, dry cloth, syringe, clean toothbrush, clean lint-free wet cloth
Cleaning the probe wipe of the open-vial module	As needed	To remove blood residue or other buildup in the probe wipe	Distilled water, dry cloth
Sterilizing the cover of the analyzer	As needed	To remove the contaminants on the cover of the analyzer	Disinfectant

Table 12-4 Replacing parts and components

Procedure	When to Replace	Tools You May Need
Replacing the reagent containers	As needed	/
Replacing the waste container	When the current waste container is full	/

Table 12-5 Replace the wearing parts

Parts Name	When to Replace	Note
Piercing probe	80,000 times	When the wear parts approach to their service life, the analyzer software will report an alarm. Contact our Customer Service Department.
Air filter	2 years	
Air dryer	4 years	
Vacuum relief valve	3 years	
Sheath fluid filter	1 year or 120,000 times	
ID1.6 Pharmed pinch sample tube	3 years	

** WARNING**

- Only parts supplied by our company can be used for equipment maintenance. For any questions, contact our Customer Service Department.
-

12.3 Replacing Reagents

** BIOHAZARD**

- All the samples, controls, calibrators, wastes and areas contacted them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab coat, glasses, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.
 - Be sure to dispose of reagents, waste, samples, consumables, etc. according to government regulations.
 - After replacing the reagent container/bag, check the tubing connected to the cap assembly and make sure it is connected firmly and not bent over. Otherwise, the operators may be exposed to biochemical hazards and incorrect measurement results may occur
 - Remove the waste container cap and replace the waste container only when the power indicator is not flickering, in order not to make the waste overflow from the container resulting in biological risks.
-

** CAUTION**

- The reagents are irritating to eyes, skin and mucous membranes. Wear proper personal protective equipment (e.g. gloves, lab coat, glasses, etc.) and follow safe laboratory procedures when handling them and the contacted areas in the laboratory.
 - If reagents come into contact with your skin or your eyes, rinse with plenty of water immediately and consult a physician if necessary.
 - When you have replaced the diluent, lyse or dye after an abnormal shutdown, run a background test to check and make sure the results meet the requirement. Otherwise an inaccurate result may occur.
 - Use the reagents, controls and calibrators specified by the manufacturer. Store and use the reagents as instructed by instructions for use of the reagents. Using controls, calibrators and reagents of other manufacturers may lead to damage to the analyzer and incorrect control, calibration and analysis results.
 - Check if the reagent sample tubes are properly connected before using the system. Otherwise the inaccurate analysis results may occur.
 - To ensure measurement accuracy, do not mix the new container of reagent with the residue in the replaced container, and prevent reagents from contamination following safe laboratory procedures.
-

Select **"Reagent"**–**"Reagent Overview"** to go to **"Reagent Overview"** screen, you can review the validity, open dates, valid days, expiration dates, and remaining volumes of the reagents used for testing.

When the reagent is insufficient, exhausted and expired or related alarm occurs, replace them timely.

Replace the reagents as follows:

1. Replace an old reagent with a new reagent.
2. Enter the reagent information.
3. Replace the old reagent in the tubing with the new reagent.

12.3.1 Replacing an Old Reagent with a New Reagent

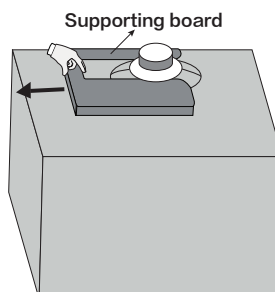
The reagents to be replaced are as follows:

- DS diluent
- Fluorescent reagents

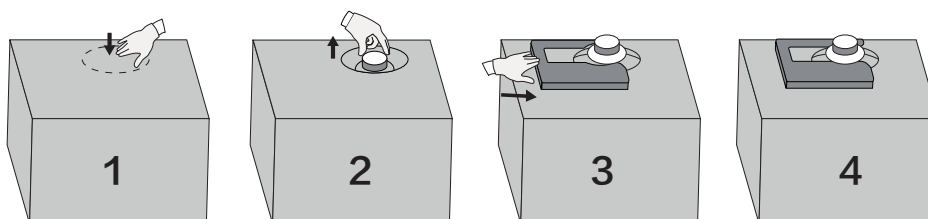
- Other reagent (except the DS diluent and fluorescent reagents)

12.3.1.1 Replacing the DS diluent container

1. Remove the cap of a new DS diluent container, and place the container next to the one to be replaced.
2. Remove the supporting board under the cap of the old container.



3. Rotate the cap counter-clockwise to open it, and then take out the cap assembly with caution.
4. Insert the pickup sample tube of the cap assembly into the new container, and then turn the cap clockwise until it is secured.
5. Install the supporting board under the new container's cap as shown below.



NOTE

- Keep the diluent container from severe shock or crashing against other object. Otherwise, the alarming would be unreliable.
- After replacing the reagent containers, check the tubing connected to the cap assembly and make sure it is not bent.

12.3.1.2 Replacing dye bags

NOTE

- After replacing the reagent container/bag, check the tubing connected to the cap assembly and make sure it is not bent over.

Before replacing the dyes, confirm that the reagent name on the reagent bag label is consistent with that on the connecting cable label of the cap assembly.

1. Open the fluorescent dye compartment door.
2. Get a new bag of fluorescent dye, open the cap and the aluminum film sealing the bag.
3. Take out the bag to be replaced along the direction of the supporting bracket.
4. Turn the cap of the old bag counterclockwise and then take out the cap assembly carefully.
5. Insert the removed pickup sample tube of the cap assembly into the new reagent bag, and then turn the cap assembly clockwise until it is secured.
6. Put the sealed new bag back on the support rack, making sure the bag is securely accommodated.
7. Cap the old reagent bag using the cap of the new bag and dispose of the old bag properly.

NOTE

- If the pickup sample tube of the cap assembly is stuck when it is taken out of the fluorescent dye bag, slightly adjust the position of the pickup sample tube and then take it out without pulling by force.
- During replacement, make sure that the pickup sample tube of the cap assembly does not reach the bottom of the reagent bag, otherwise the reagent cannot be aspirated normally.

12.3.1.3 Replacing other reagents (except DS diluent and fluorescent dyes)

1. Remove the cap of a new reagent container, and place the container next to the one to be replaced.
2. Turn the cap of the old container counterclockwise, and then take out the cap assembly with caution.
3. Insert the pickup sample tube of the cap assembly into the new container, and then turn the cap clockwise until it is secured.
4. Cap the old container with the cap of the new one and dispose of the old container properly.

12.3.2 Entering Reagent Information

Follow below steps to enter the reagent information.

1. Replace the reagents. For more information, refer to **12.3.1 Replacing an Old Reagent with a New Reagent**.
2. Select **"Reagent"**–**"Reagent Overview"** to go to **"Reagent Overview"** screen.

When a related alarm occurs	The "Reagent Setup" dialog box displays automatically.
When there is no related alarm	Select the reagents (multiple reagents can be selected at one time) to be replaced, and select "Setup" , then "Reagent Setup" dialog box displays.

√ **"Reagent Setup"** dialog box displays and the information of reagents to be replaced displays on the dialog box.

3. Enter the reagent information by one of the following ways.

If...	Then...
Barcode entry is selected	■ In the "Barcode Entry" field, enter the barcode information by a pop-up keyboard; or ■ Scan the reagent barcode labels with an external barcode scanner.
RFID entry is selected	Place the reagent container in the RFID region.

NOTE

- **RFID card is only available for ESR Solution Reagent and dyes.**
- **If you want to change the way of entering reagent information, contact our Customer Service Department.**

√ If the barcode is valid, the reagent information will be refreshed.

NOTE

- **If you enter/scan the barcode in the "Reagent Setup" dialog box but there is no related reagents information, the software will report an alarm. Check and make sure you have replaced an accurate reagent.**
4. (Optional) if more than one reagent needs to be replaced, repeat step 3 to complete the setup of all reagents.

- Replace the old reagents in the fluidic system.

If "Auto replace reagent after setup" is enabled	After the reagent setup is completed, when the count down completes, the analyzer automatically replaces the reagents in the fluidics system.
If "Auto replace reagent after setup" is not enabled	Select "Replace Reagent" to replace the old reagent in the fluidic system.

- The analyzer automatically removes the errors, loads new reagents information and replaces the old reagent in the fluidics. When you are replacing expired reagents, the analyzer will also perform the reagent priming process.

12.3.3 Replacing the Reagents in the Fluidics and Priming with Reagents

Under the following circumstances, replace the old reagents in the fluidics, or prime the fluidic sample tubes with reagents.

Program	When to Replace	Purpose
Replacing Reagents	After new reagents are installed	Replace the residue reagent in the sample tubes
Reagent Priming	- When reagent(s) get(s) contaminated - The analyzer reports there are bubbles in the WBC/RBC channels	- Replace the residue reagent in the sample tubes - Remove the bubbles in the WBC/RBC channels

Follow below instructions:

- Select **"Maintenance"** – **"Reagent"** to enter the reagent maintenance screen.
- Select the reagent to be replaced or to be primed with in the **"Replace Reagent"** or **"Prime Reagent"** area.
✓ The analyzer automatically completes the operation.

12.4 Replacing Waste Container



BIOHAZARD

- Remove the waste container cap and replace the waste container only when the power indicator is not flickering, in order not to make the waste overflow from the container.
- After replacing the reagent container/bag, check the tubing connected to the cap assembly and make sure it is connected firmly and not bent over. Otherwise, the operators may be exposed to biochemical hazards and incorrect measurement results may occur.
- If the waste is discharged using waste container, make sure the pickup sample tube of the waste container cap assembly is above the waste container, and the sample tube is smooth and not bent. If the waste is discharged directly, make sure the waste tube is at a lower position than the waste outlet on the analyzer.

- Get an empty waste container, remove the cap and place it next to the one to be replaced.
- Turn the cap counterclockwise and remove the cap assembly from the old container with caution.
- Insert the old cap assembly into the new container as vertically as possible, and secure the cap by turning it clockwise.
- Cap the old container with the cap of the new one, and then dispose of the waste properly.

12.5 When Two DS Diluent Containers are Connected at the Same Time

If your analyzer supports to connect two DS Diluent containers at the same time, when one container of the DS diluent runs out during the analysis process, the analyzer automatically switches to use the other container.

Connect the two DS Diluent containers as shown in **Figure 12-1 Using 2 DS Diluent containers at the same time.**

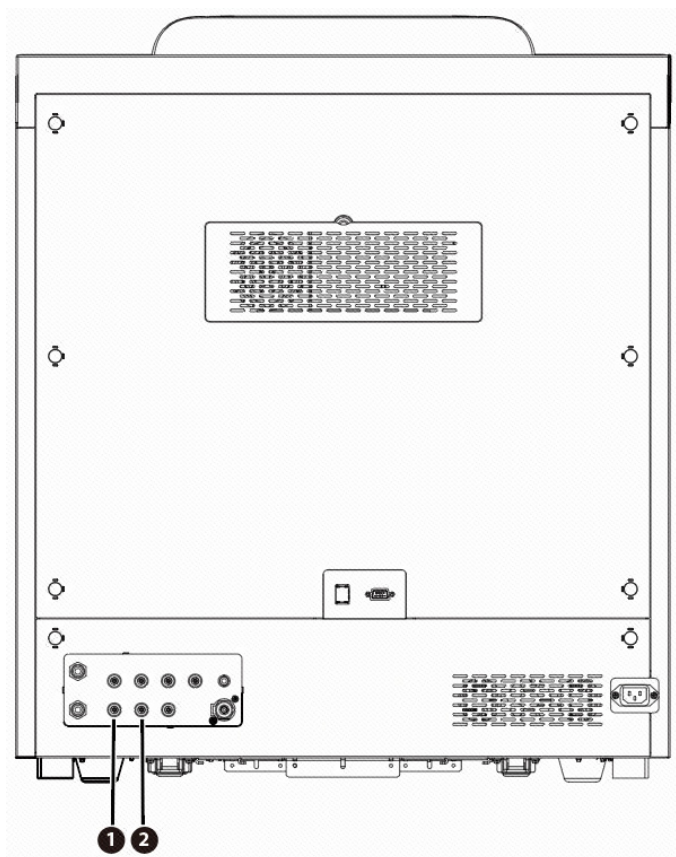


Figure 12-1 Using 2 DS Diluent containers at the same time

	Name
1	DS Diluent A Inlet
2	DS Diluent B Inlet

You can find the DS-A/DS-B labels on the sample tubes of the DS diluent reagent cap assemblies. The labels are in different colors (DS-A: yellow, DS-B purple).



Figure 12-2 DS-A/B labels on the sample tubes of the reagent cap assembly (connecting to the reagent inlets on the back panel of the analyzer)



Figure 12-3 DS-A label on the sample tubes of the reagent cap assembly (Example)

Check the status of both DS Diluent containers on the “**Reagent Overview**” screen.

Reagent Overview					
Reagent Name	Expiration Date	Open Date	Valid Days	Use before	Volume(%)
DS DILUENT A	01-01-2036	01-14-2025	60	03-14-2025	100.00%
DS DILUENT B	01-01-2036	01-14-2025	60	03-14-2025	100.00%
M-6DR DILUENT	01-01-2036	01-14-2025	90	04-13-2025	100.00%
M-6LH LYSE	01-01-2036	01-14-2025	60	03-14-2025	100.00%
M-6LN LYSE	01-01-2036	01-14-2025	60	03-14-2025	100.00%
M-6LD LYSE	01-01-2036	01-14-2025	60	03-14-2025	100.00%
M-6FN DYE	01-01-2036	11-06-2024	60	01-04-2025	91.87%
M-6FD DYE	01-01-2036	11-06-2024	60	01-04-2025	89.20%
M-6FR DYE	01-01-2036	11-06-2024	90	02-03-2025	98.37%
ESR Solution Reagent	01-01-2036	11-06-2024	90	02-03-2025	99.74%

Figure 12-4 “Reagent Overview” screen when using 2 DS Diluent containers

NOTE

- On the software “Reagent Overview” screen, the “DS Diluent A” corresponds to the diluent connecting to the inlet of DS Diluent A, and “DS Diluent B” to the diluent connecting to the inlet of DS Diluent B.

12.5.1 Replacing the used-up DS diluent container

If the analyzer connects two DS Diluent containers at the same time, when one container of the DS diluent runs out during the analysis process, the “**Reagent Overview**” utility button is highlighted in orange. The analyzer pauses the analysis, and automatically switches to use the other container.

- When the analyzer is analyzing samples under the auto-loading mode, the “**Error Information**” screen displays.
- When the analyzer is analyzing samples under the open-vial mode, a progress bar displays on the software.

In either case, the analyzer automatically switches the DS Diluent channel; and when the analyzer has successfully switched channel, it automatically resumes the analysis. In the meanwhile, the analyzer reports the **"No DS DILUENT A/No DS DILUENT B"** error, and the indicator on the analyzer lights in red. You can then replace the empty DS Diluent container.

1. Follow the instructions in **12.3.1 Replacing an Old Reagent with a New Reagent** to replace the DS Diluent containers.

NOTE

- **When replacing the DS Diluent containers, pay attention to the positions of the DS Diluent A and DS Diluent B. You can find the DS-A/B labels on the sample tubes of the reagent cap assemblies (see *Figure 12-2 DS-A/B labels on the sample tubes of the reagent cap assembly (connecting to the reagent inlets on the back panel of the analyzer)* and *Figure 12-3 DS-A label on the sample tubes of the reagent cap assembly (Example)*). Make sure you replace the correct container.**

2. Enter the **"Reagent Setup"** dialog box.

You can enter the **"Reagent Setup"** dialog box in either of the following ways:

- Select **"Reagent"** – **"Reagent Overview"** – **"Setup"**;
 - Select the reagent error message on the right lower corner on the screen to open the **"Remove Error"** dialog box, and then select **"Remove Error"**.
 ✓ The **"Reagent Setup"** dialog box displays.
3. Select the **"Barcode Entry"** edit box, and use the pop-out keyboard to enter the reagent barcode; or scan the reagent barcode labels with an external barcode scanner.
 ✓ If the barcode is valid, the reagent information will be refreshed.

NOTE

- **After you have registered the new DS Diluent information, the analyzer does not replace the reagent in the fluidics immediately. When the DS Diluent currently in use runs out, and the analyzer switches back to the newly replaced DS Diluent, the analyzer will then replace the reagent in the fluidics.**

12.5.2 Replacing both DS Diluent containers at the same time

When the DS Diluent in both containers runs out, you can replace both the DS Diluent at the same time. If the analyzer is performing analysis, the analysis process will be paused.

Follow below instructions:

1. Follow the instructions in **12.3.1 Replacing an Old Reagent with a New Reagent** to replace the DS Diluent containers.

NOTE

- **When replacing the DS Diluent containers, pay attention to the positions of the DS Diluent A and DS Diluent B. You can find the DS-A/B labels on the sample tubes of the reagent cap assemblies (see *2. Enter the "Reagent Setup" dialog box.*). Make sure you replace the correct container.**
2. Follow the instructions in **12.3.1 Replacing an Old Reagent with a New Reagent** to register the new reagent information and complete the replacement.
 - When registering reagent information on the **"Reagent Overview"** screen, you can select DS Diluent A and DS Diluent B one by one, and register the reagent information respectively; or you can select both reagents at the same time to register the information of both reagents together.
 - When you select DS Diluent A and DS Diluent B together on the **"Reagent Overview"** screen and select **"Setup"**, the **"Reagent Setup"** dialog box displays the two reagents together.
 3. Scan or enter the barcode information for DS Diluent A and for DS Diluent B.

NOTE

- **When you select to register DS Diluent A and DS Diluent B together, the first barcode you scanned or entered will be assigned to DS Diluent A, and the second barcode will be assigned to DS Diluent B.**

- On the software “Reagent Setup” screen, the “DS Diluent A” corresponds to the diluent connecting to the inlet of DS Diluent A, and “DS Diluent B” to the diluent connecting to the inlet of DS Diluent B. Make sure you scan/enter the correct barcode information.

4. Follow the instructions in **12.3.1 Replacing an Old Reagent with a New Reagent** to replace the reagent in the fluidic system.

12.6 Probe Cleanser Maintenance

12.6.1 Daily Probe Cleanser Maintenance

When it is time for daily Probe Cleanser maintenance, the analyzer will give a prompt. You can also initiate the Probe Cleanser maintenance actively.

Follow below instructions:

1. When the time for Probe Cleanser maintenance dialog box displays, select “**Yes**”.
 - ✓ The analyzer prepares for Probe Cleanser maintenance. After the preparation for Probe Cleanser maintenance completes, a dialog box displays.
2. Place the Probe Cleanser under the sample probe as instructed.
3. Press the [**Aspirate**] key to start the Probe Cleanser maintenance procedure.
 - ✓ The analyzer aspirates Probe Cleanser.
4. Remove the sample tube with Probe Cleanser when you hear two beeps.
 - ✓ The analyzer automatically completes Probe Cleanser maintenance.

12.6.2 Probe Cleanser Maintenance to Parts and Components

Follow below instructions:

1. Select “**Maintenance**” – “**Maintenance**” to enter the “**Maintenance**” screen.
2. Select the buttons of the parts and components that need Probe Cleanser maintenance.
 - ✓ The analyzer prepares for Probe Cleanser maintenance. After the preparation for Probe Cleanser maintenance completes, a dialog box displays.
3. Present the sample tube with the Probe Cleanser under the sample probe as instructed.
4. Press [**Aspirate**] key to start the Probe Cleanser maintenance procedure.
 - ✓ The analyzer aspirates Probe Cleanser.
5. Remove the sample tube with Probe Cleanser when you hear two beeps.
 - ✓ The analyzer automatically completes Probe Cleanser maintenance.

NOTE

- When the analyzer has fault or needs your attention, it beeps to alarm.

12.7 Auto-Cleaning the Parts and Components

When you need to perform the parts/components auto-cleaning process, refer to **12.2 When and Why to Perform the Maintenance**.

Follow below instructions:

1. Select “**Maintenance**” – “**Cleaning**” to enter the “**Cleaning**” screen.
2. Select the corresponding cleaning program.
 - ✓ The analyzer automatically completes the operation.

12.8 Manually Cleaning the Parts and Components

You should clean the following parts or components as required:

Table 12-6 Manually cleaning the parts and components

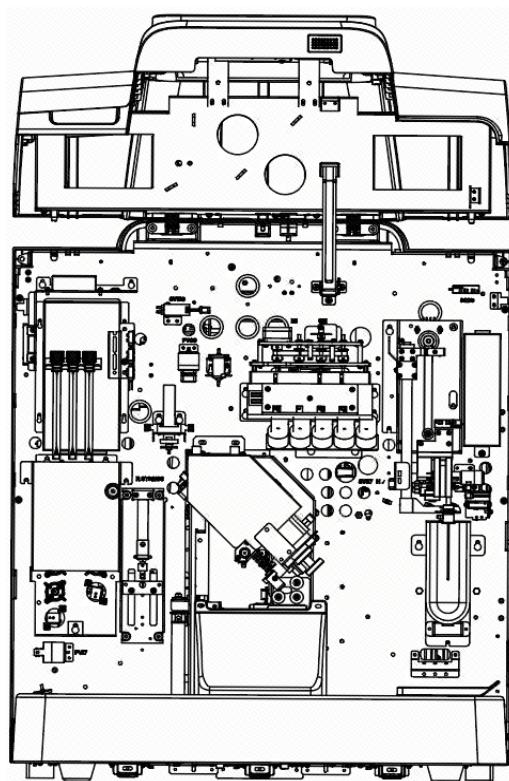
Procedure	When to Clean	Why to Clean	Consumables and Tools You May Need
Cleaning the waste collecting trays	As needed	To remove the buildup in the trays	Distilled water, dry cloth
Cleaning the SRV	Every 2 months (of operation)	To remove blood residue or other buildup in the SRV	Distilled water, dry cloth, syringe, clean toothbrush, clean lint-free wet cloth
Cleaning the probe wipe of the open-vial module	As needed	To remove blood residue or other buildup in the probe wipe	Distilled water, dry cloth
Sterilizing the cover of the analyzer	As needed	To remove the contaminants on the cover of the analyzer	Disinfectant (refer to 12.8.5 Cleaning the Analyzer Cover)

12.8.1 Opening and Closing the Front Cover

You may need to open and close the front cover in some manual cleaning procedures. Be sure to perform strictly according the standard procedures stated below.

12.8.1.1 Opening the front cover

1. Hold the sides of the front cover and lift it upwards.
2. Fix the opened front cover with the stop bar.



12.8.1.2 Closing the front cover

1. Hold the front cover steadily, and then release the stop bar.
2. Move the front cover downwards and close it.

12.8.2 Cleaning the Waste Collecting Trays

NOTE

-
- **To avoid damage to the components of the analyzer, close the front cover gently.**
 - **When removing the SRV waste collecting tray, do not make the screw thread of the connecting part on top of the sample probe loose, which may bring air in the sample probe and lead to unreliable results.**
 - **Reassemble the SRV waste collecting tray by facing the side with the groove up.**
 - **The sample probe and the SRV may contain biohazardous materials. Exercise caution to avoid direct contact with the probe and the SRV when working around them.**
 - **After the analyzer is turned off, wait at least 30 seconds to release the pressure and vacuum accumulated in the pneumatic lines. Do not perform any maintenance or replacing procedure immediately after you turn off the analyzer.**
-

12.8.2.1 Cleaning the waste collecting tray of the open-vial module

Follow the steps below:

1. Open the front cover.
2. Pull out the waste collecting tray under the open-vial sample probe.
3. Wash the tray with clean water.
4. Wipe up the water left in the tray.
5. Install the tray back properly.
6. Close the front cover.

12.8.2.2 Cleaning the SRV waste collecting tray

Follow the steps below:

1. Shut down the analyzer, and wait several minutes after power-off.
2. Open the front cover.
3. Pull out the waste collecting tray under the SRV.
4. Wash the tray with clean water.
5. Wipe up the water left in the tray.
6. Install the tray back properly.
7. Close the front cover.

12.8.2.3 Cleaning the waste collecting tray of the autoloading module

Follow the steps below:

1. Shut down the analyzer, and wait several minutes after power-off.
2. Open the front cover.
3. Pull out the waste collecting tray under the autoloading piercing unit.
4. Wash the tray with clean water.
5. Wipe up the water left in the tray.
6. Install the tray back properly.
7. Close the front cover.

12.8.3 Cleaning the SRV

NOTE

- After the analyzer is turned off, wait at least 30 seconds to release the pressure and vacuum accumulated in the pneumatic lines. Do not perform any maintenance or replacing procedure immediately after you turn off the analyzer.
- When the SRV is disassembled, reagent may leak from the sample tubes. Place a dry cloth or tissue under the SRV to absorb any reagent.
- Do not remove the rear fixed valve of the SRV.
- Do not use too much force to pull sample tubes on sides of the SRV when disassembling the SRV. Otherwise, the sample tubes may be disconnected and the leakage may be caused.
- The sample probe should be completely separated from the probe wipe when the SRV is disassembled. Otherwise, the probe may be deformed or the probe wipe may be damaged.
- The SRV is so fragile that any knock or drop may damage it. Exercise caution when disassembling/ assembling or cleaning the SRV. Do not loosen or bend the tiny steel sample tubes at sides of the SRV.
- Do not loosen or deform the sample probe when disassembling, cleaning and reassembling the SRV.

Follow the steps below:

1. Shut down the analyzer, and wait several minutes after power-off.
2. Open the front cover.
3. Remove the SRV waste collecting tray (see **12.8.2 Cleaning the Waste Collecting Trays**) and pull the probe wipe assembly down to the lowest position.
4. Remove the probe wipe from the sample probe.
5. Remove the pressure screw by turning it counterclockwise.
6. Remove the sampling valve (the outer piece) and the rotor valve (the middle piece) of the SRV respectively.
7. Inject some probe cleanser into holes and grooves of the sampling valve, rotor valve and rear fixed valve respectively. You can also clean the holes and grooves gently with a brush dampened with probe cleanser.
8. Use clean lint-free tissues dampened with probe cleanser to wipe the contact surfaces of the valves. Then wash the valves with distilled water.

NOTE

- Make sure no dust should be found in the holes and grooves and on the contact surfaces after the cleaning. Unclean surface(s) may lead to leakage from the SRV and unreliable analysis results.

9. Install the rotor valve and sampling valve back properly.

NOTE

- The valve contact surfaces must be damp when the SRV is reassembled.
- Be sure the sampling valve, rotor valve and rear fixed valve of the SRV attach well after the reassembly.
- While installing the rotor valve of the SRV, the plane part shall be placed horizontally on the top, and the metal knob shall be placed between the two stoppers. Otherwise, the SRV will not be able to work properly.

10. Install the pressure screw by turning it clockwise, and then secure it.
11. Put the sample probe through the center of the probe wipe, lift the probe wipe assembly back to the original position when the analyzer is turned off.
12. Install the SRV waste collecting tray back properly, and then close the front cover.

NOTE

- Make sure the sample probe is properly installed through the probe wipe and the probe wipe is at its original position. Otherwise the wipe may be stuck and unable to work properly

- Once reagents spill on the surface of the analyzer, wipe them off with a damp cloth or tissue as soon as possible.
- After the SRV tray is reassembled, check whether the connection between the SRV and the top of the sample probe is fixed. If not, tighten it to prevent air from coming into the probe and lead to unreliable analysis result.
- To avoid damage to the components of the analyzer, close the front cover gently.

13. Run a blank count after startup. Be sure the results are within required ranges.

12.8.4 Cleaning the Probe Wipe

NOTE

- After the analyzer is turned off, wait at least 30 seconds to release the pressure and vacuum accumulated in the pneumatic lines. Do not perform any maintenance or replacing procedure immediately after you turn off the analyzer.
- Disconnect the sample tubes by pinching the connecting ends, swaying them while pulling them outward. Do not use too much force while disconnecting the sample tubes. Otherwise, the probe wipe of the open vial sampling unit may be damaged.
- Make sure the sample probe is properly installed through the probe wipe and the probe wipe is at its original position. Otherwise the wipe may be stuck and unable to work properly.
- To avoid damage to the components of the analyzer, close the front cover gently.
- When disassembling, cleaning and reassembling the probe wipe of the open-vial module, do not use too much force. Otherwise, the probe wipe may be damaged.

1. Shut down the analyzer, and wait several minutes after power-off. And then open the front cover.
2. Pull the probe wipe assembly down to the lowest position, and then remove it from the sample probe.
3. Separate the probe wipe from the probe wipe assembly. Disconnect the tubing from the probe wipe.
4. Wash the probe wipe with clean water, wipe it dry and install it back properly. Place the probe wipe assembly to the original position when the analyzer is turned off.
5. Close the front cover.

12.8.5 Cleaning the Analyzer Cover



BIOHAZARD

- Our company does not claim the validity of the listed chemicals in infection control. For effective control of infection, consult the Infection Prevention Department of the hospital or the epidemic professionals.



CAUTION

- Perform regular cleaning on the covers of the equipment. Use the specified materials to clean the equipment only. For any damage or accidents caused by the use of materials other than those specified, our company will not provide any warranty.
- The sterilization may damage the system to some extent. It is recommended to perform sterilization only when necessary according to your laboratory protocol. Remember to clean the equipment before sterilizing.
- Do not use any cleaning agents that could cause hazards as a result of a reaction with parts of the analyzer or with materials inside the analyzer. If there is any doubt about the compatibility of the cleaning agents with parts of the equipment or with material contained in it, please contact our Customer Service Department or the local distributor.
- If you accidentally spill sample and reagents on the surface of the analyzer, clean the surface of the analyzer. Do not use material that could corrupt metal (for example, 3% hydrogen peroxide). Wear proper personal protective equipment (e.g. gloves, lab protective clothe) and follow safe laboratory procedures during cleaning.

Perform regular cleaning on the covers of the analyzer.

12.9 Performing Packing-up Procedure

If the analyzer is not to be used for a long time (over 10 days), you should perform this procedure.

Follow below instructions:

1. Select **"Maintenance"** - **"Fluidics"** to enter the **"Fluidics"** maintenance screen.
2. Select **"Pack-up"**, and follow the software instruction to complete the pack-up procedure.

12.10 Performing Touchpad Calibration

If the touchpad response to touching is not sensitive or accurate, perform the procedure to calibrate the touchpad.

NOTE

- **Do not use anything sharp on the touchscreen or strike on it.**

Follow below instructions.

1. Select **"Service"** - **"Debug & Self-Test"** - **"Touchpad Cal."** to go to **"Touchpad Cal."** screen.
 2. Select the options as needed.
 - Select **"CD Key"**, **"CD-E Key"**, **"Count Key"** or **"Mode Switch"**, and a prompt of **"Select"OK".Calibration will start after 3 seconds. Cover the indictaor completely with your finger until the calibration result displays"** will display.
 - Select **"Guard Cap."**, a prompt of **"Select"OK". Calibration will start after 3 seconds. Press and hold the corresponding key on the touchpad until the calibration result displays"** will display.
 3. Select **"OK"**.
- ✓ After the calibration is completed, the software displays **"Calibration succeeded"**.

12.11 Reviewing and Exporting Logs

The **"Log"** screen records all activities of the analyzer. It contributes significantly to searching for operation history and troubleshooting the analyzer.

The analyzer can save logs of the recent two years. If number of logs exceeds the upper limit, the latest log will overwrite the oldest one. You can browse and print logs, but cannot delete them.

Administrators and operators have different authorities:

Table 12-7 Logs

	Administrators' level	Common users' level
All Logs	Viewing all types of logs	Viewing the logs for analyzer startup and shutdown, user logging in and logging out at the operator's level.
General Logs	Viewing all operation-related logs at both administrator's and operator's levels.	Viewing the logs for analyzer startup and shutdown, user logging in and logging out at the operator's level.
Setup Adjustment	Viewing all setting adjustment logs at both administrator's and operator's levels.	Cannot review
Error Information	Viewing error information and troubleshooting information of the analyzer.	Cannot review

12.11.1 Viewing Logs

Follow below instructions:

1. Select "**Service**"–"**Log**" to go to "**Log**" screen.
2. Select a category of logs to be viewed.
3. Select "**Go to**".
- √ The dialog box displays.
4. In the "**Date**" edit box, specify the date on which the logs need to be viewed, and select "**OK**".
- √ The tab displays the logs at the specified date.

12.11.2 Exporting Logs

You may export the logs in specified time range to the USB device.

Before exporting sample records, make sure you have inserted a USB flash drive safely to The USB ports on the left side of the analyzer.



CAUTION

- **The USB ports on the left side of the analyzer shall only be used to connect the peripheral devices specified in this manual. See Section B.7 Input/output device for details about supported devices and models.**
 - **The user should ensure the data safety of the USB devices connecting to the analyzer.**
-

Follow below instructions:

1. Select "**Service**"–"**Log**" to go to "**Log**" screen.
2. Select "**Export**".
- √ The dialog box displays.
3. In the "**Start date:**" and "**End date:**" edit boxes, specify the time range in which logs need to be exported.
4. Select "**OK**".
- √ The analyzer automatically exports the logs in specified time range to the USB flash drive.
- √ After the exporting is completed, the software displays "**Export succeeded.**"

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13 Troubleshooting

13.1 Introduction

This chapter contains information that is helpful in locating and correcting problems that may occur during operation of your analyzer.

NOTE

- This chapter is not a complete service manual and is limited to problems that are readily diagnosed and/or corrected by the user of the analyzer.

13.2 Checking Analyzer Status

You can check analyzer information from the “**Status**” menu, including statistics, temperature and pressure, floater status, sensor status, voltage and current, as well as version information. To check the status information at the Status Screen is significant for you to locate and remove errors of the analyzer.

Follow the path below to review analyzer status:

Item/Software access	Status	Path	Account level requirements
Statistics	Valid Runs	Service–Status–Statistics	All
	Runs since latest initialization		
	Sample Runs		
	QC Runs		
	Valid Sample Runs		
	QC Runs		
	Calibration Runs		
	Valid Runs After Latest Startup		
	Runs after Probe Cleanser Maintain		
	Clogs in Impedance Channel		
	Background Runs		
	Wearing Parts		
Temp. & Pressure	Displays the current temperature and pressure as well as the acceptable range for various items. Out-of-range values are highlighted in red background.	Service–Status–Temp. & Pressure	Administrator's level
Floater Status	Display the full or empty status of the bathes and waste cistern.	Service–Status–Floater Status	Administrator's level
Pos. & Photocoupler	Display the motor or cylinder positions on the syringe unit and sample transportation unit; as well as the sensor status	Service–Status–Pos. & Photocoupler	Administrator's level

Item/Software access	Status	Path	Account level requirements
Voltage & Current	Display the Out-of-range values are highlighted in red background.	Service-Status-Voltage & Current	Administrator's level
Version Info.	Review the analyzer software version information	Service-Status-Version Info.	All

13.3 Error Messages and Solutions

During the operation, if error(s) is detected, the analyzer will beep and display the corresponding error message in the pop-up dialog box.

The name and troubleshooting method of the errors are displayed. Names of the errors are displayed by the order of their occurrence.

You may tap to select the error, and view its troubleshooting information in the troubleshooting box. The troubleshooting information of the first error is displayed by default. The troubleshooting information of the first error is displayed by default.

The following functions are provided:

■ Remove error

Tap the "Remove Error" button to clear all the errors that can be removed automatically. For the errors that cannot removed automatically, follow the troubleshooting method to solve them.

■ Close the error info. dialog box

Tap "**Close**" to close the dialog box, but the errors will still be displayed in the error info. area on the screen. Tap the error info. area again, the dialog box will be displayed.

Error ID	Message	Details	Solutions
0x00000001	Insufficient aspiration	Bubbles found in tubing during aspiration	1. Found bubbles in tubing while monitoring the aspiration process. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000002	Insufficient aspiration	Found the end segment of the sample not fully fill the tubing during aspiration	1. Found the end segment of the sample hanging on the tubing rather than fully filling the tubing while monitoring the aspiration process. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000003	Insufficient aspiration	Incontinuous sample in tubing or plenty of bubbles found during aspiration	1. Found bubbles or incontinuity in the aspirated sample while monitoring the aspiration process. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000004	Insufficient aspiration	Found sample not aspirated while monitoring the aspiration process.	1. Found sample not aspirated while monitoring the aspiration process. 2. Check whether the volume of sample in the tubing is higher than minimum.

Error ID	Message	Details	Solutions
0x00000005	Insufficient aspiration	Found aspirated sample insufficient while monitoring the aspiration process.	1. Found aspirated sample insufficient while monitoring the aspiration process. 2. Check whether the volume of sample in the tubing is higher than minimum.
0x00000006	Insufficient aspiration	Blood sensor 1 status abnormal	1. Blood sensor 1 status abnormal. Possible reasons: Blood sensor error; connection error between the blood sensor and sample aspiration or dividing tubing; diluent and sample not flow in the tubing as expected due to fluidic error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000007	Insufficient aspiration	Blood sensor 2 status abnormal	1. Blood sensor 2 status abnormal. Possible reasons: Blood sensor error; connection error between the blood sensor and sample aspiration or dividing tubing; diluent and sample not flow in the tubing as expected due to fluidic error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000008	Blood sensor 1 error	Blood sensor 1 self-test voltage low	1. Blood sensor 1 status abnormal. Possible reasons: Blood sensor error; connection error between the blood sensor and sample aspiration or dividing tubing; diluent and sample not flow in the tubing as expected due to fluidic error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000009	Blood sensor 1 error	Blood sensor 1 self-test voltage high	1. Blood sensor 1 status abnormal. Possible reasons: Blood sensor error; connection error between the blood sensor and sample aspiration or dividing tubing; diluent and sample not flow in the tubing as expected due to fluidic error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000010	Blood sensor 2 error	Blood sensor 2 self-test voltage low	1. Blood sensor 2 status abnormal. Possible reasons: Blood sensor error; connection error between the blood sensor and sample aspiration or dividing tubing; diluent and sample not flow in the tubing as expected due to fluidic error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00000011	Blood sensor 2 error	Blood sensor 2 self-test voltage high	1. Blood sensor 2 status abnormal. Possible reasons: Blood sensor error; connection error between the blood sensor and sample aspiration or dividing tubing; diluent and sample not flow in the tubing as expected due to fluidic error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000012	Aspiration process unidentified	Unreliable aspiration due to error in the aspiration process	1. Unidentified aspiration process found while monitoring the aspiration process. The aspiration may be unreliable. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00001001	No DS DILUENT	DS DILUENT running out or bubbles in tubing	1. Please check if there is DS DILUENT in the container, and if the pickup sample tube is placed properly. 2. If there is diluent in the container. Re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of DS DILUENT. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x00001002	No M-6LH LYSE	M-6LH LYSE running out or bubbles in tubing	1. Please check if there is M-6LH LYSE in the container, and if the pickup sample tube is placed properly. 2. If there is lyse in the container. Re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no lyse, replace with a new container of M-6LH LYSE. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x00001003	No M-6LD LYSE	M-6LD LYSE running out or bubbles in tubing	1. Please check if there is M-6LD LYSE in the container, and if the pickup sample tube is placed properly. 2. If there is lyse in the container. Re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no lyse, replace with a new container of M-6LD LYSE. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.

Error ID	Message	Details	Solutions
0x00001004	No M-6LB LYSE	M-6LB LYSE running out or bubbles in tubing	1. Please check if there is M-6LB LYSE in the container, and if the pickup sample tube is placed properly. 2. If there is lyse in the container. Re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no lyse, replace with a new container of M-6LB LYSE. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x00001005	No M-6DR DILUENT	M-6DR DILUENT running out or bubbles in tubing	1. Please check if there is M-6DR DILUENT in the container, and if the pickup sample tube is placed properly. 2. If there is diluent in the container. Re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of M-6DR DILUENT. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x00001006	No M-6LN LYSE	M-6LN LYSE running out or bubbles in tubing	1. Please check if there is M-6LN LYSE in the container, and if the pickup sample tube is placed properly. 2. If there is lyse in the container. Re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no lyse, replace with a new container of M-6LN LYSE. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x00001010	Waste container full	Waste container about to be full	1. Empty the waste container or install a new one. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00001011	DS DILUENT expired	DS DILUENT expired	1. DS DILUENT expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001012	M-6LH LYSE expired	M-6LH LYSE expired	1. M-6LH LYSE expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001013	M-6LB LYSE expired	M-6LB LYSE expired	1. M-6LB LYSE expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.

Error ID	Message	Details	Solutions
0x00001014	M-6LD LYSE expired	M-6LD LYSE expired	1. M-6LD LYSE expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001015	M-6DR DILUENT expired	M-6DR DILUENT expired	1. M-6DR DILUENT expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001016	M-6LN LYSE expired	M-6LN LYSE expired	1. M-6LN LYSE expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001017	M-6FD DYE expired	M-6FD DYE expired	1. M-6FD DYE expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001018	M-6FR DYE expired	M-6FR DYE expired	1. M-6FR DYE expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001019	M-6FN DYE expired	M-6FN DYE expired	1. M-6FN DYE expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001021	Sequence command error	Abnormal M-6LD LYSE detection command	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00001022	Sequence command error	Abnormal M-6LB LYSE detection command	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00001023	Sequence command error	Abnormal M-6LH LYSE detection command	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00001024	Sequence command error	Abnormal M-6LN LYSE detection command	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00001025	Sequence command error	Abnormal M-6DR DILUENT detection command	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00001026	Sequence command error	Abnormal DS DILUENT detection command	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00001028	Sequence command error	Diluent channel locked due to no diluent or bubbles in tubing	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00001029	Sequence command error	Diluent channel locked due to no diluent or bubbles in tubing	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00001030	Sequence command error	Abnormal reagent detection command	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00001041	Insufficient DS DILUENT	Insufficient DS DILUENT	1. No DS DILUENT. Please get a new barcode. 2. Tap the "Remove Error" button and enter the new DS DILUENT barcode. 3. If the error still exists, contact our customer service department.
0x00001042	Insufficient M-6DR DILUENT	Insufficient M-6DR DILUENT	1. No M-6DR DILUENT. Please get a new barcode. 2. Tap the "Remove Error" button and enter the new M-6DR DILUENT barcode. 3. If the error still exists, contact our customer service department.
0x00001043	Insufficient M-6LD LYSE	Insufficient M-6LD LYSE	1. No M-6LD LYSE. Please get a new barcode. 2. Tap the "Remove Error" button and enter the new M-6LD LYSE barcode. 3. If the error still exists, contact our customer service department.
0x00001044	Insufficient M-6LN LYSE	Insufficient M-6LN LYSE	1. No M-6LN LYSE. Please get a new barcode. 2. Tap the "Remove Error" button and enter the new M-6LN LYSE barcode. 3. If the error still exists, contact our customer service department.
0x00001045	Insufficient M-6LB LYSE	Insufficient M-6LB LYSE	1. No M-6LB LYSE. Please get a new barcode. 2. Tap the "Remove Error" button and enter the new M-6LB LYSE barcode. 3. If the error still exists, contact our customer service department.
0x00001046	Insufficient M-6LH LYSE	Insufficient M-6LH LYSE	1. No M-6LH LYSE. Please get a new barcode. 2. Tap the "Remove Error" button and enter the new M-6LH LYSE barcode. 3. If the error still exists, contact our customer service department.
0x00001047	Insufficient M-6FN DYE	Insufficient M-6FN DYE	1. Registered volume of M-6FN DYE for closed-reagent model runs out. 2. Click the "Remove Error" button, and input the M-6FN DYE barcode to remove the error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00001048	Insufficient M-6FR DYE	Insufficient M-6FR DYE	1. Registered volume of M-6FR DYE for closed-reagent model runs out. 2. Click the "Remove Error" button, and input the M-6FR DYE barcode to remove the error. 3. If the error still exists, contact our customer service department.
0x00001049	Insufficient M-6FD DYE	Insufficient M-6FD DYE	1. Registered volume of M-6FD DYE for closed-reagent model runs out. 2. Click the "Remove Error" button, and input the M-6FD DYE barcode to remove the error. 3. If the error still exists, contact our customer service department.
0x00001061	Prime after loading DS DILUENT	Failed to prime after loading DS DILUENT. Please prime again.	1. Failed to prime after loading DS DILUENT. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001062	Prime after loading M-6DR DILUENT	Failed to prime after loading M-6DR DILUENT. Please prime again.	1. Failed to prime after loading M-6DR DILUENT. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001063	Prime after loading M-6LD LYSE	Failed to prime after loading M-6LD LYSE. Please prime again.	1. Failed to prime after loading M-6LD LYSE. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001064	Prime after loading M-6LN LYSE	Failed to prime after loading M-6LN LYSE. Please prime again.	1. Failed to prime after loading M-6LN LYSE. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001065	Prime after loading M-6LB LYSE	Failed to prime after loading M-6LB LYSE. Please prime again.	1. Failed to prime after loading M-6LB LYSE. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001066	Prime after loading M-6LH LYSE	Failed to prime after loading M-6LH LYSE. Please prime again.	1. Failed to prime after loading M-6LH LYSE. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001067	Prime after loading M-6FN DYE	Failed to prime after loading M-6FN DYE. Please prime again.	1. Failed to prime after loading M-6FN DYE. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00001068	Prime after loading M-6FR DYE	Failed to prime after loading M-6FR DYE. Please prime again.	1. Failed to prime after loading M-6FR DYE. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001069	Prime after loading M-6FD DYE	Failed to prime after loading M-6FD DYE. Please prime again.	1. Failed to prime after loading M-6FD DYE. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001071	Replace after loading DS DILUENT	Failed to replace after loading DS DILUENT. Please replace again.	1. Failed to replace after loading DS DILUENT. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001072	Replace after loading M-6DR DILUENT	Failed to replace after loading M-6DR DILUENT. Please replace again.	1. Failed to replace after loading M-6DR DILUENT. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001073	Replace after loading M-6LD LYSE	Failed to replace after loading M-6LD LYSE. Please replace again.	1. Failed to replace after loading M-6LD LYSE. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001074	Replace after loading M-6LN LYSE	Failed to replace after loading M-6LN LYSE. Please replace again.	1. Failed to replace after loading M-6LN LYSE. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001075	Replace after loading M-6LB LYSE	Failed to replace after loading M-6LB LYSE. Please replace again.	1. Failed to replace after loading M-6LB LYSE. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001076	Replace after loading M-6LH LYSE	Failed to replace after loading M-6LH LYSE. Please replace again.	1. Failed to replace after loading M-6LH LYSE. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001077	Replace after loading M-6FN DYE	Failed to replace after loading M-6FN DYE. Please replace again.	1. Failed to replace after loading M-6FN DYE. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00001078	Replace after loading M-6FR DYE	Failed to replace after loading M-6FR DYE. Please replace again.	1. Failed to replace after loading M-6FR DYE. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001079	Replace after loading M-6FD DYE	Failed to replace after loading M-6FD DYE. Please replace again.	1. Failed to replace after loading M-6FD DYE. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001050	No M-6FD DYE	M-6FD DYE running out or bubbles in tubing	1. M-6FD DYE running out or bubbles in tubing. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001051	No M-6FR DYE	M-6FR DYE running out or bubbles in tubing	1. M-6FR DYE running out or bubbles in tubing. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001052	No M-6FN DYE	M-6FN DYE running out or bubbles in tubing	1. M-6FN DYE running out or bubbles in tubing. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001053	No M-6FD DYE	M-6FD DYE running out or bubbles in tubing	1. M-6FD DYE running out or bubbles in tubing. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001054	No M-6FR DYE	M-6FR DYE running out or bubbles in tubing	1. M-6FR DYE running out or bubbles in tubing. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001055	No M-6FN DYE	M-6FN DYE running out or bubbles in tubing	1. M-6FN DYE running out or bubbles in tubing. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.

Error ID	Message	Details	Solutions
0x00001056	DIFF fluorochrome sensor error	Self-test voltage of DIFF fluorochrome sensor out of range	1. Self-test voltage of DIFF fluorochrome sensor out of range. 2. Re-calibrate the sensor, and then tap "Remove Error". 3. If the error still exists, contact our customer service department.
0x00001057	RET fluorochrome sensor error	Self-test voltage of RET fluorochrome sensor out of range	1. Self-test voltage of RET fluorochrome sensor out of range. 2. Re-calibrate the sensor, and then tap "Remove Error". 3. If the error still exists, contact our customer service department.
0x00001058	NRBC fluorochrome sensor error	Self-test voltage of NRBC fluorochrome sensor out of range	1. Self-test voltage of NRBC fluorochrome sensor out of range. 2. Re-calibrate the sensor, and then tap "Remove Error". 3. If the error still exists, contact our customer service department.
0x00001080	No M-6 NRBC Kit	M-6 NRBC Kit runs out	1. Replace the M-6 NRBC Kit. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001081	No M-6 RET Kit	M-6 RET Kit runs out	1. Replace the M-6 RET Kit. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001082	Insufficient M-6 NRBC Kit	Insufficient M-6 NRBC Kit	1. Registered volume of M-6 NRBC Kit for closed-reagent model runs out. 2. Click the "Remove Error" button, and input the M-6 FN DYE and M-6LN LYSE barcodes to remove the error. 3. If the error still exists, contact our customer service department.
0x00001083	Insufficient M-6 RET Kit	Insufficient M-6 RET Kit	1. Registered volume of M-6 RET Kit for closed-reagent model runs out. 2. Click the "Remove Error" button, and input the M-6 FR DYE and M-6DR DILUENT barcodes to remove the error. 3. If the error still exists, contact our customer service department.
0x00001084	Prime after loading M-6 NRBC Kit	Failed to prime after loading M-6 NRBC Kit. Please prime again.	1. Failed to prime after loading M-6 NRBC Kit. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001085	Prime after loading M-6 RET Kit	Failed to prime after loading M-6 RET Kit. Please prime again.	1. Failed to prime after loading M-6 RET Kit. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00001086	Replace after loading M-6 NRBC Kit	Failed to replace after loading M-6 NRBC Kit. Please replace again.	1. Failed to replace after loading M-6 NRBC Kit. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001087	Replace after loading M-6 RET Kit	Failed to replace after loading M-6 RET Kit. Please replace again.	1. Failed to replace after loading M-6 RET Kit. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001088	M-6 NRBC Kit expired	M-6 NRBC Kit expired	1. M-6 NRBC Kit expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001089	M-6 RET Kit expired	M-6 RET Kit expired	1. M-6 RET Kit expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001091	No DS DILUENT	DS DILUENT running out or bubbles in tubing	1. Please check if there is DS DILUENT in the container, and if the pickup sample tube is placed properly. 2. If there is diluent in the container. Re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of DS DILUENT. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x00001093	No DS DILUENT A	DS DILUENT A running out or bubbles in tubing	1. Please check if there is DS DILUENT in the DS DILUENT A container, and if the pickup sample tube is placed properly. 2. If there is diluent in the container, re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of DS DILUENT. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x00001094	No DS DILUENT B	DS DILUENT B running out or bubbles in tubing	1. Please check if there is DS DILUENT in the DS DILUENT B container, and if the pickup sample tube is placed properly. 2. If there is diluent in the container, re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of DS DILUENT. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.

Error ID	Message	Details	Solutions
0x00001095	No DS DILUENT A	DS DILUENT A running out or bubbles in tubing	1. Please check if there is DS DILUENT in the DS DILUENT A container, and if the pickup sample tube is placed properly. 2. If there is diluent in the container, re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of DS DILUENT. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x00001096	No DS DILUENT A	DS DILUENT A running out or bubbles in tubing	1. Please check if there is DS DILUENT in the DS DILUENT A container, and if the pickup sample tube is placed properly. 2. If there is diluent in the container, re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of DS DILUENT. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x00001097	No DS DILUENT B	DS DILUENT B running out or bubbles in tubing	1. Please check if there is DS DILUENT in the DS DILUENT B container, and if the pickup sample tube is placed properly. 2. If there is diluent in the container, re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of DS DILUENT. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x00001098	No DS DILUENT B	DS DILUENT B running out or bubbles in tubing	1. Please check if there is DS DILUENT in the DS DILUENT B container, and if the pickup sample tube is placed properly. 2. If there is diluent in the container, re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of DS DILUENT. Scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x00001099	Insufficient DS DILUENT A	Insufficient DS DILUENT A	1. No DS DILUENT A. Please get a new barcode. 2. Tap the "Remove Error" button and enter the new DS DILUENT barcode. 3. If the error still exists, contact our customer service department.
0x00001100	Insufficient DS DILUENT A	Insufficient DS DILUENT A	1. No DS DILUENT A. Please get a new barcode. 2. Tap the "Remove Error" button and enter the new DS DILUENT barcode. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00001101	Insufficient DS DILUENT B	Insufficient DS DILUENT B	1. No DS DILUENT B. Please get a new barcode. 2. Tap the "Remove Error" button and enter the new DS DILUENT barcode. 3. If the error still exists, contact our customer service department.
0x00001102	Insufficient DS DILUENT B	Insufficient DS DILUENT B	1. No DS DILUENT B. Please get a new barcode. 2. Tap the "Remove Error" button and enter the new DS DILUENT barcode. 3. If the error still exists, contact our customer service department.
0x00001103	Replace after loading DS DILUENT A	Failed to replace after loading DS DILUENT A. Please replace again.	1. Failed to replace after loading DS DILUENT A. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001104	Replace after loading DS DILUENT B	Failed to replace after loading DS DILUENT B. Please replace again.	1. Failed to replace after loading DS DILUENT B. Please replace again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001105	DS DILUENT A expired	DS DILUENT A expired	1. DS DILUENT A expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001106	DS DILUENT A expired	DS DILUENT A expired	1. DS DILUENT A expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001107	DS DILUENT B expired	DS DILUENT B expired	1. DS DILUENT B expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001108	DS DILUENT B expired	DS DILUENT B expired	1. DS DILUENT B expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001109	Prime after loading DS DILUENT A	Failed to prime after loading DS DILUENT A. Please prime again.	1. Failed to prime after loading DS DILUENT A. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00001110	Prime after loading DS DILUENT B	Failed to prime after loading DS DILUENT B. Please prime again.	1. Failed to prime after loading DS DILUENT B. Please prime again. 2. Click the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001111	Switching to DS DILUENT A	DS DILUENT B is running out; switching to DS DILUENT A	1. Click the "Remove Error" button. The analyzer will try to resume automatically. 2. If the error still exists, contact our customer service department.
0x00001112	Switching to DS DILUENT B	DS DILUENT A is running out; switching to DS DILUENT B	1. Click the "Remove Error" button. The analyzer will try to resume automatically. 2. If the error still exists, contact our customer service department.
0x00002001	Temperature setup command error	Reaction bath temperature setting out of range	1. Temperature setup out of expected range.2. Modify the setup and enter a temperature within normal range.
0x00002002	Temperature setup command error	Flow cell temperature setting out of range	1. Temperature setup out of expected range.2. Modify the setup and enter a temperature within normal range.
0x00002003	Temperature setup command error	Sheath fluid bath temperature setting out of range	1. Temperature setup out of expected range.2. Modify the setup and enter a temperature within normal range.
0x00002004	Temperature setup command error	Optical system temperature setting out of range	1. Temperature setup out of expected range.2. Modify the setup and enter a temperature within normal range.
0x00002005	Temperature setup command error	Preheating bath temperature setting out of range	1. Temperature setup out of expected range.2. Modify the setup and enter a temperature within normal range.
0x00002011	Reaction bath temperature initialization process abnormal	Reaction bath temperature initialization process abnormal	1. Temperature of the reaction bath still not fall in the expected range after initialization, and temperature control system error is reported. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002012	Flow cell temperature initialization process abnormal	Flow cell temperature initialization process abnormal	1. Temperature of the flow cell still not fall in the expected range after initialization, and temperature control system error is reported. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002013	Sheath fluid bath temperature initialization process abnormal	Sheath fluid bath temperature initialization process abnormal	1. Temperature of the sheath fluid bath still not fall in the expected range after initialization, and temperature control system error is reported. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00002014	Optical system temperature initialization process abnormal	Optical system temperature initialization process abnormal	1. Temperature of the optical system still not fall in the expected range after initialization, and temperature control system error is reported. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002015	Preheating bath temperature initialization process abnormal	Preheating bath temperature initialization process abnormal	1. Temperature of the preheating bath still not fall in the expected range after initialization, and temperature control system error is reported. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002016	Diluent temperature initialization process abnormal	Diluent temperature initialization process abnormal	1. Temperature of diluent still not fall in the expected range after initialization, and temperature control system error is reported. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002021	Reaction bath temperature high	Reaction bath temperature abnormally high in standby	1. Temperature of the reaction bath higher than expected. The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002022	Flow cell temperature high	Flow cell temperature abnormally high in standby	1. Temperature of the flow cell higher than expected. The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002023	Sheath fluid bath temperature high	Sheath fluid bath temperature abnormally high in standby	1. Temperature of the sheath fluid bath higher than expected. The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002024	Optical system temperature high	Optical system temperature abnormally high in standby	1. Temperature of the optical system higher than expected. The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002025	Preheating bath temperature high	Preheating bath temperature abnormally high in standby	1. Temperature of the preheating bath higher than expected. The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00002026	Diluent temperature high	Diluent temperature abnormally high in standby	1. Temperature of diluent higher than expected. The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002027	Diluent heating bath temperature too high	Diluent heating bath temperature abnormally high when the analyzer is standby	1. Diluent heating bath temperature is higher than the acceptable range. There may be errors in temperature control system. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002031	Reaction bath temperature low	Reaction bath temperature abnormally low in standby	1. Temperature of the reaction bath lower than expected. The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002032	Flow cell temperature low	Flow cell temperature abnormally low in standby	1. Temperature of the flow cell lower than expected. The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002033	Sheath fluid bath temperature low	Sheath fluid bath temperature abnormally low in standby	1. Temperature of the sheath fluid bath lower than expected. The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002034	Optical system temperature low	Optical system temperature abnormally low in standby	1. Temperature of the optical system lower than expected. The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002035	Preheating bath temperature low	Preheating bath temperature abnormally low in standby	1. Temperature of the preheating bath lower than expected. The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002036	Diluent temperature low	Diluent temperature abnormally low in standby	1. Temperature of diluent lower than expected. The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00002037	Diluent heating bath temperature too low	Diluent heating bath temperature abnormally low when the analyzer is standby	1. Diluent heating bath temperature is lower than the acceptable range. There may be errors in temperature control system. 2. Tap the "Remove Error" button to remove this error; 3. If the error still exists, contact our customer service department.
0x00002041	Reaction bath temperature extremely high	Reaction bath high temperature error in standby	1. Temperature of the reaction bath higher than expected. The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002042	Flow cell temperature extremely high	Flow cell high temperature error in standby	1. Temperature of the flow cell higher than expected. The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002043	Sheath fluid bath temperature extremely high	Sheath fluid high temperature error in standby	1. Temperature of the sheath fluid bath higher than expected. The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002044	Optical system temperature extremely high	Optical system high temperature error in standby	1. Temperature of the optical system higher than expected. The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002045	Preheating bath temperature extremely high	Preheating bath high temperature error in standby	1. Temperature of the preheating bath higher than expected. The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002046	Diluent temperature extremely high	Diluent high temperature error in standby	1. Temperature of diluent higher than expected. The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002047	Diluent heating bath temperature extremely high	High Diluent heating bath temperature error when the analyzer is standby	1. Diluent heating bath temperature is higher than the acceptable range. The analyzer cannot work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00002051	Reaction bath temperature high	Reaction bath temperature abnormally high in operation	1. Temperature of the reaction bath higher than expected while working (e.g. in the process of analyzing). The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002052	Flow cell temperature high	Flow cell temperature abnormally high in operation	1. Temperature of the flow cell higher than expected while working (e.g. in the process of analyzing). The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002053	Sheath fluid bath temperature high	Sheath fluid bath temperature abnormally high in operation	1. Temperature of the sheath fluid bath higher than expected while working (e.g. in the process of analyzing). The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002054	Optical system temperature high	Optical system temperature abnormally high in operation	1. Temperature of the optical system higher than expected while working (e.g. in the process of analyzing). The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002055	Preheating bath temperature high	Preheating bath temperature abnormally high in operation	1. Temperature of the preheating bath higher than expected while working (e.g. in the process of analyzing). The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002056	Diluent temperature high	Diluent temperature abnormally high in operation	1. Temperature of diluent higher than expected while working (e.g. in the process of analyzing). The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002057	Diluent heating bath temperature too high	Diluent heating bath temperature abnormally high when the analyzer is working	1. Diluent heating bath temperature is higher than the acceptable range during sample analysis or other operations. There may be errors in temperature control system. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00002061	Reaction bath temperature low	Reaction bath temperature abnormally low in operation	1. Temperature of the reaction bath lower than expected while working (e.g. in the process of analyzing). The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002062	Flow cell temperature low	Flow cell temperature abnormally low in operation	1. Temperature of the flow cell lower than expected while working (e.g. in the process of analyzing). The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002063	Sheath fluid bath temperature low	Sheath fluid bath temperature abnormally low in operation	1. Temperature of the sheath fluid bath lower than expected while working (e.g. in the process of analyzing). The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002064	Optical system temperature low	Optical system temperature abnormally low in operation	1. Temperature of the optical system lower than expected while working (e.g. in the process of analyzing). The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002065	Preheating bath temperature low	Preheating bath temperature abnormally low in operation	1. Temperature of the preheating bath lower than expected while working (e.g. in the process of analyzing). The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002066	Diluent temperature low	Diluent temperature abnormally low in operation	1. Temperature of diluent lower than expected while working (e.g. in the process of analyzing). The temperature control system may have error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002067	Diluent heating bath temperature too low	Diluent heating bath temperature abnormally low when the analyzer is working	1. Diluent heating bath temperature is lower than the acceptable range during sample analysis or other operations. There may be errors in temperature control system. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00002071	Reaction bath temperature extremely high	Reaction bath high temperature error in operation	1. Temperature of the reaction bath higher than expected while working (e.g. in the process of analyzing). The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002072	Flow cell temperature extremely high	Flow cell high temperature error in operation	1. Temperature of the flow cell higher than expected while working (e.g. in the process of analyzing). The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002073	Sheath fluid bath temperature extremely high	Sheath fluid bath high temperature error in operation	1. Temperature of the sheath fluid bath higher than expected while working (e.g. in the process of analyzing). The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002074	Optical system temperature extremely high	Optical system high temperature error in operation	1. Temperature of the optical system higher than expected while working (e.g. in the process of analyzing). The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002075	Preheating bath temperature extremely high	Preheating bath high temperature error in operation	1. Temperature of the preheating bath higher than expected while working (e.g. in the process of analyzing). The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002076	Diluent temperature extremely high	Diluent high temperature error in operation	1. Temperature of diluent higher than expected while working (e.g. in the process of analyzing). The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002077	Diluent heating bath temperature extremely high	High Diluent heating bath temperature error when the analyzer is working	1. Diluent heating bath temperature is higher than the acceptable range during sample analysis or other operations. The analyzer cannot work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00002081	Reaction bath temperature extremely low	Reaction bath low temperature error in standby	1. Temperature of the reaction bath lower than expected. The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002082	Flow cell temperature extremely low	Flow cell low temperature error in standby	1. Temperature of the flow cell lower than expected. The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002083	Sheath fluid bath temperature extremely low	Sheath fluid low temperature error in standby	1. Temperature of the sheath fluid bath lower than expected. The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002084	Optical system temperature extremely low	Optical system low temperature error in standby	1. Temperature of the optical system lower than expected. The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002085	Preheating bath temperature extremely low	Preheating bath low temperature error in standby	1. Temperature of the preheating bath lower than expected. The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002086	Diluent temperature extremely low	Diluent low temperature error in standby	1. Temperature of diluent lower than expected. The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002087	Diluent heating bath temperature extremely low	Low Diluent heating bath temperature error when the analyzer is standby	1. Diluent heating bath temperature is lower than the acceptable range. The analyzer cannot work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002091	Reaction bath temperature extremely low	Reaction bath low temperature error in operation	1. Temperature of the reaction bath lower than expected while working (e.g. in the process of analyzing). The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00002092	Flow cell temperature extremely low	Flow cell low temperature error in operation	1. Temperature of the flow cell lower than expected while working (e.g. in the process of analyzing). The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002093	Sheath fluid bath temperature extremely low	Sheath fluid bath low temperature error in operation	1. Temperature of the sheath fluid bath lower than expected while working (e.g. in the process of analyzing). The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002094	Optical system temperature extremely low	Optical system low temperature error in operation	1. Temperature of the optical system lower than expected while working (e.g. in the process of analyzing). The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002095	Preheating bath temperature extremely low	Preheating bath low temperature error in operation	1. Temperature of the preheating bath lower than expected while working (e.g. in the process of analyzing). The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002096	Diluent temperature extremely low	Diluent low temperature error in operation	1. Temperature of diluent lower than expected while working (e.g. in the process of analyzing). The analyzer can not work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002097	Diluent heating bath temperature extremely low	Low Diluent heating bath temperature error when the analyzer is working	1. Diluent heating bath temperature is lower than the acceptable range during sample analysis or other operations. The analyzer cannot work properly. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00002101	Analyzer temperature high	Analyzer temperature abnormally high in standby	1. The working temperature of the analyzer higher than expected. The function of the analyzer will be restrained. 2. Check the working temperature of the analyzer. If it is within the expected range, contact our customer service department.
0x00002102	Analyzer temperature low	Analyzer temperature abnormally low in standby	1. The working temperature of the analyzer lower than expected. The function of the analyzer will be restrained. 2. Check the working temperature of the analyzer. If it is within the expected range, contact our customer service department.

Error ID	Message	Details	Solutions
0x00002103	Analyzer temperature extremely high	Analyzer high temperature error in standby	1. The working temperature of the analyzer higher than expected. Sample analysis may be disabled. 2. Check the working temperature of the analyzer. If it is within the expected range, contact our customer service department.
0x00002104	Analyzer temperature extremely low	Analyzer low temperature error in standby	1. The working temperature of the analyzer lower than expected. Sample analysis may be disabled. 2. Check the working temperature of the analyzer. If it is within the expected range, contact our customer service department.
0x00002111	Analyzer temperature high	Analyzer temperature abnormally high in operation	1. The working temperature of the analyzer higher than expected while working (e.g. in the process of analyzing). The function of the analyzer will be restrained. 2. Check the working temperature of the analyzer. If it is within the expected range, contact our customer service department.
0x00002112	Analyzer temperature low	Analyzer temperature abnormally low in operation	1. The working temperature of the analyzer lower than expected while working (e.g. in the process of analyzing). The function of the analyzer will be restrained. 2. Check the working temperature of the analyzer. If it is within the expected range, contact our customer service department.
0x00002113	Analyzer temperature extremely high	Analyzer high temperature error in operation	1. The working temperature of the analyzer higher than expected while working (e.g. in the process of analyzing). Sample analysis may be disabled. 2. Check the working temperature of the analyzer. If it is within the expected range, contact our customer service department.
0x00002114	Analyzer temperature extremely low	Analyzer low temperature error in operation	1. The working temperature of the analyzer lower than expected while working (e.g. in the process of analyzing). Sample analysis may be disabled. 2. Check the working temperature of the analyzer. If it is within the expected range, contact our customer service department.
0x00003001	250kPa pressure out of range	Pneumatic unit output pressure initialization abnormal	1. The 250kPa pressure not fall in normal range within expected time after turning on the pneumatic unit. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00003002	-70kPa vacuum out of range	Pneumatic unit output vacuum initialization abnormal	1. The 70kPa vacuum not fall in normal range within expected time after turning on the pneumatic unit. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00003011	250kPa pressure out of range	250kPa standby pressure abnormal	1. The 250kPa pressure out of expected range. Sample analysis may be disabled. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00003012	160kPa pressure out of range	160kPa standby pressure abnormal	1. The 160kPa pressure out of expected range. Sample analysis may be disabled. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00003013	70kPa pressure out of range	70kPa standby pressure abnormal	1. The 70kPa pressure out of expected range. Sample analysis may be disabled. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00003014	40kPa pressure out of range	40kPa standby pressure abnormal	1. The 40kPa pressure (of sheath fluid) out of expected range. Sample analysis may be disabled. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00003015	Please adjust the -40KPa vacuum	-40kPa standby vacuum abnormal	1. Open the left door of the analyzer, and find the valve labeled with "-0.04MPa"; Turn the locknut of the valve counterclockwise to loosen it; Check the pressure value in the "PS5 (-40)" column at the "Temperature&Pressure" screen, and then turn the locknut (clockwise to increase the vacuum, counterclockwise to decrease) until the value fall in the expected range $(-38 \pm 0.5) \text{Kpa}$; Secure the locknut by turning it clockwise after the adjustment is finished; 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00003016	-70kPa vacuum out of range	-70kPa standby vacuum abnormal	1. The 70kPa vacuum out of expected range. Sample analysis may be disabled. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00003021	250kPa pressure out of range	250kPa working pressure abnormal	1. The 250kPa pressure out of expected range. Sample analysis results may be affected. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00003022	160kPa pressure out of range	160kPa working pressure abnormal	1. The 160kPa pressure out of expected range. Sample analysis results may be affected. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00003023	70kPa pressure out of range	70kPa working pressure abnormal	1. The 70kPa pressure out of expected range. Sample analysis results may be affected. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00003024	40kPa pressure out of range	40kPa working pressure abnormal	1. The 40kPa pressure (of sheath fluid) out of expected range. Sample analysis results may be affected. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00003025	Please adjust the -40KPa vacuum	-40kPa working vacuum abnormal	1. Open the left door of the analyzer, and find the valve labeled with "-0.04MPa"; Turn the locknut of the valve counterclockwise to loosen it; Check the pressure value in the "PS5 (-40)" column at the "Temperature&Pressure" screen, and then turn the locknut (clockwise to increase the vacuum, counterclockwise to decrease) until the value fall in the expected range ((-38±0.5)Kpa); Secure the locknut by turning it clockwise after the adjustment is finished; 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00003026	-70kPa vacuum out of range	-70kPa working vacuum abnormal	1. The 70kPa vacuum out of expected range. Sample analysis results may be affected. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00003031	250kPa pressure out of range	Pneumatic unit output pressure abnormal	1. 250kPa pressure abnormal. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004001	System software or sequence operation command error	Undefined function code issued by system software	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004002	System software or sequence operation command error	Incorrect length issued by system software	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004003	System software or sequence operation command error	Undefined command code issued by system software	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004004	System software or sequence operation command error	Communication check code inconsistent	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00004005	System software or sequence operation command error	Incorrect end code issued by system software	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004006	System software or sequence operation command error	Undefined identification code issued by system software	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004007	System software or sequence operation command error	Previous sequence packet analyzing not completed	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004008	System software or sequence operation command error	Incomplete sequence packet issued by system software	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004009	System software or sequence operation command error	Sequence packet overlapped	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004010	System software or sequence operation command error	Incorrect sequence packet No.	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004011	System software or sequence operation command error	Incorrect frame serial number of sequence packet	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004012	System software or sequence operation command error	Incorrect sequence packet storage command	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004013	System software or sequence operation command error	No matched packet head frame of the packet end frame	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004014	System software or sequence operation command error	No matched packet head frame of the sequence frame	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004015	System software or sequence operation command error	Incorrect sequence packet command	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004016	System software or sequence operation command error	Sequence packet execution overtime	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004017	System software or sequence operation command error	Found non-sequence-packet command	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00004018	System software or sequence operation command error	Download symbol error	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004019	System software or sequence operation command error	Undefined download command	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004021	System software or sequence operation command error	Maximal write bytes over range	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004022	System software or sequence operation command error	Write data exceed the capacity of a page	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004023	Configuration chip write operation failed	Configuration chip write operation failed	1. Configuration chip write operation failed 2. Shut off the analyzer power directly, and restart the analyzer later. 3. If the error still exists, contact our customer service department.
0x00004024	Drive board EEPROM read-write failed	Drive board EEPROM read-write failed	1. Drive board EEPROM read-write failed 2. Shut off the analyzer power directly, and restart the analyzer later. 3. If the error still exists, contact our customer service department.
0x00004025	Pneumatic pressure detection board EEPROM read-write failed	Pneumatic pressure detection board EEPROM read-write failed	1. Pneumatic pressure detection board EEPROM read-write failed 2. Shut off the analyzer power directly, and restart the analyzer later. 3. If the error still exists, contact our customer service department.
0x00004030	System software or sequence operation command error	Incorrect number of system configuration items	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004031	System software or sequence operation command error	System configuration item undefined	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004101	System software or sequence operation command error	Valve No. out of range	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004201	System software or sequence operation command error	Priming or draining actions overlap	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00004202	System software or sequence operation command error	Incorrect priming or draining time parameter	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00004301	DIL cistern floater error	DIL cistern floater detection status abnormal	1. Tap the "Remove Error" button, and the analyzer will perform floater self-test to remove the error. 2. If the error still exists, contact our customer service department.
0x00004302	FCM cistern floater error	FCM cistern floater detection status abnormal	1. Tap the "Remove Error" button, and the analyzer will perform floater self-test to remove the error. 2. If the error still exists, contact our customer service department.
0x00004303	SCI cistern floater error	SCI cistern floater detection status abnormal	1. Tap the "Remove Error" button, and the analyzer will perform floater self-test to remove the error. 2. If the error still exists, contact our customer service department.
0x00004304	WC1 floater error	WC1 floater detection status abnormal	1. Tap the "Remove Error" button, and the analyzer will perform floater self-test to remove the error. 2. If the error still exists, contact our customer service department.
0x00004305	WC2 floater error	WC2 floater detection status abnormal	1. Tap the "Remove Error" button, and the analyzer will perform floater self-test to remove the error. 2. If the error still exists, contact our customer service department.
0x00004306	WC3 floater error	WC3 floater detection status abnormal	1. Tap the "Remove Error" button, and the analyzer will perform floater self-test to remove the error. 2. If the error still exists, contact our customer service department.
0x00004307	LD cistern floater error	LD cistern floater detection status abnormal	1. Tap the "Remove Error" button, and the analyzer will perform floater self-test to remove the error. 2. If the error still exists, contact our customer service department.
0x00004308	DR cistern floater error	DR cistern floater detection status abnormal	1. Tap the "Remove Error" button, and the analyzer will perform floater self-test to remove the error. 2. If the error still exists, contact our customer service department.
0x00004309	LN cistern floater error	LN cistern floater detection status abnormal	1. Tap the "Remove Error" button, and the analyzer will perform floater self-test to remove the error. 2. If the error still exists, contact our customer service department.
0x00004310	LH cistern floater error	LH cistern floater detection status abnormal	1. Tap the "Remove Error" button, and the analyzer will perform floater self-test to remove the error. 2. If the error still exists, contact our customer service department.
0x00004401	RBC syringe error	RBC syringe in action...	1. RBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00004402	RBC syringe error	RBC syringe resetting or initialization action unidentified	1. RBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004403	RBC syringe error	RBC syringe performed actions unidentified	1. RBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004404	RBC syringe error	RBC syringe performed actions unidentified	1. RBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004405	RBC syringe error	RBC syringe can not perform current action since its actions are unidentified	1. RBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004406	RBC syringe error	RBC syringe can not perform current action since its actions are unidentified	1. RBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004407	RBC syringe error	RBC syringe can not perform current action due to its excessive aspiration volume	1. RBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004408	RBC syringe error	RBC syringe can not perform current action due to its excessive dispensation volume	1. RBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004409	RBC syringe error	RBC syringe execution overtime	1. RBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004410	WBC syringe error	WBC syringe in action...	1. WBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004411	WBC syringe error	WBC syringe resetting or initialization action unidentified	1. WBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004412	WBC syringe error	WBC syringe performed actions unidentified	1. WBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00004413	WBC syringe error	WBC syringe performed actions unidentified	1. WBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004414	WBC syringe error	WBC syringe can not perform current action since its actions are unidentified	1. WBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004415	WBC syringe error	WBC syringe can not perform current action since its actions are unidentified	1. WBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004416	WBC syringe error	WBC syringe can not perform current action due to its excessive aspiration volume	1. WBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004417	WBC syringe error	WBC syringe can not perform current action due to its excessive dispensation volume	1. WBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004418	WBC syringe error	WBC syringe execution overtime	1. WBC syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004419	Whole blood aspiration syringe error	Whole blood aspiration syringe in action...	1. Whole blood aspiration syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004420	Whole blood aspiration syringe error	Whole blood aspiration syringe resetting or initialization action unidentified	1. Whole blood aspiration syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004421	Whole blood aspiration syringe error	Whole blood aspiration syringe performed actions unidentified	1. Whole blood aspiration syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004422	Whole blood aspiration syringe error	Whole blood aspiration syringe performed actions unidentified	1. Whole blood aspiration syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00004423	Whole blood aspiration syringe error	Whole blood aspiration syringe can not perform current action since its actions are unidentified	1. Whole blood aspiration syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004424	Whole blood aspiration syringe error	Whole blood aspiration syringe can not perform current action since its actions are unidentified	1. Whole blood aspiration syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004425	Whole blood aspiration syringe error	Whole blood aspiration syringe can not perform current action due to its excessive aspiration volume	1. Whole blood aspiration syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004426	Whole blood aspiration syringe error	Whole blood aspiration syringe can not perform current action due to its excessive dispensation volume	1. Whole blood aspiration syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004427	Whole blood aspiration syringe error	Whole blood aspiration syringe execution overtime	1. Whole blood aspiration syringe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004501	Probe wipe error	Probe wipe in action...	1. Probe wipe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004502	Probe wipe error	Probe wipe action unidentified	1. Probe wipe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004503	Probe wipe error	Probe wipe action overtime	1. Probe wipe action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00004601	Stirring motor error	Stirring module action unidentified. Photocoupler broken or motor action not proper.	1. Stirring module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005001	Pincher abnormal	Pincher lengthwise driving cylinder action unidentified	1. Pincher lengthwise driving cylinder action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00005007	Pincher abnormal	Piercing module vertical driving cylinder action unidentified	1. Piercing module vertical driving cylinder action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005013	Mixing module abnormal	Mixing module action unidentified	1. Mixing module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005020	Piercing module abnormal	Piercing module photocoupler status detection result not as expected (expected to be blocked)	1. Piercing module photocoupler not get expected status result (expected result: blocked). Piercing module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005021	Piercing module abnormal	Piercing module photocoupler status detection result not as expected (expected to be not blocked)	1. Piercing module photocoupler not get expected status result (expected result: not blocked). Piercing module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005022	Piercing module abnormal	Module not in expected position before piercing	1. Mixing&Piercing module not in expected position while about to pierce. Piercing module can not perform current action. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005023	Piercing module abnormal	Module not in expected position after piercing	1. Mixing&Piercing module not in expected position while about to withdraw the piercing probe. Piercing module can not perform current action. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005024	Current action of autoloader not completed	Current action of autoloader not completed	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00005025	Autoloader not properly initialized	Autoloader not properly initialized	1. Initialization not performed before resetting or autoloading. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005026	Can not read the barcode of the current sample tube	Can not read the barcode of the current sample tube	1. Can not read valid barcode information from current sample tube. 2. Disable the barcode scanning function if the barcode is not in place. 3. If this error is reported when the barcode is in place, contact our customer service department.

Error ID	Message	Details	Solutions
0x00005027	Can not read the barcode of the current sample rack	Can not read the barcode of the current sample rack	1. Can not read valid barcode information from current sample rack. 2. Disable the barcode scanning function if the barcode is not in place. 3. If this error is reported when the barcode is in place, contact our customer service department.
0x00005028	Feeding module error	Tube position counter status abnormal	1. sample rack feeding abnormal. 2. Please remove the sample rack in the feeding path, and then tap "Remove Error". 3. If the error still exists, contact our customer service department.
0x00005029	Feeding module error	Start counter status and end counter status do not match	1. sample rack feeding abnormal. 2. Please remove the sample rack in the feeding path, and then tap "Remove Error". 3. If the error still exists, contact our customer service department.
0x00005030	Can not unload since the unloading end position photocoupler is blocked	Can not unload since the unloading end position photocoupler is blocked	1. Found the unloading module end position photocoupler blocked before the unloading action of the unloading motor. Can not perform the unloading action. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005031	Feeding module blocked	Can not perform action since there is a sample rack in the feeding line	1. There is a sample rack in the feeding line. Autoloader action paused. 2. Remove the sample rack and tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005032	Unloading module error	Unloading module action unidentified	1. Unloading module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005033	Unloading module error	Unloading module action overtime	1. Unloading module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005034	sample rack not properly accommodated	Loading module action unidentified	1. Loading module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005234	sample rack not properly accommodated	Loading module action unidentified	1. Loading module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005037	sample rack not properly accommodated	Loading module action overtime	1. Loading module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00005038	Feeding module error	Feeding assisting pin action unidentified	1. Feeding assisting pin action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005041	Feeding module error	Feeding module action unidentified	1. Feeding module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005042	Feeding module error	Feeding module action overtime	1. Feeding module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005045	Rotary scanning module action unidentified	Rotary scanning module action unidentified	1. Rotary scanning module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005049	Piercing module abnormal	Piercing module photocoupler status detection result not as expected (expected to be not blocked)	1. Piercing module photocoupler not get expected status result (expected result: not blocked). Piercing module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005051	Barcode reading error	Barcode length out of limit or barcode reading overtime	1. Barcode length out of limit or barcode reading overtime. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005052	Internal barcode scanner error	Scanner self-test in startup failed	1. Scanner self-test in startup failed. Check if the barcode scanner can work properly. 2. If the error still exists, contact our customer service department.
0x00005053	Scanner not configured	Run barcode scanning function with no scanner configured	1. Run barcode scanning function with no scanner configured 2. If this error is reported when there is a scanner configured, contact our customer service department.
0x00005060	Rotary scanning module error	Rotary scanning module action overtime	1. Rotary scanning module action unidentified. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00005070	Unloading tray full	Remove sample racks or other obstacles on the unloading tray	1. Remove sample racks or other obstacles on the unloading tray. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00005080	Track moving fail	Track moving fail	1. Tap the "Remove Error" button to remove this error. 2. If the error still exists, contact our customer service department.
0x00005082	Track movement paused	Track movement paused	1. Long push the "Reset" key on the track to reset the track. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00006001	Front cover open	Front cover is open	1. Front cover open. Some functions of the analyzer are disabled. 2. Close the front cover and tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00006002	Optical system cover open	Optical system cover open	1. Optical system cover open. Some functions of the analyzer are disabled. 2. Close the optical system cover and tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00006003	Front cover open	Front cover is open	1. Front cover open. Some functions of the analyzer are disabled. 2. Close the front cover and tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00006011	D5V voltage abnormal	+5V digital voltage abnormal	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00006012	D5V voltage abnormal	+12V digital voltage abnormal	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00006013	Power supply error	+12V power voltage abnormal	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00006014	Power supply error	+24V power voltage abnormal	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00006015	Power supply error	+12V analog voltage abnormal	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00006016	Power supply error	-12V analog voltage abnormal	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00006021	Background abnormal	Background abnormal	1. Check whether the diluent is contaminated. 2. If it is not contaminated, tap the "Remove Error" button to remove the error. 3. If the error still exists, contact our customer service department.
0x00006022	Not initialized in startup	Analyzer system not initialized	1. Analyzer system not initialized as expected. Initialize the analyzer and then try to start counting again. 2. If the error still exists, contact our customer service department.
0x00006023	Serial port communication error	Serial port communication error	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00006024	System clock error	System clock error	1. Reset the system clock. 2. If the error still exists, contact our customer service department.
0x00006025	System software communication frame abnormal	System software communication frame abnormal	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00006026	System software command sending overtime	System software command sending overtime	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00006027	Automatic converting to open-vial mode failed	Automatic converting to open-vial mode failed due to other error(s). Remove the error(s) first.	1. Automatic converting to open-vial mode failed due to other error(s). 2. Remove the error(s) first, and then remove this error. The analyzer will try to convert again and resume to normal status. 3. If the error still exists, contact our customer service department.
0x00006028	Hard disk full	Removable disk full	1. Removable disk full. Install another one. 2. Contact the engineer collecting data on site if necessary.
0x00006031	Exiting standby mode failed	Exiting standby mode failed since error is found during the exiting process	1. Exiting standby mode failed since error is found during the exiting process Remove the error found in the exiting process, and then tap the "Remove Error" button to remove this error. The system will exit the standby mode and get ready for the counting. The counting function is disabled before this error is removed. 2. If the error still exists, contact our customer service department.
0x00006032	Exiting standby mode failed	Exiting standby mode failed since error is found during the exiting process	1. Exiting standby mode failed since error is found during the exiting process Remove the error found in the exiting process, and then tap the "Remove Error" button to remove this error. The system will exit the standby mode and get ready for the counting. The counting function is disabled before this error is removed. 2. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00006033	Exiting standby mode failed	Exiting standby mode failed since error is found during the exiting process	1. Exiting standby mode failed since error is found during the exiting process Remove the error found in the exiting process, and then tap the "Remove Error" button to remove this error. The system will exit the standby mode and get ready for the counting. The counting function is disabled before this error is removed. 2. If the error still exists, contact our customer service department.
0x00006034	Exiting standby mode failed	Exiting standby mode failed since error is found during the exiting process	1. Exiting standby mode failed since error is found during the exiting process Remove the error found in the exiting process, and then tap the "Remove Error" button to remove this error. The system will exit the standby mode and get ready for the counting. The counting function is disabled before this error is removed. 2. If the error still exists, contact our customer service department.
0x00006035	Exiting standby mode failed	Exiting standby mode failed since error is found during the exiting process	1. Exiting standby mode failed since error is found during the exiting process Remove the error found in the exiting process, and then tap the "Remove Error" button to remove this error. The system will exit the standby mode and get ready for the counting. The counting function is disabled before this error is removed. 2. If the error still exists, contact our customer service department.
0x00006036	Exiting standby mode failed	Exiting standby mode failed since error is found during the exiting process	1. Exiting standby mode failed since error is found during the exiting process Remove the error found in the exiting process, and then tap the "Remove Error" button to remove this error. The system will exit the standby mode and get ready for the counting. The counting function is disabled before this error is removed. 2. If the error still exists, contact our customer service department.
0x00008001	HGB abnormal	HGB blank voltage abnormal	1. HGB blank voltage abnormal. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00008002	Clogging	Aperture voltage abnormal	1. Aperture voltage abnormal. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00008003	Clogging	Impedance channel clogging	1. Impedance channel clog. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00008004	Clogging	Impedance channel signal interference	1. Impedance channel signal interference. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00008005	Laser emitter needs to be replaced	Laser PD current abnormal	1. Laser PD current abnormal. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00008006	Laser emitter needs to be replaced	Forward scatter blank voltage abnormal	1. Forward scatter blank voltage abnormal. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00008007	Fluorescent signal abnormal	Fluorescent signal abnormal	1. Fluorescent signal abnormal. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00008050	FlowCell Bubbles	FlowCell Bubbles	1. Flow Cell Bubbles 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00008020	Clogging	Micro-clog in impedance channel	1. If this error is reported repeatedly, the analyzer will try to resume automatically. 2. If the error still exists, contact our customer service department.
0x00008021	RBC abnormal	Electrical noise in impedance channel	1. Check if there is any wireless communication base near the laboratory where the analyzer is installed, or any high power instrument (e.g. large centrifuge) used near the analyzer. 2. Check if there is any large instrument which produces great vibration on the platform where the analyzer is placed. 3. Contact our customer service department, and do as instructed to solve the problem.
0x00008022	RBC abnormal	Impedance channel signal interfered	1. Check if there is any wireless communication base near the laboratory where the analyzer is installed, or any high power instrument (e.g. large centrifuge) used near the analyzer. 2. Check if there is any large instrument which produces great vibration on the platform where the analyzer is placed. 3. Contact our customer service department, and do as instructed to solve the problem.
0x00008024	RBC abnormal	Incontinuous RBC data	1. If this error is reported repeatedly, the analyzer will stop working automatically. 2. Shut off the analyzer power directly, and restart the analyzer later. 3. If the error still exists, contact our customer service department.
0x00008025	PLT abnormal	Incontinuous PLT data	1. If this error is reported repeatedly, the analyzer will stop working automatically. 2. Shut off the analyzer power directly, and restart the analyzer later. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00008026	RBC abnormal	Improper RBC data	1. If this error is reported repeatedly, the analyzer will stop working automatically. 2. Shut off the analyzer power directly, and restart the analyzer later. 3. If the error still exists, contact our customer service department.
0x00008027	PLT abnormal	Improper PLT data	1. If this error is reported repeatedly, the analyzer will stop working automatically. 2. Shut off the analyzer power directly, and restart the analyzer later. 3. If the error still exists, contact our customer service department.
0x00007001	Sample probe is about to reach the end of its service life	Sample probe is about to reach the end of its service life	1. The sample probe will reach the end of its service life after running 2000 more samples in autoloading mode, contact our customer service department for replacement. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00007002	Sample probe has reached the end of its service life	Sample probe has reached the end of its service life	1. The sample probe has reached the end of its service life, contact our customer service department for replacement. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00007003	Optical system working voltage abnormal	PMT voltage abnormal	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00007004	Optical system working voltage abnormal	PMT gain abnormal	1. Tap the "Remove Error" button to remove this error. 2. If the error still exists, shut off the analyzer power directly and restart the analyzer later. 3. If the error still exists, contact our customer service department.
0x00007005	Laser abnormal	Laser power abnormal	1. Tap the "Remove Error" button to remove this error. 2. If the error still exists, contact our customer service department.
0x00007006	Laser abnormal	Laser drive current abnormal	1. Tap the "Remove Error" button to remove this error. 2. If the error still exists, contact our customer service department.
0x00007007	FS baseline abnormal	FS baseline abnormal	1. Tap the "Remove Error" button to remove this error. 2. If the error still exists, shut off the analyzer power directly and restart the analyzer later. 3. If the error still exists, contact our customer service department.
0x00007008	SS baseline abnormal	SS baseline abnormal	1. Tap the "Remove Error" button to remove this error. 2. If the error still exists, shut off the analyzer power directly and restart the analyzer later. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00007009	FL baseline abnormal	FL baseline abnormal	1. Tap the "Remove Error" button to remove this error. 2. If the error still exists, shut off the analyzer power directly and restart the analyzer later. 3. If the error still exists, contact our customer service department.
0x00007010	Insufficient disk space	Insufficient disk space	1. The current disk space usage has exceeded 85%. 2. Delete some data at the "Table Review" screen to release the disk space. 3. The instrument will try to automatically remove error after the disk space is released. 4. If the error still exists, contact our Customer Service Department.
0x00007011	Disk space is full	Disk space is full	1. The current disk space usage has exceeded 95%. 2. Delete some data at the "Table Review" screen to release the disk space. 3. The instrument will try to automatically remove error after the disk space is released. 4. If the error still exists, contact our Customer Service Department.
0x00007012	Failed to read the rack ID	Failed to read the rack ID	1. Failed to read rack ID. Either the rack has no barcode label, the label has been stained, or the rack type is not supported. 2. Place the unprocessed samples into another sample rack and re-enter the sample IDs. 3. Tap the "Remove Error" button to remove this error. 4. If the error still exists, contact our customer service department.
0x00007020	Please perform the Probe Cleanser maintenance	Please perform the Probe Cleanser maintenance	1. Tap the "Remove Error" button, and follow the software instruction to perform Probe Cleanser Maintenance; 2. If the error still exists, contact our customer service department.
0x00007021	Please perform the Probe Cleanser maintenance	Please perform the Probe Cleanser maintenance	1. Tap the "Remove Error" button, and follow the software instruction to perform Probe Cleanser Maintenance; 2. If the error still exists, contact our customer service department.
0x00007022	Please perform the Probe Cleanser maintenance	Please perform the Probe Cleanser maintenance	1. Tap the "Remove Error" button, and follow the software instruction to perform Probe Cleanser Maintenance; 2. If the error still exists, contact our customer service department.
0x00007023	Please perform the Probe Cleanser maintenance	Please perform the Probe Cleanser maintenance	1. Tap the "Remove Error" button, and follow the software instruction to perform Probe Cleanser Maintenance; 2. If the error still exists, contact our customer service department.
0x00007024	Please perform the Probe Cleanser maintenance	Please perform the Probe Cleanser maintenance	1. Tap the "Remove Error" button, and follow the software instruction to perform Probe Cleanser Maintenance; 2. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00007025	Consecutive samples are flagged for abnormal results	Several consecutive samples are flagged for abnormal results	1. Several consecutive samples are flagged for abnormal results. Confirm sample status or contact our customer service to check analyzer status. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00007026	Scanner communication abnormal	Scanner communication abnormal	1. Tap the "Remove Error" button to remove this error. 2. If the error still exists, contact our customer service department.
0x00010001	Routine blood test QC is out of control (low level)	Routine blood test QC is out of control (low level)	1. Check the QC results and identify reasons for this out of control; 2. Redo the QC test of this level (if the analyzer is on a sample processing line, disconnect the analyzer from the sample processing line first). If the new QC results are in control, the error will be automatically removed. 3. If the error still exists, contact our customer service department.
0x00010002	Routine blood test QC is out of control (normal level)	Routine blood test QC is out of control (normal level)	1. Check the QC results and identify reasons for this out of control; 2. Redo the QC test of this level (if the analyzer is on a sample processing line, disconnect the analyzer from the sample processing line first). If the new QC results are in control, the error will be automatically removed. 3. If the error still exists, contact our customer service department.
0x00010003	Routine blood test QC is out of control (high level)	Routine blood test QC is out of control (high level)	1. Check the QC results and identify reasons for this out of control; 2. Redo the QC test of this level (if the analyzer is on a sample processing line, disconnect the analyzer from the sample processing line first). If the new QC results are in control, the error will be automatically removed. 3. If the error still exists, contact our customer service department.
0x00010004	Body fluid test QC is out of control (low level)	Body fluid test QC is out of control (low level)	1. Check the QC results and identify reasons for this out of control; 2. Redo the QC test of this level (if the analyzer is on a sample processing line, disconnect the analyzer from the sample processing line first). If the new QC results are in control, the error will be automatically removed. 3. If the error still exists, contact our customer service department.
0x00010005	Body fluid test QC is out of control (normal level)	Body fluid test QC is out of control (normal level)	1. Check the QC results and identify reasons for this out of control; 2. Redo the QC test of this level (if the analyzer is on a sample processing line, disconnect the analyzer from the sample processing line first). If the new QC results are in control, the error will be automatically removed. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00010006	Body fluid test QC is out of control (high level)	Body fluid test QC is out of control (high level)	1. Check the QC results and identify reasons for this out of control; 2. Redo the QC test of this level (if the analyzer is on a sample processing line, disconnect the analyzer from the sample processing line first). If the new QC results are in control, the error will be automatically removed. 3. If the error still exists, contact our customer service department.
0x00010007	Routine blood test QC is out of control (low level)	Routine blood test QC is out of control (low level)	1. Check the QC results and identify reasons for this out of control; 2. Redo the QC test of this level (if the analyzer is on a sample processing line, disconnect the analyzer from the sample processing line first). If the new QC results are in control, the error will be automatically removed. 3. If the error still exists, contact our customer service department.
0x00010008	Routine blood test QC is out of control (normal level)	Routine blood test QC is out of control (normal level)	1. Check the QC results and identify reasons for this out of control; 2. Redo the QC test of this level (if the analyzer is on a sample processing line, disconnect the analyzer from the sample processing line first). If the new QC results are in control, the error will be automatically removed. 3. If the error still exists, contact our customer service department.
0x00010009	Routine blood test QC is out of control (high level)	Routine blood test QC is out of control (high level)	1. Check the QC results and identify reasons for this out of control; 2. Redo the QC test of this level (if the analyzer is on a sample processing line, disconnect the analyzer from the sample processing line first). If the new QC results are in control, the error will be automatically removed. 3. If the error still exists, contact our customer service department.
0x00010010	Body fluid test QC is out of control (low level)	Body fluid test QC is out of control (low level)	1. Check the QC results and identify reasons for this out of control; 2. Redo the QC test of this level (if the analyzer is on a sample processing line, disconnect the analyzer from the sample processing line first). If the new QC results are in control, the error will be automatically removed. 3. If the error still exists, contact our customer service department.
0x00010011	Body fluid test QC is out of control (normal level)	Body fluid test QC is out of control (normal level)	1. Check the QC results and identify reasons for this out of control; 2. Redo the QC test of this level (if the analyzer is on a sample processing line, disconnect the analyzer from the sample processing line first). If the new QC results are in control, the error will be automatically removed. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00010012	Body fluid test QC is out of control (high level)	Body fluid test QC is out of control (high level)	1. Check the QC results and identify reasons for this out of control; 2. Redo the QC test of this level (if the analyzer is on a sample processing line, disconnect the analyzer from the sample processing line first). If the new QC results are in control, the error will be automatically removed. 3. If the error still exists, contact our customer service department.
0x40001	CR-30[H]-Diluent conductivity over standard	CR-30[H]-Diluent conductivity over standard	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Calibrate the diluent conductivity; 2. Tap the "Remove Error" button to remove the error; 3. If the error still exists, contact our customer service department.
0x40002	CR-30[H]-RO conductivity over standard	CR-30[H]-RO conductivity over standard	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Calibrate the RO conductivity; 2. Tap the "Remove Error" button to remove the error; 3. If the error still exists, contact our customer service department.
0x10100	CR-30[H]-Concentrated reagent cistern floater status abnormal	CR-30[H]-Concentrated reagent cistern floater status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove this error; 2. If the error still exists, contact our customer service department.
0x40003	CR-30[H]-RO cistern upper floater status abnormal	CR-30[H]-RO cistern upper floater status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove this error; 2. If the error still exists, contact our customer service department.
0x40004	CR-30[H]-RO cistern lower floater status abnormal	CR-30[H]-RO cistern lower floater status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove this error; 2. If the error still exists, contact our customer service department.
0x10101	CR-30[H]-Mixing bath 1 floater status abnormal	CR-30[H]-Mixing bath 1 floater status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove this error; 2. If the error still exists, contact our customer service department.
0x10102	CR-30[H]-Mixing bath 2 floater status abnormal	CR-30[H]-Mixing bath 2 floater status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove this error; 2. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x40005	CR-30[H]-Buffer tank upper floater status abnormal	CR-30[H]-Buffer tank upper floater status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1.Tap the "Remove Error" button to remove this error; 2.If the error still exists, contact our customer service department.
0x40006	CR-30[H]-Buffer tank mid floater status abnormal	CR-30[H]-Buffer tank mid floater status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1.Tap the "Remove Error" button to remove this error; 2.If the error still exists, contact our customer service department.
0x40007	CR-30[H]-Buffer tank lower floater status abnormal	CR-30[H]-Buffer tank lower floater status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1.Tap the "Remove Error" button to remove this error; 2.If the error still exists, contact our customer service department.
0x40008	CR-30[H]-RO quantification cistern upper electrode status abnormal	CR-30[H]-RO quantification cistern upper electrode status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1.Tap the "Remove Error" button to remove this error; 2.If the error still exists, contact our customer service department.
0x40009	CR-30[H]-RO quantification cistern lower electrode status abnormal	CR-30[H]-RO quantification cistern lower electrode status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1.Tap the "Remove Error" button to remove this error; 2.If the error still exists, contact our customer service department.
0x40010	CR-30[H]-Concentrated reagent quantification cistern electrode status abnormal	CR-30[H]-Concentrated reagent quantification cistern electrode status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1.Tap the "Remove Error" button to remove this error; 2.If the error still exists, contact our customer service department.
0x10103	CR-30[H]-Backflow	CR-30[H]-Backflow preventer detects full. Backflow occurs in vacuum channel.	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1.Tap the "Remove Error" button to remove this error; 2.If the error still exists, contact our customer service department.
0x30700	CR-30[H]-Sytsem bottom spill detection abnormal	CR-30[H]-Sytsem bottom spill detection abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1.Tap the "Remove Error" button to remove this error; 2.If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x30600	CR-30[H]-Buffer tank spill detection abnormal	CR-30[H]-Buffer tank spill detection abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove this error; 2. If the error still exists, contact our customer service department.
0x99800	CR-30[H]-Startup initiation not performed	CR-30[H]-Startup initiation not performed	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove this error; 2. If the error still exists, contact our customer service department.
0x10321	CR-30[H]-250kPa pressure out of range	CR-30[H]-250kPa standby pressure abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please check the pneumatic unit connection and power. Power it on if it is off. 2. If the pneumatic unit cannot be powered on, or the error still exists after powering on the pneumatic unit, contact our customer service department.
0x10322	CR-30[H]-70kPa pressure out of range	CR-30[H]-70kPa standby pressure abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove this error; 2. If the error still exists, contact our customer service department.
0x10323	CR-30[H]-Please adjust -40kPa pressure	CR-30[H]-40kPa standby pressure abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove this error; 2. If the error still exists, contact our customer service department.
0x10000	CR-30[H]-NO DCT diluent A	CR-30[H]-NO DCT diluent A	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please check if there is DCT diluent A, and if the pickup sample tube is placed properly. 2. If there is diluent in the container, re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of DCT diluent, scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.

Error ID	Message	Details	Solutions
0x10200	CR-30[H]-DCT diluent A expired	CR-30[H]-DCT diluent A expired	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please check if there is DCT diluent A, and if the pickup sample tube is placed properly. 2. If there is diluent in the container, re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of DCT diluent, scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x10400	CR-30[H]-DCT diluent A low volume	CR-30[H]-DCT diluent A low volume	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. No DCT diluent A. Please get a new barcode. 2. Tap the "Remove Error" button and enter the new DCT diluent barcode. 3. If the error still exists, contact our customer service department.
0x10001	CR-30[H]-NO DCT diluent B	CR-30[H]-NO DCT diluent B	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please check if there is DCT diluent B, and if the pickup sample tube is placed properly. 2. If there is diluent in the container, re-scan the current reagent barcode, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of DCT diluent, scan the new barcode, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x10201	CR-30[H]-DCT diluent B expired	CR-30[H]-DCT diluent B expired	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. DCT diluent B expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x10401	CR-30[H]-DCT diluent B low volume	CR-30[H]-DCT diluent B low volume	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. No DCT diluent B. Please get a new barcode. 2. Tap the "Remove Error" button and enter the new DCT diluent barcode. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x10202	CR-30[H]-DCT diluent in buffer tank expired	CR-30[H]-DCT diluent in buffer tank expired	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please check whether the current reagent residue is over 30%. 2. Tap the "Remove Error" button to remove the error. 3. If the error still exists, contact our customer service department.
0x99803	CR-30[H]-Not used for a long time	CR-30[H]-Not used for a long time	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1.The device has not been used for a long time and is of high risk,please contact our customer service department.
0x99801	CR-30[H]-Exiting standby status failed	CR-30[H]-Exiting standby status failed	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1.Tap the "Remove Error" button to remove this error; 2.If the error still exists, contact our customer service department.
0x99802	CR-30[H]-Exiting standby status failed	CR-30[H]-Exiting standby status failed	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1.Tap the "Remove Error" button to remove this error; 2.If the error still exists, contact our customer service department.
0x30101	CR-30[H]-RO serial port comm. abnormal	CR-30[H]-RO serial port comm. abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please contact our customer service department. 2. Wire connector is loose or the wire is abnormal. FRU: lead wire, water conductivity assembly (009-011945-00); connecting wire, RO conductivity board to mother board (009-012577-00). 3.RO conductivity assembly is abnormal. FRU: RO conductivity assembly (115-077351-00). 4.Mother board is abnormal. FRU: PCBA, mother board (051-004008-00).
0x30100	CR-30[H]-RO conductivity assembly abnormal	CR-30[H]-RO conductivity assembly abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1.Please contact our customer service department. 2.RO conductivity assembly is abnormal. FRU: RO conductivity assembly (115-077351-00).
0x30104	CR-30[H]-Water supply or RO valve abnormal	CR-30[H]-Water supply or RO valve abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please check the RO supply. If it is abnormal, restore normal supply and tap the "Remove Error" button to remove the error. 2. If the error still exists after auto removal, or the RO supply is normal, contact our customer service department.

Error ID	Message	Details	Solutions
0x30102	CR-30[H]-RO conductivity abnormal	CR-30[H]-RO conductivity abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please check the RO conductivity. If it is abnormal, supply RO under 4uS/cm and tap the "Remove Error" button to remove the error. 2. If the error still exists after auto removal, or the RO conductivity is normal, contact our customer service department.
0x30103	CR-30[H]-No RO	CR-30[H]-No RO	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please check the RO supply. If it is abnormal, restore normal supply and tap the "Remove Error" button to remove the error. 2. If the error still exists after auto removal, or the RO supply is normal, contact our customer service department.
0x30105	CR-30[H]-RO filter or valve abnormal	CR-30[H]-RO filter or valve abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please contact our customer service department. 2. RO filter is blocked. FRU: RO filter (082-000507-00). 3. RO valve is abnormal. FRU: SV18 or SV20 (082-001454-00).
0x30000	CR-30[H]-Reagent conductivity assembly abnormal	CR-30[H]-Reagent conductivity assembly abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please contact our customer service department. 2. Reagent conductivity assembly is abnormal. FRU: reagent conductivity assembly (115-077352-00).
0x30001	CR-30[H]-Reagent conductivity assembly comm. abnormal	CR-30[H]-Reagent conductivity assembly comm. abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please contact our customer service department. 2. Wire connector is loose or the wire is abnormal. FRU: lead wire, reagent conductivity assembly (009-011943-00); connecting wire, reagent conductivity board to mother board (009-012576-00). 3. Reagent conductivity assembly is abnormal. FRU: reagent conductivity assembly (115-077352-00). 4. Mother board is abnormal. FRU: PCBA, mother board (051-004008-00).
0x30002	CR-30[H]-Reagent not mixed fully	CR-30[H]-Reagent not mixed fully	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please contact our customer service department. 2. Mixing bath 1 is abnormal. FRU: mixing bath 1 assembly (115-082903-00). 3. Mixing bath 2 is abnormal. FRU: mixing bath 2 assembly (115-082916-00)

Error ID	Message	Details	Solutions
0x30003	CR-30[H]-MIX 2 pressure abnormal	CR-30[H]-MIX 2 pressure abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please contact our customer service department. 2. Please perform self-test for 70kpa pressure. 3. Pressure detection board is abnormal. FRU: PCBA, digital sensor board (051-004546-00). 4. Mother board is abnormal. FRU: PCBA, mother board (051-004008-00).
0x30004	CR-30[H]-Liquid valve error or reagent sample tube is blocked	CR-30[H]-Liquid valve error or reagent sample tube is blocked	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please contact our customer service department. 2. Liquid valve error or relevant liquid sample tubes are blocked. FRU: SV07 (115-010434-00).
0x30005	CR-30[H]-Reagent conductivity abnormal	CR-30[H]-Reagent conductivity abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Please replace with a new reagent container and scan the reagent barcode to replace the old reagent. If the error still exists after replacement, contact our customer service department. 2. Reagent conductivity assembly is abnormal. FRU: reagent conductivity assembly (115-077352-00). 3. The quantified volume is inaccurate. FRU: quantification cistern (115-082918-00).
0x10300	CR-30[H]-The pressure rises for a long time when the pneumatic unit starts pressure	CR-30[H]-The pressure rises for a long time when the pneumatic unit starts pressure	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The current of pneumatic unit pressure is low, and pressure rises for a long time. FRU: pneumatic unit. 4. Pressure rises for a long time due to air leakage of dryer. FRU: dryer (115-006311-00). 5. Pressure rises for a long time due to pressure outlet sample tube leakage. FRU: pressure pneumatic sample tube.
0x10301	CR-30[H]-The vacuum rises for a long time when the pneumatic unit starts vacuum	CR-30[H]-The vacuum rises for a long time when the pneumatic unit starts vacuum	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The current of pneumatic unit vacuum is low, and vacuum rises for a long time. FRU: pneumatic unit. 4. Vacuum rises for a long time due to vacuum relief valve error. FRU: vacuum relief valve (115-008186-00). 5. Vacuum rises for a long time due to vacuum outlet sample tube leakage. FRU: vacuum pneumatic sample tube.

Error ID	Message	Details	Solutions
0x10302	CR-30[H]-The pressure in the RO quantification cistern is low when it switches to pressure	CR-30[H]-The pressure in the RO quantification cistern is low when it switches to pressure	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The pressure is low due to RO quantification cistern valve error. FRU: GV34(115-046257-00). 4. The pressure is low due to RO quantification cistern leakage. FRU: quantification cistern assembly (FRU) (115-082918-00).
0x10303	CR-30[H]-The vacuum degree in the RO quantification cistern is high when it switches to pressure	CR-30[H]-The vacuum degree in the RO quantification cistern is high when it switches to pressure	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The vacuum degree in RO quantification cistern is high when it switches to pressure due to RO quantification cistern valve error. FRU: GV34(115-046257-00). 4. The vacuum degree in RO quantification cistern is high when it switches to pressure due to RO quantification cistern leakage. FRU: quantification cistern assembly (FRU) (115-082918-00).
0x10304	CR-30[H]-The pressure in the concentrated reagent quantification cistern is low when it switches to pressure	CR-30[H]-The pressure in the concentrated reagent quantification cistern is low when it switches to pressure	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The pressure is low due to concentrated reagent quantification cistern valve error. FRU: GV34(115-046257-00). 4. The pressure is low due to concentrated reagent quantification cistern leakage. FRU: quantification cistern assembly (FRU) (115-082918-00).
0x10305	CR-30[H]-The vacuum degree in the concentrated reagent quantification cistern is high when it switches to vacuum	CR-30[H]-The vacuum degree in the concentrated reagent quantification cistern is high when it switches to vacuum	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The vacuum degree in concentrated reagent quantification cistern is high when it switches to pressure due to concentrated reagent quantification cistern valve error. FRU: GV34(115-046257-00). 4. The vacuum degree in concentrated reagent quantification cistern is high when it switches to pressure due to concentrated reagent quantification cistern leakage. FRU: quantification cistern assembly (115-082918-00).

Error ID	Message	Details	Solutions
0x10306	CR-30[H]-The pressure in mixing bath 1 is low when it switches to pressure	CR-30[H]-The pressure in mixing bath 1 is low when it switches to pressure	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The pressure in mixing bath 1 is low when it switches to pressure due to valve GV33 leakage. FRU: GV33(115-046257-00). 4. The pressure in mixing bath 1 is low when it switches to pressure due to poor airtightness. FRU: mixing bath 1 assembly (115-077594-00).
0x10307	CR-30[H]-The atmospheric pressure in mixing bath 1 monitored by pressure sensor is 5kPa higher	CR-30[H]-The atmospheric pressure in mixing bath 1 monitored by pressure sensor is 5kPa higher	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The atmospheric pressure in mixing bath 1 monitored by pressure sensor is high due to valve GV33 leakage. FRU: GV33(115-046257-00). 4. The pressure sensor in mixing bath 1 is abnormal. FRU: PCBA, digital pressure detection board (115-077594-00).
0x10308	CR-30[H]-The pressure in mixing bath 1 monitored by pressure sensor is over 5kPa lower than that monitored by the 70kPa pressure sensor	CR-30[H]-The pressure in mixing bath 1 monitored by pressure sensor is over 5kPa lower than that monitored by the 70kPa pressure sensor	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The pressure in mixing bath 1 monitored by pressure sensor is lower than that monitored by 70kPa pressure sensor due to valve GV33 leakage. FRU: GV33(115-046257-00). 4. The 70kPa pressure sensor is abnormal. FRU: PCBA, digital pressure detection board (051-004448-00).
0x10309	CR-30[H]-The atmospheric pressure in mixing bath 1 rises too fast when it switches to pressure	CR-30[H]-The atmospheric pressure in mixing bath 1 rises too fast when it switches to pressure	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The atmospheric pressure in mixing bath 1 rises too fast when it switches to pressure, because the pneumatic sample tube is blocked. FRU: mixing bath 1 assembly (115-077594-00). 4. The atmospheric pressure in mixing bath 1 rises too fast when it switches to pressure, because the pneumatic sample tube from mixing bath 1 to MIX 1 pressure sensor is blocked. FRU: pneumatic sample tube from mixing bath 1 to MIX 1 pressure sensor.

Error ID	Message	Details	Solutions
0x10310	CR-30[H]-The pressure in mixing bath 2 is low when it switches to pressure	CR-30[H]-The pressure in mixing bath 2 is low when it switches to pressure	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The pressure in mixing bath 2 is low when it switches to pressure due to valve GV32 leakage. FRU: GV33(115-046257-00). 4. The pressure in mixing bath 2 is low when it switches to pressure due to poor airtightness. FRU: mixing bath 2 assembly (115-082916-00).
0x10311	CR-30[H]-The atmospheric pressure in mixing bath 2 monitored by pressure sensor is 5kPa higher	CR-30[H]-The atmospheric pressure in mixing bath 2 monitored by pressure sensor is 5kPa higher	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The atmospheric pressure in mixing bath 2 monitored by pressure sensor is high due to valve GV32 leakage. FRU: GV32(115-046257-00). 4. The pressure sensor in mixing bath 2 is abnormal. FRU: PCBA, digital pressure detection board (115-004510-00).
0x10312	CR-30[H]-The atmospheric pressure in mixing bath 2 monitored by pressure sensor is 5kPa lower	CR-30[H]-The atmospheric pressure in mixing bath 2 monitored by pressure sensor is 5kPa lower	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The atmospheric pressure in mixing bath 2 monitored by pressure sensor is low due to valve GV32 leakage. FRU: GV37(115-046257-00). 4. The pressure sensor in mixing bath 2 is abnormal. FRU: PCBA, digital pressure detection board (115-004510-00).
0x10313	CR-30[H]-The pressure in mixing bath 2 monitored by pressure sensor is over 5kPa lower than that monitored by the 70kPa pressure sensor	CR-30[H]-The pressure in mixing bath 2 monitored by pressure sensor is over 5kPa lower than that monitored by the 70kPa pressure sensor	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The pressure in mixing bath 2 monitored by pressure sensor is lower than that monitored by 70kPa pressure sensor due to valve GV32 leakage. FRU: GV32(115-046257-00). 4. The 70kPa pressure sensor is abnormal. FRU: PCBA, digital pressure detection board (051-004448-00).

Error ID	Message	Details	Solutions
0x10314	CR-30[H]-The atmospheric pressure in mixing bath 2 rises too fast when it switches to pressure	CR-30[H]-The atmospheric pressure in mixing bath 2 rises too fast when it switches to pressure	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The atmospheric pressure in mixing bath 2 rises too fast when it switches to pressure, because the pneumatic sample tube is blocked. FRU: mixing bath 2 assembly (FRU) (115-082916-00). 4. The atmospheric pressure in mixing bath 1 rises too fast when it switches to pressure, because the pneumatic sample tube from mixing bath 2 to MIX 2 pressure sensor is blocked. FRU: pneumatic sample tube from mixing bath 2 to MIX 2 pressure sensor.
0x10315	CR-30[H]-The vacuum degree in mixing bath 2 is low when it switches to vacuum	CR-30[H]-The vacuum degree in mixing bath 2 is low when it switches to vacuum	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The vacuum degree in mixing bath 2 is low when it switches to vacuum due to valve GV37 leakage. FRU: GV37(115-046257-00). 4. The vacuum degree in mixing bath 2 is low when it switches to pressure due to poor airtightness. FRU: mixing bath 2 assembly (FRU) (115-082916-00).
0x10316	CR-30[H]-The vacuum degree in mixing bath 2 monitored by pressure sensor is over 8kPa lower than that monitored by the -40kPa pressure sensor	CR-30[H]-The vacuum degree in mixing bath 2 monitored by pressure sensor is over 8kPa lower than that monitored by the -40kPa pressure sensor	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error. 2. If the error still exists, contact our customer service department. 3. The vacuum in mixing bath 2 monitored by pressure sensor is over 10kPa lower than that monitored by the -40kPa pressure sensor due to valve GV37 leakage. FRU: GV37(115-046257-00). 4. The -40kPa pressure sensor is abnormal. FRU: PCBA, digital pressure detection board (051-004510-00).

Error ID	Message	Details	Solutions
0x10317	CR-30[H]-The pressure in the concentrated reagent cistern is low when it switches to pressure	CR-30[H]-The pressure in the concentrated reagent cistern is low when it switches to pressure	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The pressure in concentrated reagent cistern is low when it switches to pressure due to valve GV39 leakage. FRU: GV39(115-046257-00). 4. The pressure in concentrated reagent cistern is low when it switches to pressure due to poor airtightness. FRU: concentrated reagent cistern (115-018617-00).
0x10318	CR-30[H]-The vacuum in the concentrated reagent cistern is low when it switches to vacuum	CR-30[H]-The vacuum in the concentrated reagent cistern is low when it switches to vacuum	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The vacuum in concentrated reagent cistern is low when it switches to vacuum due to valve GV35 leakage. FRU: GV35(115-046257-00). 4. The vacuum in concentrated reagent cistern is low when it switches to vacuum due to poor airtightness. FRU: concentrated reagent cistern (115-018617-00).
0x10319	CR-30[H]-The pressure in the RO cistern is low when it switches to pressure	CR-30[H]-The pressure in the RO cistern is low when it switches to pressure	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The pressure in RO cistern is low when it switches to pressure due to valve GV31 leakage. FRU: GV31(115-046257-00). 4. The pressure in RO cistern is low when it switches to pressure due to poor airtightness. FRU: RO cistern assembly (FRU) (115-018617-00).
0x10320	CR-30[H]-The vacuum in the RO cistern is low when it switches to vacuum	CR-30[H]-The vacuum in the RO cistern is low when it switches to vacuum	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The vacuum in RO cistern is low when it switches to vacuum due to valve GV36 leakage. FRU: GV36(115-046257-00). 4. The vacuum in RO cistern is low when it switches to vacuum due to poor airtightness. FRU: RO cistern assembly (FRU) (115-082917-00).

Error ID	Message	Details	Solutions
0x30200	CR-30[H]-Electrode of concentrated reagent quantification cistern conducts abnormally	CR-30[H]-Electrode of concentrated reagent quantification cistern conducts abnormally	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Liquid flow into concentrated reagent quantification cistern when it is placed still. FRU: SV16 and SV17(115-071230-00). 4. Electrode of concentrated reagent quantification cistern conducts abnormally. FRU: quantification cistern (FRU) (115-082918-00). 5. Board error. FRU:PCBA,RGB screen commutator board (051-004343-01) 6. Wire connection abnormal. FRU: connecting wire, RGB screen commutator board electrode detection (009-011727-00)
0x30201	CR-30[H]-Insufficient vacuum during the priming of concentrated reagent quantification cistern	CR-30[H]-Insufficient vacuum during the priming of concentrated reagent quantification cistern	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Check standby pressure and regulate to corresponding range. Then perform pressure self-test for pneumatic unit.
0x30202	CR-30[H]-Electrode of concentrated reagent quantification cistern fails to detect full	CR-30[H]-Electrode of concentrated reagent quantification cistern fails to detect full	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The inlet throttle sample tube of the concentrated reagent quantification cistern is blocked. FRU:T062 (M90-000026---) 4. The inlet valve of the concentrated reagent quantification cistern is blocked or cannot be opened. FRU:SV16 (115-071230-00). 5. The electrode of the concentrated reagent quantification cistern cannot detect liquid. FRU: quantification cistern assembly (FRU) (115-082918-00). 6.Board error. FRU: RGB screen commutator board (051-004343-01). 7.Wire connection abnormal. FRU: connecting wire, RGB screen commutator board electrode detection (009-011727-00).

Error ID	Message	Details	Solutions
0x30300	CR-30[H]-Low electrode of concentrated reagent quantification cistern conducts abnormally	CR-30[H]-Low electrode of concentrated reagent quantification cistern conducts abnormally	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error. 2. If the error still exists, contact our customer service department. 3. Liquid flow into the fast channel of RO quantification cistern when it is placed still. FRU:SV14 (115-077672-00) . 4. Liquid flow into the slow channel of RO quantification cistern when it is placed still. FRU: SV13 (115-071230-00). 5. The electrode of the RO quantification cistern conducts abnormally. FRU: quantification cistern assembly (FRU) (115-082918-00) 6.Board error. FRU: RGB screen commutator board (051-004343-01) 7.Wire connection abnormal. FRU: connecting wire, RGB screen commutator board electrode detection (009-011727-00)
0x30301	CR-30[H]-Up electrode of concentrated reagent quantification cistern conducts abnormally	CR-30[H]-Up electrode of concentrated reagent quantification cistern conducts abnormally	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The electrode of the RO quantification cistern conducts abnormally. FRU: quantification cistern assembly (FRU) (115-082918-00). 4. Board error. FRU: RGB screen commutator board (051-004343-01). 5. Wire connection abnormal. FRU: connecting wire, RGB screen commutator board electrode detection (009-011727-00).
0x30302	CR-30[H]-Insufficient vacuum during the priming of RO quantification cistern	CR-30[H]-Insufficient vacuum during the priming of RO quantification cistern	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Check standby pressure and regulate to corresponding range. Then perform pressure self-test for pneumatic unit.

Error ID	Message	Details	Solutions
0x30303	CR-30[H]-Low electrode of RO quantification cistern fails to detect full	CR-30[H]-Low electrode of RO quantification cistern fails to detect full	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The fast priming valve of the RO quantification cistern is blocked. FRU: SV14 (115-077672-00). 4. The low electrode of the RO quantification cistern cannot detect liquid. FRU: quantification cistern assembly (FRU) (115-082918-00). 5. Board error. FRU: RGB screen commutator board (051-004343-01). 6. Wire connection abnormal. FRU: connecting wire, RGB screen commutator board electrode detection (009-011727-00).
0x30304	CR-30[H]-Up electrode of RO quantification cistern fails to detect full	CR-30[H]-Up electrode of RO quantification cistern fails to detect full	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The slow priming valve of the RO quantification cistern is blocked. FRU: SV13 (115-071230-00). 4. The up electrode of the RO quantification cistern cannot detect liquid. FRU: quantification cistern assembly (FRU) (115-082918-00). 5. The inlet throttle sample tube of the RO quantification cistern is blocked. FRU: quantification cistern assembly (FRU) (115-082918-00). 6. Board error. FRU: RGB screen commutator board (051-004343-01). 7. Wire connection abnormal. FRU: connecting wire, RGB screen commutator board electrode detection (009-011727-00).
0x30400	CR-30[H]-Floater of mixing bath 1 fails to detect full	CR-30[H]-Floater of mixing bath 1 fails to detect full	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. No water added into mixing bath 1. FRU: SV15 (115-077672-00). 4. The floater of mixing bath 1 cannot detect full. FRU: mixing bath 1 assembly (FRU) (115-082903-00). 5. Wire connection abnormal. FRU: connecting wire, floater (009-011588-00).

Error ID	Message	Details	Solutions
0x30401	CR-30[H]-Floater of mixing bath 1 fails to detect full	CR-30[H]-Floater of mixing bath 1 fails to detect full initially	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Liquid discharge from mixing bath 1 is abnormal due to low pressure. FRU:GV33(115-046257-00). 4. Liquid discharge from mixing bath 1 is abnormal due to outlet valve error. FRU:SV05 (115-077672-00). 5. The floater of mixing bath 1 cannot detect empty. FRU: mixing bath 1 assembly (FRU)(115-082903-00). 6. Liquid discharge from mixing bath 1 is abnormal due to pressure error. Troubleshooting: check standby pressure and regulate to corresponding pressure range. Then perform pressure self-test for pneumatic unit.
0x30402	CR-30[H]-Floater of mixing bath 1 fails to detect full during maintenance	CR-30[H]-Floater of mixing bath 1 fails to detect full during maintenance	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Liquid discharge from mixing bath 1 is abnormal due to low pressure. FRU:GV33(115-046257-00). 4. Liquid discharge from mixing bath 1 is abnormal due to outlet valve error. FRU:SV05 (115-077672-00). 5. The floater of mixing bath 1 cannot detect empty. FRU: mixing bath 1 assembly (FRU)(115-082903-00). 6. Liquid discharge from mixing bath 1 is abnormal due to pressure error. Troubleshooting: check standby pressure and regulate to corresponding pressure range. Then perform pressure self-test for pneumatic unit.
0x30403	CR-30[H]-Insufficient 70kPa pressure for the emptying of mixing bath 1 during reagent preparation	CR-30[H]-Insufficient 70kPa pressure for the emptying of mixing bath 1 during reagent preparation	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Valve GV33 of mixing bath 1 is abnormal during reagent preparation. FRU: GV33(115-046257-00). 4. Liquid discharge from mixing bath 1 is abnormal due to pressure error. Troubleshooting: check standby pressure and regulate to corresponding pressure range. Then perform pressure self-test for pneumatic unit.

Error ID	Message	Details	Solutions
0x30404	CR-30[H]-Insufficient MIX 1 pressure for the emptying of mixing bath 1 during reagent preparation	CR-30[H]-Insufficient MIX 1 pressure for the emptying of mixing bath 1 during reagent preparation	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Valve GV33 of mixing bath 1 is abnormal during reagent preparation. FRU: GV33(115-046257-00). 4. Liquid discharge from mixing bath 1 is abnormal due to pressure error. Troubleshooting: check standby pressure and regulate to corresponding pressure range. Then perform pressure self-test for pneumatic unit.
0x30405	CR-30[H]-The hydraulic resistance in the emptying channel of mixing bath 1 is high during reagent preparation	CR-30[H]-The hydraulic resistance in the emptying channel of mixing bath 1 is high during reagent preparation	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Liquid discharge from mixing bath 1 is abnormal due to mixing bath 1 outlet valve error. FRU:SV05 (115-077672-00). 4. Liquid discharge from mixing bath 1 is abnormal due to discharge sample tube error. FRU: mixing bath 1 discharge sample tube.
0x30406	CR-30[H]-Floater of mixing bath 1 fails to detect full during reagent preparation	CR-30[H]-Floater of mixing bath 1 fails to detect full during reagent preparation	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The floater of mixing bath 1 cannot detect empty. FRU: mixing bath 1 assembly (FRU)(115-082903-00). 4. Liquid discharge from mixing bath 1 is abnormal due to outlet valve error. FRU: SV05(115-077672-00). 5. Valve GV33 of mixing bath 1 is abnormal during reagent preparation. FRU: GV33(115-046257-00).
0x30500	CR-30[H]-Floater of mixing bath 2 fails to detect full	CR-30[H]-Floater of mixing bath 2 fails to detect full	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Reagent in mixing bath 1 fails to be discharged to mixing bath 2. FRU: SV05 (115-077672-00) 4. Floater of mixing bath 2 cannot detect full. FRU: mixing bath 2 assembly (FRU) (115-082916-00). 5. Wire connection error. FRU: connecting wire, floater (009-011588-00).

Error ID	Message	Details	Solutions
0x30501	CR-30[H]-Floater of mixing bath 2 fails to detect full initially	CR-30[H]-Floater of mixing bath 2 fails to detect full initially	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Liquid discharge from mixing bath 2 is abnormal due to low pressure. FRU:GV33(115-046257-00). 4. Liquid discharge from mixing bath 2 is abnormal due to outlet valve error. FRU:SV05(115-077672-00). 5. The floater of mixing bath 2 cannot detect empty. FRU: mixing bath 2 assembly (FRU)(115-082903-00). 6. Liquid discharge from mixing bath 2 is abnormal due to pressure error. Troubleshooting: check standby pressure and regulate to corresponding pressure range. Then perform pressure self-test for pneumatic unit.
0x30502	CR-30[H]-Floater of mixing bath 2 fails to detect full during maintenance	CR-30[H]-Floater of mixing bath 2 fails to detect full during maintenance	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Liquid discharge from mixing bath 2 is abnormal due to low pressure. FRU:GV33(115-046257-00). 4. Liquid discharge from mixing bath 2 is abnormal due to outlet valve error. FRU:SV05(115-077672-00). 5. The floater of mixing bath 2 cannot detect empty. FRU: mixing bath 2 assembly (FRU)(115-082903-00). 6. Liquid discharge from mixing bath 2 is abnormal due to pressure error. Troubleshooting: check standby pressure and regulate to corresponding pressure range. Then perform pressure self-test for pneumatic unit.
0x30503	CR-30[H]-Insufficient 70kPa pressure for the emptying of mixing bath 2 during reagent preparation	CR-30[H]-Insufficient 70kPa pressure for the emptying of mixing bath 2 during reagent preparation	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Valve GV32 of mixing bath 2 is abnormal during reagent preparation. FRU: GV32(115-046257-00). 4. Liquid discharge from mixing bath 2 is abnormal due to pressure error. Troubleshooting: check standby pressure and regulate to corresponding pressure range. Then perform pressure self-test for pneumatic unit.

Error ID	Message	Details	Solutions
0x30504	CR-30[H]-Insufficient MIX 2 pressure for the emptying of mixing bath 2 during reagent preparation	CR-30[H]-Insufficient MIX 2 pressure for the emptying of mixing bath 2 during reagent preparation	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Valve GV32 of mixing bath 2 is abnormal during reagent preparation. FRU: GV32(115-046257-00). 4. Liquid discharge from mixing bath 2 is abnormal due to pressure error. Troubleshooting: check standby pressure and regulate to corresponding pressure range. Then perform pressure self-test for pneumatic unit.
0x30505	CR-30[H]-The hydraulic resistance in the emptying channel of mixing bath 2 is high during reagent preparation	CR-30[H]-The hydraulic resistance in the emptying channel of mixing bath 2 is high during reagent preparation	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. Liquid discharge from mixing bath 2 is abnormal due to outlet valve error. FRU:SV07 (115-077672-00). 4. Liquid discharge from mixing bath 2 is abnormal due to discharge sample tube error. FRU: mixing bath 2 discharge sample tube.
0x30506	CR-30[H]-Floater of mixing bath 2 fails to detect full during reagent preparation	CR-30[H]-Floater of mixing bath 2 fails to detect full during reagent preparation	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Tap the "Remove Error" button to remove the error; 2. If the error still exists, contact our customer service department. 3. The floater of mixing bath 2 cannot detect empty. FRU: mixing bath 2 assembly (FRU)(115-082903-00). 4. Liquid discharge from mixing bath 2 is abnormal due to outlet valve error. FRU:SV07 (115-077672-00). 5. Valve GV32 of mixing bath 2 is abnormal during reagent preparation. FRU: GV32(115-046257-00).
0x10105	CR-30[H]-DCT diluent A is going to expire	CR-30[H]-DCT diluent A is going to expire	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. DCT diluent A is going to expire. Replace it with a new one within its validity period. If you replace the reagent after it expires, the system will need to drain and prime the sample tubes, which will take a long time; 2. Tap the "Remove Error" button to remove the error; 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x10106	CR-30[H]-DCT diluent B is going to expire	CR-30[H]-DCT diluent B is going to expire	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. DCT diluent B is going to expire. Replace it with a new one within its validity period. If you replace the reagent after it expires, the system will need to drain and prime the sample tubes, which will take a long time; 2. Tap the "Remove Error" button to remove the error; 3. If the error still exists, contact our customer service department.
0x30204	CR-30[H]-Concentrated reagent quantification cistern status abnormal	CR-30[H]-Concentrated reagent quantification cistern status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Contact our customer service department. 2. Replace the throttle pipe T602, GV38, SV16, RGB board, wires or quantification cistern assembly.
0x30306	CR-30[H]-RO quantification cistern status abnormal	CR-30[H]-RO quantification cistern status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Contact our customer service department. 2. Replace the throttle pipe T077, GV34, SV13, RGB board, wires or quantification cistern assembly.
0x40018	CR-30[H]-Communication link abnormal	CR-30[H]-Communication link abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Contact our customer service department. 2. There may be problems with RGB board, wires, concentrated reagent quantification cistern, SV16 or SV17.
0x30205	CR-30[H]-Concentrated reagent quantification cistern status abnormal	CR-30[H]-Concentrated reagent quantification cistern status abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Contact our customer service department. 2. There may be problems with RGB board, motherboard, communication wires or connector.
0x30307	CR-30[H]-RO quantification cistern abnormal	CR-30[H]-RO quantification cistern abnormal	Perform procedures below on the CONCENTRATED REAGENT DELIVERY SYSTEM: 1. Contact our customer service department. 2. There may be problems with RGB board, wires, RO quantification cistern, SV13 or SV14.
0x00007013	ESR1 signal abnormal	ESR1 baseline abnormal	1.Tap the "Remove Error" button to remove the error. 2.If the error still exists, shut off the analyzer power directly, and restart the analyzer later. 3.If the error still exists, contact our customer service department.
0x00007014	ESR2 signal abnormal	ESR2 baseline abnormal	1.Tap the "Remove Error" button to remove the error. 2.If the error still exists, shut off the analyzer power directly, and restart the analyzer later. 3.If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00001007	No ESR Solution Reagent	ESR Solution Reagent running out or bubbles in the tubing.	1. Please check if there is ESR Solution Reagent in the container, and if the pickup sample tube is placed properly. 2. If there is reagent in the container, re-scan the current reagent card, and perform the replacing procedure. 3. If there is no diluent, replace with a new container of ESR Solution Reagent. Scan the new reagent card, and perform the replacing procedure. 4. If the error still exists after replacing, contact our customer service department.
0x00001059	Insufficient ESR Solution Reagent	Insufficient ESR Solution Reagent	1. Registered volume of ESR Solution Reagent for closed-reagent model runs out. 2. Tap the "Remove Error" button, and scan the reagent card to remove the error. 3. If the error still exists, contact our customer service department.?
0x00004315	ESR Solution Reagent cistern floater error	ESR Solution Reagent cistern floater detection status abnormal	1. Tap the "Remove Error" button, and the analyzer will perform floater self-test to remove the error. 2. If the error still exists, contact our customer service department.
0x00001020	ESR Solution Reagent expired	ESR Solution Reagent expired	1. ESR Solution Reagent expired. 2. Tap the "Remove Error" button and complete the replacing procedure in the pop-up dialog box. 3. If the error still exists after replacing, contact our customer service department.
0x00001060	Prime after loading ESR Solution Reagent	Failed to prime after loading ESR Solution Reagent. Please prime again.	1. Failed to prime after loading ESR Solution Reagent. Please prime again. 2. Tap the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00001070	Replace after loading ESR Solution Reagent	Failed to replace after loading ESR Solution Reagent. Please replace again.	1. Failed to replace after loading ESR Solution Reagent. Please replace again. 2. Tap the "Remove Error" button. The analyzer will try to resume automatically. 3. If the error still exists, contact our customer service department.
0x00002115	ESR assembly temperature too low	ESR assembly low temperature error in operation	1. The working temperature of ESR assembly lower than expected while working (e.g., in the process of analyzing). Sample analysis may be disabled. 2. Check the working temperature of the analyzer. If it is within the expected range, contact our customer service department.
0x00002116	ESR assembly temperature too high	ESR assembly low temperature error in operation	1. The working temperature of ESR assembly higher than expected while working (e.g., in the process of analyzing). Sample analysis may be disabled. 2. Check the working temperature of the analyzer. If it is within the expected range, contact our customer service department.
0x00001035	Sequence command error	ESR Solution Reagent channel locked due to no reagent or bubbles in tubing	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00006037	RFID board communication time out	RFID board communication time out	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00006038	ESR board communication time out	ESR board communication time out	1. Shut off the analyzer power directly, and restart the analyzer later. 2. If the error still exists, contact our customer service department.
0x00000102	Insufficient aspiration	Found the end segment of the sample not fully fill the tubing during aspiration	1. Found the end segment of the sample hanging on the tubing rather than fully filling the tubing while monitoring the aspiration process. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000103	Insufficient aspiration	Incontinuous sample in tubing or plenty of bubbles found during aspiration	1. Found bubbles or incontinuity in the aspirated sample while monitoring the aspiration process. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000104	Insufficient aspiration	Found sample not aspirated while monitoring the aspiration process.	1. Found sample not aspirated while monitoring the aspiration process. 2. Check whether the volume of sample in the tubing is higher than minimum.
0x00000105	Insufficient aspiration	Found aspirated sample insufficient while monitoring the aspiration process.	1. Found aspirated sample insufficient while monitoring the aspiration process. 2. Check whether the volume of sample in the tubing is higher than minimum.
0x00000106	Insufficient aspiration	Blood sensor 1 status abnormal	1. Blood sensor 1 status abnormal. Possible reasons: Blood sensor error; connection error between the blood sensor and sample aspiration or dividing tubing; diluent and sample not flow in the tubing as expected due to fluidic error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000107	Insufficient aspiration	Blood sensor 2 status abnormal	1. Blood sensor 2 status abnormal. Possible reasons: Blood sensor error; connection error between the blood sensor and sample aspiration or dividing tubing; diluent and sample not flow in the tubing as expected due to fluidic error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000202	Aspiration abnormal for ESR test	Found the end segment of the sample not fully fill the tubing during aspiration	1. Found the end segment of the sample hanging on the tubing rather than fully filling the tubing while monitoring the aspiration process. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.

Error ID	Message	Details	Solutions
0x00000203	Aspiration abnormal for ESR test	Incontinuous sample in tubing or plenty of bubbles found during aspiration	1. Found bubbles or incontinuity in the aspirated sample while monitoring the aspiration process. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000204	Aspiration abnormal for ESR test	Found sample not aspirated while monitoring the aspiration process.	1. Found sample not aspirated while monitoring the aspiration process. 2. Check whether the volume of sample in the tubing is higher than minimum.
0x00000205	Aspiration abnormal for ESR test	Found aspirated sample insufficient while monitoring the aspiration process.	1. Found aspirated sample insufficient while monitoring the aspiration process. 2. Check whether the volume of sample in the tubing is higher than minimum.
0x00000206	Aspiration abnormal for ESR test	Blood sensor 1 status abnormal	1. Blood sensor 1 status abnormal. Possible reasons: Blood sensor error; connection error between the blood sensor and sample aspiration or dividing tubing; diluent and sample not flow in the tubing as expected due to fluidic error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00000207	Aspiration abnormal for ESR test	Blood sensor 2 status abnormal	1. Blood sensor 2 status abnormal. Possible reasons: Blood sensor error; connection error between the blood sensor and sample aspiration or dividing tubing; diluent and sample not flow in the tubing as expected due to fluidic error. 2. Tap the "Remove Error" button to remove this error. 3. If the error still exists, contact our customer service department.
0x00007016	Blood sample sensor 3 voltage abnormal	Blood sample sensor 3 voltage abnormal	1. Select the "Remove Error" button to remove this error. 2. If the error still exists, power off the analyzer and restart the analyzer later. 3. If the error still exists, contact our Customer Service Department.

A Index

A

- Access Levels for Different Accounts 6 - 1
- accessories C - 1
- Applicable Sample Tubes B - 1
- Auto Shutdown 7 - 17

B

- Barcode 6 - 2
- barcode
 - place barcode labels 7 - 8, 9 - 5
- blood sample test-test panels 3 - 6
- Body Fluid Parameters 4 - 10

C

- calibrate
 - calibrator 3 - 24
- Calibrating with Calibrators 10 - 3
- Calibrating with Fresh blood Samples 10 - 4
- Calibrating Your Analyzer 10 - 1
- Calibration History 10 - 6
- Customizing Your Analyze 6 - 1

D

- daily operation
 - initial checks 7 - 3
 - sample preparation 7 - 5
 - body fluid samples 7 - 7
 - predilute samples 7 - 6
 - whole blood samples 7 - 5
- Deleting sample records 8 - 5
- DIFF Scattergram 3 - 6, 3 - 7
- DIFF scattergram 4 - 3

E

- Editing sample results 8 - 7
- enter standby 7 - 16
- Entering Reagent Information 12 - 6
- Erythrocyte Sedimentation Rate Measurement 4 - 10

F

- flags
 - abnormal blood cell differential or morphology 8 - 9
 - parameter flags 8 - 8

H

- HGB Measurement 4 - 6
- Histogram
 - RBC Histogram 3 - 5, 3 - 7

I

- installation requirements
 - environment requirements 5 - 2
 - power requirements 5 - 1
 - space requirements 5 - 1

L

- log
 - export logs 12 - 17
- logs
 - review logs 12 - 16

M

- maintenance and service
 - auto cleaning 12 - 11
 - daily Probe Cleanser maintenance 12 - 11
 - frequency and purposes 12 - 2
 - manual cleaning 12 - 12
 - parts and components 12 - 2
 - Probe Cleanser maintenance to parts and components 12 - 11
 - replace and prime with reagents 12 - 7
 - replace the waste containers 12 - 7
 - replacing parts and components 12 - 3
- Manual Calibration 10 - 2

O

- operating your analyzer 7 - 1

P

- parameters 3 - 1, B - 2
 - blood sample test histograms 3 - 5
 - blood sample test RUO parameters 3 - 3
 - blood sample test scattergrams 3 - 6
 - body fluid sample test histograms 3 - 7
 - body fluid sample test panel 3 - 7
 - body fluid sample test report parameters 3 - 6
 - body fluid sample test RUO parameters 3 - 6
 - body fluid sample test scattergram 3 - 7
- pasting barcode labels 9 - 5
- Preparation before Calibration 10 - 1
- prepare to ship 12 - 16
- Probe cleanser maintenance 12 - 11
- product overview
 - accessories 3 - 14
 - modules and components 3 - 7
 - structure and components 3 - 7

Q

- QC
 - review QC results
 - QC table 9 - 8, 9 - 14
 - run controls on the count screen 9 - 6
 - run controls on the QC count screen 9 - 5
 - set up QC files 9 - 1
 - X-B QC
 - run X-B QC program 9 - 13
 - sample validity 9 - 12
 - X-B QC rules 9 - 12
- quality control
 - controls 3 - 23

R

- RBC/PLT Measurement 4 - 7
- reagent connection
 - connecting fluorescent dyes 5 - 6
 - connecting lysates and diluent 5 - 7
- Reagents B - 1
- reagents 3 - 23
 - calibrator 3 - 24
 - M-6DR Diluent
 - M-6FD DYE 3 - 23
 - replacing and priming 12 - 2, 12 - 7
- Reference F - 1
- Reviewing reference results 8 - 8
- Reviewing RUO parameters results 8 - 6
- Reviewing special information 8 - 7

S

- sample analysis
 - autoloading analysis 7 - 13
 - auto re-exam 7 - 15
 - stop count 7 - 15
 - closed sample tube/open-vial mode
 - OV-BF sample analysis 7 - 11
 - OV-WB sample analysis 7 - 10
 - set analysis order 7 - 9
 - STAT 7 - 16
- sample result review
 - calculate CV 8 - 5
 - export sample records 8 - 4
 - graph review 8 - 6
 - review traceability information 8 - 7
 - table review 8 - 1
 - validate sample results 8 - 4
- Saving Settings after Changes 6 - 1
- Setting up Transmission Mode 6 - 2
- Shutdown 7 - 16
 - normal shutdown 7 - 16
- shutdown the analyzer
 - normal shutdown 7 - 16
- software setup
 - auto standby 7 - 16
 - auto startup/shutdown 6 - 18, 7 - 4
 - communication 6 - 1
 - flag alarm sensitivity 6 - 14
 - laboratory information 6 - 4
 - open-vial/closed-sample tube analysis 6 - 10
 - user accounts and passwords 6 - 4
- specification
 - carryover B - 9
 - correlation B - 9
 - sample volumes required for each analysis B - 4
- specification minimum sample volumes B - 4
- specifications
 - background/blank count requirements B - 5
 - deviation of reading B - 10
 - display ranges for major parameters B - 5
 - linearity range B - 6
 - repeatability B - 7
- startup
 - auto startup 7 - 4
 - normal startup procedure 7 - 4
- status indicator 3 - 8
- system principles 4 - 1

T

- troubleshooting
 - check analyzer status 13 - 1
 - Troubleshooting 13 - 1

U

- Using the Printing Function 11 - 1

V

- Verifying Calibration Factors 10 - 6

W

- When to Calibrate 10 - 1
- WNB Scattergram 3 - 6

B

Technical Information

B.1 Reagents

For the reagents, controls, and calibrators that can be used on this analyzer, refer to **3.8 Reagents, Controls and Calibrators**.

NOTE

- For any questions about the reagents, controls and calibrators configuration, consult your sales representative.

B.2 Recommended Sample Tube Types

NOTE

- Some sample tubes may not adapt to the sample rack (e.g. sample tubes with unique caps), in such case, contact our Customer Service Department.

The analyzer supports the following sample tube types:

Table B-1 Sample Tube specifications for autoloading sampling modes

Applicable sample tubes	Supported presentation mode	Note
1-13mm (external diameter)×65-78mm/80-83mm (height without cap/height with cap)	AL-WB	The height of evacuated blood collection sample tube with cap should be no larger than 83mm.

Table B-2 Recommended sample tubes for autoloading sampling modes

Manufacturer	Product	Product No.	Sample Tube material	Anticoagul-ant	Volume (mL)	External diameter × Length (without cap) (mm)	Presentation mode
BD	Vacutainer Plus	367835	Plastic	K ₂ EDTA	3	13×75	AL
BD	Vacutainer Plus	367841	Plastic	K ₂ EDTA	2	13×75	AL
S-Monovette	Sarstedt	04.1901.001	Plastic	K ₃ EDTA	1	13×65	AL
Guangzhou Yangpu	Anticoagulant evacuated blood collection sample tube	80715	Glass	K ₂ EDTA	2	12.6×75	AL
Jiangsu Kangjian	Anticoagulant sample tube for routine blood test	/	Plastic	K ₂ EDTA	2	12×75	AL
Jiangsu Kangjian	Anticoagulant sample tube for routine blood test	/	Plastic	K ₃ EDTA	3	13×75	AL

B.3 Parameters

Table B-3 Blood sample test report parameters

Groups	Name	Abbreviation
WBC group	White Blood Cell count	WBC
	Basophils number	Bas#
	Basophils percentage	Bas%
	Neutrophils number	Neu#
	Neutrophils percentage	Neu%
	Eosinophils number	Eos#
	Eosinophils percentage	Eos%
	Lymphocytes number	Lym#
	Lymphocytes percentage	Lym%
	Monocytes number	Mon#
	Monocytes percentage	Mon%
	Immature Granulocyte number	IMG#
	Immature Granulocyte percentage	IMG%
	DIFF scattergram, mean neutrophil distribution-side scatter intensity	Neu-X
	DIFF scattergram, mean neutrophil distribution-side fluorescent light intensity	Neu-Y
	DIFF scattergram, mean monocyte distribution-side scatter intensity	Mon-X
RET group	Reticulocyte percentage	RET%
	Reticulocyte number	RET#
	Reticulocyte Hemoglobin Expression	RHE
	Immature reticulocyte fraction	IRF
	Low fluorescent ratio	LFR
	Middle fluorescent ratio	MFR
	High fluorescent ratio	HFR
	Mean corpuscular volume of reticulocyte	MCVr
	Mean corpuscular hemoglobin of reticulocyte	MCHr
	Expression of MCHr - MCHm	DELTA-CH
RBC group	Red Blood Cell count	RBC
	Hemoglobin Concentration	HGB
	Mean Corpuscular Volume	MCV
	Mean Corpuscular Hemoglobin	MCH
	Mean Corpuscular Hemoglobin Concentration	MCHC

Groups	Name	Abbreviation
	Mean hemoglobin distribution width of red blood cell	HDW
	Percentage of mature red blood cell with a low corpuscular hemoglobin	HYPO-CH%
	Percentage of mature red blood cell with a high corpuscular hemoglobin	HYPER-CH%
	Mean corpuscular hemoglobin of mature red blood cell	MCHm
	Red Blood Cell Distribution Width - Coefficient of Variation	RDW-CV
	Red Blood Cell Distribution Width - Standard Deviation	RDW-SD
	Hematocrit	HCT
	Microcyte percentage	Micro%
	Macrocyte percentage	Macro%
	Nucleated red blood cell number	NRBC#
	Nucleated red blood cell percentage	NRBC%
Platelet group	Platelet count	PLT
	Optical platelet count	PLT-O
	Platelet count hybrid	PLT-H
	Platelet count- impedance	PLT-I
	Mean Platelet Volume	MPV
	Platelet Distribution Width	PDW
	Plateletcrit	PCT
	Immature Platelet Fraction	IPF
	Platelet-large cell ratio	P-LCR
	Platelet-large cell count	P-LCC
	Immature platelet count	IPF#
ESR	Erythrocyte sedimentation rate	ESR

NOTE

- Zhang et al.[8] reported that the area under the receiver operating characteristic curve (AUC) of Neu-X, Neu-Y, and Mon-X for the diagnosis of sepsis was 0.751 (sensitivity 76.1%, specificity 58.2%), 0.877 (87.3%, 72.1%) and 0.791 (77.6%, 65.9%) respectively. Among these three parameters, the Neu-Y had the best diagnostic performance in the diagnosis of sepsis.

Table B-4 Body fluid sample parameters

Groups	Parameter Name	Abbreviation
WBC group (6)	White Blood Cell count-body fluid	WBC-BF
	Total nucleated cell counts-body fluid	TC-BF#
	Mononuclear cell number	MN#
	Mononuclear cell percentage	MN%
	Polymorphonuclear cell number	PMN#
	Polymorphonuclear cell percentage	PMN%
RBC group (1)	Red Blood Cell count-body fluid	RBC-BF

B.4 Sampling Features

B.4.1 Sample Volumes Required for Each Analysis

Table B-5 Sample volumes required for each analysis

Presentation Mode	Required Volume
AL-WB (for tests except ESR tests)	≤200 µL
AL-WB (for ESR/ CBC+ESR/ CD+ESR/ CDR+ESR tests)	≤400 µL
OV-WB	≤150 µL
OV-BF	≤150 µL
OV-PD	≤40 µL

NOTE

- For certain countries such as Italy, France, German and Japan, the sample volume required for each ESR analysis is no more than 600 µL. For more information, contact Our Customer Service Department.

B.4.2 Minimum Sample Volume

To attain accurate analysis results, make sure the sample volume under various analysis modes meets the following requirements.

Table B-6 Minimum sample volume under various analysis modes

Presentation Mode	Required Volume
AL-WB (for tests except ESR tests)	≥1 mL
AL-WB (for ESR/ CBC+ESR/ CD+ESR/ CDR+ESR tests)	≥2 mL
OV-WB	≥0.5 mL
OV-BF	≥0.5 mL

B.4.3 Throughput

Table B-7 Throughput

Presentation Mode	Sample Type	Test Panels	Throughput (tests /hour)
Auto loading	Whole blood	CBC	200
		CD	200
		CDR	120
		CR	120
		Ret	120
		ESR	120
		CBC+ESR	120
		CD+ESR	120
		CDR+ESR	85
Open-vial	Whole blood	CD	120
	Body fluid	CD	40

NOTE

- For certain countries such as Italy, France, German and Japan, the whole blood samples throughput for ESR analysis (ESR, CBC+ESR and CD+ESR tests) is 70 tests/hour and that for CDR+ESR tests is 60 tests/hour; For more information, contact our Customer Service Department.

B.5 Performance Specifications

B.5.1 Background/Blank Count Requirements

Perform the background/blank test using the diluent on the analyzer for 3 times. The test results should meet the requirements shown in table below.

Reference: CLSI. *Validation, Verification and Quality Assurance of Automated Hematology Analyzers; Approved Standard-Second Edition*. CLSI document H26-A2. Wayne, PA: Clinical and Laboratory Standards Institute; 2010.

Table B-8 Background/blank count requirements for blood samples

Parameters	Acceptable Range
WBC	$\leq 0.10 \times 10^9 / L$
RBC	$\leq 0.02 \times 10^{12} / L$
HGB	$\leq 1 \text{ g/L}$
PLT	$\leq 3 \times 10^9 / L$
PLT-I	$\leq 3 \times 10^9 / L$
PLT-H	$\leq 3 \times 10^9 / L$
PLT-O	$\leq 3 \times 10^9 / L$

Table B-9 Background/blank count requirements for body fluid samples

Parameters	Acceptable Range
WBC-BF/ TC-BF#	$\leq 0.001 \times 10^9 / L$
RBC-BF	$\leq 0.003 \times 10^{12} / L$

B.5.2 Linearity Range

The linearity range refers to the analyzer's ability to provide results that are directly proportional to the concentration of the analyte in the sample within a specific range. To assess this, known levels of the analyte, either by formulation or relative to each other (not necessarily know absolute), can be used for measurement. When the system's results are plotted against these values, the extent to which the resulting curve approximates a straight line is a measure of the system's linearity.

Reference: CLSI. *Evaluation of Linearity of Quantitative Measurement Procedures*. 2nd ed. CLSI guideline EP06. Clinical and Laboratory Standards Institute; 2020.

Table B-10 Linearity requirements for blood samples

Parameters	Range	Deviation Range (Whole Blood Mode)	Deviation Range (Predilute Mode)	Correlation r
WBC	$(0.00-100.00) \times 10^9 / L$	$\pm 0.20 \times 10^9 / L$ or $\pm 2\%$	$\pm 0.50 \times 10^9 / L$ or $\pm 5\%$	≥ 0.990
	$(100.01-350.00) \times 10^9 / L$	$\pm 6\%$	$\pm 6\%$	≥ 0.990
	$(350.01-500.00) \times 10^9 / L$	$\pm 11\%$	$\pm 11\%$	≥ 0.990
RBC	$(0.00-8.60) \times 10^{12} / L$	$\pm 0.03 \times 10^{12} / L$ or $\pm 2\%$	$\pm 0.05 \times 10^{12} / L$ or $\pm 5\%$	≥ 0.990
HGB	$(0-260) g / L$	$\pm 2g / L$ or $\pm 2\%$	$\pm 2g / L$ or $\pm 3\%$	≥ 0.990
HCT	$(0.0-75.0)\%$	$\pm 1.0 HCT$ or $\pm 2\%$	$\pm 2.0 HCT$ or $\pm 4\%$	/
PLT	$(0-1000) \times 10^9 / L$	$\pm 10 \times 10^9 / L$ or $\pm 5\%$	$\pm 10 \times 10^9 / L$ or $\pm 10\%$	≥ 0.990
	$(1001-5000) \times 10^9 / L$	$\pm 6\%$	$\pm 10\%$	≥ 0.990
RET%	$(0.00-30.00)\%$	$\pm 0.3 RET\%$ or $\pm 20\%$	/	/
RET#	$(0.0000-0.8000) \times 10^{12} / L$	$\pm 0.0150 \times 10^{12} / L$ or $\pm 20\%$	/	/
NRBC%	$(0.00-600.00) / 100WBC$	$\pm 2.00 NRBC\%$ or $\pm 10\%$	/	/
NRBC#	$(0.000-20.000) \times 10^9 / L$	$\pm 0.200 \times 10^9 / L$ or $\pm 10\%$	/	/

Table B-11 Linearity requirements for body fluid samples

Parameters	Range	Deviation Range
WBC-BF/TC-BF#	$(0.000-0.050) \times 10^9 / L$	$\pm 0.010 \times 10^9 / L$
	$(0.051-1.000) \times 10^9 / L$	$\pm 20\%$
	$(1.001-10.000) \times 10^9 / L$	$\pm 20\%$
RBC-BF	$(0.000-0.100) \times 10^{12} / L$	$\pm 0.010 \times 10^{12} / L$ or $\pm 5\%$
	$(0.101-5.000) \times 10^{12} / L$	$\pm 0.030 \times 10^{12} / L$ or $\pm 2\%$

B.5.3 Repeatability

The analyzer can output the repeatability parameters after it runs the same sample test for at least 10 times. The repeatability is described according to CLSI EP05-A3.

Reference: CLSI EP05-A3. *Evaluation of Precision of Quantitative Measurement Procedures; Approved Guideline-Third Edition*.

Table B-12 Repeatability requirements for blood

Measurement parameters	Measuring ranges	Whole blood mode repeatability CV% / absolute deviation (d/SD)	Predilute mode repeatability CV% / absolute deviation (d/SD)
WBC	$(3.50-4.50) \times 10^9/L$	$\leq 3.0\%$	$\leq 4.0\%$
	$\geq 4.51 \times 10^9/L$	$\leq 2.5\%$	$\leq 3.5\%$
RBC	$\geq 3.50 \times 10^{12}/L$	$\leq 1.5\%$	$\leq 2.0\%$
HGB	$(110-180) g/L$	$\leq 1.0\%$, Range ≤ 3	$\leq 2.0\%$
MCV	$(80.0-100.0) fL$	$\leq 1.0\%$	$\leq 3.0\%$
HCT	$(30.0-50.0)\%$	$\leq 1.5\%$	$\leq 3.0\%$
MCH	/	$\leq 1.5\%$	$\leq 4.5\%$
MCHC	/	$\leq 1.5\%$	$\leq 6.0\%$
RDW-SD	/	$\leq 2.0\%$	$\leq 6.0\%$
RDW-CV	/	$\leq 2.0\%$	$\leq 6.0\%$
PLT	$\geq 100 \times 10^9/L$	$\leq 4.0\%$	$\leq 8.0\%$
		$\leq 2.5\%$ applicable to CDR/PLT-8X, PLT-O and CD/PLT-O tests	
	$(20-100) \times 10^9/L$	$\leq 8.0\%$ applicable to CD and CDR tests	/
		$\leq 5.0\%$ applicable to CDR/PLT-8X, PLT-O and CD/PLT-O tests	/
	$\leq 20 \times 10^9/L$	$\leq 1.5(SD)$ applicable to CDR/PLT-8X, PLT-O and CD/PLT-O tests	/
PDW	$PLT \geq 100 \times 10^9/L$	$\leq 10.0\%$	$\leq 20.0\%$
MPV	$PLT \geq 100 \times 10^9/L$	$\leq 3.0\%$	$\leq 8.0\%$
P-LCR	$PLT \geq 100 \times 10^9/L$	$\leq 15.0\%$	$\leq 36.0\%$
P-LCC	$PLT \geq 100 \times 10^9/L$	$\leq 15.0\%$	$\leq 36.0\%$
PCT	$PLT \geq 100 \times 10^9/L$	$\leq 5.0\%$	$\leq 12.0\%$
Neu%	Neu% $\geq 30.0\%$ WBC $\geq 4.00 \times 10^9/L$	$\leq 6.0\%$	$\leq 12.0\%$
Lym%	Lym% $\geq 15.0\%$ WBC $\geq 4.00 \times 10^9/L$	$\leq 6.0\%$	$\leq 12.0\%$
Mon%	Mon% $\geq 5.0\%$ WBC $\geq 4.00 \times 10^9/L$	$\leq 16.0\%$	$\leq 32.0\%$
Eos%	WBC $\geq 4.00 \times 10^9/L$	$\leq 20.0\%$ or $\pm 1.5\%(d)$	$\leq 40.0\%$ or $\pm 3.0\%(d)$
Bas%	WBC $\geq 4.00 \times 10^9/L$	$\leq 30.0\%$ or $\pm 1.0\%(d)$	$\leq 60.0\%$ or $\pm 2.0\%(d)$
NRBC%	WBC $\geq 4.00 \times 10^9/L$	$\leq 20.0\%$ or $\pm 1.50\%(d)$	$\leq 50.0\%$ or $\pm 3.00\%(d)$
Neu#	$\geq 1.20 \times 10^9/L$	$\leq 6.0\%$	$\leq 12.0\%$
Lym#	$\geq 0.60 \times 10^9/L$	$\leq 6.0\%$	$\leq 12.0\%$
Mon#	$\geq 0.20 \times 10^9/L$	$\leq 16.0\%$	$\leq 32.0\%$
Eos#	WBC $\geq 4.00 \times 10^9/L$	$\leq 20.0\%$ or $\pm 0.12 \times 10^9/L(d)$	$\leq 40.0\%$ or $\pm 0.24 \times 10^9/L(d)$

Measurement parameters	Measuring ranges	Whole blood mode repeatability CV% / absolute deviation (d/SD)	Predilute mode repeatability CV% / absolute deviation (d/SD)
Bas#	WBC $\geq 4.00 \times 10^9/L$	$\leq 30.0\%$ or $\pm 0.06 \times 10^9/L(d)$	$\leq 60.0\%$ or $\pm 0.12 \times 10^9/L(d)$
IMG%	WBC $\geq 4.00 \times 10^9/L$, IMG% $\geq 2\%$	$\leq 25.0\%$ or $\pm 1.5\%(d)$	$\leq 75.0\%$ or $\pm 4.5\%(d)$
IMG#	$\geq 0.10 \times 10^9/L$	$\leq 25.0\%$ or $\pm 0.12 \times 10^9/L(d)$	$\leq 75.0\%$ or $\pm 0.36 \times 10^9/L(d)$
NRBC#	WBC $\geq 4.00 \times 10^9/L$	$\leq 20.0\%$ or $\pm 0.12 \times 10^9/L(d)$	$\leq 50.0\%$ or $\pm 0.25 \times 10^9/L(d)$
RET#	RBC $\geq 3.00 \times 10^{12}/L$, RET%:1.00%–4.00%	$\leq 15\%$	$\leq 30\%$
RET%	RBC $\geq 3.00 \times 10^{12}/L$, RET%: 1.00%–4.00%	$\leq 15\%$	$\leq 30\%$
RHE	RET# $\geq 0.02 \times 10^{12}/L$	$\leq 5\%$	/
LFR	RBC $\geq 3.00 \times 10^{12}/L$, RET%:1.00%–4.00% LFR $\geq 20\%$	$\leq 30\%$	/
MFR	RBC $\geq 3.00 \times 10^{12}/L$, RET%:1.00%–4.00% MFR $\geq 20\%$	$\leq 50\%$	/
HFR	RBC $\geq 3.00 \times 10^{12}/L$, RET%:1.00%–4.00%	$\leq 100\%$ or $\pm 2.0\%(d)$	/
IRF	RBC $\geq 3.00 \times 10^{12}/L$, RET%:1.00%–4.00% IRF $\geq 20\%$	$\leq 30\%$	/
IPF	PLT $\geq 50 \times 10^9/L$, IPF $\geq 3\%$	$\leq 25\%$	$\leq 40\%$
MCVr	MCVr:100.0fL–120.0fL RET# $\geq 0.02 \times 10^{12}/L$	$\leq 5\%$	/
MCHr	RET# $\geq 0.02 \times 10^{12}/L$	$\leq 2\%$	/
HDW	/	$\leq 5\%$	/
DELTA-CH	DELTA-CH $\geq 1pg$	$\leq 25\%$	/
HYPO-CH%	/	$\leq 25.0\%$ or $\pm 1.0\%(d)$	/
HYPER-CH%	/	$\leq 25.0\%$ or $\pm 1.0\%(d)$	/
Micro%	/	$\leq 15.0\%$ or $\pm 1.0\%(d)$	/
Macro%	/	$\leq 15.0\%$ or $\pm 1.0\%(d)$	/
MCHm	/	$\leq 1.5\%$	/
IPF#	PLT $\geq 50 \times 10^9/L$, IPF $\geq 3\%$	$\leq 25.0\%$	$\leq 40\%$
Neu-X	/	$\leq 5\%$	/
Neu-Y	/	$\leq 5\%$	/
Mon-X	/	$\leq 5\%$	/
ESR	0–15mm/h	$\leq 1.35(SD)$	/
	$> 15mm/h$	$\leq 9\%(CV)$	/

1. Absolute deviation d = MAX |Measured value - measured mean|
2. Range = Maximum measurement value – Minimum measurement value

Table B-13 Repeatability requirements for body fluid samples

Measurement parameters	Measuring ranges	Repeatability CV% / absolute deviation (d)
WBC-BF/TC-BF#	$(0.015-0.100) \times 10^9/L$	$\leq 30\%$
RBC-BF	$(0.003-0.050) \times 10^{12}/L$	$\leq 40\%$ or $\text{Range} \leq 7000/\mu L$

B.5.4 Carryover

The carryover is described according to CLSI Standard H26-A2.

Reference: CLSI. *Validation, Verification and Quality Assurance of Automated Hematology Analyzers; Approved Standard-Second Edition*. CLSI document H26-A2. Wayne, PA: Clinical and Laboratory Standards Institute; 2010.

■ For whole blood samples and pre-diluent samples

The carryover is calculated by testing a parameter from a high level blood sample for three times followed by a parameter from a low level blood sample.

The carryover is calculated as follows:

$$\text{Carryover} = \frac{|\text{1st Low} - \text{3rd Low}|}{\text{3rd High} - \text{3rd Low}} \times 100\%$$

Table B-14 Carryover requirements for blood samples

Parameters	Carryover
WBC	$\leq 1.0\%$
RBC	$\leq 1.0\%$
HGB	$\leq 1.0\%$
HCT	$\leq 1.0\%$
PLT	$\leq 1.0\%$
PLT-I	$\leq 1.0\%$
PLT-H	$\leq 1.0\%$
PLT-O	$\leq 1.0\%$

■ For body fluid samples

The carryover is calculated by testing a parameter from a high level body fluid sample for three times followed by a parameter from a low level body fluid sample or diluent.

Table B-15 Carryover requirements for body fluid samples

Measurement parameters	Carryover
WBC-BF/TC-BF#	$\leq 0.3\%$ or $\leq 0.001 \times 10^9/L$
RBC-BF	$\leq 0.3\%$ or $\leq 0.003 \times 10^{12}/L$

B.5.5 Correlation

The correlation is evaluated by comparing the results from the BC-6800Plus and BC-7600 series analyzers.

To determine whether the tested analyzer (BC-6800Plus analyzer) meets the requirements, samples covering the entire measuring range are analyzed on both the BC-6800Plus and BC-7600 series analyzers. The tested analyzer is considered to meet the standards if its results fall within the specified limits shown in the table below.

Reference: CLSI. *Measurement Procedure Comparison and Bias Estimation Using Patient Samples*. 3rd ed. CLSI guideline EP09c. Wayne, PA: Clinical and Laboratory Standards Institute; 2018.

■ For whole blood sample tests

Table B-16 Correlation requirements for blood samples

Parameters	Correlation with Reference Instrument	Deviation Range
WBC	≥ 0.99	$\pm 0.2 \times 10^9/\text{L}$ or $\pm 3\%$
Neu%	≥ 0.90	± 3.0 Neu% or $\pm 10\%$
Lym%	≥ 0.90	± 3.0 Lym% or $\pm 10\%$
Mon%	≥ 0.75	± 2.0 Mon% or $\pm 10\%$
Eos%	≥ 0.80	± 1.0 Eos% or $\pm 10\%$
Bas%	≥ 0.50	± 1.0 Bas% or $\pm 10\%$
RBC	≥ 0.99	$\pm 0.03 \times 10^{12}/\text{L}$ or $\pm 2\%$
HGB	≥ 0.98	± 2 g/L or $\pm 2\%$
MCV	≥ 0.98	± 2.0 fL or $\pm 3\%$
HCT	≥ 0.95	± 1.0 HCT or $\pm 3\%$
RDW-CV	≥ 0.85	± 0.5 RDW-CV or $\pm 5\%$
RDW-SD	≥ 0.85	± 3.0 fL or $\pm 5\%$
PLT	≥ 0.95	$\pm 10 \times 10^9/\text{L}$ or $\pm 5\%$
MPV	≥ 0.80	± 1.0 fL or $\pm 5\%$
NRBC%	≥ 0.90	± 2.0 NRBC% or $\pm 10\%$
IMG%	≥ 0.80	± 0.5 IMG% or $\pm 10\%$
RET%	≥ 0.90	± 0.3 RET% or $\pm 20\%$
RET#	≥ 0.90	$\pm 0.015 \times 10^{12}/\text{L}$ or $\pm 20\%$
RHE	≥ 0.90	± 3 pg/ $\pm 10\%$
IPF	≥ 0.80	± 2 IPF or $\pm 30\%$
MCVr	≥ 0.80	± 10 fL or $\pm 20\%$
MCHr	≥ 0.85	± 5 pg or $\pm 5\%$
HDW	≥ 0.70	± 15 g/L or $\pm 50\%$
DELTA-CH	≥ 0.80	± 2 pg or $\pm 40\%$
HYPO-CH%	≥ 0.70	± 1 HYPO-CH% or $\pm 50\%$
HYPER-CH%	≥ 0.70	± 0.5 HYPER-CH% or $\pm 100\%$
Micro%	≥ 0.90	± 1 Micro% or $\pm 40\%$
Macro%	≥ 0.90	± 0.3 Macro% or $\pm 40\%$
MCHm	≥ 0.80	± 2 pg or $\pm 5\%$
ESR	≥ 0.90	± 8 mm/h or $\pm 30\%$
Neu-X	≥ 0.80	$\pm 10\%$
Neu-Y	≥ 0.80	$\pm 20\%$
Mon-X	≥ 0.60	$\pm 40\%$
IRF	≥ 0.80	± 10 IRF or $\pm 30\%$
LFR	≥ 0.80	± 10 LFR or $\pm 10\%$
MFR	≥ 0.60	± 6 MFR or $\pm 40\%$
HFR	≥ 0.60	± 3 HFR or $\pm 100\%$

Parameters	Correlation with Reference Instrument	Deviation Range
PDW	≥0.80	±4 PDW or ±20%
PCT	≥0.80	±0.2 PCT or ±10%
P-LCR	≥0.80	±5 P-LCR or ±30%
P-LCC	≥0.80	±10×10 ⁹ /L or ±30%
IPF#	≥0.80	±4×10 ⁹ /L or ±50%

■ For body fluid sample tests

Table B-17 Correlation requirements for body fluid samples

Parameters	Correlation with Reference Instrument (Correlation Coefficient or Slope)
WBC-BF/TC-BF#	≥0.90 and slope between 0.7-1.3
RBC-BF	≥0.80 and slope between 0.7-1.3
MN#	≥0.90 and slope between 0.5-1.5
MN%	≥0.70 and slope between 0.5-1.5
PMN#	≥0.90 and slope between 0.5-1.5
PMN%	≥0.70 and slope between 0.5-1.5

B.5.6 Accuracy

B.5.6.1 Comparing with the Reference Method

Analyze 3 fresh blood samples with assigned-values (derived by reference method) on an analyzer within its calibration validity period. Calculate the deviation between the test results and the target values of each sample using the formula as shown below.

$$B_i = [(X_i - T) / T] \times 100\%$$

Whereas:

- B_i stands for relative deviation;
- X_i stands for test results;
- T stands for target value of fresh blood sample;

The results meet the requirements if the deviations fall within the limits shown in table below.

Reference: CLSI. *Measurement Procedure Comparison and Bias Estimation Using Patient Samples*. 3rd ed. CLSI guideline EP09c. Wayne, PA: Clinical and Laboratory Standards Institute; 2018.

Table B-18 Accuracy requirements

Parameters	Measuring Range	Deviation Range
WBC	3.50×10 ⁹ /L–9.50×10 ⁹ /L	±10.0%
RBC	3.80×10 ¹² /L–5.80×10 ¹² /L	±6.0%
HGB	115g/L–175 g/L	±6.0%
PLT	125×10 ⁹ /L–350×10 ⁹ /L	±20.0%
HCT/MCV	35.0%–50.0% (HCT) or 82.0 fL–100.0 fL (MCV)	±9.0 (HCT) or ±7.0 fL (MCV)

B.5.6.2 Comparing with the Reference Instrument

Analyze at least 100 samples on the tested analyzer and on a reference instrument, and the difference between the measured values and values measured by a reference instrument shall meet the requirements as indicated in the table below.

Reference: CLSI. *Measurement Procedure Comparison and Bias Estimation Using Patient Samples*. 3rd ed. CLSI guideline EP09c. Wayne, PA: Clinical and Laboratory Standards Institute; 2018.

Table B-19 Accuracy requirements for whole blood samples

Parameters	Correlation	Deviation Range
WBC	≥0.99	±0.2×10 ⁹ /L or ±3%
Neu%	≥0.90	±3.0 Neu% or ±10%
Lym%	≥0.90	±3.0 Lym% or ±10%
Mon%	≥0.75	±2.0 Mon% or ±10%
Eos%	≥0.80	±1.0 Eos% or ±10%
Bas%	≥0.50	±1.0 Bas% or ±10%
IMG%	≥0.80	±0.5 IMG% or ±10%
RBC	≥0.99	±0.03×10 ¹² /L or ±2%
HGB	≥0.98	±2 g/L or ±2%
HCT	≥0.95	±1.0 HCT or ±3%
MCV	≥0.98	±2.0 fL or ±3%
RDW-CV	≥0.85	±0.5 RDW-CV or ±5%
RDW-SD	≥0.85	±3.0 fL or ±5%
NRBC%	≥0.90	±2.0 NRBC% or ±10%
PLT	≥0.95	±10×10 ⁹ /L or ±5%
MPV	≥0.80	±1.0 fL or ±5%
PCT	≥0.80	±0.2 PCT or ±10%
IPF	≥0.80	±2 IPF or ±30%
IPF#	≥0.80	±4×10 ⁹ /L or ±50%
PLT-O	≥0.95	±10×10 ⁹ /L or ±5%
PLT-H	≥0.95	±10×10 ⁹ /L or ±5%
PLT-I	≥0.95	±10×10 ⁹ /L or ±5%
RET%	≥0.90	±0.3 RET% or ±20%
RET#	≥0.90	±0.015×10 ¹² /L or ±20%
RHE	≥0.90	±3 pg or ±10%
IRF	≥0.80	±10 IRF or ±30%
LFR	≥0.80	±10 LFR or ±10%
MFR	≥0.60	±6 MFR or ±40%
HFR	≥0.60	±3 HFR or ±100%
HDW	≥ 0.70	±15 g/L or ±50%
HYPO-CH%	≥ 0.70	±1 HYPO-CH% or ±50%
HYPER-CH%	≥ 0.70	±0.5 HYPER-CH% or ±100%

Parameters	Correlation	Deviation Range
Micro%	≥ 0.90	± 1 Micro% or $\pm 40\%$
Macro%	≥ 0.90	± 0.3 Macro% or $\pm 40\%$
MCVr	≥ 0.80	± 10 fL or $\pm 20\%$
Mon%	≥ 0.75	± 2.0 Mon% or $\pm 10\%$
Neu-X	≥ 0.80	$\pm 10\%$
Neu-Y	≥ 0.80	$\pm 20\%$
Mon-X	≥ 0.60	$\pm 40\%$
PDW	≥ 0.80	± 4 PDW or $\pm 20\%$
P-LCC	≥ 0.80	$\pm 10 \times 10^9$ /L or $\pm 30\%$
P-LCR	≥ 0.80	± 5 P-LCR or $\pm 30\%$
DELTA-CH	≥ 0.80	± 2 pg or $\pm 40\%$
MCHr	≥ 0.85	± 5 pg or $\pm 5\%$
MCHm	≥ 0.80	± 2 pg or $\pm 5\%$
ESR	≥ 0.90	± 8.0 mm/h or $\pm 30\%$

Table B-20 Accuracy requirements for body fluid samples

Parameters	Correlation	Deviation Range
WBC-BF	≥ 0.90	Slope between 0.7–1.3
RBC-BF	≥ 0.80	Slope between 0.7–1.3
TC-BF#	≥ 0.90	Slope between 0.7–1.3
MN#	≥ 0.90	Slope between 0.5–1.5
PMN#	≥ 0.90	Slope between 0.5–1.5
MN%	≥ 0.70	Slope between 0.5–1.5
PMN%	≥ 0.70	Slope between 0.5–1.5

B.5.7 Precision

The precision is indicated as coefficients of variation (CV) shown as table below, when controls samples are measured according to CLSI EP05-A3.

Reference: CLSI EP05-A3. *Evaluation of Precision of Quantitative Measurement Procedures; Approved Guideline-Third Edition.*

Table B-21 Precision requirements for controls(blood sample parameters)

Parameters	CV (Low-Level Control)	CV (Normal-Level Control)	CV (High-Level Control)
WBC	$\leq 10\%$	$\leq 6\%$	$\leq 5\%$
Neu%	$\leq 15\%$	$\leq 10\%$	$\leq 10\%$
Lym%	$\leq 20\%$	$\leq 10\%$	$\leq 10\%$
Mon%	$\leq 40\%$	$\leq 30\%$	$\leq 20\%$
Eos%	$\leq 40\%$	$\leq 30\%$	$\leq 30\%$
Baso%	$\leq 50\%$	$\leq 50\%$	$\leq 50\%$

Parameters	CV (Low-Level Control)	CV (Normal-Level Control)	CV (High-Level Control)
NRBC%	≤ 40%	≤ 40%	≤ 40%
Neu#	≤ 15%	≤ 10%	≤ 10%
Lym#	≤ 20%	≤ 10%	≤ 10%
Mon#	≤ 40%	≤ 30%	≤ 20%
Eos#	≤ 40%	≤ 30%	≤ 30%
Baso#	≤ 50%	≤ 50%	≤ 50%
NRBC#	≤ 40%	≤ 40%	≤ 40%
IMG%	≤ 20%	≤ 10%	≤ 10%
IMG#	≤ 20%	≤ 10%	≤ 10%
Neu-X	≤ 20%	≤ 15%	≤ 15%
Neu-Y	≤ 30%	≤ 30%	≤ 30%
Mon-X	≤ 40%	≤ 40%	≤ 40%
RBC	≤ 4%	≤ 4%	≤ 4%
HGB	≤ 4%	≤ 3%	≤ 3%
HCT	≤ 5%	≤ 5%	≤ 5%
MCV	≤ 5%	≤ 5%	≤ 5%
MCH	≤ 6%	≤ 6%	≤ 6%
MCHC	≤ 10%	≤ 10%	≤ 10%
RDW-CV	≤ 6%	≤ 6%	≤ 6%
RDW-SD	≤ 6%	≤ 6%	≤ 6%
Macro%	≤ 30%	≤ 30%	≤ 30%
Micro%	≤ 30%	≤ 30%	≤ 30%
PLT	≤ 20%	≤ 10%	≤ 10%
MPV	≤ 10%	≤ 5%	≤ 5%
PDW	≤ 10%	≤ 5%	≤ 5%
PCT	≤ 30%	≤ 15%	≤ 15%
P-LCR	≤ 30%	≤ 15%	≤ 15%
P-LCC	≤ 30%	≤ 15%	≤ 15%
PLT-I	≤ 20%	≤ 10%	≤ 10%
PLT-H	≤ 20%	≤ 10%	≤ 10%
IPF-D	≤ 30%	≤ 30%	≤ 30%
ESR	≤ 15%	≤ 10%	/
RET%	≤ 20%	≤ 20%	≤ 20%
RET#	≤ 20%	≤ 20%	≤ 20%
IRF	≤ 40%	≤ 40%	≤ 40%
LFR	≤ 30%	≤ 30%	≤ 30%
MFR	≤ 30%	≤ 30%	≤ 30%
HFR	≤ 30%	≤ 30%	≤ 30%

Parameters	CV (Low-Level Control)	CV (Normal-Level Control)	CV (High-Level Control)
RHE	≤ 10%	≤ 10%	≤ 10%
IPF	≤ 40%	≤ 40%	≤ 40%
IPF#	≤ 20%	≤ 20%	≤ 20%
PLT-O	≤ 20%	≤ 10%	≤ 10%
MCHr	≤ 10%	≤ 10%	≤ 10%
HDW	≤ 20%	≤ 20%	≤ 20%
MCVr	≤ 10%	≤ 10%	≤ 10%
MCHm	≤ 10%	≤ 10%	≤ 10%
DELTA-CH	≤ 40%	≤ 40%	≤ 40%
HYPER-CH%	≤ 50%	≤ 50%	≤ 50%
HYPO-CH%	≤ 50%	≤ 50%	≤ 50%

Table B-22 Precision requirements for controls (body fluid sample parameters)

Parameters	CV (Low-Level Control)	CV (Normal-Level Control)	CV (High-Level Control)
WBC-BF	≤ 35%	≤ 25%	≤ 15%
TC-BF#	≤ 35%	≤ 25%	≤ 15%
PMN%	≤ 60%	≤ 60%	≤ 60%
MN%	≤ 60%	≤ 60%	≤ 60%
PMN#	≤ 60%	≤ 60%	≤ 60%
MN#	≤ 60%	≤ 60%	≤ 60%
RBC-BF	≤ 35%	≤ 25%	≤ 15%

B.5.8 Mode Comparison

To perform comparison tests between different modes, the samples are analyzed on the analyzer with different sample modes.

A linear regression analysis was performed to assess the comparability of various analysis modes on one BC-6800Plus analyzer.

Reference: CLSI. *Validation, Verification and Quality Assurance of Automated Hematology Analyzers; Approved Standard-Second Edition*. CLSI document H26-A2. Wayne, PA: Clinical and Laboratory Standards Institute; 2010.

Table B-23 Comparison requirements

Parameters	Deviation Range between OV-WB Mode and AL-WB Mode	Deviation Range between OV-WB Mode and OV-PD Mode
WBC	$\pm 0.2 \times 10^9/\text{L}$ or $\pm 3\%$	$\pm 0.8 \times 10^9/\text{L}$ or $\pm 10\%$
Neu%	± 3.0 Neu% or $\pm 10\%$	± 9.0 Neu%
Lym%	± 3.0 Lym% or $\pm 10\%$	± 9.0 Lym%
Mon%	± 2.0 Mon% or $\pm 10\%$	± 6.0 Mon%
Eos%	± 1.0 Eos% or $\pm 10\%$	± 3.0 Eos%
Bas%	± 1.0 Bas% or $\pm 10\%$	± 3.0 Bas%
RBC	$\pm 0.03 \times 10^{12}/\text{L}$ or $\pm 2\%$	$\pm 0.20 \times 10^{12}/\text{L}$ or $\pm 4\%$

Parameters	Deviation Range between OV-WB Mode and AL-WB Mode	Deviation Range between OV-WB Mode and OV-PD Mode
HGB	±2 g/L or ±2%	±6 g/L or ±4%
MCV	±2.0 fL or ±2%	±4.0 fL or ±4%
HCT	±1.0 HCT or ±3%	±0.6 HCT or ±4%
RDW-CV	±0.5 RDW-CV or ±5%	±1.0 RDW-CV or ±5%
RDW-SD	±3.0 fL or ±5%	±1.5 fL or ±5%
PLT	±10×10 ⁹ /L or ±5%	±30×10 ⁹ /L or ±14%
MPV	±1.0 fL or ±5%	±1.0 fL or ±10%
IPF	±0.5 IPF or ±30%	±2 IPF or ±80%
RET%	±0.3 RET% or ±20%	±0.45 RET% or ±30%
IRF	±10.0 IRF or ±30%	±10.0 IRF or ±30%
RHE	±3.0 pg or ±10%	±3.0 pg or ±10%
IMG%	±0.5 IMG% or ±10%	±1.0 IMG% or ±10%
NRBC%	±2.0 NRBC% or ±10%	±2.0 NRBC% or ±10%
Neu-X	±10%	±10%
Neu-Y	±20%	±20%
Mon-X	±40%	±40%
LFR	±10 LFR or ±10%	±10 LFR or ±10%
MFR	±6 MFR or ±40%	±6 MFR or ±40%
HFR	±3 HFR or ±100%	±3 HFR or ±100%
MCVr	±20 fL or 20%	±20 fL or 20%
DELTA-CH	±2 pg or 40%	±2 pg or 40%
MCHr	±5 pg or ±5%	±5 pg or ±5%
HDW	±15 g/L or ±50%	±15 g/L or ±50%
MCHm	±2 pg or ±5%	±2 pg or ±5%
HYPO-CH%	±1 HYPO-CH% or ±50%	±1 HYPO-CH% or ±50%
HYPER-CH%	±0.5 HYPER-CH% or ±100%	±0.5 HYPER-CH% or ±100%
Micro%	±1 Micro% or ±40%	±1 Micro% or ±40%
Macro%	±0.3 Macro% or ±40%	±0.3 Macro% or ±40%
PDW	±4 PDW or ±20%	±4 PDW or ±20%
PCT	±0.2 PCT or ±10%	±0.2 PCT or ±10%
P-LCR	±5 P-LCR or ±30%	±5 P-LCR or ±30%
P-LCC	±10×10 ⁹ /L or ±30%	±10×10 ⁹ /L or ±30%
IPF#	±4×10 ⁹ /L or ±50%	±4×10 ⁹ /L or ±50%
ESR	±8 mm/h or ±30%	/

B.5.9 Sample Stability

For samples to be tested, the sample stability is evaluated by comparing the results of the first analysis (within two hours after collection) with the results from samples stored at controlled room temperature (18°C–26°C) for 24 hours and refrigerated temperature (2°C–8°C) for 48 hours.

Reference: *International Council for Standardization in Hematology, Writing Group; ICSH guidelines for the evaluation of blood cell analysers including those used for differential leucocyte and reticulocyte counting. Int J Lab Hematol. 2014 Dec;36(6):613-27.*

Table B-24 Sample stability

Parameters	Deviation Range
WBC	±5%
Neu%	±5 Neu%
Lym%	±4 Lym%
Mon%	±3 Mon%
Eos%	±2 Eos%
Bas%	±1 Bas%
RBC	±3%
HGB	±3%
MCV	±6%
HCT	±7%
PLT	±10%
MPV	±15%
RET%	±0.3 RET% or ±20%
RHE	±8%
NRBC%	±0.5 NRBC%
IPF	±30.0% or ±4.0 IPF% (PLT ≥ 100 × 10 ⁹ /L, IPF ≥ 2.0%)
IMG%	±2 IMG%
Neu-X	±15%
Neu-Y	±15%
Mon-X	±15%
IRF	±10 IRF or ±30%
LFR	±10 LFR or ±30%
MFR	±10 MFR or ±30%
HFR	±5 HFR
MCVr	±8% (RET# ≥ 0.01 × 10 ¹² /L)
DELTA-CH	MCVr ≤ ±8%; MCHr ≤ ±8% RET# ≥ 0.01 × 10 ¹² /L
MCHr	±8% (RET# ≥ 0.01 × 10 ¹² /L)
HDW	±4 g/L or ±20%
MCHm	±8%
HYPO-CH%	±2 HYPO-CH%
HYPER-CH%	±1 HYPER-CH%

Parameters	Deviation Range
Micro%	±2 Micro% or ±30% (stored at room temperature for 8 hours and refrigerated temperature for 48 hours)
Macro%	±2 Macro% or ±30% (stored at room temperature for 8 hours and refrigerated temperature for 48 hours)
PDW	±5%
PCT	±20%
P-LCR	±10 P-LCR or ±40%
P-LCC	±20 P-LCC or ±40%
IPF #	±10 ×10 ⁹ /L or ±30%
RET#	±0.015×10 ¹² /L or ±30%
ESR	±3.0 mm/h or ±15% (stored at room temperature for 6 hours and refrigerated temperature for 24 hours)

NOTE

- If the sample is taken out from a refrigerator, place it at room temperature for 30 minutes to warm it up. And then mix it before analyzing it.
- If you collect or handle the sample improperly, the values may not fall within the above range.

B.5.10 Interference Study

Interfering substances studies for bilirubin (conjugated), bilirubin (unconjugated), and triglyceride were performed on the BC-6800Plus analyzers.

When tested at or below the specified concentrations, the following substances do not present significant interference to the results.

Reference: CLSI. *Interference Testing in Clinical Chemistry*. 3rd ed. CLSI guideline EP07. Wayne, PA: Clinical and Laboratory Standards Institute; 2018.

Table B-25 Interference study

Parameter	Bilirubin (conjugated) Interference (mg/dL)	Bilirubin (unconjugated) Interference (mg/dL)	Triglyceride Interference (mg/dL)
WBC	45	45	2041
RBC	45	45	2041
HGB	45	45	729
HCT	45	45	2041
RET%	45	45	1799
RET#	45	45	1553
IRF	45	45	2041
RHE	45	45	2041
PLT	45	45	2041
IPF	45	45	2041

B.5.10.1 Allowable Relative Deviation Range

The allowable relative deviation range refers to the allowable bias for the parameters tested. If the bias falls out of the allowable relative deviation range, this indicates an interference presents in the result.

For all parameters (except for RET%, RET#, IRF, RHE, and IPF) listed in the table below, the concentration indicating no presence of significant interference is determined by the allowable deviation range which derives from biological variation (CV%) of CLSI document H26-A2.

For RET%,RET#, IRF, RHE and IPF, the deviation range is determined by the display value limits of CV (normal-level control) corresponding to the normal range for human blood.

Table B-26 Allowable relative deviation range

Parameters	Units	Allowable Relative Deviation Range
WBC	$\times 10^9/L$	$\pm 19.6\%$
RBC	$\times 10^{12}/L$	$\pm 6.1\%$
HGB	g/L	$\pm 6.6\%$
HCT	%	$\pm 6.4\%$
RET%	%	$\pm 10\%$
RET#	$\times 10^{12}/L$	$\pm 10\%$
IRF	%	$\pm 40\%$
RHE	g/L	$\pm 10\%$
PLT	$\times 10^9/L$	$\pm 21.9\%$
IPF	%	$\pm 40\%$

B.5.11 Analytical Measuring Intervals

The analytical measuring interval refers to the range of measured quantities within which accurate measurements can be made with specified precision.

Specifically, the lower limit of linearity serves as the lower boundary of the measuring interval; while the upper limit of linearity serves as the upper boundary of the measuring interval.

Table B-27 Analytical measuring intervals

Parameter	Units	Analytical Measuring Interval	Display Range
WBC	$\times 10^9/L$	0.00–500.00	0.00–999.99
Neu%	%	N/A	0.0–100.0
Lym%	%	N/A	0.0–100.0
Mon%	%	N/A	0.0–100.0
Eos%	%	N/A	0.0–100.0
Bas%	%	N/A	0.0–100.0
IMG%	%	N/A	0.0–100.0
Neu#	$\times 10^9/L$	N/A	0.00–999.99
Lym#	$\times 10^9/L$	N/A	0.00–999.99
Mon#	$\times 10^9/L$	N/A	0.00–999.99
Eos#	$\times 10^9/L$	N/A	0.00–999.99
Bas#	$\times 10^9/L$	N/A	0.00–999.99
IMG#	$\times 10^9/L$	N/A	0.00–999.99
Neu-X	/	N/A	0.0–9999.9
Neu-Y	/	N/A	0.0–9999.9
Mon-X	/	N/A	0.0–9999.9

Parameter	Units	Analytical Measuring Interval	Display Range
RET%	%	N/A	0.0–100.0
RET#	$\times 10^{12}/L$	0.0000–0.8000	0.00–999.99
RHE	g/L	N/A	0.0–999.9
IRF	%	N/A	0.0–100.0
LFR	%	N/A	0.0–100.0
MFR	%	N/A	0.0–100.0
HFR	%	N/A	0.0–100.0
MCVr	fL	N/A	0.0–999.9
DELTA-CH	pg	N/A	–999.9–999.9
MCHr	pg	N/A	0.0–999.9
RBC	$\times 10^{12}/L$	0.00–8.60	0.00–99.99
HGB	g/L	0–260	0–999
MCV	fL	N/A	0.0–999.9
MCH	pg	N/A	0.0–999.9
MCHC	g/L	N/A	0–9999
RDW-CV	%	N/A	0.0–100.0
RDW-SD	fL	N/A	0.0–999.9
HCT	%	0.0–75.0	0.0–100.0
NRBC%	%	N/A	0.000–9999.999
NRBC#	$\times 10^9/L$	0.00–20.00	0.00–9999.99
HDW	g/L	N/A	0.0–999.9
MCHm	pg	N/A	0.0–999.9
HYPO-CH%	%	N/A	0.0–100.0
HYPER-CH%	%	N/A	0.0–100.0
Micro%	%	N/A	0.0–100.0
Macro%	%	N/A	0.0–100.0
PLT	$\times 10^9/L$	0–5000	0–9999
MPV	fL	N/A	0–25
PDW	/	N/A	0.0–99.9
PCT	%	N/A	0.0–100.0
P-LCR	%	N/A	0.0–100.0
P-LCC	$\times 10^9/L$	N/A	0–5000
IPF	%	N/A	0.0–100.0
IPF #	$\times 10^9/L$	N/A	0–100
PLT-O	$\times 10^9/L$	0–5000	0–9999
PLT-H	$\times 10^9/L$	0–5000	0–9999
PLT-I	$\times 10^9/L$	0–5000	0–9999
ESR	mm/h	0.00–140.00	0.00–999.99

Parameter	Units	Analytical Measuring Interval	Display Range
WBC-BF	$\times 10^9/L$	0.000–10.000	0.000–999.999
RBC-BF	$\times 10^{12}/L$	0.000–5.000	0.000–99.999
TC-BF#	$\times 10^9/L$	0.000–10.000	0.000–999.999
MN#	$\times 10^9/L$	N/A	0.000–999.999
PMN#	$\times 10^9/L$	N/A	0.000–999.999
MN%	%	N/A	0.0–100.0
PMN%	%	N/A	0.0–100.0

NOTE

- The display ranges indicate the allowable ranges shown in the software.

B.5.12 Flags of Sensitivity and Specificity

The general purpose of our auto hematology analyzer is to screen and report normal samples and to identify the abnormalities. Samples which fail to meet the conditions defined by the analyzer or criteria established by the laboratory are flagged. The table below lists the data collected from different sites.

Reference: International Council for Standardization in Hematology, Writing Group; ICSH guidelines for the evaluation of blood cell analyzers including those used for differential leucocyte and reticulocyte counting. *Int J Lab Hematol.* 2014 Dec;36(6):613-27.

Table B-28 Flags analysis-BC-6800Plus compared to BC-7600 series

Sites	Models	Agreement (Overall Efficiency)	Sensitivity	Specificity	PV% Positive	PV% Negative
Site 1 N=323	- BC-6800Plus - BC-7800	93.8%	90.8%	95.6%	92.3%	94.7%
Site 2 N=482	- BC-6800Plus - BC-7900	94.6%	95.3%	94.0%	93.7%	95.5%
Combined Sites N=805	- BC-6800Plus - BC-7800 and BC-7900	94.3%	93.8%	94.7%	93.2%	95.1%
Agreement Specifications	≥90%					
Note: Representative data or results in individual laboratories may vary.						

Calculations

Sensitivity indicates the analyzer's ability to output positive results which are also regarded as positive results by using a reference method, or the ability to identify abnormal samples.

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100\%$$

Specificity indicates the analyzer's ability to output negative results which are also regarded as negative results by using a reference method, or the ability to identify normal samples.

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100\%$$

Predicative value of a positive result means the proportion of samples with positive test results which are actually positive.

$$\text{PV\% Positive} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100\%$$

Predicative value of a negative result means the proportion of samples with negative test results which are actually negative.

$$\text{PV\% Negative} = \frac{\text{TN}}{\text{TN} + \text{FN}} \times 100\%$$

Efficiency (or Agreement) indicates the general performance of the analyzer in identifying normal and abnormal samples.

$$\text{Efficiency} = \frac{\text{TP} + \text{TN}}{\text{Total number}} \times 100\%$$

B.5.13 Reference Intervals

Reference intervals were obtained from normal population. Each parameter is calculated for 95% confidence intervals. The table below shows the normal population reference ranges.

Reference: CLSI. *Defining, Establishing, and Verifying Reference Intervals in the Clinical Laboratory; Approved Guideline-Third Edition*. CLSI document C28-A3. Wayne, PA: Clinical and Laboratory Standards Institute; 2008.

Table B-29 Reference intervals

Parameter	Gender	Units	Range
WBC	Male/Female	$\times 10^9/\text{L}$	3.56-9.88
Neu%	Male/Female	%	39.4-69.3
Lym%	Male/Female	%	19.4-52.9
Mon%	Male/Female	%	3.4-10.4
Eos%	Male/Female	%	0.7-9.3
Baso%	Male/Female	%	0.1-1.5
NRBC%	Male/Female	%	0.0-0.0
Neu#	Male/Female	$\times 10^9/\text{L}$	1.88-6.27
Lym#	Male/Female	$\times 10^9/\text{L}$	1.03-3.51
Mon#	Male/Female	$\times 10^9/\text{L}$	0.16-0.70
Eos#	Male/Female	$\times 10^9/\text{L}$	0.04-0.54
Baso#	Male/Female	$\times 10^9/\text{L}$	0.01-0.08
NRBC#	Male/Female	$\times 10^9/\text{L}$	0.000-0.000
IMG%	Male/Female	%	0.0-0.7
IMG#	Male/Female	$\times 10^9/\text{L}$	0.00-0.05
RBC	Male	$\times 10^{12}/\text{L}$	4.4-5.56
	Female		3.99-5.05
HCT	Male	%	42.2-52.2
	Female		35.8-46.6
HGB	Male	g/L	136-164
	Female		110-149
MCV	Male/Female	fL	75.8-102.3
MCH	Male/Female	pg	23.3-32.9

Parameter	Gender	Units	Range
MCHC	Male/Female	g/L	304.2-335.6
RDW-CV	Male/Female	%	11.7-15.3
RDW-SD	Male/Female	fL	36.5-46.4
RET%	Male/Female	%	0.8-3.4
RET#	Male/Female	$\times 10^{12}/L$	0.0399-0.1799
IRF	Male/Female	%	2.6-22.7
LFR	Male/Female	%	77.3-97.4
MFR	Male/Female	%	2.6-17.0
HFR	Male/Female	%	0.0-7.7
RHE	Male/Female	pg	25.3-36.5
PLT	Male/Female	$\times 10^9/L$	106-362
PLT-I	Male/Female	$\times 10^9/L$	109-347
PLT-O	Male/Female	$\times 10^9/L$	106-362
PLT-H	Male/Female	$\times 10^9/L$	92-337
MPV	Male/Female	fL	7.8-12.8
PDW	Male/Female	/	15.3-17.0
PCT	Male/Female	%	0.141-0.371
P-LCR	Male/Female	%	11.26-44.96
P-LCC	Male/Female	%	28-105
IPF	Male/Female	%	0.4-5.8
IPF#	Male/Female	$\times 10^9/L$	1.1-14.1
ESR	Male	mm/h	0-13.45
	Female		0-21.65
IPF-D	Male/Female	%	0.9-4.5
Macro%	Male/Female	%	1.6-3.2
Micro%	Male/Female	%	1.0-7.1
Neu-X	Male/Female	/	293.8-455.4
Neu-Y	Male/Female	/	416.9-530.8
Mon-X	Male/Female	/	164.0-224.6
MCHr	Male/Female	pg	24.9-37.3
HDW	Male/Female	g/L	12.8-29.8
MCVr	Male/Female	fL	81.7-111.6
MCHm	Male/Female	pg	22.5-34.7
DELTA-CH	Male/Female	pg	0.0-3.6
HYPER-CH%	Male/Female	%	0.0-2.9
HYPO-CH%	Male/Female	%	0.1-1.0

NOTE

- Representative results in individual laboratories may vary.

- According to the CLSI Document C28-A3c *“Defining, Establishing, and Verifying Reference Intervals in the Clinical Laboratory; Approved Guideline-Third Edition”*, since the blood cell analysis results are affected by factors such as gender, age, diet, geographical location, and race, it is recommended that clinical laboratories should establish expected reference ranges that comply with the local environment based on the general patient population.

B.5.14 Limitations

B.5.14.1 For Blood Routine Tests

Table B-30 Limitations

Parameter	Limitations
WBC	Where the following is present, the WBC may be reported falsely low: • WBC agglutination Where the following is present, the WBC may be reported falsely high: • Cryoglobulin • Fibrin • Lyse resistant RBC • Giant platelet • Platelet clump
RBC	Where the following is present, the RBC may be reported falsely low: • RBC cold agglutination • Microcytosis • RBC fragment Where the following is present, the RBC may be reported falsely high: • Giant platelets
HGB	Where the following is present, the HGB result may be reported falsely low: • Lyse resistant RBC • RBC cold agglutination Where the following is present, the HGB result may be reported falsely high: • Leukocytosis (> 100000/μL) • Hyperlipidemia
MCV	Where the following is present, the MCV result may be reported falsely low: • Giant platelet Where the following is present, the MCV result may be reported falsely high: • RBC fragments • Microcytosis • RBC agglutination • Spherocyte

Parameter	Limitations
PLT	Where the following is present, the PLT count may be reported falsely low: • Platelet clump • Giant platelet • Leukocytosis Where the following is present, the PLT count may be reported falsely high: • Microcytosis • RBC fragment • Cryoglobulin • Hyperlipidemia
RET#	Where the following is present, the RET count may be reported falsely high: • Howell-Jolly body • Giant Platelet • RBC agglutination • Malaria
NRBC#	Where the following is present, the NRBC count may be reported falsely high: • Lyse resistant RBC • Malaria

B.5.14.2 ESR Tests

Erythrocyte sedimentation rate (ESR) is not only a non-specific inflammation index, but also one of indicators of erythrocyte aggregation. The analyzer measures the erythrocyte sedimentation rate by detecting the signals from erythrocyte aggregation process. However, the analyzer has limitations when analyzing the following samples:

1. Samples from patients with plasma cell diseases (e.g., multiple myeloma). In these samples, the erythrocytes may have already accumulated to rouleaux formation; therefore the analyzer may not be able to detect the process of erythrocyte aggregation, and may produce wrong results. The analyzer will give an alarm and may shield the results. To ensure accurate ESR results of these samples, it is recommended to use the traditional Westphal method to measure such samples.
2. Samples in which red blood cell agglutination has occurred (e.g., patient's blood containing cold agglutinin, and red blood cells aggregates in vitro cold environment). For such samples, the analyzer may not be able to detect the process of erythrocyte aggregation, and may produce wrong results. The analyzer will give an alarm and may shield the results. To ensure accurate ESR results of these samples, it is recommended to use the traditional Westphal method to measure such samples.
3. Other samples that may have rouleaux formation, blood coagulation, and abnormal erythrocyte morphology. The analyzer may give wrong results. Doctors should report based on the comprehensive judgment taking in consideration of analyzer-provided flags and prompts, as well as other clinical information of the patients.
4. Presence of hyperlipidemia or leukocytosis may result in falsely high ESR result.

B.6 Input/Output Devices

The analyzer works with computers and printers.

Also, the analyzer can be connected to a USB flash drive, keyboard, mouse, and barcode scanner, as required.

WARNING

- **Be sure to use the specified devices only.**
- **External equipment connected to the analyzer and digital interfaces must be authorized and complied with relevant safety and EMC standards (e.g. IEC 60950 Safety of Information Technology Equipment Standard and CISPR 22 EMC of Information Technology Equipment Standard (Class B)). Any persons who connects additional equipment to the signal input or output ports and configures an IVD system, is responsible for ensuring that the system works normally and complies within the**

safety and EMC requirements. If you have any questions, consult the technical service department of your local representative.

B.6.1 Computer

- CPU: Intel(R) i5 or above
- RAM: 4GB or above
- Hard disk: 500GB or above
- USB ports: at least 4 USB ports
- Network adapter: at least 2 adapters
- Operating system: Microsoft Windows 10 64bit
- Recommended screen resolution: 1920×1080
- Support VGA, DP display interface

B.6.2 Keyboard (Optional)

USB keyboard

B.6.3 Mouse (Optional)

USB mouse.

B.6.4 External bar-code scanner (optional)

External USB barcode scanner (hand-held)

B.6.5 Printer (optional)

B.6.6 USB Device (Optional)

B.7 Interfaces

NOTE

- **The USB ports on the left side of the analyzer shall only be used to connect the peripheral devices specified in this manual. See Section B Input/Output Devices for details about supported devices and models.**

-
- 1 network port
 - 4 USB ports

B.8 Electronic Interface Specifications

NOTE

- **Connection of the electronic interface to an IT network that includes other equipment could result in previously unidentified risks to patients, users or third parties; the responsible organization should identify, analyse, evaluate and control these risks.**

	Specifications	
Communication formats (protocol) and relevant standards	USB ports	Type A interface, complied with USB 2.0 standard TCP/IP protocol bottom layer. DICOM/HL7 protocol application layer.
	Network ports	RJ45 interface, supporting wired network 10 M/100 M/1000M, and complied with technical standard IEEE802.3. NTP/SNTP Calibration protocol of TCP/IP.
Time synchronization	■ When the analyzer is not connected to the labXpert software, the service personnel enters the time information on the main unit; ■ When the analyzer is connected to the labXpert software, the main unit synchronizes time with the labXpert software every hour.	

B.9 Power supply

	Voltage	Frequency	Power
Analyzer	a.c. 110V/115V $\pm 10\%$	50/60Hz $\pm 2\text{Hz}$	500VA
	a.c. 220V/230V $\pm 10\%$	50/60Hz $\pm 2\text{Hz}$	500VA
Compressor	a.c. 110V/115V $\pm 10\%$	60Hz $\pm 2\text{Hz}$	600VA
	a.c. 220V/230V $\pm 10\%$	50Hz $\pm 2\text{Hz}$	450VA
		60Hz $\pm 2\text{Hz}$	300VA

B.10 Fuse

WARNING

- **Only install fuses of specified specification on the system.**

Pneumatic unit fuse specification:

	Voltage	Fuse	Power
Pneumatic unit	a.c. 110V/115V $\pm 10\%$	60Hz $\pm 2\text{Hz}$	T6.3AL 250V
	a.c. 220V/230V $\pm 10\%$	50Hz $\pm 2\text{Hz}$	T3.15AH250V
		60Hz $\pm 2\text{Hz}$	T3.15AH250V

B.11 EMC Description

The IVD device complies with the EMC standard IEC 61326-1 and IEC 61326-2-6.



WARNING

- The IVD MEDICAL EQUIPMENT complies with the emission and immunity requirements described in this part of IEC 61326.
- This equipment is not intended for use in residential environments and may not provide adequate protection to radio reception in such environments.
- This equipment is designed for use in a PROFESSIONAL HEALTHCARE FACILITY ENVIRONMENT. It is likely to perform incorrectly if used in a HOME HEALTHCARE ENVIRONMENT. If it is suspected that performance is affected by electromagnetic interference, correct operation may be restored by increasing the distance between the equipment and the source of the interference.
- The electromagnetic environment should be evaluated prior to operation of the device.
- Do not use this device in proximity to sources of strong electromagnetic radiation (e.g. unshielded intentional RF sources), as these can interfere with proper operation.

NOTE

- It is the manufacturer's responsibility to provide equipment electromagnetic compatibility information to the customer or user.
- It is the user's responsibility to ensure that a compatible electromagnetic environment for the equipment can be maintained in order that the equipment can be maintained in order that the device will perform as intended.
- The calculation formula to determine the separation distance between an IVD MEDICAL EQUIPMENT and a mobile phone is given by $d = 6/E \cdot \sqrt{P}$, where d is the minimum separation distance in metres, P is the maximum power in watts, and E is the immunity test level in V/m.

B.12 Sound Level

The maximum sound pressure level is not greater than 80dB(A), the daily noise exposure level ($L_{EX,8h}$) is not greater than 80dB(A), and the peak sound pressure level ($L_{pC,peak}$) is not greater than 135dB(C).

NOTE

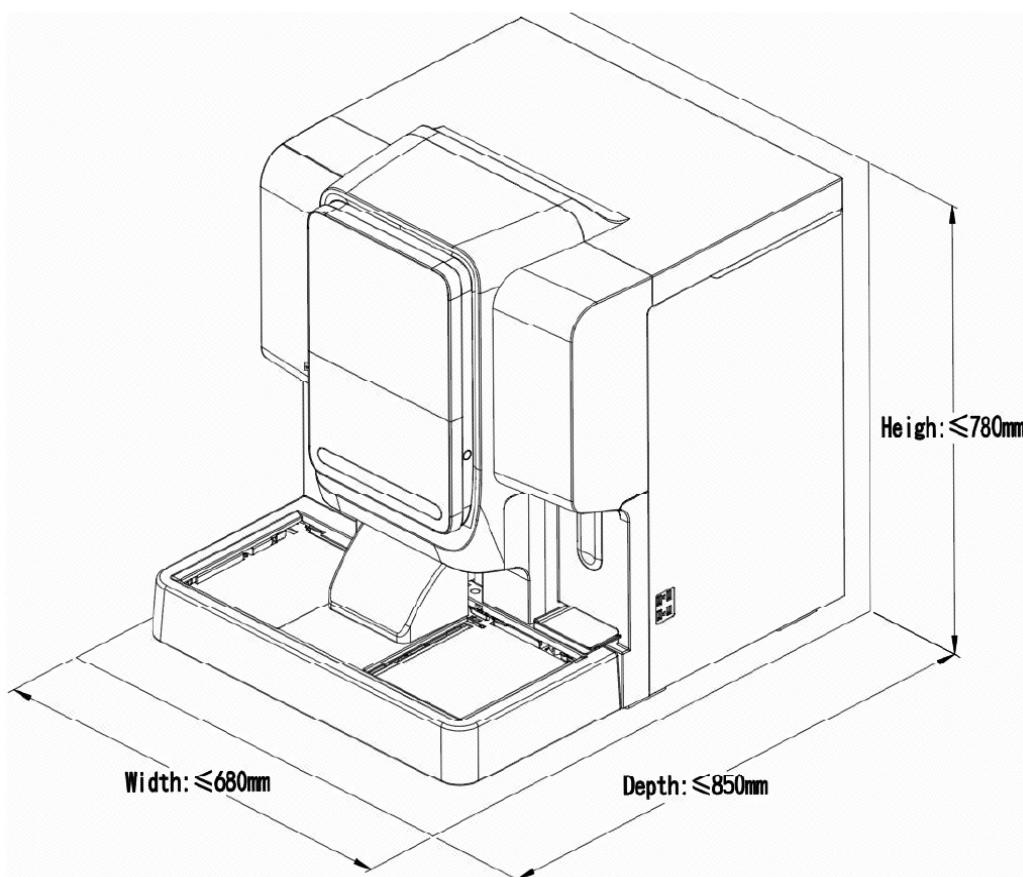
- Be sure to use and store the analyzer in the specified environment.

B.13 Environment Requirements

	Working Environment	Transportation and Storage Environment
Ambient Temperature	15°C–32°C	-10°C– 40°C
Relative Humidity	30%–85%	10%–90%
Atmospheric pressure	70kPa–106kPa ^{Note 1}	50kPa–106kPa

Note: When converted to altitude, the acceptable range is -400m–3,000m

B.14 Dimensions and Weight



	Main Unit (Analyzer)	Pneumatic Unit
Width (autoloader included)	≤ 680 mm	≤ 310 mm
Depth (autoloader included)	≤ 850 mm	≤ 480 mm
Height (autoloader included)	≤ 780 mm	≤430 mm
Weight (autoloader included)	≤ 150 kg	≤ 20 kg

B.15 Contraindications

None

B.16 Barcode Specifications

The analyzer can read barcodes that are stuck to the test sample tubes. The sample ID read from the barcode will be stored and used as the only identification of the sample.

The barcodes used with the analyzer shall meet the specifications stated in this section.

1. Supported barcode types

All code types and check digit supported by the analyzer are listed as follows.

Supported Barcode Types

Code Type	Check Digit	Number of Digits
CODE128	Self-checking (check digit is always included)	No more than 20 digits (sample ID)

Code Type	Check Digit	Number of Digits
CODE93	Self-checking (check digit is always included)	No more than 20 digits (sample ID)
UPC/EAN/JAN	Self-checking (check digit is always included)	Fixed length: 8 or 13 digits
ITF	Not use check digit	No more than 20 digits (sample ID)
	Use check digit	No more than 19 digits (sample ID)+1 digit (check digit) = no more than 20 digits
CODE39	Not use check digit	No more than 20 digits (sample ID)
	Use check digit	No more than 19 digits (sample ID)+1 digit (check digit) = no more than 20 digits
CODABAR	Not use check digit	No more than 20 digits (sample ID)
	Use check digit	No more than 19 digits (sample ID)+1 digit (check digit) = no more than 20 digits

NOTE

- The manufacturer suggests you use check digit to reduce the possibility of misreading.

2. Barcode label dimensions

Barcode height: $A \geq 10\text{mm}$

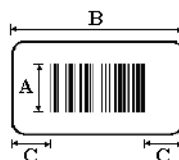
Label width: $B \leq 50\text{mm}$

Clear area width: $C \geq 5\text{mm}$

Wide-to-narrow ratio: between 2.5:1 and 3.0:1

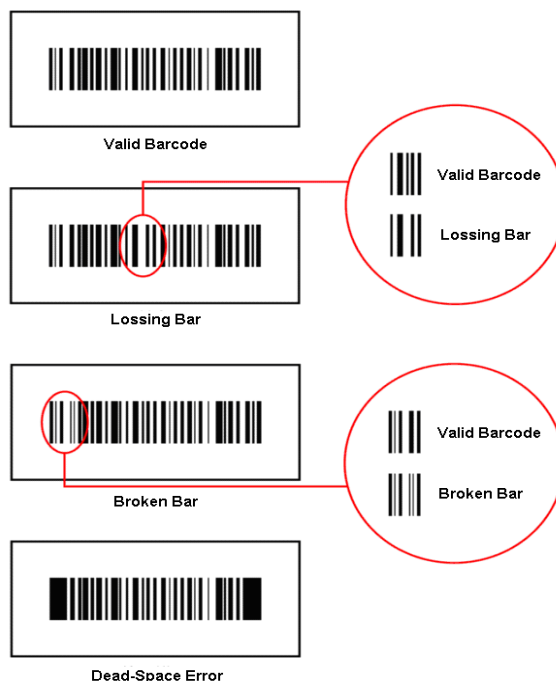
Width of the narrowest bar: above 0.152mm

Code quality: According to ANSI MH10.8M standard, the code quality shall be Level C or above.



3. Samples of valid and invalid barcode labels:

Use of invalid barcode labels shown in the following figures will increase the possibility of misreading. To ensure good readability, use valid labels.





4. Where to place the barcode label

To ensure good readability of the barcode, you must place the label right on the region as shown in Figure B-1, and place the label correctly as shown in Figure B-2

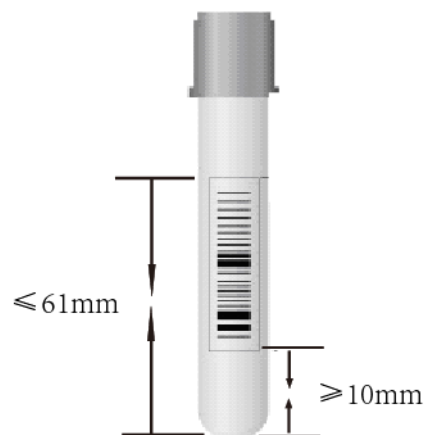


Figure B-1 Where to place the barcode label

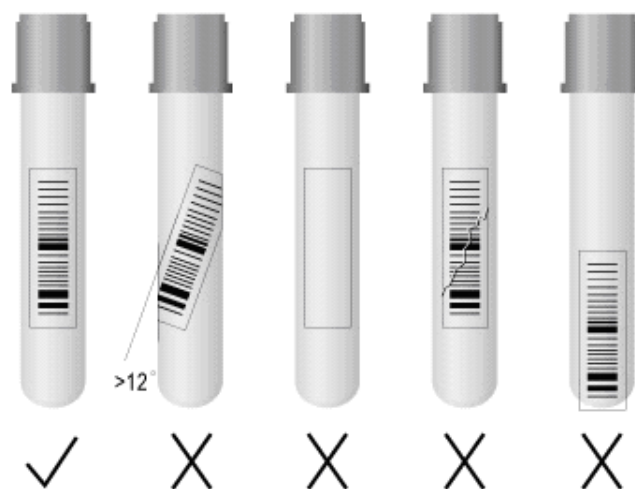


Figure B-2 How to place the barcode label

5. Selection of barcode label paper

It is recommended to use the matte label paper instead of the glossy label paper to print labels.

B.17 Safety Classification

Level of transient overvoltage: Category II.

Rated pollution degree: 2.

B.18 Software Version

Main unit software version: 01

C Accessories

NOTE

- **Actual accessories are subject to your product configurations.**
- **After unpacking the package, check intactness according to the packing list. If any component is lost, contact our Customer Service Department or your local distributor.**
- **For information on the configured/optional accessories, consult your sales representative.**

Table C-1 Packing list

	Name
1	Main unit
2	Waste tube assm (6800 without sensor)
3	Handle for Transportation
4	HP LaserJet Printer
5	Network cable (analyzer to PC)
6	BC-6800Plus Operator's Manual (EN)-IVDR
7	M-68LB CAP Component for Lyse
8	FQC Inspection Record
9	Safety Inspection Record
10	Cable, power (EU)
11	Laboratory Data Managt. SW CD
12	Laboratory Data Management Software (USB flash drive)
13	Handheld Barcode Scanner
14	BC-6800Plus Instruction for Use (EN)
15	Mindray DMS Operator's Manual (EN)-IVDR
16	M-68DR CAP Component for RET Diluent
17	Control Wire of Pneumatic System
18	20L reagent soft bottle (Japan)
19	Pneumatic System(220V)
20	Transparent PU Tube 4*6
21	The M3X8 combination Bolt
22	20L reagent soft bottle cap (Japan)
23	Cross recessed pan head screw M4X8
24	M-68LN CAP Component for Lyse
25	Waste Cap assm
26	M-68LH CAP Component for Lyse
27	FUSE Time-lag 250V 3.15AD5X20
28	M-68D CAP Component for Diluent

	Name
29	lever
30	Waste Cap Assembly (WC_W)
31	Diluent Assembly holder
32	M-68LD CAP Component for Lyse
33	sample rack Assembly (CAL 8000)
34	autoloader
35	autoloader beam
36	shield
37	salver
38	L bracket

D

Transmission

The LIS/HIS function of the analyzer enables the communication between the analyzer and the PC in laboratory through Ethernet, including sending analysis results to and receiving worklist from PC.

There is one type of communication protocol involved in the LIS/HIS communication process of the analyzer: HL7 protocol. For details about the connection control, and the introduction, message definition and examples, please contact Mindray Customer Service Department or your local distributor.

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E Radio Regulatory Compliance

RF Parameters

Radio Devices	RF ID
Operating Frequency	13.56MHz
Modulation mode	ASK
RF output power/Transmitter H-field	≤60 dBuA/m@10m



The radio device used in this product as well as our product is in compliance with the essential requirements and other relevant provisions of Directive 2014/53/EU.

Declaration of Conformity –V2.0

Declaration of Conformity

Manufacturer: Shenzhen Mindray Bio-Medical Electronics Co., Ltd.
Mindray Building, Keji 12th Road South, High-Tech Industrial
Park, Nanshan, 518057, Shenzhen, P. R. China

EC-Representative: Shanghai International Holding Corp. GmbH (Europe)
Eiffestraße 80 20537 Hamburg, Germany

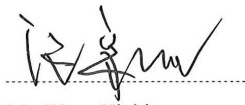
Product: Auto Hematology Analyzer

Model: BC-6600Plus, BC-6700Plus, BC-6800Plus

We herewith declare under our sole responsibility that the above mentioned products meet the provisions of the Council Directive 2014/53/EU concerning radio equipment. All supporting documentation is retained under the premises of the manufacturer.

Standards Applied:

☒ EN IEC 61326-1: 2021	☒ EN 50364: 2018
☒ ETSI EN 301 489-1 V2.2.3	☒ ETSI EN 301 489-3 V2.3.2
☒ ETSI EN 300 330 V2.1.1	

Start of CE-Marking: 2017-09-28**Place, Date of Issue:** Shenzhen, 2025-12-11**Signature:****Name of Authorized Signatory:** Mr. Wang Xinbing**Position Held in Company:** Deputy Director, Technical Regulation Department

F

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G Glossary

A

ACK

Acknowledge is a part of communication protocol, which replies to the message transmitted by the analyzer during the analysis.

Auto Startup/Shutdown

When the function is enabled, the analyzer will be shutdown/start up at the specified time.

Auto-Loading

For sample analysis in auto-loading mode, place the sample tubes in the sample rack, and then place the sample racks in the autoloader. The analyzer automatically transports the sample into the aspiration position, executes aspiration and starts analysis. The analyzer can analyze multiple samples at one time in auto-loading mode.

C

CV

Coefficient of Variation

E

ESR

Erythrocyte sedimentation rate

L

L-J QC

L-J QC is named after S. Levey and E.R. Jennings, who in 1950 introduced statistical control into the clinical laboratories

LIS

Laboratory Information System

M

Message Alarm

The message area displays sample errors other than the analyzer error.

P

PLT Histogram

Platelet Histogram

PLT-H Histogram

Platelet Hybrid Histogram

PLT-O Scattergram

Optical Platelet Scattergram

R

RBC Histogram

Red Blood Cell Histogram

RBC Scattergram

Red Blood Cell Scattergram

RBC VHF Scattergram

Red blood cell volume hemoglobin concentration fluorescence signal scattergram

RET-EXT Scattergram

Reticulocyte - Extension Scattergram

RUO

Research Use Only

S

Sample Rack

Used to hold the sample tubes

STAT

Emergent situation that a sample needs to be analyzed first when in autoloading mode.

W

WNB Scattergram

Basophils and Nucleated Red Blood Cells Scattergram

USB

Universal Serial Bus

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H Maintenance Logs

NOTE

- Users are advised to make a maintenance checklist suitable for their work environment.
- For information on the maintenance procedure, see *12 Servicing Your Analyzer*.

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Daily Maintenance Items

Date	Probe Cleanser maintenance	Cover Cleaning	Cleaning of Probe Wipe, Floating Blood Barrier, and Blood Bracket	Date	Probe Cleanser maintenance	Cover Cleaning	Cleaning of Probe Wipe, Floating Blood Barrier, and Blood Bracket
1				17			Cleaning of Probe Wipe, Floating Blood Barrier, and Blood Bracket
2				18			
3				19			
4				20			
5				21			
6				22			
7				23			
8				24			
9				25			
10				26			
11				27			
12				28			
13				29			
14				30			
15				31			
16							

